

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Marta Coll and colleagues’ 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri’s 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers’ decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____

Which choice most logically completes the text?

Answer

- A. Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri’s census.
- B. some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri but treated as indicative of distinct species by Coll and colleagues.
- C. Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical species of microorganism than Coll and colleagues were.
- D. the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and colleagues underestimating the number of microorganism species.

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically completes the text’s discussion of the different counts of species in the Mediterranean Sea. The text states that Coll and colleagues reported almost double the number of species that Bianchi and Morri reported in their study ten years earlier. According to the text, this difference can only be partly attributed to new invertebrate species being described in the years between the two studies, which means there must be an additional factor that made Coll and colleagues’ count so much higher than Bianchi and Morri’s count. The text goes on to explain that factor: researchers have a relatively poor understanding of microorganisms’ morphological variability, or the differences in microorganisms’ structure and form. This poor understanding makes it hard to classify microorganisms by species and means that researchers’ decisions about classifying microorganisms can have a large effect on the overall species counts that researchers report. Additionally, the text says that the two censuses reported similar numbers of vertebrate, plant, and algal species, which means that the difference in overall species did not come from differences in those categories. Given all this information, it most logically follows that Coll and colleagues may have treated some of the differences among microorganisms as indicative of the microorganisms being different species, whereas Bianchi and Morri treated those differences as variations within species, resulting in Coll and colleagues reporting many more species than Bianchi and Morri did.

Choice A is incorrect because the text explicitly addresses this issue by stating that the description of new invertebrate species in the years between the two studies can explain only part of the difference in the number of species reported by the studies. The focus of the text is on explaining the difference between Coll and colleagues’ count and Bianchi and Morri’s count that cannot be accounted for by the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri’s study. Choice C is incorrect because nothing in the text suggests that Bianchi and Morri may have been less sensitive to how much the form and structure of microorganisms vary within the same species than Coll and colleagues were. If Bianchi and Morri had been less sensitive to within-species variation than Coll and colleagues were, Bianchi and Morri would likely have reported more species than Coll and colleagues did, since less sensitivity to within-species variation would

lead researchers to classify as different species microorganisms that more sensitive researchers would classify as variations within the same species. The text indicates, however, that Bianchi and Morri reported far fewer species than Coll and colleagues did; since the text also excludes other explanations for this difference, it suggests that in fact Bianchi and Morri were more sensitive to within-species variation than Coll and colleagues were, leading Bianchi and Morri to report fewer overall species. Choice D is incorrect because the text is focused on explaining why Coll and colleagues reported many more species than Bianchi and Morri did, and an underestimate of the number of microorganism species by Coll and colleagues would not explain that difference—it would suggest, in fact, that the difference in the number of species should have been even larger.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

One recognized social norm of gift giving is that the time spent obtaining a gift will be viewed as a reflection of the gift’s thoughtfulness. Marketing experts Farnoush Reshadi, Julian Givi, and Gopal Das addressed this view in their studies of norms specifically surrounding the giving of gift cards, noting that while recipients tend to view digital gift cards (which can be purchased online from anywhere and often can be redeemed online as well) as superior to physical gift cards (which sometimes must be purchased in person and may only be redeemable in person) in terms of usage, 94.8 percent of participants surveyed indicated that it is more socially acceptable to give a physical gift card to a recipient. This finding suggests that _____

Which choice most logically completes the text?

Answer

- A. gift givers likely overestimate the amount of effort required to use digital gift cards and thus mistakenly assume gift recipients will view them as less desirable than physical gift cards.
- B. physical gift cards are likely preferred by gift recipients because the tangible nature of those cards offers a greater psychological sense of ownership than digital gift cards do.
- C. physical gift cards are likely less desirable to gift recipients than digital gift cards are because of the perception that physical gift cards require unnecessary effort to obtain.
- D. gift givers likely perceive digital gift cards as requiring relatively low effort to obtain and thus wrongly assume gift recipients will appreciate them less than they do physical gift cards.

Correct Answer: D

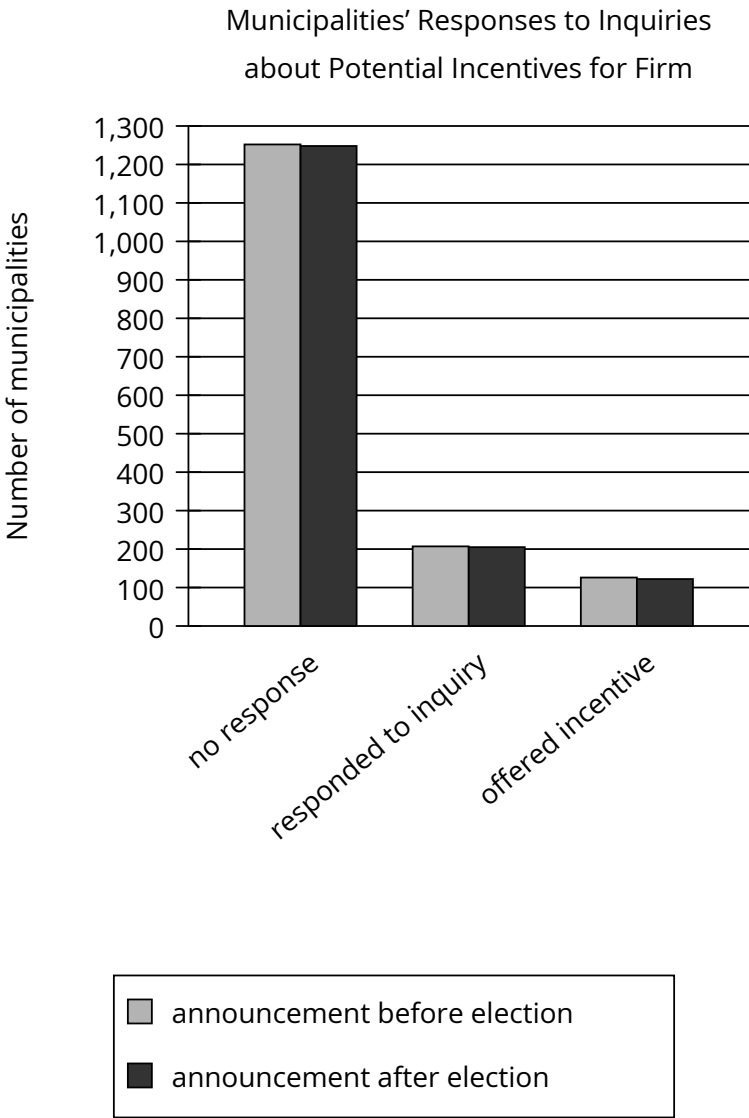
Rationale

Choice D is the best answer because it most logically completes the text’s discussion of perceptions of digital versus physical gift cards. The text begins by explaining that the perception of "the time spent obtaining a gift...as a reflection of the gift’s thoughtfulness" is a social norm of gift giving. The text then explains that although those who receive digital gift cards view them as easier to use than physical gift cards, a marketing study nonetheless showed that 94.8% of participants found physical gift cards more "socially acceptable" to give. The text specifically contrasts the ease with which digital gift cards "can be purchased online from anywhere" with the fact that physical gift cards "sometimes must be purchased in person"—suggesting the greater difficulty of obtaining physical cards. Given the text’s initial premise that gift-giving norms equate the thoughtfulness of a gift with the effort involved in acquiring that gift, it is reasonable to infer that people perceive digital gift cards as requiring less effort to obtain and thus assume recipients will appreciate them less, even though recipients actually prefer gift cards in the more usable digital format.

Choice A is incorrect. Although the text does discuss recipients’ preference of digital versus physical gift cards and the relative ease with which the two formats can be used, it doesn’t consider the misconceptions that gift givers may have of these factors. Moreover, the text establishes that recipients regard digital gift cards as easier to use and therefore preferable to physical gift cards. Choice B is incorrect because the text doesn’t consider whether recipients of gift cards feel a sense of ownership toward them, nor does the text touch on the greater tangibility of physical versus digital gift cards. Instead, the text contrasts the two formats of gift cards in terms of their respective usability and the difficulty involved in acquiring them and discusses how those factors influence people’s perceptions of the two formats. Choice C is incorrect because it contradicts the text, which explains that recipients regard digital gift cards as superior to physical ones because they are easier to use than physical cards, not because physical gift cards require greater effort to obtain than digital gift cards do. Moreover, the text doesn’t characterize the effort required to obtain physical gift cards as "unnecessary."

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



In the United States, firms often seek incentives from municipal governments to expand to those municipalities. A team of political scientists hypothesized that municipalities are much more likely to respond to firms and offer incentives if expansions can be announced in time to benefit local elected officials than if they can't. The team contacted officials in thousands of municipalities, inquiring about incentives for a firm looking to expand and indicating that the firm would announce its expansion on a date either just before or just after the next election.

Which choice best describes data from the graph that weaken the team's hypothesis?

Answer

- A. A large majority of the municipalities that received an inquiry mentioning plans for an announcement before the next election didn't respond to the inquiry.
- B. The proportion of municipalities that responded to the inquiry or offered incentives didn't substantially differ across the announcement timing conditions.

C. Only around half the municipalities that responded to inquiries mentioning plans for an announcement before the next election offered incentives.

D. Of the municipalities that received an inquiry mentioning plans for an announcement date after the next election, more than 1,200 didn't respond and only around 100 offered incentives.

Correct Answer: B

Rationale

Choice B is the best answer. The text explains that firms in the United States often look to municipal governments (local governing bodies) for incentives to expand the firms into those areas. The text then indicates that a team of political scientists hypothesized that those governing bodies are much more likely to respond and offer incentives when the expansion (if there's going to be one) can be announced before an election, since announcing a firm coming to their area can benefit local elected officials. However, the graph—which presents the results of the political scientists' inquiries to municipalities about incentives for a firm in two timing scenarios—shows that the proportion of municipalities that responded to the inquiry or offered incentives didn't substantially differ based on timing. According to the graph, about 200 municipalities responded to the inquiry and about 100 offered an incentive when the announcement could be made before an election (represented by the gray bars) and about 200 municipalities responded to the inquiry and about 100 offered an incentive when the announcement would be made after an election (represented by the black bars). These data weaken the team's hypothesis by demonstrating that municipalities are *not* much more likely to respond and offer incentives when the timing means they could announce an expansion before an election.

Choice A is incorrect. Although the graph does show that a large number of municipalities didn't respond to inquiries in the scenario in which an expansion announcement could be made before an election (represented by the gray bars), the text indicates that the team hypothesized that there would be a substantial difference in municipalities' actions based on timing, and information about only one timing scenario doesn't reveal anything about whether such a difference occurred. Choice C is incorrect. The graph does show that in the scenario in which an expansion announcement could be made before an election (represented by the gray bars), about 200 municipalities responded to inquiries and about 100 offered an incentive, but the text indicates that the team hypothesized that there would be a substantial difference in municipalities' actions based on timing, and information about only one timing scenario doesn't reveal anything about whether such a difference occurred. Choice D is incorrect. Although this choice accurately describes data from the graph for the scenario in which an expansion announcement would be made after an election (represented by the black bars), the text indicates that the team hypothesized that there would be a substantial difference in municipalities' actions based on timing, and information about only one timing scenario doesn't reveal anything about whether such a difference occurred.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The most recent iteration of the immersive theater experience *Sleep No More*, which premiered in New York City in 2011, transforms its performance space—a five-story warehouse—into a 1930s-era hotel. Audience members, who wander through the labyrinthine venue at their own pace and follow the actors as they play out simultaneous, interweaving narrative loops, confront the impossibility of experiencing the production in its entirety. The play’s refusal of narrative coherence thus hinges on the sense of spatial fragmentation that the venue’s immense and intricate layout generates.

What does the text most strongly suggest about *Sleep No More*’s use of its performance space?

Answer

- A. The choice of a New York City venue likely enabled the play’s creators to experiment with the use of theatrical space in a way that venues from earlier productions could not.
- B. Audience members likely find the experience of the play disappointing because they generally cannot make their way through the entire venue.
- C. The production’s dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.
- D. Audience members who navigate the space according to a recommended itinerary will likely have a better grasp of the play’s narrative than audience members who depart from that itinerary.

Correct Answer: C

Rationale

Choice C is the best answer. The text says that the production’s use of its large, winding space has a very specific effect on the audience. Given that the space itself is so important to creating this effect, it would be difficult to reproduce the production in a different space.

Choice A is incorrect. The fact that the venue is in New York City isn’t connected to the experimental nature of the performance. It’s the size of the venue, not its location in New York, that affects the theatrical experience. Choice B is incorrect. The text never suggests that audience members are disappointed because they can’t see the entire production. In fact, it suggests that that’s an important part of the experience. Choice D is incorrect. The text doesn’t mention a recommended itinerary for audience members.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

The interiors of many temples in the ancient Middle East needed to satisfy a precise set of acoustic demands: the sounds of chants and hymns should travel with clarity, while profound silences should be fully felt and appreciated. In a research paper, a student claims that the users of one such temple were aware of how the materials that were used within the structure could affect sound quality and that they deliberately applied this knowledge to influence how sound was experienced in the space.

Which quotation from a work by a historian would most directly support the student’s claim?

Answer

- A. “Many researchers believe that the central chamber of the temple had a high ceiling, a feature that has since become essential to the acoustic design of modern concert halls.”
- B. “The innermost room of the temple was likely among the quietest spaces in the interior of the temple.”
- C. “The acoustic environment of the temple was best suited for music that eschewed ornamentation in favor of simple melodies, harmonies, and rhythms.”
- D. “During special occasions, curtains were placed inside the temple to minimize reverberation and confine the sound to designated locations.”

Correct Answer: D

Rationale

Choice D is the best answer because it provides a quotation that would most directly support the student’s claim that the temple’s users were aware of how materials used in the temple affected sound and deliberately applied that knowledge to shape the sonic experience of people in the space. The quotation indicates that curtains would, at times, be installed specifically “to minimize reverberation and confine the sound,” thereby illustrating that the users of the temple had knowledge of the acoustic properties of materials within the temple (how curtains dampen sound) and employed this knowledge to affect the experience of sound within the space.

Choice A is incorrect. Although the quotation describes a feature of the temple’s structure (the high ceiling), it makes no mention of the materials used, the materials’ acoustic properties, or temple users’ awareness of how materials impact sound quality. Choice B is incorrect because it doesn’t address any materials that were used in the temple, let alone their acoustic properties or temple users’ awareness of how materials impact sound quality. Choice C is incorrect because it doesn’t address any materials that were used in the temple, let alone their acoustic properties or temple users’ awareness of how materials impact sound quality.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Although many transposons, DNA sequences that move within an organism’s genome through shuffling or duplication, have become corrupted and inactive over time, those from the long interspersed nuclear elements (LINE) family appear to remain active in the genomes of some species. In humans, they are functionally important within the hippocampus, a brain structure that supports complex cognitive processes. When the results of molecular analysis of two species of octopus—an animal known for its intelligence—were announced in 2022, the confirmation of a LINE transposon in *Octopus vulgaris* and *Octopus bimaculoides* genomes prompted researchers to hypothesize that that transposon family is tied to a species’ capacity for advanced cognition.

Which finding, if true, would most directly support the researchers’ hypothesis?

Answer

- A. The LINE transposon in *O. vulgaris* and *O. bimaculoides* genomes is active in an octopus brain structure that functions similarly to the human hippocampus.
- B. The human genome contains multiple transposons from the LINE family that are all primarily active in the hippocampus.
- C. A consistent number of copies of LINE transposons is present across the genomes of most octopus species, with few known corruptions.
- D. *O. vulgaris* and *O. bimaculoides* have smaller brains than humans do relative to body size, but their genomes contain sequences from a wider variety of transposon families.

Correct Answer: A

Rationale

Choice A is the best answer. The text says that LINE transposons are important in the human hippocampus, which supports complex cognition. If the LINE transposon found in octopuses is active in a similar part of their brain, that would suggest that LINE transposons support complex cognition in octopuses too, which in turn supports the hypothesis that LINE transposons are linked to advanced cognition in general.

Choice B is incorrect. This choice doesn’t support the hypothesis. It doesn’t include anything about how LINE transposons function in species other than humans. Choice C is incorrect. This choice doesn’t support the hypothesis. It doesn’t include anything about how the LINE transposon in octopuses might support advanced cognition. Choice D is incorrect. This choice doesn’t support the hypothesis. It doesn’t include anything about how the LINE transposon in octopuses might support advanced cognition.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

To understand how temperature change affects microorganism-mediated cycling of soil nutrients in alpine ecosystems, Eva Kaštovská et al. collected plant-soil cores in the Tatra Mountains at elevations around 2,100 meters and transplanted them to elevations of 1,700–1,800 meters, where the mean air temperature was warmer by 2°C. Microorganism-mediated nutrient cycling was accelerated in the transplanted cores; crucially, microorganism community composition was unchanged, allowing Kaštovská et al. to attribute the acceleration to temperature-induced increases in microorganism activity.

It can most reasonably be inferred from the text that the finding about the microorganism community composition was important for which reason?

Answer

- A. It provided preliminary evidence that microorganism-mediated nutrient cycling was accelerated in the transplanted cores.
- B. It suggested that temperature-induced changes in microorganism activity may be occurring at increasingly high elevations.
- C. It ruled out a potential alternative explanation for the acceleration in microorganism-mediated nutrient cycling.
- D. It clarified that microorganism activity levels in the plant-soil cores varied depending on which microorganisms comprised the community.

Correct Answer: C

Rationale

Choice C is the best answer because it accurately describes why the finding about the microorganism community composition was important. The text describes an experiment by Eva Kaštovská and her team in which they collected plant-soil cores at one elevation and transplanted them to sites at a lower elevation, where the mean air temperature was warmer. Kaštovská and her team observed that microorganism-mediated nutrient cycling was accelerated in the transplanted cores and that "crucially, microorganism community composition was unchanged," which allowed the team to attribute the acceleration to changes in microorganism activity brought about by the difference in temperature. This strongly implies that the team wouldn't have been able to make that attribution otherwise, meaning that a change in microorganism composition represented another possible explanation for the acceleration that had to be ruled out.

Choice A is incorrect. Although the text says microorganism-mediated cycling of soil nutrients increased in the transplanted cores, this is unrelated to what's important about the finding that the microorganism composition didn't change—that it allowed the team to attribute the change in activity solely to the change in temperature. Choice B is incorrect. Although the text compares activity in one core at two different elevations, the text doesn't address changes in activity at various elevations over time. Choice D is incorrect. Although different microorganisms likely exhibit different levels of activity, the text indicates that there was no change in microorganism composition, and there is nothing in the text about different microorganisms having different activity levels.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Jean-Bernard Caron and colleagues recently discovered a cache of jellyfish fossils in the Burgess Shale, a site in the Canadian Rockies that is rich in fossils from the Cambrian period (over 500 million years ago). Caron and colleagues claim that these are the oldest jellyfish fossils ever discovered. In the past twenty years, two sites in China and the United States have yielded fossils of a similar age that some experts believe are most likely jellyfish due to their shapes and the appearance of projecting tentacles. But Caron and colleagues argue that the apparent tentacles are in fact the comb rows of ctenophores, gelatinous animals that are only distantly related to jellyfish.

Which statement, if true, would most directly weaken the claim by Caron and colleagues about the fossils found in China and the United States?

Answer

- A. Sites in the Canadian Rockies from later periods than the Cambrian period have yielded fossils that have been conclusively identified as ctenophore fossils.
- B. The fossils found in China and the United States are so poorly preserved that though they cannot be conclusively identified as jellyfish, they cannot be conclusively identified as ctenophores either.
- C. While ctenophore fossils have been discovered in China and the United States, they have never been discovered in the Burgess Shale.
- D. The fossils discovered by Caron and colleagues in the Burgess Shale were better preserved than the fossils discovered by other researchers in China and the United States.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a statement that, if true, would most directly weaken Caron and colleagues’ claim that the apparent tentacles in the Chinese and American fossils are actually ctenophore comb rows. If the fossils are so poorly preserved that they cannot be conclusively identified as either organism, neither the claim that they are jellyfish nor, as Caron claims, that they are ctenophores would be supported.

Choice A is incorrect. Caron’s claim is that fossils from the US and China are ctenophores, not jellyfish. These fossils are said to be "of a similar age" to the Cambrian fossils found in the Canadian Rockies. And nothing in the text or this choice suggests that the presence or absence of ctenophores after the Cambrian would have any bearing on whether the Cambrian fossils from the US and China are ctenophores. Choice C is incorrect. Caron’s claim is that fossils from the US and China are ctenophores, not jellyfish. Nothing in the text suggests that the presence or absence of ctenophores in the Burgess Shale (in Canada) would affect whether the fossils found in the US and China are ctenophores. Choice D is incorrect. Caron’s claim is that fossils from the US and China are ctenophores, not jellyfish. Although fossil quality is a plausible issue for the research described in the text, nothing in the text or this choice suggests that the fossils from US and China would have been too poorly preserved for proper identification.

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Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Many literary theorists distinguish between *fabula*, a narrative’s content, and *syuzhet*, a narrative’s arrangement and presentation of events. In the film *The Godfather Part II*, the *fabula* is the story of the Corleone family, and the *syuzhet* is the presentation of the story as it alternates between two timelines in 1901 and 1958. But literary theorist Mikhail Bakhtin maintained that *fabula* and *syuzhet* are insufficient to completely describe a narrative—he held that systematic categorizations of artistic phenomena discount the subtle way in which meaning is created by interactions between the artist, the work, and the audience.

Which choice best states the main idea of the text?

Answer

- A. Literary theorist Mikhail Bakhtin argued that there are important characteristics of narratives that are not fully encompassed by two concepts that other theorists have used to analyze narratives.
- B. Literary theorist Mikhail Bakhtin claimed that meaning is not inherent in a narrative but is created when an audience encounters a narrative so that narratives are interpreted differently by different people.
- C. The storytelling methods used in *The Godfather Part II* may seem unusually complicated, but they can be easily understood when two concepts from literary theory are utilized.
- D. Narratives that are told out of chronological order are more difficult for audiences to understand than are narratives presented chronologically.

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main idea of the text. The text begins by explaining that many literary theorists rely on the concepts of *fabula* (a narrative’s content) and *syuzhet* (a narrative’s arrangement and presentation of events) and illustrates these concepts by explaining how they can be applied to the film *The Godfather Part II*. The text then discusses how Mikhail Bakhtin, a literary theorist, argued that *fabula* and *syuzhet* can’t fully describe a narrative, since systematic categorizations such as these fail to account for all the ways in which interactions between the artist, the work, and the audience produce meaning. Thus, the main idea is that Bakhtin argued that there are important characteristics of narratives that are not fully encompassed by two concepts that other theorists have used to analyze narratives.

Choice B is incorrect because according to the text, Mikhail Bakhtin believed that meaning was created through the interactions of the artist, narrative, and audience, not simply through the interaction between the audience and narrative; moreover, the text doesn’t address whether Bakhtin focused on the ways in which different people interpret narratives differently. Choice C is incorrect. Although the text implies that the storytelling methods used in *The Godfather Part II* are complicated, it discusses the film only to illustrate how the concepts of *fabula* and *syuzhet* may be applied to a narrative. The film’s storytelling methods aren’t the primary focus of the text. Choice D is incorrect. The text discusses *The Godfather Part II*, whose narrative doesn’t adhere to a single chronological order, only to illustrate the concepts of *fabula* (a narrative’s content) and *syuzhet* (a narrative’s arrangement and presentation of events). The primary focus of this text isn’t the structure of this film or of other narratives that are told out of chronological order; moreover, the text doesn’t consider whether such structures make it harder for audiences to understand narratives.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

The ancient Greek concept of “mimesis,” a term used in the works of Plato, Aristotle, and other Greek philosophers in discussions of representational art—visual, performance, or literary art that aims to depict the real world—is a foundational concept of the Western philosophy of aesthetics. Mimesis is typically translated as “imitation” in modern editions of ancient Greek texts, but scholar Stephen Halliwell warns that this is overly reductive: “imitation” implies that art merely copies—and is thus by definition entirely derivative of—a reality that exists outside and prior to the work of art, and translating “mimesis” thusly obscures the multifaceted ways in which the ancient Greeks understood the relationship between art and reality.

Which statement, if true, would most directly support the claim by Halliwell presented in the text?

Answer

- A. One of the earliest appearances of mimesis’s root word, *mimos*, can be found in an ancient Greek tragedy in reference to dramatic impersonation, and the *mim-* root came to be generally associated with the musical and poetic arts by the fifth century BCE.
- B. Both Plato’s and Aristotle’s theorizations of mimesis examine the psychological effects that works of art induce in the viewer or listener.
- C. Although several of Plato’s earliest philosophical works discuss aesthetic ideas, the term “mimesis” doesn’t appear in Plato’s discussions of art until *Cratylus*, a relatively late work.
- D. Although Plato’s writings typically characterize representational art as an inferior reflection of the physical world, Aristotle suggests that mimesis can refer to art’s capacity to envision hypothetical conditions that could, but don’t yet, exist.

Correct Answer: D

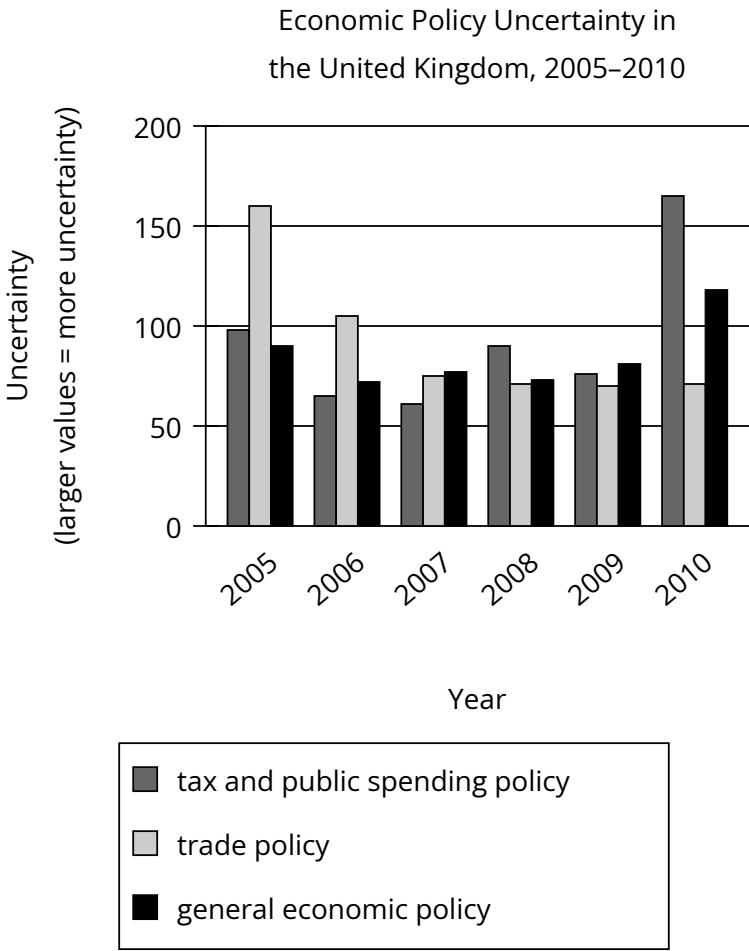
Rationale

Choice D is the best answer because it presents a statement that would support Halliwell’s claim that “imitation” is an overly simplified translation of the word “mimesis” in the context of ancient Greek philosophical discussions of representational art and that, because it suggests a view of art as merely copying things that exist in reality, the translation obscures the fact that Plato, Aristotle, and other ancient Greeks understood the relationship between art and reality in varied ways. If Plato’s works tend to treat representational art as an inferior or inadequate reflection of the physical world (consistent with the idea of mere imitation), while Aristotle’s works suggest that the term “mimesis” can refer to art’s ability to show conditions that don’t currently exist but could exist (going beyond imitation into novel creation), that evidence would support the claim that the term “mimesis” means more than just “imitation” and that ancient Greek philosophers held different views of the relationship between art and reality.

Choice A is incorrect because information about the root word *mimos* first being used in drama and the root *mim-* coming to be associated with music and poetry wouldn’t indicate anything about the meaning of the specific term “mimesis” as it was used by ancient Greek philosophers in discussions of representational art; thus, the information would have no bearing on Halliwell’s claim that a common translation of the specific term is an oversimplification. Choice B is incorrect because the issue of art’s psychological effects on audiences gets at how people respond to works of art instead of how art itself is related to reality, so the idea that Plato and Aristotle both addressed such effects in their considerations of mimesis wouldn’t have any bearing on Halliwell’s claim that a common translation of that term oversimplifies the relationship between art and reality. Further, this idea would touch on one general similarity in approaches to mimesis instead of illustrating the multifaceted, or varied, ways Halliwell claims ancient Greek philosophers understood art’s relation to reality. Choice C is incorrect because the idea that Plato discussed aesthetics in his early works but didn’t use the term “mimesis” until later doesn’t give any indication of how Plato or any other ancient Greek philosophers understood the relation between art and reality, so it would have no bearing on Halliwell’s claim that a common translation of the term “mimesis” is oversimplified and fails to reflect the varied ways the philosophers understood that relationship.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



High levels of public uncertainty about which economic policies a country will adopt can make planning difficult for businesses, but measures of such uncertainty have not tended to be very detailed. Recently, however, economist Sandile Hlatshwayo analyzed trends in news reports to derive measures not only for general economic policy uncertainty but also for uncertainty related to specific areas of economic policy, like tax or trade policy. One revelation of her work is that a general measure may not fully reflect uncertainty about specific areas of policy, as in the case of the United Kingdom, where general economic policy uncertainty _____

Which choice most effectively uses data from the graph to illustrate the claim?

Answer

- A. aligned closely with uncertainty about tax and public spending policy in 2005 but differed from uncertainty about tax and public spending policy by a large amount in 2009.
- B. was substantially lower than uncertainty about tax and public spending policy each year from 2005 to 2010.
- C. reached its highest level between 2005 and 2010 in the same year that uncertainty about trade policy and tax and public spending policy reached their lowest levels.
- D. was substantially lower than uncertainty about trade policy in 2005 and substantially higher than uncertainty about trade policy in 2010.

Correct Answer: D

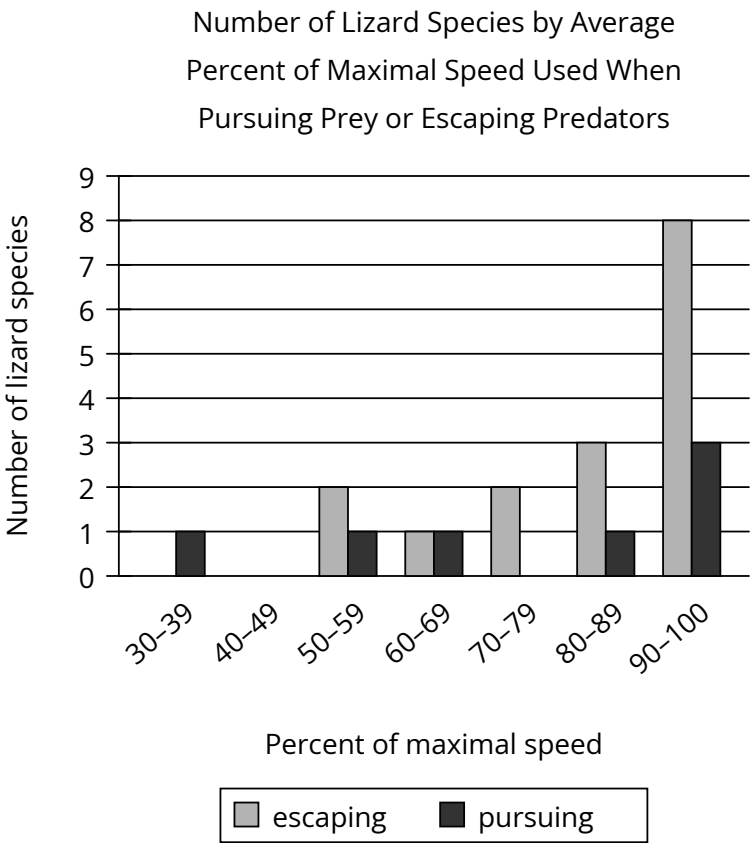
Rationale

Choice D is the best answer because it uses data from the graph to effectively illustrate the text's claim about general economic policy uncertainty in the United Kingdom. The graph presents values for economic policy uncertainty in tax and public spending policy, trade policy, and general economic policy in the UK from 2005 to 2010. The graph shows that in 2005, the value for general economic policy uncertainty (approximately 90) was substantially lower than the value for uncertainty about trade policy specifically (approximately 160). It also shows that in 2010, the value for general economic policy uncertainty (approximately 120) was substantially higher than the value for uncertainty about trade policy (approximately 70). The substantial differences between these values in 2005 and 2010 support the claim that a general measure may not fully reflect uncertainty about specific areas of policy.

Choice A is incorrect because the graph shows that the level of general economic policy uncertainty was similar to the level of uncertainty about tax and public spending policy in both 2005 (with values of approximately 90 and 100, respectively) and 2009 (with values of approximately 80 and 75, respectively). Choice B is incorrect because the graph shows that general economic policy uncertainty was higher than uncertainty about tax and public spending policy in 2006, 2007, and 2009, not that it was lower each year from 2005 to 2010. Choice C is incorrect because the graph shows that general economic policy uncertainty reached its highest level in 2010, which was when uncertainty about tax and public spending policy also reached its highest level, not its lowest level.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



It may seem that the optimal strategy for an animal pursuing prey or escaping predators is to move at maximal speed, but the energy expense of exploiting full speed capacity can disfavor such a strategy even in escape contexts, as evidenced by the fact that _____

Which choice most effectively uses data from the graph to complete the text?

- Answer**
- A. most lizard species use about the same percentage of their maximal speed when escaping predation as they do when pursuing prey.
 - B. multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.
 - C. more lizard species use, on average, 90%–100% of their maximal speed while escaping predation than use any other percentage of their maximal speed.
 - D. at least 4 lizard species use, on average, less than 100% of their maximal speed while pursuing prey.

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the graph that complete the text’s discussion of lizard species’ use of maximal speed when escaping predators. According to the text, moving at maximal speed (the highest speed possible) requires so much energy that it is not always an effective strategy for animals, even when they are escaping predators. The graph displays data on the average percent of maximal

speed used by lizard species while either escaping predators or pursuing prey. The graph categorizes the data for both pursuing and escaping by the number of species using 30%–39% of maximal speed, 40%–49% of maximal speed, 50%–59% of maximal speed, 60%–69% of maximal speed, 70%–79% of maximal speed, 80%–89% of maximal speed, and 90%–100% of maximal speed, respectively. In the graph, there is at least one species in each of the following percent categories for maximal speed while escaping predators: 50%–59%, 60%–69%, 70%–79%, and 80%–89%. Thus, the data in the graph show that multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.

Choice A is incorrect because the data in the graph isn't organized in such a way that a comparison of the percentage of maximal speed used when escaping predation with the percentage used when pursuing prey is possible at the level of individual species. Choice C is incorrect. It is true that in the graph, the percent category with the largest number of species using maximal speed while escaping predators is 90%–100% (8 species total). However, these data don't complete the text, which is concerned instead with how animals are discouraged from using maximal speed even when escaping predators because of the amount of energy required to use it. Choice D is incorrect because these data from the graph pertain to maximal speed while pursuing prey and therefore don't complete the text's discussion of lizard species' use of maximal speed when escaping predators.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Almost all works of fiction contain references to the progression of time, including the time of day when events in a story take place. In a 2020 study, Allen Kim, Charuta Pethe, and Steven Skiena claim that an observable pattern in such references reflects a shift in human behavior prompted by the spread of electric lighting in the late nineteenth century. The researchers drew this conclusion from an analysis of more than 50,000 novels spanning many centuries and cultures, using software to recognize and tally both specific time references—that is, clock phrases, such as 7 a.m. or 2:30 p.m.—and implied ones, such as mentions of meals typically associated with a particular time of day.

Which finding from the study, if true, would most directly support the researchers’ conclusion?

Answer

- A. Novels published after the year 1800 include the clock phrase 10 a.m. less often than novels published before the year 1800 do.
- B. Novels published after 1880 contain significantly more references to activities occurring after 10 p.m. than do novels from earlier periods.
- C. Among novels published in the nineteenth century, implied time references become steadily more common than clock phrases as publication dates approach 1900.
- D. The time references of noon (12 p.m.) and midnight (12 a.m.) are used with roughly the same frequency in the novels.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the researchers’ conclusion that an observable pattern in time references in novels reflects a shift in human behavior prompted by the spread of electric lighting in the late nineteenth century. If novels published after 1880 contain significantly more references to activities occurring after 10 p.m. than novels from earlier periods do, this would suggest a change in human behavior and daily routines enabled by the availability of electric lighting. Before electric lighting—which provided illumination more easily than other available forms of light—many activities ceased after nightfall, so references to late-night activities would be less common in earlier novels. An increase in such references after 1880 would align with the researchers’ conclusion, reflecting an increase in late-night activities made possible by electric lighting.

Choice A is incorrect because a decrease in references to 10 a.m. after the year 1800 would not support the researchers’ conclusion involving a shift in human behavior prompted by the spread of electric lighting toward the end of the 1800s. The time of 10 a.m. is in the morning and, in most places, characterized by daylight, so a change in references to that time would not be clearly linked to the impact of electric lighting. Choice C is incorrect because while an increase in implied time references relative to clock phrases in nineteenth-century novels could suggest a change in writing style or conventions, it does not directly support the conclusion involving a shift in human behavior prompted by the spread of electric lighting. The text indicates that the researchers’ conclusion is based on the content of the time references themselves, not the phrasing used. Choice D is incorrect. If references to noon and midnight are used with roughly the same frequency in all the novels analyzed by the researchers, this would reflect a lack of change in human behavior with regard to time and therefore would not support the researchers’ conclusion involving a shift in human behavior that occurred in response to the spread of electric lighting.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Although Eastern North Pacific (ENP) gray whales generally migrate between their wintering waters along the coast of Mexico and their foraging waters in the Arctic, a subset of this population—known as the Pacific Coast Feeding Group (PCFG)—forages along the coastlines of Northern California (USA) and British Columbia (Canada) instead. Interestingly, individuals in this subset reach smaller maximum sizes than other ENP whales do, despite having similar pre-maximum growth rates. Researchers hypothesize that this difference may be an adaptation to distinct resource opportunities in the PCFG foraging range.

Which finding, if true, would most directly support the researchers’ claim regarding the size of PCFG whales?

Answer

- A. The average body size of PCFG whales observed along the coasts of Northern California and British Columbia has remained relatively steady in recent decades, while the average body size of ENP whales in the main group has slightly decreased.
- B. When present along the coasts of Northern California and British Columbia, PCFG whales tend to forage in rocky kelp beds at shallow depths inaccessible to whales as large as those in the ENP main group.
- C. When foraging along the coasts of Northern California and British Columbia, PCFG whales are in closer proximity to major ports and urban populations than ENP whales in the main group are when foraging in Arctic waters.
- D. Certain crustacean prey species available along the coasts of Northern California and British Columbia where PCFG whales tend to forage are not available in the Arctic waters where ENP whales in the main group forage.

Correct Answer: B

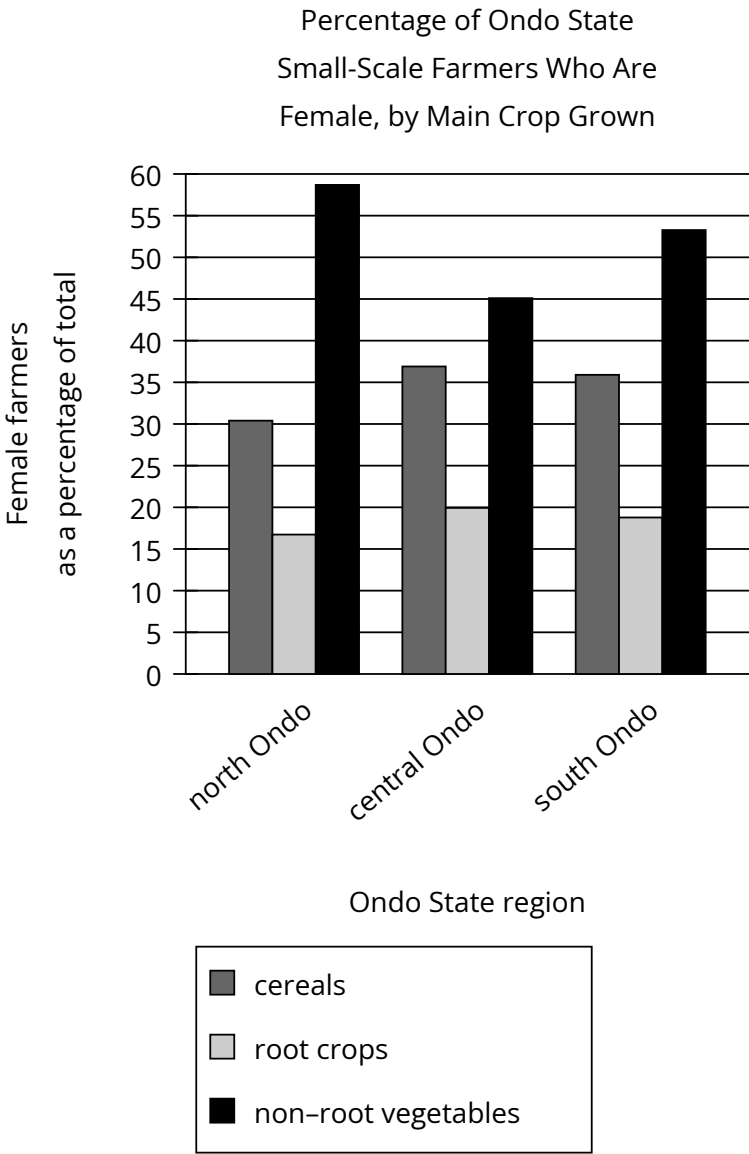
Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the researchers’ hypothesis that the smaller maximum size of PCFG whales is an adaptation to resource opportunities particular to their foraging range. The text explains that PCFG whales, a subgroup of ENP gray whales, forage along the coasts of Northern California and British Columbia rather than in the Arctic, where the main ENP group forages. The text goes on to state that PCFG whales reach smaller maximum sizes than other ENP gray whales despite having similar pre-maximum growth rates. If PCFG whales tend to forage in shallow, rocky kelp beds that whales as large as those in the main group cannot access, that would indicate that smaller body size confers a direct advantage in this foraging range—namely, access to a food source that larger whales cannot reach.

Choice A is incorrect because a finding describing recent trends in average body size for both groups of whales wouldn’t address whether the smaller maximum size of PCFG whales reflects an adaptation to their foraging environment. Recent stability or change in average size doesn’t speak to why the size difference between PCFG whales and the main ENP group exists. Choice C is incorrect because a finding about PCFG whales’ proximity to ports and urban populations wouldn’t address the resources available to PCFG whales or the relationship between body size and foraging. Proximity to human activity is irrelevant to a hypothesis about body size being an adaptation to resource opportunities. Choice D is incorrect. Although a finding about a certain prey species being available in the PCFG whales’ foraging waters but not in the main group of ENP whales’ foraging waters would establish that the PCFG foraging range offers distinct prey species, it wouldn’t connect the availability of those prey species to body size in a way that would explain the PCFG whales’ smaller size. Without indicating that the available prey in the PCFG range favors smaller whales, the finding falls short of supporting the hypothesis that smaller size is an adaptation to resource opportunities.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Geographer Adebayo Oluwole Eludoyin and his colleagues surveyed small-scale farmers in three locations in Ondo State, Nigeria—which has mountainous terrain in the north, an urbanized center, and coastal terrain in the south—to learn more about their practices, like the types of crops they mainly cultivated. In some regions, female farmers were found to be especially prominent in the cultivation of specific types of crops and even constituted the majority of farmers who cultivated those crops; for instance, _____

Which choice most effectively uses data from the graph to complete the example?

Answer

- A. most of the farmers who mainly cultivated cereals and most of the farmers who mainly cultivated non-root vegetables in south Ondo were women.
- B. more women in central Ondo mainly cultivated root crops than mainly cultivated cereals.
- C. most of the farmers who mainly cultivated non-root vegetables in north and south Ondo were women.

D. a relatively equal proportion of women across the three regions of Ondo mainly cultivated cereals.

Correct Answer: C

Rationale

Choice C is the best answer because it uses data from the graph to effectively complete the example of Eludoyin and his colleagues' findings concerning female farmers in some regions of Ondo State, Nigeria. The graph presents values for the percentage of Ondo State small-scale farmers who are female, by type of crop and region. The graph shows that of the farmers mainly cultivating non-root vegetables, approximately 57% in north Ondo and approximately 54% in south Ondo are female; in other words, most of those farmers are female, which exemplifies the idea that female farmers make up the majority (more than half) of the farmers cultivating specific types of crops in some regions.

Choice A is incorrect because it inaccurately cites data from the graph: the graph shows that in south Ondo, most of the farmers mainly cultivating non-root vegetables are women (approximately 54%), but that only about 35% (less than half) of the farmers mainly cultivating cereals are women. Choice B is incorrect because it inaccurately cites data from the graph: the graph shows that more women in central Ondo mainly cultivate cereals than mainly cultivate root crops (approximately 36% and 20%, respectively). Additionally, it doesn't effectively complete the example because the graph shows that female farmers don't make up the majority (more than half) of the farmers for any type of crop in central Ondo. Choice D is incorrect because it doesn't effectively complete the example; it simply states that a relatively equal proportion of women across the three regions mainly cultivate cereals, which doesn't address the value for that proportion and thus doesn't show that a majority (more than half) of the farmers cultivating certain crops are female.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

While attending school in New York City in the 1980s, Okwui Enwezor encountered few works by African artists in exhibitions, despite New York’s reputation as one of the best places to view contemporary art from around the world. According to an arts journalist, later in his career as a renowned curator and art historian, Enwezor sought to remedy this deficiency, not by focusing solely on modern African artists, but by showing how their work fits into the larger context of global modern art and art history.

Which finding, if true, would most directly support the journalist’s claim?

Answer

- A. As curator of the Haus der Kunst in Munich, Germany, Enwezor organized a retrospective of Ghanaian sculptor El Anatsui’s work entitled *El Anatsui: Triumphant Scale*, one of the largest art exhibitions devoted to a Black artist in Europe’s history.
- B. In the exhibition *Postwar: Art Between the Pacific and the Atlantic, 1945–1965*, Enwezor and cocurator Katy Siegel brought works by African artists such as Malangatana Ngwenya together with pieces by major figures from other countries, like US artist Andy Warhol and Mexico’s David Siqueiros.
- C. Enwezor’s work as curator of the 2001 exhibition *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* showed how African movements for independence from European colonial powers following the Second World War profoundly influenced work by African artists of the period, such as Kamala Ibrahim Ishaq and Thomas Mukarobgwa.
- D. Enwezor organized the exhibition *In/sight: African Photographers, 1940 to the Present* not to emphasize a particular aesthetic trend but to demonstrate the broad range of ways in which African artists have approached the medium of photography.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the arts journalist’s claim about Enwezor’s work as a curator and art historian. In the text, the arts journalist asserts that Enwezor wished not just to focus on modern African artists but also to show “how their work fits into the larger context of global modern art and art history,” or how their work relates to artistic developments and work by other artists elsewhere in the world. The description of *Postwar: Art Between the Pacific and the Atlantic, 1945–1965* indicates that Enwezor and Siegel’s exhibition brought works by African artists together with works by artists from other countries, thus supporting the arts journalist’s claim that Enwezor sought to show works by African artists in a context of global modern art and art history.

Choice A is incorrect because it describes a retrospective that wouldn’t support the arts journalist’s claim that Enwezor wanted to show how works by modern African artists fit into the larger context of global modern art and art history. The description of *El Anatsui: Triumphant Scale* indicates that the retrospective focused only on the work of a single African artist, El Anatsui. The description doesn’t suggest that the exhibition showed how El Anatsui’s works fit into a global artistic context. Choice C is incorrect because it describes an exhibition that wouldn’t support the arts journalist’s claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* indicates that the exhibition showed how African artists were influenced by movements for independence from European colonial powers following the Second World War. Although this suggests that Enwezor intended the exhibition to place works by African artists in a political context, it doesn’t indicate that the works were placed in a global artistic context. Choice D is incorrect because it describes an exhibition that wouldn’t support the arts journalist’s claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *In/sight: African Photographers, 1940 to the Present* indicates that the exhibition was intended to reveal the broad range of approaches taken by African photographers, not that the exhibition showed how photography by African artists fits into a global artistic context.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers’ reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____

Which choice most logically completes the text?

Answer

- A. find the disruptions more irritating as time went on.
- B. rate the listening experience as more negative than those whose listening experience was uninterrupted.
- C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.
- D. perceive the volume of the construction noise as growing softer over time.

Correct Answer: B

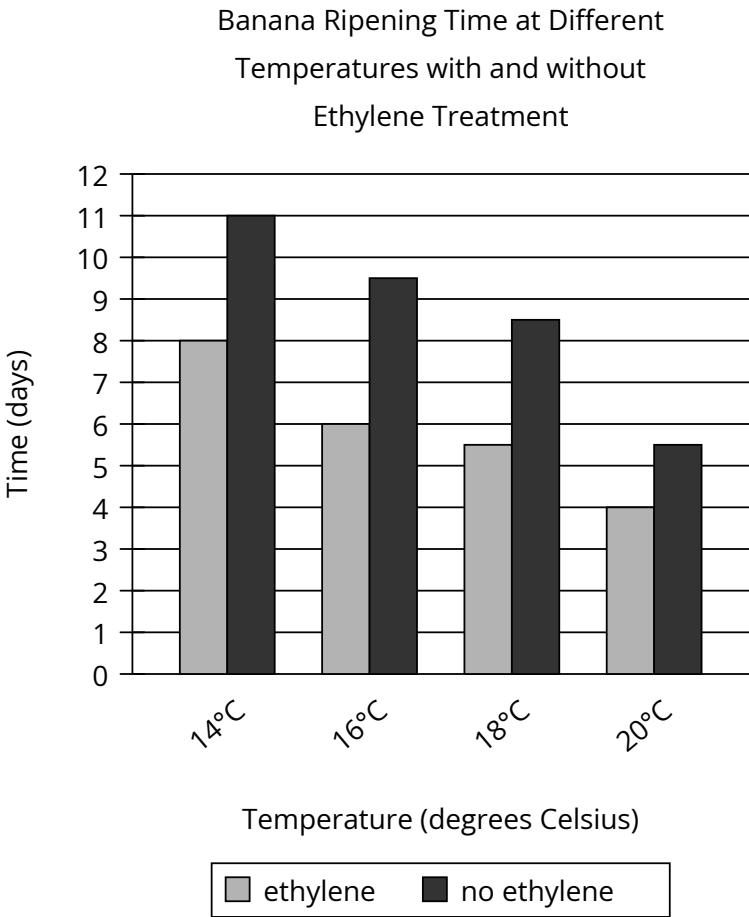
Rationale

Choice B is the best answer. It most logically completes the text. The text tells us that disruptions to an enjoyable experience increase viewers’ enjoyment. It also says that researchers suspect the opposite is true for disruptions to an unpleasant experience. Thus, we can infer that the researchers expect to find that the interrupted unpleasant experience was worse for listeners than the uninterrupted unpleasant experience.

Choice A is incorrect. It doesn’t logically complete the text. The text never makes any claims about how irritating the disruptions themselves are perceived to be. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences. Choice C is incorrect. It doesn’t logically complete the text. The text never makes any claims about how long any experience is perceived to be. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences. Choice D is incorrect. It doesn’t logically complete the text. The text never makes any claims about how interruptions affect the perceived volume of the unpleasant or pleasant experience. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



A student is conducting an experiment to test the effect of temperature and ethylene treatment on the ripening speed of bananas. The student treated some bananas with ethylene while leaving others untreated, then allowed both types of bananas to ripen at one of four different temperatures. Comparing the data for bananas with and without ethylene, the student concluded that _____

Which choice most effectively uses data from the graph to complete the student’s conclusion?

- Answer**
- A. 20°C is the ideal temperature at which to store bananas to slow ripening time.
 - B. for those bananas that were not treated with ethylene, differences in temperature were not associated with absolute differences in ripening time.
 - C. bananas treated with ethylene ripen faster at 14°C and 16°C than at 18°C and 20°C.
 - D. ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses data from the graph to complete the student's conclusion about banana ripening time with and without ethylene at different temperatures. The graph shows that at 20°C, the gap between the two bars showing ripening times for ethylene-treated bananas and untreated bananas crosses fewer than 2 gridlines (from about 4 days for ethylene-treated bananas to about 5.5 days for untreated bananas). Meanwhile, the graph shows that at 14°C, 16°C, and 18°C, the gap between the bars crosses more than 2 gridlines (from about 8 days for treated bananas to about 11 days for untreated bananas at 14°C; from about 6 days for treated bananas to about 9.5 days for untreated bananas at 16°C; and from about 5.5 days for treated bananas to about 8.5 days for untreated bananas at 18°C). Since the gap between the bars at each of these temperatures crosses more than 2 gridlines, and since each of these gaps is larger than the gap between the bars at 20°C, it can be concluded that ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Choice A is incorrect because the graph shows that ethylene-treated bananas stored at 20°C ripen more quickly than ethylene-treated bananas stored at the other temperatures do (about 4 days at 20°C vs. about 5.5, 6, and 8 days at 18°C, 16°C, and 14°C, respectively) and that untreated bananas stored at 20°C ripen more quickly than untreated bananas stored at the other temperatures do (about 5.5 days at 20°C vs. about 8.5, 9.5, and 11 days at 18°C, 16°C, and 14°C, respectively). The information in the graph therefore indicates that storing bananas at 20°C speeds up ripening time relative to storing bananas at the other temperatures shown, not that this storage temperature slows ripening time. Choice B is incorrect because the graph shows that as temperature increases, the ripening time of untreated bananas decreases, from about 11 days at 14°C to about 5.5 days at 20°C, with no exceptions to this trend. The graph therefore shows that differences in temperature were associated with absolute differences in ripening time, not that there was no association between differences in temperature and differences in ripening time. Choice C is incorrect because the graph shows that ripening times of ethylene-treated bananas at 14°C and 16°C were about 8 and 6 days, respectively, which is greater than, not less than, ripening times of ethylene-treated bananas at 18°C and 20°C, which were about 5.5 and 4 days, respectively. In other words, bananas treated with ethylene ripen more slowly, not faster, at 14°C and 16°C than at 18°C and 20°C.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Paleontologists searching for signs of ancient life have found many fossilized specimens of prehistoric human ancestors, including several from the Pleistocene era discovered in a geological formation in the Minatogawa quarry in Japan. However, to study the emergence of the earliest multicellular organisms to appear on Earth, researchers must turn elsewhere, such as to the Ediacaran geological formation at Mistaken Point in Canada. A UNESCO World Heritage Site, the 146-hectare reserve contains more than 10,000 fossils that together document a critical moment in evolutionary history.

What does the text indicate about the geological formation at Mistaken Point?

Answer

- A. It holds a greater number of fossils but from a smaller variety of species than the formation in the Minatogawa quarry does.
- B. It has provided evidence that the earliest human species may have emerged before the Pleistocene era.
- C. It is widely considered by paleontologists to be the most valuable source of information about prehistoric life forms.
- D. It contains specimens from an older time period than those found in the formation in the Minatogawa quarry.

Correct Answer: D

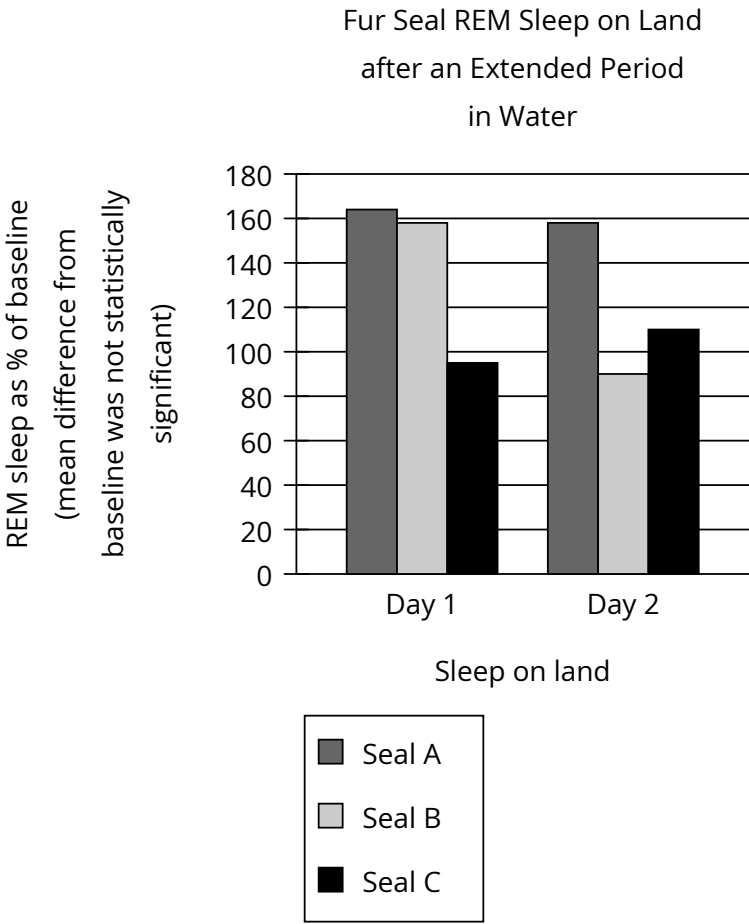
Rationale

Choice D is the best answer. The text says that the formation at Mistaken Point contains fossils of “the earliest multicellular organisms,” which implies that these fossils are from an older time period than the fossils of “prehistoric human ancestors” found in the Minatogawa quarry.

Choice A is incorrect. The text says that the formation at Mistake Point contains “more than 10,000 fossils,” but it doesn’t compare this number to the number of fossils in the Minatogawa quarry. It also doesn’t say anything about the variety of species in either formation. Choice B is incorrect. The text says that the formation at Mistaken Point contains fossils of “the earliest multicellular organisms,” but it never says that the site contains early human fossils too. Rather, the early human fossils mentioned in the text were found in the formation at Minatogawa quarry. Choice C is incorrect. The text says that the fossils at Mistaken Point “document a critical moment in evolutionary history,” but it never says that Mistaken Point is the most valuable source of information about prehistoric life forms.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Research suggests that REM sleep in animals is homeostatically regulated: animals compensate for periods of REM sleep deprivation by increasing subsequent REM sleep. When on land, fur seals get enough REM sleep, but during the weeks they’re in the water, they get almost none. In a study of fur seals’ sleep habits, researchers recorded the REM sleep (as a percentage of baseline) of fur seals once they had returned to land. They concluded that REM sleep may not be homeostatically regulated in fur seals, citing as evidence the fact that the seals in the study

Which choice most effectively uses data from the graph to complete the text?

- Answer
- A. didn’t consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.
 - B. showed no significant difference between REM sleep after returning to land and REM sleep while in the water.
 - C. didn’t show significantly less REM sleep during the second day after returning to land than they did during the first day.
 - D. showed no significant differences from one another in baseline levels of REM sleep.

Correct Answer: A

Rationale

Choice A is the best answer. If REM sleep were homeostatically regulated in fur seals, then all the seals would compensate with REM levels significantly over baseline after going weeks without REM. We'd also expect the seals to maintain those elevated REM levels for some time. Since seals B and C return very quickly to baseline REM levels, this suggests that REM sleep in fur seals may not be regulated homeostatically.

Choice B is incorrect. The graph doesn't depict REM sleep while in the water for the seals in the study. Additionally, we're told fur seals get no REM sleep while in the water, which is significantly different to the values shown in the graph for after they return to land. Choice C is incorrect. This doesn't support the conclusion. If REM sleep were homeostatically regulated in fur seals, then we'd suspect the seals to sustain REM levels well above baseline for a prolonged period in order to compensate for weeks of REM deprivation while in the water. Whether or not there's a reduction in REM sleep from day 1 to day 2 doesn't tell us how REM sleep on those days relates to baseline, which is where our focus should be. Choice D is incorrect. The y-axis of this graph doesn't depict baseline levels of REM sleep, but rather shows REM sleep as a percent of baseline.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Many of William Shakespeare’s tragedies address broad themes that still appeal to today’s audiences. For instance, *Romeo and Juliet*, which is set in the Italy of Shakespeare’s time, tackles the themes of parents versus children and love versus hate, and the play continues to be read and produced widely around the world. But understanding Shakespeare’s so-called history plays can require a knowledge of several centuries of English history. Consequently, _____

Which choice most logically completes the text?

Answer

- A. many theatergoers and readers today are likely to find Shakespeare’s history plays less engaging than the tragedies.
- B. some of Shakespeare’s tragedies are more relevant to today’s audiences than twentieth-century plays.
- C. *Romeo and Juliet* is the most thematically accessible of all Shakespeare’s tragedies.
- D. experts in English history tend to prefer Shakespeare’s history plays to his other works.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of the relative appeal of different kinds of plays by Shakespeare to today’s audiences. According to the text, Shakespeare’s tragedies address broad themes that continue to appeal to today’s audiences. Indeed, the text suggests that these themes are timeless, as illustrated by the example of *Romeo and Juliet*, which the text states is still read and widely performed despite being set in the Italy of Shakespeare’s time. In contrast, the text indicates that audiences and readers may need to be familiar with several centuries of English history in order to understand Shakespeare’s history plays. Because many theatergoers and readers are unlikely to possess such extensive historical knowledge, it follows that they are likely to find Shakespeare’s history plays less engaging than his more accessible tragedies.

Choice B is incorrect because the text never introduces a comparison between Shakespeare’s tragedies and twentieth-century plays, only between Shakespeare’s tragedies and his history plays. Since twentieth-century plays aren’t mentioned, there’s no basis in the text for the idea that some of Shakespeare’s tragedies are more relevant than twentieth-century plays to today’s audiences. Choice C is incorrect. Although the text indicates that *Romeo and Juliet* is thematically accessible to today’s audiences, it doesn’t suggest that *Romeo and Juliet* is more accessible than Shakespeare’s other tragedies. Rather, the text presents *Romeo and Juliet* as an example to support the idea that Shakespeare’s tragedies hold continued appeal for today’s readers and theatergoers. Choice D is incorrect. Although experts in English history would likely possess the knowledge needed to understand Shakespeare’s history plays, the text never mentions such experts or suggests that they would enjoy the history plays more than Shakespeare’s other works.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Three Studies' Estimated Average Velocity of LMC		
Researchers	Study year	Estimated average velocity
Murai and Fujimoto	1980	344 km/s
Kallivayalil and colleagues	2006	378 km/s
Gardiner and colleagues	1994	297 km/s

In 2006, Nitya Kallivayalil and colleagues calculated the most accurate estimate yet of the average velocity (in kilometers per second) of the Large Magellanic Cloud (LMC) galaxy. Before the 2006 study, estimates of the average velocity were low enough for the LMC to maintain an orbit around the Milky Way galaxy, but according to an analysis by Gurtina Besla and colleagues, the estimated velocity from the 2006 study is too high for the LMC to maintain such an orbit. Therefore, if Besla and colleagues are correct, the maximum average velocity for the LMC that would allow it to maintain orbit around the Milky Way is likely _____

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. above 344 km/s but below 378 km/s.
- B. above 297 km/s but below 344 km/s.
- C. above 378 km/s.
- D. below 297 km/s.

Correct Answer: A

Rationale

Choice A is the best answer because it uses data from the table to give the range of velocities for the LMC from the 1980 value (344 km/s) to the 2006 value (378 km/s), thereby effectively completing the text. The text indicates that before 2006, all the estimated velocities of the LMC were within the range necessary to maintain orbit around the Milky Way galaxy. It then indicates that, according to Besla and colleagues, the 2006 estimate of 378 km/s is too high to maintain that orbit. This strongly implies that if the 1980 value (344 km/s) is below the orbital threshold, and if Besla and colleagues are correct that the 2006 value (378 km/s) is above that threshold, the maximum orbital velocity for the LMC must be somewhere in the range from above 344 km/s to below 378 km/s.

Choice B is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first estimate that exceeded the velocity needed to maintain orbit around the Milky Way. Thus, the 1980 estimate of 344 km/s and the 1994 estimate of 297 km/s must both be below the maximum possible orbital velocity for the LMC. Choice C is incorrect. The text states that Besla and colleagues’ analysis found that the velocity from the 2006 study (378 km/s) was too high for the LMC to maintain orbit around the Milky Way. Therefore, if a velocity of 378 km/s is too high, an even higher velocity will also be too high. Choice D is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first to exceed the velocity range required to maintain orbit around the Milky Way. Thus, the 1994 estimate of 297 km/s must be below the maximum possible orbital velocity for the LMC.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Mosasosaurs were large marine reptiles that lived in the Late Cretaceous period, approximately 100 million to 66 million years ago. Celina Suarez, Alberto Pérez-Huerta, and T. Lynn Harrell Jr. examined oxygen-18 isotopes in mosasaur tooth enamel in order to calculate likely mosasaur body temperatures and determined that mosasaurs were endothermic—that is, they used internal metabolic processes to maintain a stable body temperature in a variety of ambient temperatures. Suarez, Pérez-Huerta, and Harrell claim that endothermy would have enabled mosasaurs to include relatively cold polar waters in their range.

Which finding, if true, would most directly support Suarez, Pérez-Huerta, and Harrell’s claim?

Answer

- A. Mosasaurs’ likely body temperatures are easier to determine from tooth enamel oxygen-18 isotope data than the body temperatures of nonendothermic Late Cretaceous marine reptiles are.
- B. Fossils of both mosasaurs and nonendothermic marine reptiles have been found in roughly equal numbers in regions known to be near the poles during the Late Cretaceous, though in lower concentrations than elsewhere.
- C. Several mosasaur fossils have been found in regions known to be near the poles during the Late Cretaceous, while relatively few fossils of nonendothermic marine reptiles have been found in those locations.
- D. During the Late Cretaceous, seawater temperatures were likely higher throughout mosasaurs’ range, including near the poles, than seawater temperatures at those same latitudes are today.

Correct Answer: C

Rationale

Choice C is the best answer because it presents the finding that, if true, would best support Suarez, Pérez-Huerta, and Harrell’s claim about mosasaurs. The text states that Suarez, Pérez-Huerta, and Harrell’s research on mosasaur tooth enamel led them to conclude that mosasaurs were endothermic, which means that they could live in waters at many different temperatures and still maintain a stable body temperature. The researchers claim that endothermy enabled mosasaurs to live in relatively cold waters near the poles. If several mosasaur fossils have been found in areas that were near the poles during the period when mosasaurs were alive and fossils of nonendothermic marine reptiles are rare in such locations, that would support the researchers’ claim: it would show that mosasaurs inhabited polar waters but nonendothermic marine mammals tended not to, suggesting that endothermy may have been the characteristic that enabled mosasaurs to include polar waters in their range.

Choice A is incorrect because finding that it’s easier to determine mosasaur body temperatures from tooth enamel data than it is to determine nonendothermic reptile body temperatures wouldn’t support the researchers’ claim. Whether one research process is more difficult than another indicates nothing about the results of those processes and therefore is irrelevant to the issue of where mosasaurs lived and what enabled them to live in those locations. Choice B is incorrect because finding roughly equal numbers of mosasaur and nonendothermic marine reptile fossils in areas that were near the poles in the Late Cretaceous would suggest that endothermy didn’t give mosasaurs any particular advantage when it came to expanding their range to include relatively cold polar waters, thereby weakening the researchers’ claim rather than supporting it. Choice D is incorrect because finding that the temperature of seawater in the Late Cretaceous was warmer than seawater today wouldn’t weaken the researchers’ claim. Seawater in the Late Cretaceous could have been warmer than seawater today but still cold enough for endothermy to be advantageous to mosasaurs, so this finding wouldn’t provide enough information to either support or weaken the researchers’ claim.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The *Mammillaria* cactus *M. boolii* occurs naturally only in the state of Sonora in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like *M. boolii*, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. If this view is correct, the single-species approach to the conservation of *M. boolii* could thus _____

Which choice most logically completes the text?

Answer

- A. lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.
- B. allow conservationists to better consider how climatic changes affecting Sonora may reduce the number of species competing with *M. boolii*.
- C. erroneously shift the focus of conservation efforts away from *M. boolii* itself.
- D. fail to consider the ways in which the survival of *M. boolii* may be influenced by changes in the populations of other species that inhabit Sonora.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of conservation approaches for the *Mammillaria* cactus *M. boolii*. The text establishes that *M. boolii* only grows naturally in the state of Sonora in Mexico, which makes it particularly vulnerable to extinction. The text then contrasts two approaches to conservation: the traditional single-species approach that individually focuses on at-risk species and a newer ecosystem-based approach that considers the interactions between climate, terrain, and various species in a geographical area. According to the text, conservationists have recently argued that this ecosystem-based approach may lead to better outcomes for all species in a location. If this ecosystem view is correct, then the single-species approach to conserving *M. boolii* would likely fail to consider how the cactus’s survival depends on its interactions with other species in Sonora’s ecosystem.

Choice A is incorrect because the text doesn’t address the distribution of *Mammillaria* species besides *M. boolii* throughout Mexico or discuss how that distribution affects survival. The text focuses specifically on *M. boolii* in Sonora and different approaches to its conservation. Choice B is incorrect because the text doesn’t suggest that climatic changes in Sonora would reduce competition for *M. boolii* or that conservation efforts are focused on understanding this specific dynamic. In fact, the text implies that an ecosystem-based approach would consider climate among many other factors but doesn’t specify how climatic changes might affect competition between species. Choice C is incorrect because the text doesn’t suggest that the single-species approach would shift the focus of conservation efforts away from *M. boolii* itself—rather, it suggests that this approach might be too narrowly focused on *M. boolii* alone without considering the broader ecosystem-related factors that affect its survival.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Economists have demonstrated that herding—in which individuals converge on the same action by imitating the economic decisions of one or more peers—occurs among investors. Investigating whether herding also occurs among nations in matters of economic policy, Maru Etta-Nkwelle examined changes in the foreign currency reserves of emerging economies in various world regions in the years following the widespread financial crisis of 1997. Etta-Nkwelle posited that herding would manifest in regional peer countries converging on the same foreign currency accumulation rate regardless of each country’s observable circumstances. She found that Asian countries showed evidence of herding but African countries did not.

Based on the text, which finding, if true, would best account for the discrepancy Etta-Nkwelle observed?

Answer

- A. The majority of the currency held in reserves by the African countries was foreign currency, whereas the majority of the currency held in reserves by the Asian countries was domestic currency.
- B. Most of the African countries belonged to international economic unions that set individual foreign-currency accumulation policies for each member, whereas the Asian countries were largely free to set their own policies.
- C. The 1997 financial crisis resulted in immediate declines in many of the African countries’ foreign currency reserves, whereas the crisis had few short-term effects on the foreign currency reserves of most of the Asian countries.
- D. The African countries tended to accumulate a variety of foreign currencies for their reserves, whereas the Asian countries tended to favor a single foreign currency for their reserves.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would best account for the discrepancy Etta-Nkwelle observed: Asian countries showed evidence of herding (converging on the same foreign currency accumulation rate as their regional peers), while African countries did not. The text defines herding as individuals converging on the same action by imitating their peers, which implies that the individuals choose to do so rather than being compelled to follow a particular course of action. This choice indicates that most of the African countries belonged to economic unions that dictated such monetary policies, whereas Asian countries were free to choose, and the similarity among the Asian countries suggests that herding can account for that similarity. The finding thus identifies a relevant difference between the two regions that accounts for the discrepancy.

Choice A is incorrect. Although this choice is related to reserves of foreign currency, the fact that African and Asian countries had different mixes of foreign and domestic currency reserves does not address the rate of foreign currency accumulation or whether herding is a contributing factor determining the rate of accumulation. Choice C is incorrect. This choice does not address currency accumulation. And although it describes differences in the currency reserves of African and Asian countries shortly after the 1997 financial crisis, it does not address the rate of foreign currency accumulation nor how similar the nations were in this regard to their neighboring countries. Choice D is incorrect. Although this choice discusses the variety of foreign currencies accumulated by African countries as compared with Asian countries, it does not discuss the rate of foreign currency accumulation nor whether herding is a factor determining the rate of accumulation.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

In the early nineteenth century, some Euro-American farmers in the northeastern United States used agricultural techniques developed by the Haudenosaunee (Iroquois) people centuries earlier, but it seems that few of those farmers had actually seen Haudenosaunee farms firsthand. Barring the possibility of several farmers of the same era independently developing techniques that the Haudenosaunee people had already invented, these facts most strongly suggest that _____

Which choice most logically completes the text?

Answer

- A. those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.
- B. the crops typically cultivated by Euro-American farmers in the northeastern United States were not well suited to Haudenosaunee farming techniques.
- C. Haudenosaunee farming techniques were widely used in regions outside the northeastern United States.
- D. Euro-American farmers only began to recognize the benefits of Haudenosaunee farming techniques late in the nineteenth century.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of Euro-American farmers’ use of Haudenosaunee agricultural techniques. According to the text, some Euro-American farmers were using these techniques in the early nineteenth century despite few of the farmers having seen Haudenosaunee farms. One explanation for these facts might be that the farmers developed techniques on their own that already had been invented centuries earlier by the Haudenosaunee people, but the text explicitly bars, or rules out, this explanation. If Euro-American farmers didn’t learn these techniques from direct observation of Haudenosaunee practices and didn’t invent the techniques independently, then the most logical explanation is that they learned the techniques from other people who were more directly influenced by Haudenosaunee practices than the farmers themselves were. Once they learned about Haudenosaunee agricultural practices, Euro-American farmers could then apply those practices to their own farming.

Choice B is incorrect because the fact that some Euro-American farmers in the northeastern United States were using Haudenosaunee techniques suggests that the techniques were likely useful for the crops the farmers raised, not that the crops typically cultivated by the farmers were not well suited to Haudenosaunee farming techniques. If the farmers’ crops were ill suited to the techniques, it’s unlikely that the farmers would have used those techniques. Choice C is incorrect because the text indicates only that Haudenosaunee agricultural techniques were used by Euro-American farmers in the northeastern United States, not that these techniques were widely used outside this region. Choice D is incorrect because the text states that some Euro-American farmers were using Haudenosaunee farming techniques early in the nineteenth century. This suggests that some Euro-American farmers were beginning to recognize the benefits of these techniques near the start of the century, not that such farmers only began to recognize the benefits of the techniques much later.

Question ID: 1a2b29c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is adapted from María Cristina Mena’s 1914 short story “The Vine-Leaf.”

It is a saying in the capital of Mexico that Dr. Malsufrido carries more family secrets under his hat than any archbishop.

The doctor’s hat is, appropriately enough, uncommonly capacious, rising very high, and sinking so low that it seems to be supported by his ears and eyebrows, and it has a furry look, as if it had been brushed the wrong way, which is perhaps what happens to it if it is ever brushed at all. When the doctor takes it off, the family secrets do not fly out like a flock of parrots, but remain nicely bottled up beneath a dome of old and highly polished ivory.

Based on the text, how do people in the capital of Mexico most likely regard Dr. Malsufrido?

Answer

- A. Many have come to tolerate him despite his disheveled appearance.
- B. Few feel concerned that he will divulge their confidences.
- C. Some dislike how freely he discusses his own family.
- D. Most would be unimpressed by him were it not for his professional expertise.

Correct Answer: B

Rationale

Choice B is the best answer. The text describes a saying in the capital that Malsufrido keeps more secrets than an archbishop. It also says that when he takes off his hat, “the family secrets do not fly out... but remain nicely bottled up,” suggesting that he will not betray his confidences.

Choice A is incorrect. This choice doesn’t reflect the text. While his hat is large and appears to have been brushed in the wrong direction, Dr. Malsufrido does not seem to be regarded as ill-dressed or disheveled. Choice C is incorrect. This choice is the opposite of what the text says. The secrets of families (his and others’) remain “bottled up” in his head. Choice D is incorrect. This choice isn’t supported by the text. His professional expertise is not discussed in the passage.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Linguist Deborah Tannen has cautioned against framing contentious issues in terms of two highly competitive perspectives, such as pro versus con. According to Tannen, this debate-driven approach can strip issues of their complexity and, when used in front of an audience, can be less informative than the presentation of multiple perspectives in a noncompetitive format. To test Tannen’s hypothesis, students conducted a study in which they showed participants one of three different versions of local news commentary about the same issue. Each version featured a debate between two commentators with opposing views, a panel of three commentators with various views, or a single commentator.

Which finding from the students’ study, if true, would most strongly support Tannen’s hypothesis?

Answer

- A. On average, participants perceived commentators in the debate as more knowledgeable about the issue than commentators in the panel.
- B. On average, participants perceived commentators in the panel as more knowledgeable about the issue than the single commentator.
- C. On average, participants who watched the panel correctly answered more questions about the issue than those who watched the debate or the single commentator did.
- D. On average, participants who watched the single commentator correctly answered more questions about the issue than those who watched the debate did.

Correct Answer: C

Rationale

Choice C is the best answer because it presents the finding that, if true, would most strongly support Tannen’s hypothesis. According to the text, Tannen’s hypothesis is that multiple perspectives presented in a noncompetitive format is more informative than a debate between opposing viewpoints is. If participants who saw a panel of three commentators with various views about an issue answered more questions about the issue correctly than did participants who saw a debate, that would support Tannen’s hypothesis since it would show that participants who heard multiple varied perspectives were better informed than were participants who heard a debate between opposing viewpoints.

Choice A is incorrect because finding that participants perceived commentators in the debate as more knowledgeable than commentators in the panel is irrelevant to Tannen’s hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants’ perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists. Choice B is incorrect because finding that participants perceived commentators in the panel as more knowledgeable than a single commentator is irrelevant to Tannen’s hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants’ perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists, and Tannen’s hypothesis says nothing about how informative single commentators are. Choice D is incorrect because finding that participants who watched a single commentator answered more questions correctly than participants who watched the debate did wouldn’t be relevant to Tannen’s hypothesis, which is that hearing multiple varying perspectives is more informative than hearing a debate. Tannen’s hypothesis says nothing about how informative single commentators are.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Some geologists have proposed designating the period from 1950 to the present as a new geological epoch (the Anthropocene) characterized by human impact on Earth, but they have struggled to identify reliable stratigraphic markers of the epoch’s onset. Inta Dimante-Deimantovica and a research team investigated whether the initial appearance of primary microplastics—invented and mass-produced for industrial and other purposes, beginning around the middle of the twentieth century—in sedimentary layers could serve this role. The researchers analyzed European lake sediment profiles from the late eighteenth century to the present. Microplastics were present in all layers, likely because certain microplastic shapes enabled rapid downward migration. The researchers therefore concluded that _____

Which choice most logically completes the text?

Answer

- A. the lowest sedimentary layer in which microplastics are found cannot be treated as indicative of the chronological beginning of the Anthropocene.
- B. the presence of microplastics in sediment dating to the late eighteenth century casts doubt on the appropriateness of designating 1950 as the onset of the Anthropocene.
- C. microplastics are not prevalent enough in sediment to serve as a reliable sign of the human impact characteristic of the Anthropocene.
- D. using the earliest presence of microplastics as a stratigraphic marker of the beginning of the Anthropocene is likely to be viable in some locations but is not viable in Europe.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of microplastics as a proposed stratigraphic marker for the onset of the Anthropocene. The text indicates that although primary microplastics (used for industrial and other purposes) were invented and mass-produced beginning only around the mid-twentieth century, researchers found microplastics throughout European lake sediment profiles dating from the late eighteenth century to the present. According to the text, microplastics were in all layers of sediment likely because microplastics of certain shapes can rapidly migrate downward—that is, they can move into sedimentary layers that predate their manufacture. Since it can’t be assumed that the date of a layer in which microplastics are found indicates when the microplastics were first deposited, it follows that the lowest sedimentary layer containing microplastics can’t be treated as indicative of the chronological beginning of the Anthropocene.

Choice B is incorrect. The text explains that microplastics appeared in layers of sediment long predating 1950 likely because certain shapes allowed them to rapidly migrate downward—that is, the particles physically moved into older layers. This finding doesn’t cast doubt on the appropriateness of considering 1950 as the onset of the Anthropocene; it doesn’t suggest a flaw in thinking that humans’ impact on Earth shifted around 1950 but simply suggests that the position of microplastics in sediment isn’t useful as an indicator of such a shift. Choice C is incorrect because the issue presented in the text isn’t a lack of prevalence of microplastics (and though no quantity is indicated, microplastics were found across all the layers) but rather the inability to determine when microplastics were originally deposited, since the particles can migrate rapidly into older sediment layers. Choice D is incorrect because the text indicates that the likely cause of microplastics being present in all layers of the sediment profiles is a property of microplastic particles themselves (certain shapes allowing them to rapidly move downward into older layers), not a characteristic of European lake sediment or anything else specific to Europe. There’s no reason to think the issue of particle migration wouldn’t occur in other locations.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In 1534 CE, King Henry VIII of England split with the Catholic Church and declared himself head of the Church of England, in part because Pope Clement VII refused to annul his marriage to Catherine of Aragon. Two years later, Henry VIII introduced a policy titled the Dissolution of the Monasteries that by 1540 had resulted in the closure of all Catholic monasteries in England and the confiscation of their estates. Some historians assert that the enactment of the policy was primarily motivated by perceived financial opportunities.

Which quotation from a scholarly article best supports the assertion of the historians mentioned in the text?

Answer

- A. “At the time of the Dissolution of the Monasteries, about 2 percent of the adult male population of England were monks; by 1690, the proportion of the adult male population who were monks was less than 1 percent.”
- B. “A contemporary description of the Dissolution of the Monasteries, Michael Sherbrook’s *Falle of the Religious Howses*, recounts witness testimony that monks were allowed to keep the contents of their cells and that the monastery timber was purchased by local yeomen.”
- C. “In 1535, the year before enacting the Dissolution of the Monasteries, Henry commissioned a survey of the value of church holdings in England—the work, performed by sheriffs, bishops, and magistrates, began that January and was swiftly completed by the summer.”
- D. “The October 1536 revolt known as the Pilgrimage of Grace had several economic motives: high food prices due to a poor harvest the prior year; the Dissolution of the Monasteries, which closed reliable sources of food and shelter for many; and rents and taxes throughout Northern England that were not merely high but predatory.”

Correct Answer: C

Rationale

Choice C is the best answer. The fact that Henry VIII commissioned a survey of church holdings just before enacting the Dissolution of the Monasteries suggests that he was interested in the potential profits from claiming their assets. This supports the historians’ assertion.

Choice A is incorrect. This choice describes a potential effect of the Dissolution of the Monasteries. The decrease in the proportion of monks in England isn’t relevant to the question of Henry VIII’s motivations. Choice B is incorrect. Details about how the monks were treated during the dissolution don’t address Henry VIII’s motivations for enacting the policy in the first place. That the monks could keep the content of their cells (their rooms) and sell off the timber they’d harvested has no impact on the value of the monasteries’ estates—their land holdings. Choice D is incorrect. This choice mentions one impact that the Dissolution of the Monasteries contributed to, two years after it happened. But it doesn’t help explain why Henry VIII might have wanted to enact the policy in the first place.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Like many other genera of wild bees, bumblebees have in recent decades experienced population collapse caused by, among other factors, habitat destruction and climate variation. Bumblebees are also one of the most researched bee genera, second only to honeybees. As a result, ecologists have gained much of their insight about wild-bee declines from bumblebees. In a 2021 paper, zoologist Guillaume Ghisbain notes that bumblebees are among the relatively few wild-bee genera that display social behaviors and dietary generalism (ability to obtain nectar and pollen from a diversity of plant species), two traits that are associated with increased resilience to some specific environmental changes. Ghisbain therefore contends that _____

Which choice most logically completes the text?

Answer

- A. although bumblebees and many other wild bees have experienced similar population declines in the past, compared with other wild bees, bumblebees are likely at greater risk of being harmed by climate variation than by habitat destruction.
- B. although bumblebees have been more extensively studied than most wild bees, researchers should not use bumblebees to draw conclusions about the decline of other wild bees, even ones with feeding patterns and levels of sociability that are similar to those of bumblebees.
- C. because bumblebees and other bees with generalist diets are less negatively affected by environmental stress than bees with specialized diets are, they are less likely to experience major population changes in the future than bees with specialized diets are.
- D. because the responses of bumblebees and other wild bees to environmental threats are not always comparable, researchers need to exercise caution when extrapolating information about wild-bee population declines from bumblebees.

Correct Answer: D

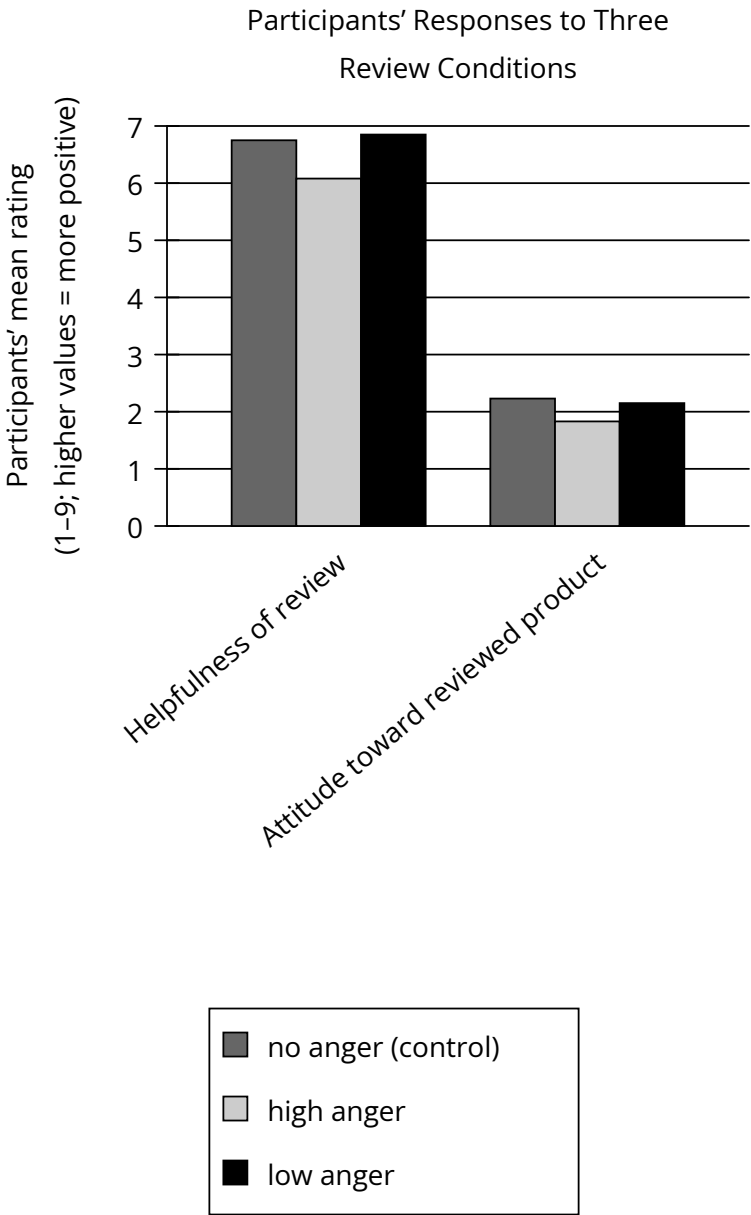
Rationale

Choice D is the best answer because it most logically completes the text’s discussion of wild-bee declines. The text establishes that bumblebees and other wild bees have been experiencing population collapse as a result of habitat destruction, climate variation, and other factors. The text then indicates that because bumblebees are very extensively researched, ecologists rely heavily on findings about them to understand wild-bee declines in general. However, the text then introduces Ghisbain’s observation that unlike most other wild-bee genera, bumblebees have certain traits (social behaviors and dietary generalism) linked to increased resilience to certain environmental changes. In other words, bumblebees aren’t necessarily representative of wild bees as a whole because they are likely more tolerant of some pressures. Therefore, it logically follows that Ghisbain would urge researchers to exercise caution when using bumblebee data to draw conclusions about other wild-bee population declines, since bumblebees and other wild bees don’t always respond comparably to environmental threats.

Choice A is incorrect because the text doesn’t indicate that climate variation is more of a threat to bumblebees than habitat destruction is or that it is a bigger threat to bumblebees than it is to other wild bees. The text simply indicates that climate variation and habitat destruction are among the factors that have caused population collapses for bumblebees and other wild-bee genera. Choice B is incorrect. As it is presented in the text, Ghisbain notes that bumblebees may respond to environmental changes differently than many other wild-bee genera do; this suggests that bumblebees shouldn’t be treated as representative of all wild bees, but there’s no reason to think that Ghisbain would go so far as to assert that bumblebee data shouldn’t be used to draw conclusions about other wild-bee genera that are similar to bumblebees in their dietary and social traits, even if there are relatively few. The text suggests caution in extrapolation, not a complete rejection of using bumblebee research. Choice C is incorrect. Although the text indicates that dietary generalism is linked to increased resilience to specific environmental changes (that is, to some but not all environmental stresses), it doesn’t suggest that bumblebees and other bees with this trait are less likely than bees with specialized diets to experience major population changes. In fact, the text makes it clear that like many other wild-bee genera—most of which don’t display dietary generalism—bumblebees are already experiencing population collapse.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



To understand how expressions of anger in reviews of products affect readers of those reviews, business scholar Dezhi Yin and colleagues measured study participants' responses to three versions of the same negative review—a control review expressing no anger, a review expressing a high degree of anger, and a review expressing a low degree of anger. Reviewing the data, a student concludes that the mere presence of anger in a review may not negatively affect readers' perceptions of the review, but a high degree of anger in a review does worsen readers' perceptions of the review.

Which choice best describes data from the graph that support the students' conclusion?

Answer

- A. On average, participants' ratings of the helpfulness of the review were substantially higher than were participants' ratings of the reviewed product regardless of which type of review participants had seen.

- B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.
- C. Participants who saw the low-anger review rated the review as slightly more helpful than participants who saw the control review did, but participants' attitude toward the reviewed product was slightly worse when participants saw the low-anger review than when they saw the no-anger review.
- D. Compared with participants who saw the low-anger review, participants who saw the high-anger review rated the review as less helpful and had a less positive attitude toward the reviewed product.

Correct Answer: B

Rationale

Choice B is the best answer. The claim is that low anger does not negatively affect readers' perceptions of the review, while high anger does negatively affect readers' perceptions of the review. This choice accurately expresses the supporting data from the "helpfulness of review" part of the graph: that low-anger reviews were rated as slightly more helpful than no-anger reviews, while high-anger reviews were rated as less helpful than no-anger reviews.

Choice A is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' ratings of the reviewed product are not relevant. Choice C is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' attitude towards the reviewed product is not relevant. Choice D is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' attitude towards the reviewed product is not relevant.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Number and Origin of Clamshell Tools Found at Different Levels Below the Surface in Neanderthal Cave

Depth of tools found below surface in cave (meters)	Clamshells that Neanderthals collected from the beach	Clamshells that Neanderthals harvested from the seafloor
3–4	99	33
6–7	1	0
4–5	2	0
2–3	7	0
5–6	18	7

Studying tools unearthed at a cave site on the western coast of Italy, archaeologist Paola Villa and colleagues have determined that prehistoric Neanderthal groups fashioned them from shells of clams that they harvested from the seafloor while wading or diving or that washed up on the beach. Clamshells become thin and eroded as they wash up on the beach, while those on the seafloor are smooth and sturdy, so the research team suspects that Neanderthals prized the tools made with seafloor shells. However, the team also concluded that those tools were likely more challenging to obtain, noting that _____

Which choice most effectively uses data from the table to support the research team’s conclusion?

Answer

- A. at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that shells were easier to collect from the beach than to harvest from the seafloor.
- B. the highest number of tools were at a depth of 3–4 meters below the surface, which suggests that the Neanderthal population at the site was highest during the related period of time.
- C. at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that Neanderthals preferred to use clamshells from the beach because of their durability.
- D. the higher number of tools at depths of 5–6 meters below the surface in the cave than at depths of 4–5 meters below the surface suggests that the size of clam populations changed over time.

Correct Answer: A

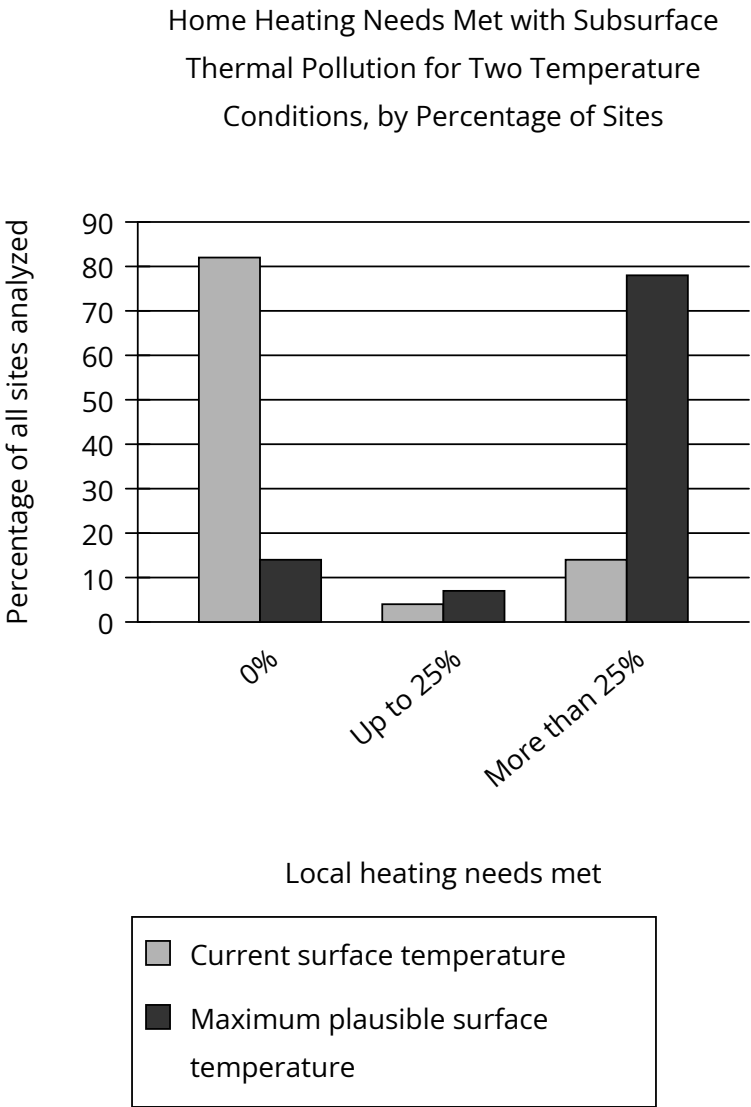
Rationale

Choice A is the best answer because it most effectively uses data from the table to support the researchers’ conclusion about the harvesting of clamshells by Neanderthals for use as tools. The text explains that Neanderthals used clamshells to make tools and that the sturdiest, and therefore the most desirable, shells for this purpose are found on the seafloor, not on the beach. However, the researchers also concluded that the clamshell tools made from shells from the seafloor are rarer than those made from shells from the beach. Meanwhile the table shows that at each depth, the number of tools made from shells from the beach exceeds the number made from the more desirable shells from the seafloor. The fact that the more desirable shells are less common suggests that it was significantly more difficult to harvest shells from the seafloor than from the beach.

Choice B is incorrect because knowing which depth represents the period of time with the highest Neanderthal population does not help answer the question of why the Neanderthals consistently made more tools from the less desirable shells from the beach than they made from the more desirable shells from the seafloor. Choice C is incorrect because it claims that the beach shells are more durable than the seafloor shells, which contradicts the text's description of shells from the seafloor as smoother and sturdier than shells from the beach. Choice D is incorrect because knowing which depth has the most artifacts or whether the clam population fluctuated does not help explain why tools made from the less desirable shells from the beach outnumber tools made from the more desirable shells from the seafloor.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Urbanization, industrialization, and the warming climate create thermal pollution (excess heat) in the shallow subsurface soil. Susanne A. Benz and colleagues analyzed thousands of sites on three continents under one scenario in which surface temperature remains at the current level and under another in which the surface reaches the maximum plausible temperature. They then categorized each site according to the percentage of local home heating needs that could be met using this excess subsurface heat. The team concluded that if surface temperature approaches the maximum plausible level, the percentage of sites where thermal pollution could feasibly contribute to meeting home heating needs will increase.

Which choice best describes data in the graph that support Benz and colleagues’ conclusion?

Answer

- A. Under both temperature conditions, less than 10% of sites were in the up-to-25% group, but at the maximum plausible surface temperature, almost 80% of sites could have all their local heating needs met by thermal pollution.

- B. At current surface temperatures, more than 80% of the sites have no need for supplemental local home heating from subsurface thermal pollution, but at the maximum plausible surface temperature, more than 70% of sites exhibit significantly greater home heating needs.
- C. At current surface temperatures, more than 80% of sites can meet, at most, 25% of local home heating needs with subsurface thermal pollution, but at the maximum plausible surface temperature, more than 80% of sites can meet greater than 25% of local home heating needs.
- D. At current surface temperatures, more than 80% of the sites cannot use subsurface thermal pollution to meet any portion of local home heating needs, but at the maximum plausible surface temperature, that percentage drops below 20%.

Correct Answer: D

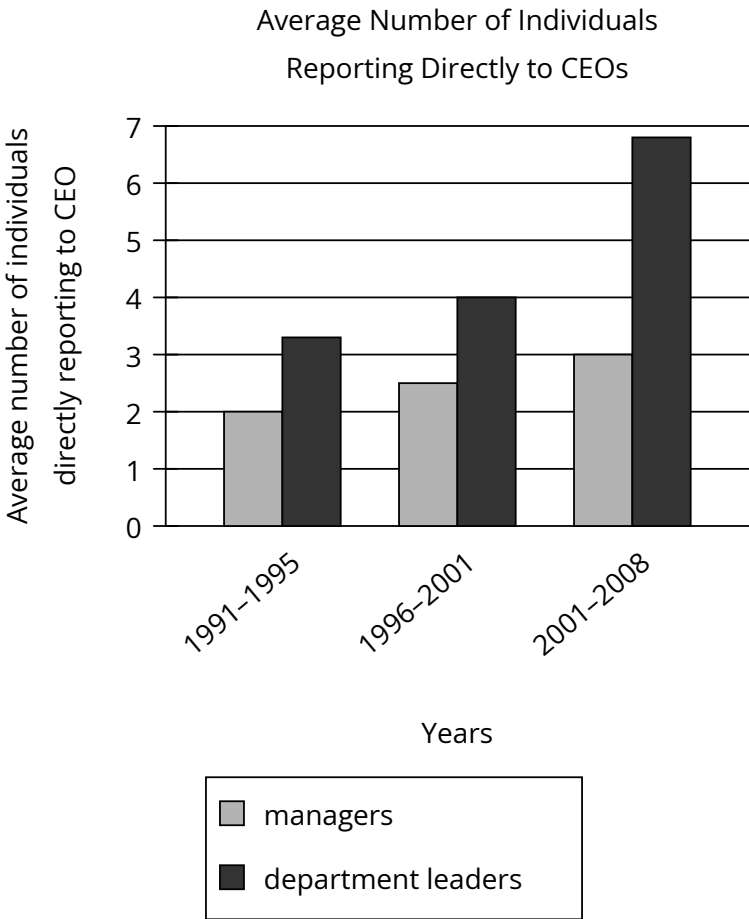
Rationale

Choice D is the best answer. The researchers concluded that as we approach maximum plausible surface temperatures, there will be a larger percentage of sites where thermal pollution could contribute to meeting home heating needs. By showing that only a small percentage of homes can currently use thermal pollution for home heating, and that this percentage would grow much larger at maximum plausible surface temperatures, this choice supports the researchers' conclusion.

Choice A is incorrect. We do not know how many sites could have all (i.e., 100%) of their local heating needs met by thermal pollution, as the graph only classifies sites by whether "0%," "Up to 25%," and "More than 25%" of heating needs could be met. Choice B is incorrect. The graph is not depicting need for supplemental heating from thermal pollution, but rather potential to use thermal pollution for supplemental heating. Choice C is incorrect. The graph indicates that, at current surface temperatures, less than 10% of sites can meet 25% of local home heating needs and that more than 80% of sites cannot meet any local home heating needs.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Considering a large sample of companies, economics experts Maria Guadalupe, Julie Wulf, and Raghuram Rajan assessed the number of managers and leaders from different departments who reported directly to a chief executive officer (CEO). According to the researchers, the findings suggest that across the years analyzed, there was a growing interest among CEOs in connecting with more departments in their companies.

Which choice best describes data from the graph that support the researchers’ conclusion?

Answer

- A. The average numbers of managers and department leaders reporting directly to their CEO didn’t fluctuate from the 1991–1995 period to the 2001–2008 period.
- B. The average number of managers reporting directly to their CEO was highest in the 1996–2001 period.
- C. The average number of department leaders reporting directly to their CEO was greater than the average number of managers reporting directly to their CEO in each of the three periods studied.
- D. The average number of department leaders reporting directly to their CEO rose over the three periods studied.

Correct Answer: D

Rationale

Choice D is the best answer because it describes data from the graph that support the researchers' conclusion that there is a growing interest among CEOs in connecting with more departments in their companies. The graph shows the average number of individuals reporting directly to CEOs during three different time periods: the individuals are divided into managers and department leaders. The average number of department leaders directly reporting to their CEO during the 1991–1995 period was slightly more than three, during the 1996–2001 period it was four, and during the 2001–2008 period it was almost seven. Thus, the average number of department leaders reporting directly to their CEO rose over the three periods studied, which suggests that CEOs were connecting with more departments.

Choice A is incorrect because the average number of managers and department leaders reporting directly to their CEO rose for both categories between the 1991–1995 and 2001–2008 periods; thus, it isn't true that the average numbers didn't fluctuate. Choice B is incorrect because the average number of managers reporting directly to their CEO was highest in the 2001–2008 period, not in the 1996–2001 period. Choice C is incorrect. Although it correctly describes a feature of the graph, the observation that more department leaders than managers are reporting to CEOs does not by itself address the question of whether CEOs are connecting with more departments over time—to address that question, one needs to know whether the number of department leaders reporting to CEOs is increasing over time.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In superfluorescence, electrical charges known as dipoles emit light in synchronized bursts so intense that they are visible to the eye. Until recently, this phenomenon has only been observed at extremely cold temperatures because dipoles cannot synchronize at higher temperatures. But in a study, Melike Biliroglu and colleagues observed superfluorescence at room temperature in thin films made of perovskite and other similarly crystalline materials; the researchers propose that the formation of shock-absorbing quasiparticles called polarons in the material protects dipoles from thermal interference.

Based on the text, how are polarons believed to be involved in the superfluorescence observed in Biliroglu and colleagues’ study?

Answer

- A. Polarons enable superfluorescent bursts to cross from one crystalline material to another.
- B. Polarons allow for the dipoles to synchronize despite higher temperatures.
- C. Polarons accelerate the dipoles’ release of superfluorescent bursts.
- D. Polarons decrease the intensity of the superfluorescent burst.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately describes Biliroglu and colleagues’ claim about how the polarons function in relation to superfluorescence. The text indicates that “until recently,” superfluorescence (intense, synchronized bursts of light emitted by dipoles) has solely been observed at very cold temperatures. However, it also states that, recently, Biliroglu and colleagues report observing the phenomenon at room temperature. They achieved this using “thin films made of perovskite and other similarly crystalline materials,” which the researchers claim allows for the formation of polarons. They also suggest that these polarons might absorb the thermal shocks that typically disrupt dipole synchronization at warmer temperatures. Thus, based on the text, Biliroglu and colleagues believe that polarons help dipoles synchronize at temperatures well above those at which superfluorescence had previously been observed.

Choice A is incorrect because the text doesn’t address the prospect of a superfluorescent burst moving between crystalline materials or any other mediums. Choice C is incorrect because the text’s discussion of polarons is about how they might enable superfluorescence at higher temperatures than those at which it had previously been observed. Rather than suggesting that polarons speed up superfluorescent bursts, the text suggests that no superfluorescence can occur at room temperature in the absence of polarons. Thus, the text indicates that polarons make superfluorescent bursts more likely to occur at higher temperatures than those at which it had previously been observed, not that polarons accelerate the bursts. Choice D is incorrect because the text’s discussion of polarons is about how they might enable superfluorescence at higher temperatures than those at which it had previously been observed. In the absence of polarons, the text suggests there would be no superfluorescence at room temperature. Thus, rather than decrease the intensity of superfluorescent bursts, polarons make them more likely to occur under certain circumstances.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

O Pioneers! is a 1913 novel by Willa Cather. In the novel, Cather depicts Alexandra Bergson as a person who takes comfort in understanding the world around her: _____

Which quotation from *O Pioneers!* most effectively illustrates the claim?

Answer

- A. “She looked fixedly up the bleak street as if she were gathering her strength to face something, as if she were trying with all her might to grasp a situation which, no matter how painful, must be met and dealt with somehow.”
- B. “She had never known before how much the country meant to her. The chirping of the insects down in the long grass had been like the sweetest music. She had felt as if her heart were hiding down there, somewhere, with the quail and the plover and all the little wild things that crooned or buzzed in the sun. Under the long shaggy ridges, she felt the future stirring.”
- C. “Alexandra drove off alone. The rattle of her wagon was lost in the howling of the wind, but her lantern, held firmly between her feet, made a moving point of light along the highway, going deeper and deeper into the dark country.”
- D. “Alexandra drew her shawl closer about her and stood leaning against the frame of the mill, looking at the stars which glittered so keenly through the frosty autumn air. She always loved to watch them, to think of their vastness and distance, and of their ordered march. It fortified her to reflect upon the great operations of nature, and when she thought of the law that lay behind them, she felt a sense of personal security.”

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses a quotation from *O Pioneers!* to illustrate the claim that Alexandra Bergson takes comfort in understanding the world around her. In the quotation, Alexandra is described as enjoying looking at the stars and feeling a “sense of personal security” when she contemplates nature’s order and its governing laws. This suggests that Alexandra takes comfort in understanding the world around her.

Choice A is incorrect because the quotation expresses how Alexandra Bergson attempts to meet difficult situations with determination, not how she takes comfort in understanding the world around her. Choice B is incorrect because the quotation expresses “how much the country meant to” Alexandra Bergson, not how she takes comfort in understanding the world around her. In detailing some of the wildlife surrounding Alexandra, the quotation conveys that nature is important to her but not necessarily that it gives her comfort. Choice C is incorrect because the quotation describes Alexandra driving her wagon down a highway at night; it doesn’t describe how she takes comfort in understanding the world around her or address how she’s feeling as she drives off.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The domestic sweet potato (*Ipomoea batatas*) descends from a wild plant native to South America. It also populates the Polynesian Islands, where evidence confirms that Native Hawaiians and other Indigenous peoples were cultivating the plant centuries before seafaring first occurred over the thousands of miles of ocean separating them from South America. To explain how the sweet potato was first introduced in Polynesia, botanist Pablo Muñoz-Rodríguez and colleagues analyzed the DNA of numerous varieties of the plant, concluding that Polynesian varieties diverged from South American ones over 100,000 years ago. Given that Polynesia was peopled only in the last three thousand years, the team concluded that _____

Which choice most logically completes the text?

Answer

- A. the cultivation of the sweet potato in Polynesia likely predates its cultivation in South America.
- B. Polynesian peoples likely acquired the sweet potato from South American peoples only within the last three thousand years.
- C. human activity likely played no role in the introduction of the sweet potato in Polynesia.
- D. Polynesian sweet potato varieties likely descend from a single South American variety that was domesticated, not wild.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of the sweet potato in Polynesia. The text indicates that the sweet potato is found in Polynesia but originated in South America, and that the sweet potato was being cultivated by Native Hawaiians and other Indigenous peoples in Polynesia long before sea voyages between South America and Polynesia began. The text goes on to note that research by Muñoz-Rodríguez and colleagues has established that the Polynesian varieties of sweet potato split from South American varieties more than 100,000 years ago, which is thousands of years before humans settled in Polynesia. If Polynesian peoples were cultivating the sweet potato before sea voyages between Polynesia and South America began, and if Polynesian varieties of sweet potato diverged from South American varieties well before people were in Polynesia, it can reasonably be concluded that humans didn’t play a role in bringing the sweet potato to Polynesia.

Choice A is incorrect. The text doesn’t provide any information about when the sweet potato began to be cultivated in South America, so there’s no support for the conclusion that cultivation began in Polynesia before it began in South America. Choice B is incorrect because the text indicates that the sweet potato was being cultivated in Polynesia long before sea journeys between Polynesia and South America began. Therefore, it wouldn’t be reasonable to conclude that Polynesian peoples acquired the sweet potato from South American peoples. Additionally, the text indicates that the Polynesian varieties of sweet potato diverged from the South American varieties thousands of years before people settled in Polynesia, which suggests that the sweet potato was already present in Polynesia when people arrived. Choice D is incorrect because the text states that the domestic sweet potato, which is found in Polynesia, descends from a wild South American plant, not from a domesticated South American plant. The only people that the text describes as cultivating the sweet potato are Native Hawaiians and other Indigenous peoples of Polynesia.

Question ID: 8e6a96f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Duckweed is a small freshwater plant that is often exposed to zinc pollution. Sofia Vámos and colleagues collected samples of four duckweed ecotypes (genetically and geographically distinct populations within a species), along with water from each ecotype’s habitat. Hypothesizing that each ecotype is adapted to its local conditions in ways that bolster its growth and resistance to pollutants, the researchers grew each ecotype in all four water samples and with three levels of zinc (none, low, high). (The researchers did not replicate local differences in light or temperature.) They found that the ecotypes grew equally well in all four water samples and that adding zinc consistently enhanced growth, regardless of concentration, suggesting that _____

Which choice most logically completes the text?

Answer

- A. while the ecotypes are genetically and geographically distinct, those differences do not represent adaptations to local environmental conditions.
- B. there may not be significant differences in the water that each ecotype inhabits, but there are significant differences in each ecotype’s resistance to zinc pollution.
- C. if each ecotype is indeed locally adapted as the researchers hypothesized, those adaptations are to other environmental conditions than the water each ecotype inhabits.
- D. although the researchers’ hypothesis does not appear to be supported, this may be because the levels of zinc exposure the plants in the experiment received did not match their exposure in their natural environments.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of Vámos and colleagues’ experiment with four duckweed ecotypes. According to the text, the researchers hypothesized that each ecotype is adapted to its local habitat in ways that help it grow more effectively and better tolerate pollutants. To test this, they grew each ecotype in water collected from all four habitats and exposed them to three different zinc levels. If the hypothesis were correct, each ecotype would be expected to thrive more in water from its own habitat than in the other waters and to respond to zinc levels differently depending on the level of zinc pollution in its home environment. However, the results showed that every ecotype grew just as well in every water sample, and that zinc boosted growth the same way across all ecotypes no matter the concentration. In other words, the experiment found no evidence that these ecotypes have specifically adapted to their local water or their local zinc exposure. It follows, then, that if each ecotype is actually locally adapted, it is adapted not to the water conditions in its habitat but to other environmental factors—such as local light or temperature levels, which the text notes the researchers didn’t account for in their experiment.

Choice A is incorrect because the findings presented in the text suggest only that the ecotypes’ genetic and geographic differences don’t represent adaptations to local water conditions and zinc levels, not that the differences don’t represent adaptations to *any* local environmental conditions. Indeed, the text notes that the researchers didn’t account for light or temperature levels in each habitat, suggesting the possibility that the ecotypes are adapted to those or other local environmental factors. Choice B is incorrect because it contradicts the findings presented in the text. The text states that zinc consistently enhanced growth regardless of concentration, meaning the ecotypes all responded similarly to zinc. This uniform response indicates there *aren’t* significant differences in their resistance to zinc pollution. Choice D is incorrect because the findings presented in the text don’t demonstrate that the researchers’ hypothesis of local adaptation isn’t supported. That the ecotypes grew equally well in all of the water samples and equally benefited from added zinc at all levels, regardless of zinc levels in their natural environments, suggests only that the ecotypes aren’t adapted to their local water conditions and zinc levels; these findings don’t rule out the possibility that the ecotypes are adapted to other local factors in their habitats, such as light or temperature (which the text notes the researchers didn’t examine).

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Neural networks are computer models intended to reflect the organization of human brains and are often used in studies of brain function. According to an analysis of 11,000 such networks, Rylan Schaeffer and colleagues advise caution when drawing conclusions about brains from observations of neural networks. They found that when attempting to mimic grid cells (brain cells used in navigation), while 90% of the networks could accomplish navigation-related tasks, only about 10% of those exhibited any behaviors similar to those of grid cells. But even this approximation of grid-cell activity has less to do with similarity between the neural networks and biological brains than it does with the rules programmed into the networks.

Which finding, if true, would most directly support the claim in the underlined sentence?

Answer

- A. The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.
- B. The networks that do not exhibit behaviors like those of grid cells were nonetheless programmed with rules that had proven useful in earlier neural-network studies.
- C. Neural networks can often accomplish tasks that biological brains do, but they are typically programmed with rules to model multiple types of brain cells simultaneously.
- D. Once a neural network is programmed, it is trained on certain tasks to see if it can independently arrive at processes that are similar to those performed by biological brains.

Correct Answer: A

Rationale

Choice A is the best answer. While many networks can perform navigation tasks, or even mimic grid cells, it doesn't mean they're actually behaving like biological brains—this finding suggests that the rules that govern neural network behavior are completely unlike the way real brains work.

Choice B is incorrect. Although it mentions the rules that are programmed into the networks, this finding wouldn't clarify whether or not these rules have anything to do with the function of biological brains. Choice C is incorrect. This choice suggests that neural networks are modeled after multiple types of brain cells, which sidesteps the question of whether these rule-based networks are genuinely similar to biological brains. Choice D is incorrect. This choice doesn't address the key point of the claim, which is that the apparent similarity between neural networks and biological brains is only due to the rules programmed into the networks. It focuses on training tasks, not the originally programmed rules.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The morphological novelty of echinoderms—marine invertebrates with radial symmetry, usually starlike, around a central point—impedes comparisons with most other animals, in which bilateral symmetry on an anterior-posterior (head to tail) axis through a trunk is typical. Particularly puzzling are sea stars, thought to have evolved a headless layout from a known bilateral origin. Applying genomic knowledge of *Saccoglossus kowalevskii* acorn worms (close relatives of sea stars, and thus expected to have similar markers for corresponding anatomical regions) to the body patterning genes of *Patiria miniata* sea stars, Laurent Formery et al. observed activity only in anterior genes across *P. miniata*'s entire body and some posterior genes limited to the edges, suggesting that _____

Which choice most logically completes the text?

Answer

- A. despite the greater prevalence of anterior genes in sea stars’ genetic makeup, posterior genes active at the body’s perimeter are primarily responsible for the starlike layout that distinguishes sea stars’ radial symmetry from that of other echinoderms.
- B. contrary to the belief that they evolved from early ancestors with the bilateral form typical of many other animals, sea stars instead originated with an atypical body layout that was neither bilaterally nor radially symmetrical.
- C. although the two species are closely related, there is only minimal correspondence in the genetic markers for head, tail, and trunk region development in *P. miniata* sea stars and *S. kowalevskii* acorn worms.
- D. rather than undergoing changes resulting in the eventual elimination of a head region in their radial body plan, as previously assumed, sea stars’ morphology evolved to completely lack a trunk and consist primarily of a head region.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of the morphology (form and structure) of sea stars, a type of echinoderm. The text indicates that echinoderms have radially symmetrical body plans (symmetrical around a central point, usually in the form of a star), whereas most animals have bilaterally symmetrical body plans (symmetrical along an axis running from head to tail through a trunk). According to the text, sea stars are unusual echinoderms because, despite their radial body plan, they descended from known bilateral ancestors. This shift in body plan was thought to be a process of losing the genetic markers associated with the head region. The text explains that by comparing the genes of one sea star species (*P. miniata*) to those of a close relative, the acorn worm, researchers determined that instead, anterior (head) genes are active across the sea star’s entire body, posterior (tail) genes are active in limited, peripheral locations of the body, and no trunk-related genes are active. This finding strongly suggests that, rather than becoming "headless" as they evolved from a bilateral ancestor, sea stars developed a body plan consisting almost entirely of a head region with a minimal tail region and no trunk region present.

Choice A is incorrect because the text doesn’t identify how any particular region of sea stars’ bodies influences the layout of sea stars’ radial symmetry. Moreover, the text indicates that the radial symmetry of echinoderms is "usually starlike," not that a starlike layout distinguishes sea stars from other echinoderms. Choice B is incorrect because the text doesn’t suggest that the idea that sea stars evolved from an ancestor with bilateral symmetry is incorrect (describing the bilateral origin as "known") and doesn’t address any body plans other than those with radial or bilateral symmetry. The text strongly suggests that rather than revealing something about sea stars’ origin, Formery et al.’s findings contradict the assumption that the current body plan of sea stars is "headless." Choice C is incorrect because the text suggests that Formery et al. were able to make determinations about *P. miniata* sea stars’ body plan based on the comparability of genetic markers between *P. miniata* and *S. kowalevskii* acorn worms. The text indicates only that little or no activity was observed in certain types of genes associated with body development in *P. miniata*, not that those genes turned out to largely differ from body-development genes in *S. kowalevskii*.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Born in Chile in 1917, artist and ethnomusicologist Violeta Parra was a pioneer in the *nueva canción chilena* (Chilean New Song) movement that emerged in the late 1950s and then spread throughout Latin America, Portugal, and Spain as *nueva canción*. Parra traveled all over Chile compiling extensive records of authentic folk music as well as recipes, proverbs, and other facets of cultural history. These records formed the foundation for the early movement’s revival of traditional Chilean folk forms in new songs that represented modern realities of the working class and strongly advocated for social change. As the movement spread beyond Chile, the breadth of musical traditions incorporated into its foundation also expanded.

Which detail about songs associated with *nueva canción*, if true, would best illustrate the underlined claim?

Answer

- A. Many feature political commentary addressing contemporary issues that stemmed from shared experiences of European colonization in Latin American countries.
- B. Many demonstrate the stylistic influence of *corrido*, a genre of narrative songs from Mexico that had come to be characterized by political themes in the early 1800s.
- C. Many were written with parts meant to be played on the *quena*, a traditional flute used across Andean countries, including Chile.
- D. Many were produced by Argentinian artists in the late 1950s, with others by artists in additional Latin American countries first emerging soon after.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a detail that, if true, would most effectively illustrate the underlined claim that as *nueva canción* spread beyond Chile, the breadth of musical traditions incorporated into its foundation also expanded. The text explains that the movement was initially founded on traditional Chilean folk forms compiled by Parra, so if many *nueva canción* songs demonstrate the stylistic influence of *corrido*—a Mexican narrative song genre—that would show that a musical tradition originating outside Chile became part of the movement’s foundation, directly illustrating the text’s claim about how the breadth of *nueva canción* expanded.

Choice A is incorrect because the finding addresses lyrical content—political commentary on issues stemming from European colonization—rather than musical traditions. Even if *nueva canción* songs across different countries share political themes, that doesn’t indicate that new musical forms or styles were incorporated into the movement’s foundation. Thematic content is distinct from the musical traditions that songs draw on stylistically. Choice C is incorrect because the *quena* is described as a “traditional flute used across Andean countries, including Chile,” meaning it’s part of the Chilean folk tradition that underpinned *nueva canción* at its origin. Use of the *quena* therefore wouldn’t illustrate the incorporation of musical traditions beyond those that founded the early movement in Chile. Choice D is incorrect because the finding establishes only that artists in additional Latin American countries produced *nueva canción* songs, not that those artists incorporated musical traditions from their own countries into the movement. Artists in other countries could still have been drawing on the same Chilean folk forms that originally founded the movement, so geographic spread alone doesn’t illustrate an expansion of musical traditions.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

One theory behind human bipedalism speculates that it originated in a mostly ground-based ancestor that practiced four-legged “knuckle-walking,” like chimpanzees and gorillas do today, and eventually evolved into moving upright on two legs. But recently, researchers observed orangutans, another relative of humans, standing on two legs on tree branches and using their arms for balance while they reached for fruits. These observations may suggest that _____

Which choice most logically completes the text?

Answer

- A. bipedalism evolved because it was advantageous to a tree-dwelling ancestor of humans.
- B. bipedalism must have evolved simultaneously with knuckle-walking and tree-climbing.
- C. moving between the ground and the trees would have been difficult without bipedalism.
- D. a knuckle-walking human ancestor could have easily moved bipedally in trees.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of the evolution of bipedalism in humans. According to the text, one potential explanation for humans walking upright on two legs is that the behavior evolved from an ancestor that mostly stayed on the ground and walked on four limbs, as modern chimpanzees and gorillas do. However, the finding that orangutans, also a relative of humans, sometimes stand on two legs in trees while using their arms to balance and reach for fruits suggests another possible explanation: perhaps a tree-dwelling ancestor of humans began moving on two legs because it offered an advantage, such as access to certain foods.

Choice B is incorrect because the finding that modern orangutans (a relative of humans) sometimes stand on two legs in trees doesn’t offer any insight into when either bipedalism or tree-climbing behavior emerged in human ancestors. Additionally, the text indicates that one theory is that bipedalism evolved from a mostly ground-based ancestor that was already practicing knuckle-walking, not that bipedalism and knuckle-walking developed at the same time. Choice C is incorrect because the finding that orangutans (a relative of humans) sometimes stand on two legs in trees doesn’t offer any insight into how difficult it would’ve been to move between the ground and the trees without bipedalism; there’s no suggestion that climbing or moving in trees depends on the ability to walk on two legs rather than four, even if that ability might be helpful in certain circumstances. Choice D is incorrect because the finding that orangutans (a relative of humans) sometimes stand on two legs in trees doesn’t suggest that a knuckle-walking human ancestor could’ve easily moved on two legs in trees. Although the text indicates that bipedalism may have evolved from a human ancestor that mostly stayed on the ground and walked on four limbs, it gives no indication of how easy it would’ve been for such an ancestor to move bipedally in trees.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

As media consumption has become increasingly multiplatform and socially mediated, active news acquisition has diminished in favor of an attitude known as “news finds me” (NFM), in which people passively rely on their social networks and ambient media environments for information about current events. Homero Gil de Zúñiga and Trevor Diehl examined data on a representative group of adults in the United States to determine participants’ strength of NFM attitude, political knowledge, and political interest. Although no major election took place sufficiently near the study for Gil de Zúñiga and Diehl to identify causality between NFM and voting behavior, they did posit that NFM may reduce voting probability through an indirect effect.

Which finding, if true, would most directly support the idea advanced by Gil de Zúñiga and Diehl?

Answer

- A. NFM attitude tends to increase in strength as major elections approach, and people are significantly more likely to vote in major elections than in minor elections.
- B. NFM attitude has a strong negative effect on political knowledge and interest, and there is known to be a strong positive correlation between political knowledge and interest and the likelihood of voting.
- C. Political interest is known to have a strong positive effect on likelihood of voting but shows only a weak positive effect on political knowledge, and NFM attitude shows little correlation with either political knowledge or political interest.
- D. The likelihood of voting increases as political knowledge increases, and the relationship between NFM attitude and political knowledge tends to strengthen as the size of people’s social networks increases.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the idea advanced by Homero Gil de Zúñiga and Trevor Diehl that NFM (“news finds me”) attitude may reduce voting probability through an indirect effect. The text describes NFM as an attitude that has lowered people’s interest in actively acquiring news and introduces Gil de Zúñiga and Diehl’s study on the effects of NFM on people in the United States’ political knowledge and interest. The text goes on to say that despite the fact that the study didn’t occur near a major election, Gil de Zúñiga and Diehl still conclude that NFM may reduce voting probability. If the likelihood, or probability, that a person will vote is linked to that person’s level of political knowledge and interest, that would suggest that negatively affecting a person’s level of political knowledge and interest would also negatively affect how likely that person is to vote. Thus, if NFM attitude has a negative effect on political knowledge and interest, then it would also likely reduce voting probability.

Choice A is incorrect because the finding that NFM attitude increases as major elections approach wouldn’t address the effect of NFM on the likelihood of voting, which is the idea advanced by Gil de Zúñiga and Diehl. Moreover, although the text mentions that there were no major elections that occurred near the time of the study, it doesn’t discern between major and minor elections when discussing voting probability. Choice C is incorrect because finding that NFM attitude shows little correlation with either political knowledge or political interest would undermine, not support, Gil de Zúñiga and Diehl’s idea that NFM may reduce voting probability because it suggests that NFM has no effect on political knowledge or political interest. Choice D is incorrect. Although Gil de Zúñiga and Diehl’s idea would be supported by the finding that the likelihood of voting increases as political knowledge increases, nothing in the text suggests that the researchers’ idea hinges on the size of people’s social networks.

Question ID: 0c622cfb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Although it’s clear that Mars once had liquid water on its surface, astronomers have debated whether the evidence of ancient water reflects a prolonged phase of warm, wet conditions—the so-called wet and warm scenario—or a brief period of melting in an otherwise consistently frozen environment. Researchers Benjamin T. Cardenas and Michael P. Lamb recently added to this debate by using data from NASA and the Mars Orbiter Laser Altimeter to map the topography of what is now a large basin in Mars’s northern hemisphere. Cardenas and Lamb concluded that the wet and warm scenario is likely correct.

Which finding about the basin, if true, would most directly support Cardenas and Lamb’s conclusion?

Answer

- A. Its dimensions and shape indicate that it is unlikely to have formed as the result of an asteroid or comet impact.
- B. It has features suggesting that it once held an ocean that underwent gradual sea-level changes over an extended time.
- C. Its physical characteristics are most consistent with it having formed as a result of a massive but short-lived influx of liquid water.
- D. It is surrounded by channels that could have been formed either by running water or by flowing lava.

Correct Answer: B

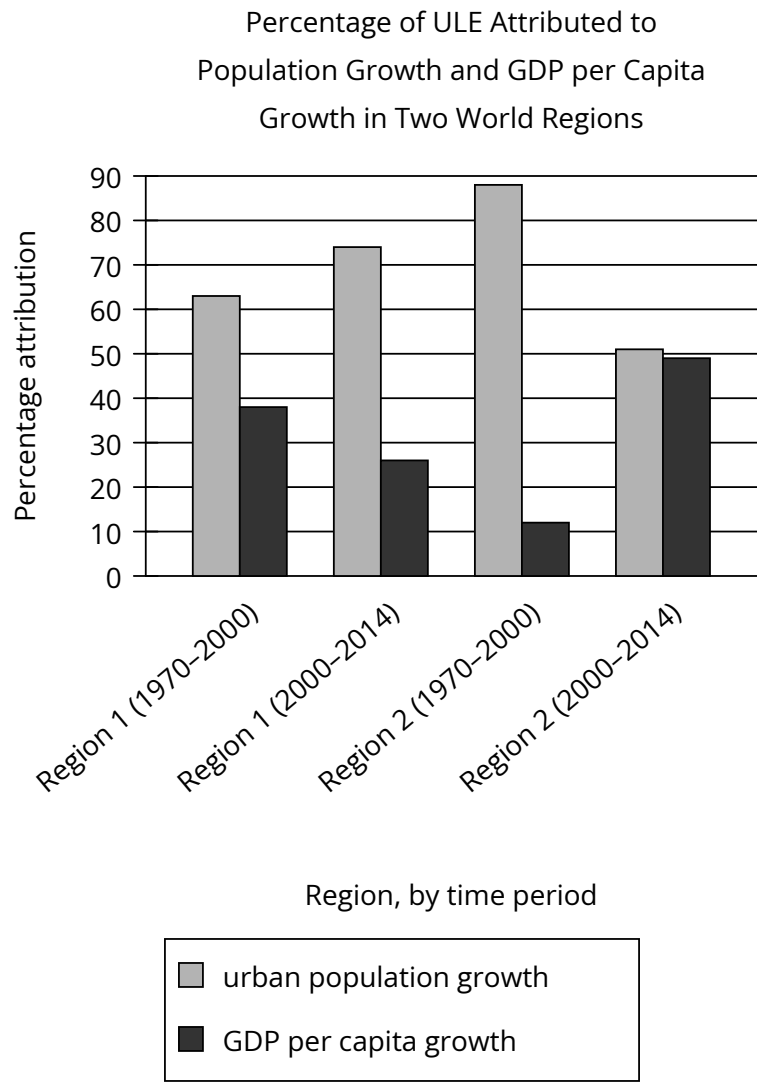
Rationale

Choice B is the best answer. This choice provides direct support for the researchers’ conclusion. If the basin once held an ocean of liquid water for "an extended time," that supports the "wet and warm scenario" on Mars, which involved a "prolonged" period of temperate conditions (and not just a "brief period of melting").

Choice A is incorrect. This choice doesn’t provide support for the researchers’ conclusions. The dimensions of the basin have little bearing on whether or not it was filled with liquid water, and for how long, and that’s the evidence that would support the "wet and warm" theory. Similarly, whether or not the basin was formed by an asteroid or a comet is irrelevant to the question of whether or not there was water in the basin for a long period. Choice C is incorrect. This choice does not provide support for the researchers’ conclusions, but rather the opposite. A "massive but short lived influx" of liquid water is not the same as a "prolonged phase of warm, wet conditions." It more reflects the opposing theory, a "brief period of melting." Choice D is incorrect. This choice does not support Cardenas and Lamb’s conclusion. Both theories about liquid water on Mars ("wet and warm" and "brief period of melting") involve flowing water, but lava isn’t mentioned at all in the text. Therefore, this choice doesn’t provide exclusive support for either theory.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



In a study of urban physical expansion, Richa Mahtta et al. conducted a meta-analysis of more than 300 cities worldwide to determine whether urban land expansion (ULE) was more strongly influenced by urban population growth or by growth in gross domestic product (GDP) per capita, a measure of economic activity. Because efficient national government is necessary to provide urban services and infrastructure that attract economic investment, Mahtta et al. propose that absent other factors, the importance of GDP per capita growth to ULE would likely increase relative to the importance of population growth as governments become more efficient. If true, this suggests the possibility that _____

Which choice most effectively uses data from the graph to complete the statement?

Answer

- A. national governments of countries in Region 1 experienced declines in efficiency in the period from 2000 to 2014, relative to the period from 1970 to 2000.
- B. countries in Region 1 experienced a slower rate of economic growth in the period from 2000 to 2014 than countries in Region 2 did, despite increasing national government efficiency in Region 1.

- C. national governments of most countries in Region 2 became more efficient in the period from 2000 to 2014 than they had been in the period from 1970 to 2000, but those of several countries in this region did not.
- D. national governments of countries in Region 1 and in Region 2 generally became more efficient in the period from 2000 to 2014 than they had been in the period from 1970 to 2000, but at different rates.

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the graph to complete the statement about Mahtta et al.'s proposal regarding factors that affect urban land expansion (ULE). According to the text, ULE is influenced by urban population growth and by gross domestic product (GDP) growth per capita. Reasoning that efficient national governments provide urban services and infrastructure needed to attract economic investment, Mahtta et al. suggest that, as governments become more efficient at providing urban services and infrastructure, GDP growth per capita will account for more ULE and urban population growth will account for less. But according to the graph, Region 1 saw an increase in the percentage attributed to urban population growth from 1970–2000 (between 60 and 65%) to 2000–2014 (between 70 and 75%) and a decrease in the percentage attributed to GDP growth per capita from 1970–2000 (between 35 and 40%) to 2000–2014 (about 25%). Because the percentage attributed to GDP growth per capita decreased (the opposite of what Mahtta et al. claimed would happen if the governments had become more efficient), the data suggest that the governments of Region 1 became less efficient at providing urban services and infrastructure over that period.

Choice B is incorrect. Neither the graph nor the text gives the regions' relative levels of economic growth or what effect Mahtta et al. would expect such growth to have. Furthermore, Mahtta et al.'s proposal suggests that Region 1's decline in the percentage of ULE attributed to GDP growth per capita from 1970–2000 (between 35 and 40%) to 2000–2014 (about 25%) would suggest decreasing, not increasing, government efficiency over this time. Choice C is incorrect. Neither the text nor the graph provides information about the relative efficiencies of different governments in Region 2. Choice D is incorrect. Mahtta et al.'s proposal suggests that more efficient governments will have a higher percentage of their ULE driven by GDP growth per capita and a lower percentage driven by urban population growth. For Region 2, the percentage of ULE attributed to GDP growth per capita increased from 1970–2000 (between 10 and 15%) to 2000–2014 (between 45 and 50%), but the opposite is true for Region 1, which saw the percentage of ULE attributed to GDP growth per capita decline over the same period. Thus, whereas the data suggest governments in Region 2 became more efficient, the data for Region 1 suggest that those governments became less efficient, not more.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Paleontologist Lucas E. Fiorelli and colleagues have reported the discovery at a mine in Brazil of several egg clutches, partially preserved single eggs, and egg shells from the Late Cretaceous period. The researchers have concluded that the area was once a nesting and breeding site for titanosaurs, a group of sauropod dinosaurs. The finding is significant given the previous lack of known nesting sites in northern regions of South America, which led many paleontologists to assume that titanosaurs migrated south to lay eggs.

What does the text most strongly suggest about the site discovered by the researchers?

Answer

- A. It is the earliest known example of a titanosaur nesting and breeding site.
- B. It was very difficult to excavate given that it was discovered in a mine.
- C. It may have been occupied by other sauropods in addition to titanosaurs.
- D. It is farther north than any other nesting site discovered in South America.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a statement about the site discovered by the researchers that is supported by the text. The text discusses Fiorelli and colleagues’ discovery of egg clutches, single eggs, and eggshells in a Brazilian mine. According to the text, the presence of these eggs, which are from the Late Cretaceous period, led the researchers to conclude that the location was once a nesting and breeding site for titanosaurs. The text then explains that the finding is important because of the “previous lack of known nesting sites in northern regions of South America.” If there haven’t been any other discoveries of a nesting site in South America’s northern regions and the site in the Brazilian mine is the first, then the text strongly suggests that the site is farther north than other nesting sites that have been discovered in South America.

Choice A is incorrect because the text doesn’t suggest that the site discovered by Fiorelli and colleagues is the earliest titanosaur nesting and breeding site known to paleontologists but rather that it’s the first nesting site found in northern regions of South America. Moreover, the text doesn’t suggest how the timeline of the newly discovered site compares with that of other titanosaur nesting and breeding sites. Choice B is incorrect because there is no mention in the text about any difficulties that Fiorelli and colleagues faced when they were excavating the nesting and breeding site in the Brazilian mine. Choice C is incorrect because the text doesn’t support the idea that the nesting and breeding site in the Brazilian mine was occupied by sauropods other than titanosaurs. The text simply mentions that titanosaurs are sauropod dinosaurs and presents the researchers’ conclusion that the site they discovered was for titanosaurs.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Within higher education, studying philosophy requires that students be conversant with the field’s foundational texts and historical figures. By contrast, *doing* philosophy within or beyond the academy demands the creative, self-directed application of acquired expertise to enduring questions about the nature of existence and knowledge. While both approaches engage with influential figures, those who do philosophy treat such figures as vital interlocutors who facilitate new insights rather than as ossified authorities who, though relevant to the present, primarily represent the discipline’s past.

Based on the text, which choice best describes the relationship between doing philosophy and studying philosophy?

Answer

- A. Doing philosophy helps students formulate concrete solutions to practical issues, whereas studying philosophy prioritizes engagement with historical arguments in the field.
- B. Doing philosophy involves developing novel ideas through imagined dialogue with past philosophers based on knowledge of those philosophers’ views acquired by studying philosophy.
- C. Doing philosophy requires students to challenge the ideas articulated by past philosophers, especially when these ideas are broadly accepted by other people studying philosophy.
- D. Doing philosophy represents a departure from the norms that govern scholarly inquiry, whereas studying philosophy requires conforming to these norms.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately describes the relationship between doing philosophy and studying philosophy as presented in the text. According to the text, studying philosophy requires being familiar with foundational texts and historical figures, while doing philosophy demands “the creative, self-directed application of acquired expertise”—a phrase implying that doing philosophy builds on knowledge gained through study. The text adds that those who do philosophy treat influential figures as “vital interlocutors who facilitate new insights” (that is, as dialogue partners whose ideas help generate new thinking). In other words, doing philosophy involves developing novel ideas through engagement with the ideas of past philosophers, knowledge of which can be acquired only by studying philosophy.

Choice A is incorrect. While the text does state that studying philosophy requires engagement with historical texts and figures, the text describes doing philosophy as addressing “enduring questions about the nature of existence and knowledge,” which are abstract philosophical questions, not practical issues. Rather than formulating concrete solutions, doing philosophy involves the creative application of expertise to these broad, enduring questions. Choice C is incorrect because the text doesn’t characterize doing philosophy as requiring students to challenge past philosophers’ ideas. Instead, the text suggests that those who do philosophy engage creatively with historical figures, which can, but doesn’t necessarily entail, opposition. Additionally, the text doesn’t claim that the ideas of past philosophers are broadly accepted by those studying philosophy. Choice D is incorrect because the text doesn’t discuss norms of scholarly inquiry or characterize either approach in terms of conforming to or departing from such norms. Rather, the text notes that doing philosophy can occur “within or beyond the academy,” which suggests compatibility with scholarly settings rather than an inevitable departure from them.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Declining fishing technology costs and overexploitation of near-shore fishing grounds have made isolated oceanic reefs, which often lack regulatory protections, increasingly attractive to commercial and sport fishers. A team led by Octavio Aburto-Oropeza surveyed the biomass density and species composition of two isolated reefs: Alacranes, a protected (fishing prohibited) reef 135 kilometers from the Yucatan Peninsula, and Bajos del Norte, an unprotected reef 25 kilometers farther out to sea. Species at the highest level of the trophic pyramid constituted 34% of the biomass at Alacranes and 10% of the biomass at Bajos del Norte. Aburto-Oropeza and colleagues attribute this difference to the two reefs’ difference in regulatory status.

Which finding, if true, would most directly support Aburto-Oropeza and colleagues’ explanation?

Answer

- A. Total biomass at Alacranes is much greater than total biomass at Bajos del Norte, though the reefs’ biomass densities are similar.
- B. Some of the species that compose the highest trophic level at Bajos del Norte are not found at Alacranes.
- C. It is somewhat more expensive for commercial and sport fishers to reach Bajos del Norte than it is to reach Alacranes.
- D. Commercial and sport fishers tend to disproportionately remove species at the highest trophic level.

Correct Answer: D

Rationale

Choice D is the best answer because it provides the finding that, if true, would most directly support Aburto-Oropeza and colleagues’ explanation that species composition differs between Alacranes and Bajos del Norte because of a difference in the regulation of fishing at the two reefs. The text explains that isolated oceanic reefs are often unprotected and are increasingly attracting commercial and sport fishing activity. According to the text, when Aburto-Oropeza and colleagues surveyed Alacranes, an isolated reef where fishing is prohibited, and Bajos del Norte, an isolated reef that is unprotected, they found that species at the highest trophic level (that is, top predators) make up a smaller proportion of the total mass of all living organisms at Bajos del Norte (10%) than they do at Alacranes (34%). If it were true that commercial and sport fishers tend to disproportionately remove top predators—that is, that they catch them at a higher rate than they catch other species—this would support the explanation that the allowance of fishing at Bajos del Norte and the prohibition of fishing at Alacranes has led to a lower proportion of top predator species remaining in the total biomass at Bajos del Norte.

Choice A is incorrect. To support the researchers’ claim that the role of fishing regulations accounts for Alacranes having a greater proportion of species at the highest trophic level (top predators) than Bajos del Norte does, a finding would need to help explain how fishing affects the presence of top predator species in particular; the finding that the two reefs have similar biomass densities (that is, about the same mass of living organisms per unit) and that Alacranes has a much greater total biomass (total mass of living organisms) wouldn’t reveal anything about the species composition at each reef or help explain why the *proportion* of top predators in the total biomass at each reef differs. This finding might simply suggest that Alacranes is a larger reef, with more units at that density. Choice B is incorrect because the finding that there are species at the highest trophic level (top predators) at Bajos del Norte that aren’t found at Alacranes would reveal only that there are some different species in the reefs’ total biomasses (total mass of living organisms) but wouldn’t explain why that’s the case; this finding wouldn’t be clearly connected to the role of fishing regulations, which is what, according to the researchers, accounts for the reefs having different proportions of top predators in their total masses. Choice C is incorrect. To support the researchers’ claim that the role of fishing regulations accounts for Alacranes having a greater proportion of species at the highest trophic level (top predators) than Bajos del Norte does, a finding would need to help explain how fishing affects the presence of top predator species in particular, but different costs of reaching the two reefs wouldn’t affect fishing behavior at all because fishing isn’t even allowed on one of the reefs.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In the 1980s, many musicians and journalists in the English-speaking world began to draw attention to music from around the globe—such as mbaqanga from South Africa and quan họ from Vietnam—that can't be easily categorized according to British or North American popular music genres, typically referring to such music as “world music.” While some scholars have welcomed this development for bringing diverse musical forms to prominence in countries where they'd previously been overlooked, musicologist Su Zheng claims that the concept of world music homogenizes highly distinct traditions by reducing them all to a single category.

Which finding about mbaqanga and quan họ, if true, would most directly support Zheng’s claim?

Answer

- A. Mbaqanga and quan họ developed independently of each other and have little in common musically.
- B. Mbaqanga is significantly more popular in the English-speaking world than quan họ is.
- C. Mbaqanga and quan họ are now performed by a diverse array of musicians with no direct connections to South Africa or Vietnam.
- D. Mbaqanga and quan họ are highly distinct from British and North American popular music genres but similar to each other.

Correct Answer: A

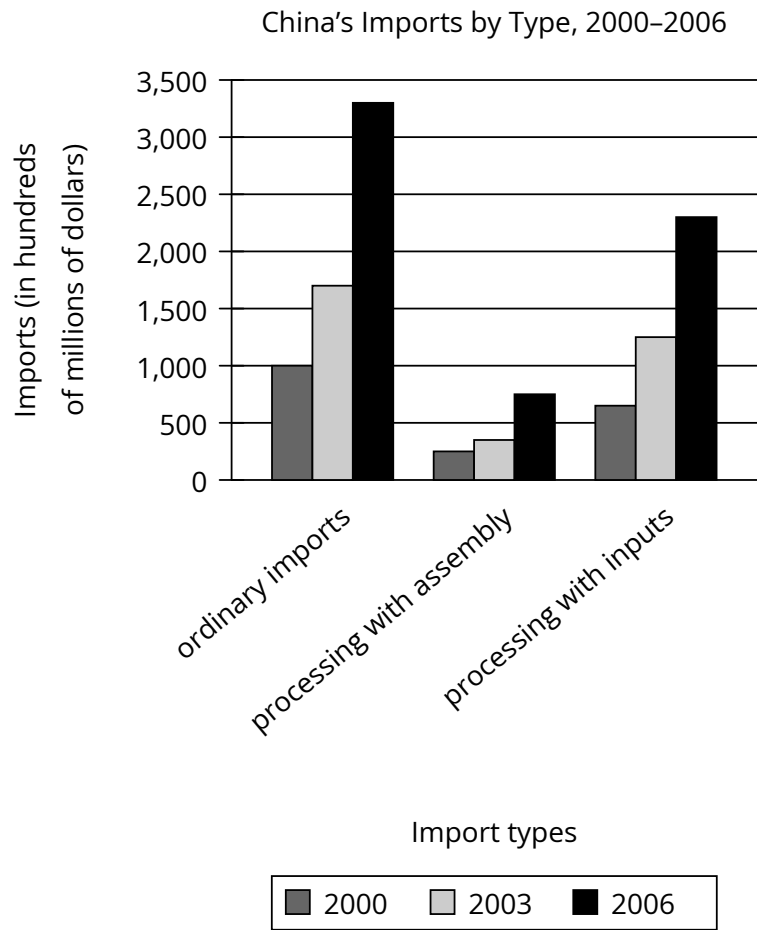
Rationale

Choice A is the best answer. Zheng’s claim is that the idea of world music “homogenizes” (meaning makes similar) distinct kinds of music by reducing them to one category. In other words, Zheng thinks the concept of world music is a harmful oversimplification of diverse musical forms. To support this claim, we need evidence that these musical traditions are so different from one another that they should not fall into the same category. If it’s true that mbaqanga and quan họ developed separately and have little in common musically, then it wouldn’t make sense to lump them into the same category.

Choice B is incorrect. If true, this wouldn’t affect the claim. To support the claim, we need evidence that these musical traditions are so different from one another that they should not fall into the same category. A difference in popularity doesn’t necessarily mean that the two musical traditions shouldn’t be categorized together: instead, we need to know if the music itself is similar or different. Choice C is incorrect. If true, this wouldn’t affect the claim. To support the claim, we need evidence that these musical traditions are so different from each other that they should not fall into the same category. This choice doesn’t do that. Choice D is incorrect. If true, this would actually weaken the claim. Zheng thinks it’s reductive or oversimplifying to put distinct musical traditions into a single category. But if mbaqanga and quan họ are similar to each other, then it would make sense to put them in the same category.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



A student is researching the Chinese government's 1992 shift to a market economy that emphasizes trade liberalization. One means of trade liberalization involves expanding from ordinary imports into an emphasis on processing imports, which have two types: processing with assembly (in which a firm obtains raw materials from a foreign trading partner without payment and sells the final goods to that partner, charging for assembly) and processing with inputs (in which a firm expends capital to buy raw materials from a trading partner, processes them into final goods, and sells those goods to whichever trading partner it chooses). The student asserts that while initial efforts at trade liberalization were shaped by Chinese firms' limited capital, this situation resolved during the 2000s.

Which choice best describes data from the graph that support the student's assertion?

Answer

- A. Processing imports with inputs were greater than both ordinary imports and processing imports with assembly in 2006.
- B. From 2000 to 2006, processing imports with inputs rose much more sharply than processing imports with assembly did.
- C. From 2000 to 2006, neither processing imports with inputs nor processing imports with assembly were greater than ordinary imports.
- D. Processing imports with assembly were greater in 2006 than processing imports with inputs in 2000.

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the graph that best support the student's assertion that initial efforts at trade liberalization in China were shaped by firms having limited capital (assets available for use) and that this situation resolved during the 2000s. The text explains that an approach to trade liberalization involves engaging in processing imports, one type of which doesn't require payment to a trade partner (processing with assembly) and one type of which requires upfront payment to a trade partner for raw materials (processing with inputs). The graph, which presents China's imports for ordinary imports and both types of processing imports in the years 2000, 2003, and 2006, shows that while processing imports with assembly rose from about 250 hundred million dollars in 2000 to about 750 hundred million dollars in 2006, processing imports with inputs rose much more sharply, increasing from approximately 650 hundred million dollars in 2000 to about 2,300 hundred million dollars in 2006. Because processing with inputs requires firms to pay for materials (expending capital) and processing with assembly doesn't, the sharper rise in processing imports with inputs suggests that Chinese firms' assets—and thus their ability to engage in that type of processing imports—were relatively limited in (and before) 2000 and then substantially increased from 2000 to 2006. In other words, the data suggest that the situation of having limited capital resolved during the 2000s.

Choice A is incorrect because the graph indicates that ordinary imports were greater than both types of processing imports in 2006, not that processing imports with inputs were greater than ordinary imports and processing imports with assembly that year. Choice C is incorrect because the observation that ordinary imports were greater than both types of processing imports in 2000, 2003, and 2006 doesn't address a change within any type of imports from 2000 to 2006, and an indication of a change in that period that might be related to the availability of assets is needed to support the assertion that the situation of having limited capital resolved during the 2000s. Choice D is incorrect because the fact that processing imports with assembly were greater at the end of the period from 2000 to 2006 than processing imports with inputs were at the start of the same period doesn't address a change within either type of imports during the period, and an indication of such a change that might be related to the availability of assets is needed to support the assertion that the situation of having limited capital resolved during the 2000s.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

One challenge when researching whether holding elected office changes a person’s behavior is the problem of ensuring that the experiment has an appropriate control group. To reveal the effect of holding office, researchers must compare people who hold elected office with people who do not hold office but who are otherwise similar to the office-holders. Since researchers are unable to control which politicians win elections, they therefore _____

Which choice most logically completes the text?

Answer

- A. struggle to find valid data about the behavior of politicians who do not currently hold office.
- B. can only conduct valid studies with people who have previously held office rather than people who presently hold office.
- C. should select a control group of people who differ from office-holders in several significant ways.
- D. will find it difficult to identify a group of people who can function as an appropriate control group for their studies.

Correct Answer: D

Rationale

Choice D is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the challenge researchers face when studying the effects of holding elected office on a person’s behavior. The text explains that it’s hard for researchers to test for the effects that elected office has on people because finding people to serve as a control group is difficult. The text indicates that a control group needs to be made up of people who share characteristics of the group being tested but don’t have the variable being tested (in this case, holding elected office). Because researchers aren’t able to influence who wins elections, they’re also unable to determine who would serve as an appropriately similar member of a control group. Thus, it logically follows that researchers will find it difficult to identify a group of people who can function as an appropriate control group for their studies.

Choice A is incorrect because the text focuses on the struggle to put together a control group for experiments; it doesn’t suggest that finding information about politicians’ behavior is difficult. Choice B is incorrect because the experiments mentioned in the text are testing the effects of holding elected office on a person’s behavior. Studying people who have already held elected office wouldn’t provide an opportunity to note any behavioral changes that the position might cause. Choice C is incorrect because the text defines people in a control group as those "who are otherwise similar to the office-holders"; selecting people who differ from the office-holders wouldn’t fit the criteria for an appropriate control group.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Having demonstrated that the power conversion efficiency of new perovskite photovoltaic cells is nearly indistinguishable from that of cells recycled multiple times, Xun Xiao and colleagues evaluated the levelized cost of electricity (LCOE)—a measure of cost per produced electricity unit that incorporates materials acquisition costs as well as operating costs—for power plants utilizing each type of cell. With a modeled 15-year cell lifetime, the LCOE of plants using new cells was 4.99 cents per kilowatt-hour, whereas that of plants using cells recycled three times was 4.05 cents per kilowatt-hour. Moreover, the LCOE differential grew from 18.8% to 31.3% when the modeled cell lifetime was reduced by two-thirds.

Based on the text, which choice, if true, most coherently and directly accounts for the observation presented in the underlined sentence?

Answer

- A. Cells can be recycled multiple times without a significant decrease in power conversion efficiency, and thus recycling cells more frequently does not change the LCOE for plants using recycled cells.
- B. Although potentially less expensive than producing new cells, recycling cells does incur costs, and thus reducing cell lifetimes entails greater cumulative costs.
- C. Securing new cell materials is more expensive than recycling cell materials, and thus more frequent cell replacement entails greater disparity in cumulative costs.
- D. As the modeled cell lifetime decreases, plant operating costs increase, and thus the LCOE for plants using new cells diverges from that of plants using recycled cells.

Correct Answer: C

Rationale

Choice C is the best answer because it presents information that, if true, would coherently and directly account for the researchers’ observation about the LCOE differential between plants using new cells and plants using recycled cells. The text explains that the researchers found that the LCOE—a cost-per-unit electricity measure that includes the cost of acquiring the materials—is higher for plants using new cells than for plants using cells recycled three times, with the gap between them growing from 18.8% to 31.3% when the researchers reduced the modeled cell lifetime by two-thirds (that is, from 15 years to 5 years). If obtaining new cell materials were more expensive than recycling preexisting cell materials, then plants using new cells would incur greater materials costs than plants using recycled cells when cells need to be replaced, and the greater frequency of cell replacement entailed by a shorter cell lifetime would force plants using new cells to incur that greater cost more often—widening the gap between those plants and plants using recycled cells.

Choice A is incorrect because the information that efficiencies are similar between new and recycled cells (which the text already establishes) and that frequency of recycling doesn’t affect the LCOE for plants using recycled cells wouldn’t address the effect of shorter cell lifetimes on the LCOE of plants using new cells. Information about that effect would be needed to account for the observation that the gap between the LCOEs for the two types of plants widens when cell lifetimes are shortened. Choice B is incorrect. Although the choice does point to the possibility of a scenario that could explain the growth in the LCOE differential (“potentially less expensive than producing new cells”), it presents that information as a concession (“although”) to its central assertion, which is that the cumulative costs would increase if cells had to be replaced more frequently than on the original 15-year timeline. This choice doesn’t indicate whether the increased cumulative costs in question are those of recycled-cell plants or both types of plants; either way, an increase in cumulative costs isn’t a coherent or direct explanation for why the difference in costs between new-cell plants and recycled-cell plants increases when the cell lifetime is reduced by two-thirds. A coherent and direct explanation for growth in the cost difference between the two types of plants must focus on a disparate effect on the plants’ costs; choice B, however, presents information that could be but is not necessarily relevant to a disparate effect and then doesn’t draw the relevant conclusion

from it, instead drawing a conclusion about cumulative costs that doesn't entail a growing differential. Choice D is incorrect because even if plant operating costs increase as cell lifetime decreases, that would be true for plants using new cells and for plants using recycled cells and thus wouldn't be responsible for a divergence; therefore, the finding wouldn't coherently and directly explain the observation that the gap between the LCOEs of the two types of plants widens as cell lifetimes are reduced.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



A sociology student is reading an essay on the median age of first marriage in Western countries throughout the twentieth century. The author of the essay cites factors common to these countries that the author believes caused an increase in the median age of first marriage, such as new technologies that shortened the time needed for domestic chores, making two-person households less necessary and living alone more viable. The student asserts that beyond these factors there must be additional ones specific to particular Western countries that influenced the increase of age at first marriage.

Which choice most effectively uses data from the graph that support the student’s assertion?

Answer

- A. Between 1970 and 2000, the median age of first marriage rose more sharply for men in England and Wales than it did for men in the United States.
- B. The median age of first marriage for men in England and Wales was lower in 1970 than in 1950 or 1990.
- C. Between 1900 and 2000, the median age of first marriage for women in England and Wales was consistently higher than for women in the United States, as was the case for men.

D. In England and Wales, the median age of first marriage was consistently higher for men than for women between 1900 and 2000, but this was not always the case in the United States.

Correct Answer: A

Rationale

Choice A is the best answer. The student concluded that country-specific factors influence changes to median age of first marriage. This choice describes a time period when the rate of change differed between countries, suggesting that country-specific factors may have played a role in these changes.

Choice B is incorrect. The student concluded that country-specific factors influence changes to median age of first marriage. However, this choice doesn't provide any contrasts between countries and thus doesn't support the idea of country-specific factors influencing median age of first marriage. Choice C is incorrect. The students' conclusion is about changes that occurred during the 20th century. This choice provides broad information about the century as a whole, so it doesn't give insight into how median age at first marriage changed over time. Choice D is incorrect. This choice misreads the graph. The median age of first marriage was consistently higher for men than for women in the United States during the time period depicted.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Elizabeth Asiedu has identified a negative correlation between the share of developing countries’ economies derived from natural-resource extraction and those countries’ receipts of foreign investment. This may appear counterintuitive—resource extraction requires initial investments (in extractive technology, for instance) at scales best met by multinational corporations—but Asiedu notes that natural-resource industries’ boom-bust cycle can destabilize local currencies and increase developing countries’ vulnerability to external shocks, creating levels of uncertainty to which foreign investors are typically averse.

Which choice best states the main idea of the text?

Answer

- A. Although it may seem surprising that foreign investment declines in developing countries as natural-resource extraction makes up a larger share of those countries’ economies, that decline happens because resource extraction requires initial investments too large for foreign investors to supply.
- B. Although developing countries tend to become less dependent on foreign investment as natural-resource industries make up a larger share of their economies, this change may not occur if the boom-bust cycle of those industries destabilizes local currencies or increases countries’ vulnerability to external shocks.
- C. Although one might expect that foreign investment would increase as natural-resource extraction makes up a larger share of developing countries’ economies, the opposite happens because heavy reliance on natural resources can lead to unattractive conditions for investors.
- D. Although foreign investors tend to avoid initial investments in natural-resource industries in developing countries, foreign investment may increase significantly as those industries stabilize and the risks associated with them decline.

Correct Answer: C

Rationale

Choice C is the best answer because it accurately states the main idea of the text. According to the text, contrary to what some might expect, foreign investment is typically lower in developing countries whose economies are more dependent on natural-resource extraction. The text explains that high reliance on natural-resource extraction can subject a developing country to economic shocks that can destabilize the local currency and introduce economic uncertainty that tends to keep investors away. In other words, although we may think otherwise, foreign investors are less willing to invest in projects in developing countries whose economies are heavily dependent on natural-resource extraction because those economies tend to exhibit instability that investors want to avoid.

Choice A is incorrect. The text does indicate that foreign investment is typically lower in developing countries whose economies are more dependent on natural-resource extraction; the text further indicates that natural-resource extraction requires substantial initial investments (to acquire things like required technologies) for which there are fewer investors willing to participate at this stage than one might think. But the text does not implicate the cost of these initial investments as a reason why foreign investment is less widely available than some might think.

Choice B is incorrect. The text indicates that greater dependence on natural-resource extraction makes a developing country less appealing to foreign investors because of associated economic instability. Rather than arguing that the goal of developing countries is to become less dependent on foreign investment, as the phrasing of choice B suggests, the text focuses only on why foreign investors become less involved with such countries, which suggests that more investment would be preferable. Choice D is incorrect. Although the text indicates that natural-resource extraction requires substantial initial investments (to acquire things like required technologies) and that there are fewer likely investors willing to participate at this stage than one might think, the text does not address what investors are likely to do over time as the industry stabilizes itself.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Sense and Sensibility is an 1811 novel by Jane Austen. In the novel, Austen describes Marianne Dashwood’s ability to persuade others of the rightness of her artistic judgments, as is evident when Marianne visits with John Willoughby, a potential suitor: _____

Which quotation from *Sense and Sensibility* most effectively illustrates the claim?

Answer

- A. “Above all, when she heard him declare, that of music and dancing he was passionately fond, she gave him such a look of approbation as secured the largest share of his discourse to herself for the rest of his stay.”
- B. “Their taste was strikingly alike. The same books, the same passages were idolized by each—or if any difference appeared, any objection arose, it lasted no longer than till the force of her arguments and the brightness of her eyes could be displayed.”
- C. “It was only necessary to mention any favourite amusement to engage her to talk. She could not be silent when such points were introduced, and she had neither shyness nor reserve in their discussion.”
- D. “They speedily discovered that their enjoyment of dancing and music was mutual, and that it arose from a general conformity of judgment in all that related to either. Encouraged by this to a further examination of his opinions, she proceeded to question him on the subject of books.”

Correct Answer: B

Rationale

Choice B is the best answer. By showing that "any difference" in taste was quickly overcome by "the force of [Marianne's] arguments," this choice effectively demonstrates Marianne’s "ability to persuade others."

Choice A is incorrect. This choice doesn’t effectively illustrate the claim. This choice shows that Marianne and John share an interest in music and dancing, but it doesn’t provide evidence of Marianne’s "ability to persuade others." Choice C is incorrect. This choice doesn’t effectively illustrate the claim. This choice shows that Marianne enjoys talking about her interests, but it doesn’t provide evidence of Marianne’s "ability to persuade others." Choice D is incorrect. This choice doesn’t effectively illustrate the claim. This choice shows that Marianne and John share many interests and generally agree on music and dancing, but it doesn’t provide evidence of Marianne’s "ability to persuade others."

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

A social species, chickens will cry out to one another in warning if they sense a hawk or other predator nearby. But if alone, a chicken will remain silent so as not to attract the predator’s attention. Sonja Hillemacher decided to use this behavior to determine whether individual chickens possess a capacity to recognize themselves visually when reflected in a mirror (a common standard for animal intelligence). In the first condition of her study, the subject could see its reflection, but no other chickens were present. In a second condition, another chicken was visible to the subject. Hillemacher presented an image of a hawk to the subject in both conditions, reasoning that if chickens lacked a capacity for visual self-recognition, then _____

Which choice most logically completes the text?

Answer

- A. neither study condition would elicit an audible response from the subject.
- B. the subject likely would cry out a warning in both study conditions.
- C. only the first study condition would elicit an audible response from the subject.
- D. the subject would fail to distinguish its reflection from the image of the hawk.

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of Hillemacher’s study of visual self-recognition in chickens. According to the text, when a predator is nearby, chickens cry out in warning if another chicken is present but remain silent if alone to avoid attracting the predator’s attention. The text explains that Hillemacher set up one condition in which a subject was alone and could see its reflection in a mirror and another condition in which the subject could see another chicken, and then presented an image of a hawk (a predator). It follows that if chickens aren’t capable of recognizing themselves visually in a mirror, the subject likely would cry out a warning in both conditions; in the first condition because it would perceive its reflection as another chicken—that is, it would think another chicken is present—and in the second condition because it would see that another chicken is actually present.

Choice A is incorrect because the text suggests that regardless of chickens’ ability to visually recognize themselves in mirrors, the subject in the second condition would cry out a warning because another chicken is actually present. Further, if chickens do lack visual self-recognition, the subject in the first condition likely would also cry out because it would think its reflection is another chicken. Choice C is incorrect. The text does suggest that if chickens lack the ability to visually recognize themselves in mirrors, the subject would cry out a warning in the first condition because it would think its reflection is another chicken. However, the text also indicates that regardless of their capacity of self-recognition, chickens cry out a warning when other chickens are present, and another chicken is present in the second condition. Therefore, both conditions—not just the first one—likely would elicit an audible response from the subject. Choice D is incorrect because the text indicates that the study was focused on determining whether chickens recognize themselves visually in a mirror or instead perceive their reflections as other chickens; there’s no suggestion that a lack of self-recognition would cause a chicken to confuse its reflection with the image of a hawk.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht’s influence can be seen in English playwright Caryl Churchill’s 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____

Which choice most logically completes the text?

Answer

- A. focus on the characters’ beliefs about social and political issues as revealed by the characters’ actions.
- B. reflect on social and political phenomena not directly related to patriarchy and colonialism.
- C. recognize pertinent social and political parallels between Germany during Brecht’s time and England at the time when Churchill was writing *Cloud 9*.
- D. be dispassionate as they think critically about the social and political questions raised by the play.

Correct Answer: D

Rationale

Choice D is the best answer. Churchill’s play was influenced by Brecht’s belief that theater should elicit an intellectual rather than an emotional response from audiences, making them think about social and political issues that also exist outside of the play. Therefore, it makes sense that Churchill would strive to have audiences think dispassionately (i.e., without emotion) and critically about the social and political questions raised by the play.

Choice A is incorrect. The passage mentions that Churchill was influenced by Brecht’s belief that plays should provoke audience members “to consider social and political realities that extend beyond the characters and events depicted onstage,” so there’s no basis for an inference about audience members deeply engaging with characters’ beliefs and actions. Choice B is incorrect. Reflecting on things that aren’t related to patriarchy and colonialism wouldn’t “engage audiences in an interrogation of patriarchy and colonialism.” That creates a confusing contradiction. Choice C is incorrect. No mention is made of social and political issues in Germany during Brecht’s time, so there’s no basis for this inference.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In episodes of high heat, corals expel symbiotic zooxanthellae from their tissues, leading to coral bleaching and heightened coral mortality. Some researchers have proposed a mitigation strategy of identifying, rearing, and disseminating corals that have evolved strong resistance to heat, but as Nia S. Walker et al. note, if genes conferring such resistance were exclusively advantageous, they should have already propagated through coral populations. Indeed, Walker et al. show that strongly heat-resistant corals display much less growth after heat stress than do moderately heat-resistant corals.

Based on the text, Walker et al. would most likely agree with which statement about the strategy some researchers have proposed?

Answer

- A. It would involve promoting a trait that protects corals against high heat but that is associated with reduced thriving after episodes of high heat.
- B. It would favor a trait that is advantageous now but would be disadvantageous if episodes of high heat become as infrequent as has been predicted.
- C. It would benefit corals that are frequently exposed to episodes of high heat but would harm corals that are frequently exposed to episodes of moderate heat.
- D. It would require manipulating a trait that is poorly understood and that may be linked to increased coral mortality during episodes of high heat.

Correct Answer: A

Rationale

Choice A is the best answer because it presents a statement about the proposed strategy that is consistent with Walker et al.'s position. The strategy involves identifying and disseminating corals with strong heat resistance as a means of mitigating coral mortality during episodes of high heat. According to the text, Walker et al. note that if this trait were "exclusively advantageous," it would have already spread throughout coral populations—implying that the trait carries some evolutionary cost. The text then provides evidence of this cost: Walker et al. have found that strongly heat-resistant corals display much less growth after heat stress (indicative of reduced thriving) than do moderately heat-resistant corals. Given the apparent trade-off between strong heat resistance and growth after heat stress, Walker et al. would be likely to agree that the strategy entails promoting a trait that protects against high heat but is associated with reduced thriving afterward.

Choice B is incorrect because the text doesn't suggest that episodes of high heat are predicted to become infrequent. Also, the text establishes that Walker et al. found that the trait of high heat tolerance is associated with a significant trade-off—reduced growth after heat stress. Given the cost that accompanies strong heat resistance, Walker et al. wouldn't be likely to agree that the trait is advantageous now. Choice C is incorrect because the text doesn't discuss the effects of moderate heat on corals or distinguish between corals that are exposed to different levels of heat. The text also doesn't address the frequency of corals' exposure to heat. Therefore, there is no basis for concluding that Walker et al. would agree that the strategy benefits corals that are frequently exposed to high heat but harms those that are frequently exposed to moderate heat. Choice D is incorrect because the text doesn't indicate that heat resistance in corals is poorly understood. On the contrary, Walker et al. present a specific finding about this trait, linking reduced growth after heat stress to strong heat resistance. In addition, the text indicates that strong heat resistance protects corals during episodes of high heat; while their research shows that heat resistance is linked to reduced growth after such episodes, Walker et al. likely wouldn't agree that it is linked to increased coral mortality during such episodes.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Several scholars have argued that conditions in England in the late ninth through early eleventh centuries—namely, burgeoning literacy amid running conflicts between England’s Anglo-Saxon kingdoms and Danish invaders—were especially conducive to the production of the Old English epic poem *Beowulf*, and they have dated the poem’s composition accordingly. It is not inconceivable that *Beowulf* emerged from such a context, but privileging contextual fit over the linguistic evidence of an eighth- or even seventh-century composition requires a level of justification that thus far has not been presented.

Which choice best states the main idea of the text?

Answer

- A. Although there are some grounds for believing that *Beowulf* was composed between the late ninth and early eleventh centuries, advocates for that view tend to rely on evidence that has been called into question by advocates for an earlier date.
- B. Although several scholars have dated *Beowulf* to the late ninth through early eleventh centuries, others have argued that doing so privileges a controversial interpretation of the social conditions of the period.
- C. Although *Beowulf* fits well with the historical context of England in the late ninth through early eleventh centuries, it fits equally well with the historical context of England in the seventh and eighth centuries.
- D. Although the claim of a late ninth- through early eleventh-century composition date for *Beowulf* has some plausibility, advocates for the claim have not compellingly addressed evidence suggesting an earlier date.

Correct Answer: D

Rationale

Choice D is the best answer because it most accurately states the main idea of the text. The text states that some scholars have dated the composition of *Beowulf* to the late ninth through early eleventh centuries due to the poem’s fit with that period’s historical context. The text goes on to say that while it is “not inconceivable that *Beowulf* emerged from such a context”—that is, it is possible that *Beowulf* was composed during the late ninth through eleventh centuries—there is linguistic evidence that the poem was composed earlier, in the seventh or eighth century. According to the text, favoring the historical context over the linguistic evidence requires justification that scholars have not yet supplied. In other words, the text suggests that scholars who favor the later composition date need to explain why the poem’s fit with historical context should take precedence over the linguistic evidence, but they have not yet done so. Thus, the main idea of the text is that while there is some plausibility to the later composition date, advocates for the later date have not compellingly addressed evidence suggestive of an earlier date.

Choice A is incorrect because the text says that scholars who date the poem to the late ninth through early eleventh centuries have failed to account for the linguistic evidence that the poem may have been composed earlier, not that the evidence those scholars cite in favor of their view is unreliable or that anyone has cast doubt on that evidence. In other words, the text does not suggest that there are problems with the evidence cited by advocates of the later composition date, only that there is other evidence of an earlier composition date that those advocates need to consider. Choice B is incorrect because nothing in the text suggests that those scholars who date the poem to the late ninth through early eleventh centuries are giving priority to a controversial view of the social conditions at that time. The text makes no reference to any controversy about how scholars interpret that historical period. Instead, the text suggests that scholars who date the poem on the basis of its fit with the historical context of England in the late ninth through early eleventh centuries have failed to account for linguistic evidence that the poem may have been composed earlier. Choice C is incorrect because the text says nothing about how well the poem fits the historical context of England in the seventh and eighth centuries, let alone that it fits that historical context as well as it fits the historical context of the late ninth through early eleventh centuries. Rather, the text says that there is linguistic evidence that the poem may have been composed in the seventh or eighth century.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Employment by Sector in France and the United States, 1800–2012 (% of total employment)

Year	Agriculture in France	Manufacturing in France	Services in France	Agriculture in US	Manufacturing in US	Services in US
1800	64	22	14	68	18	13
1900	43	29	28	41	28	31
1950	32	33	35	14	33	53
2012	3	21	76	2	18	80

Rows in table may not add up to 100 due to rounding.

Over the past two hundred years, the percentage of the population employed in the agricultural sector has declined in both France and the United States, while employment in the service sector (which includes jobs in retail, consulting, real estate, etc.) has risen. However, this transition happened at very different rates in the two countries. This can be seen most clearly by comparing the employment by sector in both countries in _____

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. 1900 with the employment by sector in 1950.
- B. 1800 with the employment by sector in 2012.
- C. 1900 with the employment by sector in 2012.
- D. 1800 with the employment by sector in 1900.

Correct Answer: A

Rationale

Choice A is the best answer because it presents data from the table that most effectively complete the statement about the rates at which employment shifted in France and the United States. The text states that over the last two hundred years employment in the agricultural sector has declined while employment in the service sector has risen in both France and the US, and the data from the table reflect these trends. The text asserts, however, that the transition from agriculture to services “happened at very different rates in the two countries.” This assertion is best supported by a comparison of data from 1900 and 1950: the table shows that in those years, employment in agriculture went from 43% to 32% in France (a decline of 11 percentage points) and from 41% to 14% in the US (a decline of 27 percentage points), and that employment in services went from 28% to 35% in France (an increase of 7 percentage points) and from 31% to 53% in the US (an increase of 22 percentage points). In other words, the rate of change was greater in the US than in France for both sectors.

Choice B is incorrect because comparing the data for 1800 and 2012 would suggest a similar rate of change in the two countries, not very different rates: employment in agriculture went from 64% in 1800 to 3% in 2012 in France, which is close to the change from 68% in 1800 to 2% in 2012 in the US, while employment in services went from 14% in 1800 to 76% in 2012 in France, which is close to the change from 13% in 1800 to 80% in 2012 in the US. Choice C is incorrect because comparing the data for 1900 and 2012 would suggest a similar rate of change in the two countries rather than very different rates: employment in agriculture went from 43% in 1900 to 3% in 2012 in France, which is close to the change from 41% in 1900 to 2% in 2012 in the US, while employment in services went from 28% in 1900 to 76% in 2012 in France, which is close to the

change from 31% in 1900 to 80% in 2012 in the US. Choice D is incorrect because comparing the data for 1800 and 1900 would suggest a similar rate of change in the two countries, not very different rates: employment in agriculture went from 64% in 1800 to 43% in 1900 in France, which is fairly close to the change from 68% in 1800 to 41% in 1900 in the US, while employment in services went from 14% in 1800 to 28% in 1900 in France, which is close to the change from 13% in 1800 to 31% in 1900 in the US.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Archaeologist Petra Vaiglova, anthropologist Xinyi Liu, and their colleagues investigated the domestication of farm animals in China during the Bronze Age (approximately 2000 to 1000 BCE). By analyzing the chemical composition of the bones of sheep, goats, and cattle from this era, the team determined that wild plants made up the bulk of sheep’s and goats’ diets, while the cattle’s diet consisted largely of millet, a crop cultivated by humans. The team concluded that cattle were likely raised closer to human settlements, whereas sheep and goats were allowed to roam farther away.

Which finding, if true, would most strongly support the team’s conclusion?

Answer

- A. Analysis of the animal bones showed that the cattle’s diet also consisted of wheat, which humans widely cultivated in China during the Bronze Age.
- B. Further investigation of sheep and goat bones revealed that their diets consisted of small portions of millet as well.
- C. Cattle’s diets generally require larger amounts of food and a greater variety of nutrients than do sheep’s and goats’ diets.
- D. The diets of sheep, goats, and cattle were found to vary based on what the farmers in each Bronze Age settlement could grow.

Correct Answer: A

Rationale

Choice A is the best answer because it presents a finding that, if true, would most strongly support the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were. The text explains that Vaiglova, Liu, and their colleagues analyzed the chemical composition of sheep, goat, and cattle bones from the Bronze Age in China in order to investigate the animals’ domestication, or their adaptation from a wild state to a state in which they existed in close connection with humans. According to the text, the team’s analysis showed that sheep and goats of the era fed largely on wild plants, whereas cattle fed on millet—importantly, a crop cultivated by humans. If analysis of the animal bones shows that the cattle’s diet also consisted of wheat, another crop cultivated by humans in China during the Bronze Age, the finding would support the team’s conclusion by offering additional evidence that cattle during this era fed on human-grown crops—and, by extension, that humans raised cattle relatively close to the settlements where they grew these crops, leaving goats and sheep to roam farther away in areas with wild vegetation, uncultivated by humans.

Choice B is incorrect because if it were true that sheep’s and goats’ diets consisted of small portions of millet, which the text states was a crop cultivated by humans, the finding would suggest that sheep and goats were raised relatively close to human settlements, weakening the team’s conclusion that cattle were likely raised closer to those settlements than sheep and goats were. Choice C is incorrect because the finding that cattle generally require more food and nutrients than do sheep and goats wouldn’t support the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were. Nothing in the text suggests that cattle were incapable of obtaining sufficient food and nutrients without access to human-grown crops. Hence, even if cattle’s diets are found to have different requirements than the diets of sheep and goats, the cattle could have met those requirements from food located far from human settlements. Choice D is incorrect because if it were true that the diets of sheep, goats, and cattle varied based on what the farmers in each Bronze Age settlement could grow, the finding would weaken the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were, suggesting instead that all three types of animals were raised close enough to human settlements to feed on those settlements’ crops.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Archaeologists and historians used to believe that the Maya civilization during its Classic period (roughly 250–900) lacked agricultural marketplaces. One reason for this belief was that these scholars misunderstood the ecology of the regions the Maya inhabited. Marketplaces typically emerge because different individuals or groups want to trade resources they control for resources they don't control. Scholars seriously underestimated the ecological diversity of the Maya landscape and thus assumed that _____

Which choice most logically completes the text?

Answer

- A. marketplaces likely would not have attracted many traders from outside the regions controlled by the Maya.
- B. farming practices would have been largely the same throughout Maya lands even if the crops people produced varied significantly.
- C. marketplaces would not have enabled Maya people to acquire many products different from those they already produced.
- D. farmers would trade agricultural products only if they had already produced enough to meet their own needs.

Correct Answer: C

Rationale

Choice C is the best answer because it presents the conclusion that most logically follows from the text's discussion of scholars' understanding of Maya ecology and agricultural marketplaces. The text indicates that scholars used to believe that during the Classic period, the Maya civilization didn't have agricultural marketplaces. According to the text, scholars held this view because they misunderstood the ecology of areas where the Maya lived. The text points out that people tend to create marketplaces in order to acquire resources they don't otherwise control. Agricultural marketplaces would have allowed farmers who produced one type of crop to trade that crop for other types of crops that they didn't produce. The text goes on to say, however, that scholars underestimated the ecological diversity of the Maya areas, meaning that scholars thought that the Maya landscape produced a smaller range of resources than it actually produced. Taken together, then, this information suggests that scholars assumed that marketplaces wouldn't have allowed Maya people to acquire products different from the products they already produced: that is, if everyone produced the same array of crops, as scholars mistakenly believed, then there wouldn't have been any need for marketplaces where people could trade those crops.

Choice A is incorrect because the text doesn't say anything about trade between the Maya and people from outside the regions controlled by the Maya. Although scholars' mistaken belief that the Maya lands weren't very ecologically diverse would give those scholars a reason to think that the Maya didn't have marketplaces, it wouldn't lead scholars to assume that traders from outside Maya lands were uninterested in acquiring resources produced by the Maya. Even if the Maya actually did produce only a small array of resources throughout their lands, there is no reason to believe from the text that people outside Maya lands also produced these same resources and thus would have no need to trade with the Maya people. Choice B is incorrect because the text indicates that scholars underestimated the ecological diversity of the Maya lands, which suggests that they mistakenly believed that the Maya produced a relatively small array of resources throughout their territory, not that the crops the Maya produced varied significantly throughout the Maya lands. Although the scholars might have assumed that a lack of ecological diversity suggests that Maya farming practices were largely the same everywhere, the text does not support that they also assumed there was a lot of variation in the crops that Maya people produced. In fact, the text states that marketplaces emerge when people want to obtain resources they don't already control. If it were the case that scholars assumed that the crops Maya people produced varied significantly, this would have led them to conclude that Maya people likely established marketplaces so they could trade for resources they didn't already possess, not that the Maya civilization lacked marketplaces. Choice D is incorrect because nothing in the text suggests that scholars assumed that farmers wouldn't trade their agricultural products unless they had already met their own needs with those products. Instead, the text says that scholars thought that the Maya lands produced a smaller array of resources than they actually did, which the text suggests led scholars to assume that the Maya

didn't have any need for marketplaces. The scholars' mistaken belief has no bearing on the issue of whether farmers met their own needs before trading their products.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In their 2022 paper, Christos Dimopoulos et al., having granted that the existence of antigravity—in which antimatter and matter repel rather than attract each other—lacked affirmative experimental support, rightly argued that such antigravity was worth considering on theoretical grounds given that evidence against it was similarly lacking. But a 2023 report by an international team of researchers details the first direct ballistic observations of antihydrogen atoms under gravity inside a CERN particle accelerator. Corresponding most closely to predictions under gravitational attraction, these observations were thoroughly inconsistent with antigravity.

Which choice best states the main idea of the text?

Answer

- A. Antihydrogen ballistics observations were conducted at CERN to test specific conclusions about antigravity presented in the 2022 paper by Dimopoulos et al.
- B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report’s findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.
- C. The theoretical approach represented in the 2022 paper by Dimopoulos et al. assumed that unambiguous proof of antigravity would not be achievable, but the results in the 2023 report undermine that assumption.
- D. Before 2023, researchers’ inordinate focus on theoretical considerations hindered the development of the experimental regimen for direct antihydrogen ballistics observations.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately states the main idea of the text. The text begins by acknowledging that in 2022, Dimopoulos et al. were justified in arguing for antigravity on theoretical grounds because although there was no experimental evidence supporting its existence at that time, there was no evidence contradicting its existence either. The text then introduces the 2023 report describing an experiment that produced observations "thoroughly inconsistent with antigravity." Thus, the main idea is that while the study of antigravity was theoretically justified before the 2023 report (due to lack of evidence against it), the report’s findings undermine the rationale for such theoretical consideration since there is now evidence against this hypothetical phenomenon.

Choice A is incorrect. Although antihydrogen ballistics observations were conducted at CERN, the text provides no evidence that they were conducted to test conclusions in Dimopoulos et al.’s 2022 paper. Rather, the text presents the 2023 report as new evidence that contradicts the existence of the hypothetical phenomenon that Dimopoulos et al. had considered. Choice C is incorrect because the text doesn’t suggest that Dimopoulos et al. assumed that unambiguous proof of antigravity would be unattainable. Rather, the text states that they argued antigravity was worth considering theoretically because evidence against it was lacking at that time. At the same time, the results in the 2023 report do not undermine but strengthen the idea that the proof of antigravity might not be achievable. Choice D is incorrect because the text doesn’t suggest that there was an "inordinate focus on theoretical considerations" before 2023 or that such a focus hindered the development of experimental methods. In fact, the text describes Dimopoulos et al.’s position regarding the theoretical consideration of antigravity as "rightly argued," indicating approval of their approach given the absence of experimental evidence at that time.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The alpaca was domesticated by Indigenous peoples in the Andes about 7,000 years ago. But which wild species did it descend from, the vicuña or guanaco? A research team led by Ruiwen Fan may have solved the mystery, concluding that the alpaca is the domesticated form of the vicuña but that the modern alpaca gets only 64 percent of its genetic material from its wild ancestor. The rest comes from the domesticated llama. The llama, meanwhile, gets 95.5 percent of its genetic material from its own wild ancestor, the guanaco, and the rest from the alpaca. The llama and alpaca apparently interbred widely for only a handful of generations between 400 and 600 years ago. Assuming that the findings of Fan’s team are valid, it can be inferred that _____

Which choice most logically completes the text?

Answer

- A. modern llama populations have a greater degree of genetic diversity, on average, than modern alpaca populations do.
- B. the domestication process of the alpaca may have involved some introduction of genetic material from the llama.
- C. the period of interbreeding resulted in a greater genetic difference between alpacas and their wild ancestors than between llamas and their wild ancestors.
- D. if they were subjected to genetic testing, modern populations of guanacos and vicuñas would likely show traces of ancient interbreeding as well.

Correct Answer: C

Rationale

Choice C is the best answer because it presents the inference that most logically follows from the text’s discussion of the genetic material of alpacas. The text states that modern alpacas get only 64 percent of their genetic material from their wild ancestor (the vicuña), while llamas get 95.5 percent of their genetic material from their wild ancestor (the guanaco). These data imply that relative to llamas, alpacas have experienced a much greater genetic shift away from their wild ancestors (36 percent of their genetic material derives from another source, compared to 4.5 percent of llamas’ genetic material). Moreover, the text establishes that the remainder of alpacas’ genetic material derives from llamas, and the remainder of llamas’ genetic material derives from alpacas, reflecting a period of interbreeding. Taken together, these details support the inference that the period of interbreeding had a more significant impact on the genetic material of alpacas than on that of llamas, thereby reducing the percentage of alpacas’ genetic material that derives from wild ancestors and placing alpacas at a greater remove from these ancestors.

Choice A is incorrect because the text doesn’t provide information about genetic diversity within modern llama and alpaca populations. While the text discusses the percentage of genetic material each domesticated species inherited from its wild ancestor and from interbreeding, it does not address the level of genetic variation within each species. Therefore, it isn’t logical to infer that modern llama populations have a greater degree of genetic diversity than modern alpaca populations do. Choice B is incorrect because it contradicts the chronology presented in the text. The text states that alpacas were domesticated about 7,000 years ago, but the interbreeding with llamas occurred “between 400 and 600 years ago.” The genetic mixing happened thousands of years after domestication, not during it; in addition, the text does not provide details about what the process of domestication might entail. Therefore, it isn’t logical to infer that llama genetic material was introduced during the domestication process of alpacas. Choice D is incorrect because though the text notes that guanacos and vicuñas contributed to the genetic material of alpacas and llamas, it only discusses alpacas’ and llamas’ genetic composition. Therefore, the text does not support the inference that modern populations of guanacos and vicuñas would likely show traces of ancient interbreeding.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Poetry in Classical Nahuatl, the language of the Aztec Empire, relies on *difrasismo*, or a parallel noun construction that conventionally operates as a single metaphor. For example, the common difrasismo *in cuauhtli in ocelotl* (literally, “the eagle, the jaguar”) signifies “warrior.” The device’s function is both formal—providing structure to lines of verse—and ritual: semantic relations among the two nouns and the concept they signify can be tenuous, as in the previous example, such that difrasismos are often only intelligible according to the conceptual associations observed in Aztec ceremonial culture.

Which statement about the difrasismo *in cuauhtli in ocelotl* is most strongly supported by the text?

Answer

- A. Its metaphorical significance derives from the semantic equivalence of the two nouns constituting the difrasismo.
- B. Its unintelligibility may cause its formal function within a line of verse to go unnoticed by present-day readers.
- C. Its apparent obscurity can be resolved when considered in the proper cultural context.
- D. Its frequency in Classical Nahuatl poetry confirms its intelligibility to the Aztec audience.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a statement about the difrasismo *in cuauhtli in ocelotl* that is directly supported by the text. The text begins by describing difrasismo, a device used in Classical Nahuatl poetry. The text then mentions the device’s two functions: a formal one (giving structure to lines of verse) and a ritualistic one. The text indicates that the relation between the words in a difrasismo may appear tenuous without the additional information supplied by Aztec ceremonial culture but that the meaning becomes intelligible in the context of that information. Therefore, the difrasismo’s apparent obscurity can be resolved when considered in the proper cultural context.

Choice A is incorrect because the text doesn’t indicate that the two nouns used in a difrasismo are semantically equivalent; instead, the text indicates that the two nouns used in a difrasismo make up a single metaphor whose meaning is often intelligible only in the context of information supplied by Aztec ceremonial culture. Choice B is incorrect because the text doesn’t indicate that there’s a relationship between the formal function of the difrasismo and the difrasismo’s intelligibility. Additionally, the text suggests that present-day readers who are familiar with Aztec ceremonial culture wouldn’t find the difrasismo to be unintelligible. Choice D is incorrect because the text doesn’t indicate that the frequency of difrasismo’s use in Classical Nahuatl is a necessary feature of intelligibility: the text indicates that an infrequently used difrasismo would presumably also be intelligible to members of an Aztec audience who are sufficiently familiar with Aztec ceremonial culture.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The ratio of methane to other atmospheric constituents—represented by a measure called the methane mole fraction—influences a variety of meteorological phenomena, notably precipitation and humidity. For Titan, Saturn’s largest moon, the observational data that exist are too sparse and discrepant to fully constrain the range of the methane mole fraction at various atmospheric levels. Juan Lora and colleagues point out that outputs of the IPSL atmospheric model of Titan, which track closely to observations in some respects, reflect how the model’s developers responded to this challenge: by prescribing a uniform methane mole fraction for the lowest level of the atmosphere. It is therefore important to note that _____

Which choice most logically completes the text?

Answer

- A. some disagreements between the model’s simulations of Titan’s precipitation and humidity and the moon’s actual precipitation and humidity are to be expected.
- B. further observations of Titan may clarify the moon’s methane mole fraction sufficiently for the model to employ a single value rather than a range.
- C. even though the model’s outputs sometimes agree with observational data, Titan’s real methane mole fraction is likely higher than the methane mole fraction used in the model.
- D. inconsistencies across the model’s simulations of Titan’s precipitation and humidity could be attributable to variations in the moon’s methane mole fraction.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of modeling the atmosphere of Titan. The text establishes that the methane mole fraction (the ratio of methane to other atmospheric components) influences precipitation, humidity, and other meteorological phenomena and that observational data for Titan are too lacking (“too sparse and discrepant”) to fully determine the range of this measure across the moon’s various atmospheric levels. The text indicates that because of this limitation, researchers developed an atmospheric model (IPSL) that uses a simplified approach: it applies a uniform methane mole fraction for the lowest atmospheric level. Since it is very unlikely that the actual methane distribution across atmospheric levels on Titan is uniform, and since the methane mole fraction directly affects precipitation and humidity, it logically follows that the model’s simplified treatment of the methane mole fraction would lead to some disagreements between the model’s simulations and Titan’s actual precipitation and humidity.

Choice B is incorrect because the text makes it clear that the model *already* uses a single value for Titan’s methane mole fraction instead of a range. Further, the text suggests that the single value is a simplification of what is likely a more complex reality, so there’s no reason to assume that a single value would be more desirable than a range. Choice C is incorrect. The text strongly suggests that Titan’s actual methane mole fraction differs from the uniform value used in the model, even though some of the model’s outputs align closely with observations, but nothing in the text indicates how the actual value would differ. There’s no reason to assume that the actual methane mole fraction is higher than what the model uses. The “lowest level” mentioned in the text refers to the atmosphere and not Titan’s methane mole fraction. Choice D is incorrect because the text makes it clear that the model *doesn’t* reflect the actual variations in the methane mole fraction across Titan’s atmospheric levels, since observational data are too limited. Because the model uniformly applies a set methane mole fraction, no inconsistencies in the simulations would be attributable to actual variations in the moon’s methane mole fraction.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Modern dog breeds are largely the result of 160 years of owners crossbreeding certain dogs in order to select for particular physical appearances. Owners often say that some breeds are also more likely than others to have particular personality traits—basset hounds are affectionate; boxers are easy to train—but Kathleen Morrill and colleagues found through a combination of owner surveys and DNA sequencing of 2,000 dogs that while physical traits are predictably heritable among purebred dogs, behavior varies widely among dogs of the same breed.

Which choice best states the main idea of the text?

Answer

- A. Dog breeds would not exist without many years of human intervention in dogs’ reproduction.
- B. Research fails to confirm a commonly held belief about dog breeds and behavior.
- C. The dog breeds most popular among owners have often changed over the past 160 years.
- D. A study of dog breeds is notable for its usage of both opinion surveys and DNA sequencing.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately states the main idea of the text. The text indicates that dog owners typically claim that some dog breeds are “more likely than others to have particular personality traits.” In other words, the text points out that a commonly held belief about dog breeds is that their personality traits are heritable. The text then states that Kathleen Morrill and colleagues undertook research about dog trait heritability and found that “behavior varies widely among dogs of the same breed.” Because Morrill and colleagues found evidence for variability rather than consistency in the behavior of dogs of the same breed, the statement that research fails to uphold a commonly held belief about dog breeds and behavior accurately reflects the main idea of the text.

Choice A is incorrect. Although the text mentions that humans have long intervened in dogs’ reproduction by intentionally crossbreeding certain dogs, it doesn’t argue that such intervention is essential to the existence of dog breeds. Choice C is incorrect because the text doesn’t discuss the popularity of any dog breeds; breeds are mentioned as having certain traits, but the text says nothing about the popularity of these breeds or traits. Choice D is incorrect. Although the text briefly mentions that Morrill and colleagues conducted a study about dog traits using both surveys and DNA sequencing, this is not the main focus of the text. The text concerns the study’s results about the heritability of dog traits, not the particular methodology used by Morrill and colleagues.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Scientists studying Mars long thought the history of its crust was relatively simple. One reason for this is that geologic and climate data collected by a spacecraft showed that the crust was largely composed of basalt, likely as a result of intense volcanic activity that brought about a magma ocean, which then cooled to form the planet’s surface. A study led by Valerie Payré focused on additional information—further analysis of data collected by the spacecraft and infrared wavelengths detected from Mars’s surface—that revealed the presence of surprisingly high concentrations of silica in certain regions on Mars. Since a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica, Payré concluded that _____

Which choice most logically completes the text?

Answer

- A. the information about silica concentrations collected by the spacecraft is likely more reliable than the silica information gleaned from infrared wavelengths detected from Mars’s surface.
- B. high silica concentrations on Mars likely formed from a different process than that which formed the crusts of other planets.
- C. having a clearer understanding of the composition of Mars’s crust and the processes by which it formed will provide more insight into how Earth’s crust formed.
- D. Mars’s crust likely formed as a result of other major geological events in addition to the cooling of a magma ocean.

Correct Answer: D

Rationale

Choice D is the best answer. Cooling magma would create basalt, but “a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica.” Since Mars’s crust does contain large amounts of silica, it is unlikely that Mars’s crust was formed exclusively by cooling magma. Therefore, there were likely other major geological events that created the high silica concentrations.

Choice A is incorrect. Although the passage discusses these two methods of collecting data about Mars’s crust, it never compares their reliability, so there’s no basis for this inference. Choice B is incorrect. The passage never mentions anything about the crusts of other planets, so there’s no basis for this inference. Choice C is incorrect. The passage never mentions Earth’s crust, so there’s no basis for this inference.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

For many years, the only existing fossil evidence of mixopterid eurypterids—an extinct family of large aquatic arthropods known as sea scorpions and related to modern arachnids and horseshoe crabs—came from four species living on the paleocontinent of Laurussia. In a discovery that expands our understanding of the geographical distribution of mixopterids, paleontologist Bo Wang and others have identified fossilized remains of a new mixopterid species, *Terropterus xiushanensis*, that lived over 400 million years ago on the paleocontinent of Gondwana.

According to the text, why was Wang and his team’s discovery of the *Terropterus xiushanensis* fossil significant?

Answer

- A. The fossil constitutes the first evidence found by scientists that mixopterids lived more than 400 million years ago.
- B. The fossil helps establish that mixopterids are more closely related to modern arachnids and horseshoe crabs than previously thought.
- C. The fossil helps establish a more accurate timeline of the evolution of mixopterids on the paleocontinents of Laurussia and Gondwana.
- D. The fossil constitutes the first evidence found by scientists that mixopterids existed outside the paleocontinent of Laurussia.

Correct Answer: D

Rationale

Choice D is the best answer because it states why Wang and his team’s discovery of the *Terropterus xiushanensis* fossil was significant. The text explains that up until Wang and his team’s discovery, the only fossil evidence of mixopterids came from the paleocontinent of Laurussia. Wang and his team, however, identified fossil remains of a mixopterid species from the paleocontinent Gondwana. Therefore, the team’s discovery was significant because the fossil remains of a mixopterid species were outside of the paleocontinent Laurussia.

Choice A is incorrect. Although the text states that Wang and his team identified fossilized remains of a mixopterid species that lived more than 400 million years ago, it doesn’t indicate that mixopterid fossils previously found by scientists dated to a more recent period than that. Choice B is incorrect. Although the text states that mixopterids are related to modern arachnids and horseshoe crabs, it doesn’t suggest that the fossil discovered by Wang and his team confirmed that this relationship is closer than scientists had previously thought. Choice C is incorrect because the team’s fossil established the presence of mixopterids on Gondwana, not on Laurussia. Moreover, the text only discusses the fossil in relation to the geographical distribution of mixopterids, not in relation to their evolution.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Archaeologists have held that the Casarabe culture, which emerged in the southwestern Amazon basin in the first millennium CE, was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness. Recently, however, archaeologist Heiko Prümers and colleagues conducted a study of the region using remote-sensing technology that enabled them to create three-dimensional images of the jungle-covered landscape from above, and the researchers concluded that the Casarabe people developed a form of urbanism in the Amazon basin.

Which finding about the remote-sensing images, if true, would most directly support Prümers and colleagues’ conclusion?

Answer

- A. They show shapes consistent with widely separated settlements of roughly equal small size surrounded by uncultivated jungle.
- B. They show shapes consistent with long-distance footpaths running from Casarabe territories to large cities outside the region inhabited by the Casarabe people.
- C. They show shapes consistent with scattered small farms created by clearing jungle areas near sources of fresh water.
- D. They show shapes consistent with monumental platforms and dense central settlements linked to smaller settlements by a system of canals and roadways.

Correct Answer: D

Rationale

Choice D is the best answer. This finding, if true, would support the archaeologists’ conclusion. Dense central settlements linked to smaller ones would provide evidence of cities and suburbs—in other words, “a form of urbanism.”

Choice A is incorrect. This finding, if true, would weaken the archaeologists’ conclusion. Widely separated, small settlements with jungle in between would support the long-held belief that the Casarabe culture “was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness.” Choice B is incorrect. This choice wouldn’t support the researchers’ conclusion. These large cities are located outside Casarabe territory, which doesn’t show evidence of Casarabe urbanism. Choice C is incorrect. This finding wouldn’t support the archaeologists’ conclusion. Scattered small farms in jungle clearings are not good evidence to support the existence of cities (“a form of urbanism”).

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

In classical Greek and Roman mythology, female characters are typically cast as either villains lacking in psychological depth or passive victims who are marginal to these stories, which usually focus on the exploits of male characters. Recently, a subgenre has emerged in which writers reimagine these stories from the perspectives of their female characters, giving them agency and complex motivations. Purists argue that such efforts represent a distinctively modern tendency to impose our own values on past civilizations, obscuring those civilizations’ beliefs. Defenders of the subgenre counter that reimaginings of the myths for new cultural contexts are almost as old as the myths themselves, suggesting that _____

Which choice most logically completes the text?

Answer

- A. bringing female perspectives to the forefront is not indicative of a novel attitude regarding fidelity to Greek and Roman myths’ ideologies.
- B. modern writers’ foregrounding of female characters is chiefly motivated by a desire to counterbalance the primacy of male perspectives among earlier adaptations of Greek and Roman myths.
- C. purists are overlooking a long tradition of adapting Greek and Roman myths to focus on female characters.
- D. the complex motivations given to female characters in modern retellings of Greek and Roman myths reflect a recent shift toward psychological depth in fictional representation.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of a literary subgenre that reimagines Greek and Roman myths in a more modern cultural context—namely, the greater visibility and psychological depth of female characters. According to the text, purists object that this subgenre imposes modern values on past civilizations, while the subgenre’s defenders argue that reimagining old myths for new cultural contexts is “almost as old as the myths themselves.” If this argument in defense of the subgenre is accurate, then bringing female perspectives to the forefront of myth is reflective of a long-established attitude regarding writers’ freedom to recontextualize the ideologies of Greek and Roman myths instead of remaining absolutely faithful to them.

Choice B is incorrect because the text suggests that modern writers who foreground female characters in their adaptations of Greek and Roman myths are motivated by a desire to counterbalance the primacy of male perspectives in the myths themselves, not in earlier adaptations of those myths. The text doesn’t indicate that the modern writers are responding to earlier adaptations and only discusses those adaptations in the context of the argument that recontextualizing Greek and Roman myths is a very old practice. Choice C is incorrect because the defenders’ claim concerns the general practice of reimagining myths for new cultural contexts, not specifically a tradition of focusing on female characters. The text doesn’t indicate that earlier reimaginings centered on female characters, so it wouldn’t be logical to conclude that purists are overlooking a long tradition of adapting myths to focus on female characters specifically. Choice D is incorrect. While the text supports the claim that the modern retellings of Greek and Roman myths have given complex motivations to the female characters, it doesn’t associate this development with a trend toward the portrayal of greater psychological depth in fiction more broadly or claim that such a trend exists. Moreover, the claim that there is an association wouldn’t follow logically from the last sentence of the text, which asserts that reimagining myths in new cultural contexts is an established practice.

Question ID: 3f4ab688

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In a research paper, a student criticizes some historians of modern African politics, claiming that they have evaluated Patrice Lumumba, the first prime minister of what is now the Democratic Republic of the Congo, primarily as a symbol rather than in terms of his actions.

Which quotation from a work by a historian would best illustrate the student’s claim?

Answer

- A. “Lumumba is a difficult figure to evaluate due to the starkly conflicting opinions he inspired during his life and continues to inspire today.”
- B. “The available information makes it clear that Lumumba’s political beliefs and values were largely consistent throughout his career.”
- C. “Lumumba’s practical accomplishments can be passed over quickly; it is mainly as the personification of Congolese independence that he warrants scholarly attention.”
- D. “Many questions remain about Lumumba’s ultimate vision for an independent Congo; without new evidence coming to light, these questions are likely to remain unanswered.”

Correct Answer: C

Rationale

Choice C is the best answer because it illustrates the student’s claim about some historians viewing Lumumba primarily as a symbol. This quotation argues that Lumumba “warrants” (or deserves) “scholarly attention” as a symbol and not for his “practical accomplishments”—that is, his actions as prime minister—which “can be passed over quickly,” or dismissed as being of comparatively little importance. Thus, the quotation expresses the view that the student criticizes some historians for holding.

Choice A is incorrect. Although this quotation touches on the difficulty of evaluating Lumumba’s legacy, it doesn’t address how historians of modern African politics view him as a symbol. Choice B is incorrect. While this quotation mentions Lumumba’s political beliefs, it doesn’t discuss historians viewing him as a symbol. Choice D is incorrect. This quotation touches on Lumumba’s vision for his country, but it doesn’t discuss historians viewing him as a symbol.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The Hubble Space Telescope (HST) is projected to maintain operation until at least 2030, but it has already revolutionized high-resolution imaging of solar-system bodies in visible and ultraviolet (UV) light wavelengths, notwithstanding that only about 6% of the bodies imaged by the HST are within the solar system. NASA researcher Cindy L. Young and colleagues assert that a new space telescope dedicated exclusively to solar-system observations would permit an extensive survey of minor solar-system bodies and long-term UV observation to discern how solar-system bodies change over time. Young and colleagues’ recommendation therefore implies that the HST _____

Which choice most logically completes the text?

Answer

- A. will likely continue to be used primarily to observe objects outside the solar system.
- B. will no longer be used to observe solar system objects if the telescope recommended by Young and colleagues is deployed.
- C. can be modified to observe the features of solar system objects that are of interest to Young and colleagues.
- D. lacks the sensors to observe the wavelengths of light needed to discern how solar system bodies change over time.

Correct Answer: A

Rationale

Choice A is the best answer. The HST will operate until at least 2030, but it’s only observing stuff inside our solar system 6% of the time. If we could get a different telescope to observe stuff inside our solar system 100% of the time and take more extensive images of certain things, then the HST could continue to be used mainly for observing stuff outside the solar system.

Choice B is incorrect. This inference is too strong to be supported by the text. Even if the new telescope is deployed, the HST might still be used as it’s being used now. Based on the text, the new telescope would just be used for more extensive and long-term imaging of solar system bodies, which doesn’t necessarily overlap with the HST. Choice C is incorrect. This inference isn’t supported. The text never mentions the possibility of modifying the HST, so there is no basis to make this inference. Rather, the researchers suggest using a different telescope to more closely observe certain objects. Choice D is incorrect. This inference is too strong to be supported. The text doesn’t give us enough info to assume that the HST lacks any particular sensors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

In 2022, Crystal Reeck and colleagues studied whether the decision-making modes that guide consumers influence their choice between nonenvironmentally friendly standard electricity plans and environmentally friendly green plans that cap electricity usage. Study participants who self-reported using either an Affect Mode or Role Mode—which prioritize choices that have a stronger positive emotional or social impact, respectively—were more likely to select a green plan. Conversely, participants using a Calculation Mode—which aims to minimize both financial cost and personal inconvenience—were more likely to select a standard plan, even when the green option was cheaper. This finding suggests that participants using a Calculation Mode _____

Which choice most logically completes the text?

Answer

- A. were equally unlikely to factor the financial savings of the green plan into their decision-making as were participants using either the Affect or Role Modes.
- B. may have been less strongly motivated to appear socially responsible with their choice of plan than they realized.
- C. may have determined that the green plan imposed additional burdens on them that were not sufficiently offset by the potential financial savings.
- D. were less likely to believe that the green plan was truly cost-effective than were participants using either the Affect or Role Modes.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of the influence of decision-making modes on consumers’ choices of different electricity plans. The Affect Mode describes choices based on the decision-maker’s preferred emotional outcome, the Role Mode describes choices based on the decision-maker’s social motivations, and the Calculation Mode describes choices based on minimizing financial costs and maximizing the decision-maker’s convenience. Thus, if those using a Calculation Mode disfavor the green plan, it would be due to some financial or convenience burden the green plan imposes that the other doesn’t. The text indicates that the green plan had a lower financial cost but was nonetheless rejected by participants using the Calculation Mode. It therefore follows that the green plan likely imposed a convenience burden that outweighed potential financial savings for these participants.

Choice A is incorrect. According to the text, the Affect Mode involves the decision-maker’s preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker’s convenience. Thus, unlike the Affect or Role Modes, the Calculation Mode would almost certainly involve taking price differences into account. Choice B is incorrect. According to the text, the Affect Mode involves the decision-maker’s preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker’s convenience. Thus, there is no reason to think that participants using the Calculation Mode at any point took social responsibility into account, let alone think that they discovered a disparity between their expectations and reality. Choice D is incorrect. According to the text, the Affect Mode involves the decision-maker’s preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker’s convenience. Thus, the text doesn’t indicate that cost-effectiveness is a factor for participants using the Affect or Role Modes, and there is no basis to compare their beliefs in the truthfulness of the green-plan description with those of the Calculation Mode users.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Geoglyphs are large-scale designs of lines or shapes created in a natural landscape. The Nazca Lines were created in the Nazca Desert in Peru by several Indigenous civilizations over a period of many centuries. Peruvian archaeologist Johny Isla specializes in these geoglyphs. At a German exhibit about the Nazca Lines, he saw an old photograph of a large geoglyph of a whalelike figure and was surprised that he didn't recognize it. Isla returned to Peru and used a drone to search a wide area, looking for the figure from the air. This approach suggests that Isla thought that if he hadn't already seen it, the whalelike geoglyph _____

Which choice most logically completes the text?

Answer

- A. must represent a species of whale that went extinct before there were any people in Peru.
- B. is actually located in Germany, not Peru, and isn't part of the Nazca Lines at all.
- C. is probably in a location Isla hadn't ever come across while on the ground.
- D. was almost certainly created a long time after the other Nazca Lines geoglyphs were created.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text's discussion of Johny Isla and the whalelike geoglyph. The text indicates that the German exhibit about the Nazca Lines included a photograph showing a whalelike geoglyph that Isla hadn't known about before attending the exhibit, even though Isla "specializes in" Nazca Lines geoglyphs. Given his expertise, and his surprise at being unfamiliar with the whale glyph, the text strongly suggests that Isla believed he would have noticed the glyph if he had been to its location. Thus, the text implies that the whalelike geoglyph is likely in a location Isla had not previously been to.

Choice A is incorrect because the text doesn't address either the species of whale that the geoglyph is meant to represent or its relationship to the earliest humans in the area that is now Peru. Choice B is incorrect. Although the text indicates that the photograph of the whalelike geoglyph was on display at a "German exhibit," that exhibit was specifically "about the Nazca Lines," which the text indicates are located in Peru. Choice D is incorrect. Although the text does indicate that the glyphs were created "over a period of many centuries," the text doesn't address when in that period of time any particular glyphs were created.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Some *Astyanax mexicanus*, a river-dwelling fish found in northeast Mexico, have colonized caves in the region. Although there is little genetic difference between river and cave *A. mexicanus* and all members of the species can emit the same sounds, biologist Carole Hyacinthe and colleagues found that the context and significance of those sounds vary by location—e.g., the click that river-dwelling *A. mexicanus* use to signal aggression is used by cave dwellers when foraging—and the acoustic properties of cave fish sounds show some cave-specific variations as well. Hyacinthe and colleagues note that differences in sonic communication could accumulate to the point of inhibiting interbreeding among fish from different locations, suggesting that _____

Which choice most logically completes the text?

Answer

- A. although *A. mexicanus* living in rivers are genetically similar to those living in caves, river fish rely on sonic communication less than cave fish do.
- B. although *A. mexicanus* is a single species at present, it could be in the process of splitting into distinct populations with different characteristics.
- C. although all *A. mexicanus* emit sounds, the fish living in rivers produce some sounds that the fish living in caves do not, and vice versa.
- D. although *A. mexicanus* from different locations can interbreed currently, river fish and cave fish are sufficiently genetically distinct that they can be considered separate species.

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text’s discussion of *A. mexicanus*. According to the text, *A. mexicanus*, a river-dwelling fish species, has colonized caves. The fish that live in rivers and those that live in caves show no significant genetic differences and are all capable of making the same sounds. The text indicates, however, that Hyacinthe and colleagues found that sounds that the river fish use in a particular context and for a particular purpose are used in a different context and for a different purpose by the cave fish. Additionally, the sounds made by cave fish show some differences depending on the specific cave the fish inhabit. The text goes on to state that Hyacinthe and colleagues have noted that differences in how the fish use sound to communicate could eventually become so great that they prevent fish from different locations from interbreeding. In other words, the river fish might eventually only breed with other river fish (with whom they share characteristics regarding sonic communication that they do not share with cave fish), while the cave fish might only breed with other cave fish for a similar reason. In context, this observation suggests that even though the fish are a single species right now, they could be in the process of splitting into distinct populations with different characteristics.

Choice A is incorrect because there is no information in the text suggesting that the river fish are less reliant on sonic communication than the cave fish are. Although the text does indicate that the river fish and cave fish are genetically similar, the text describes both groups as using sonic communication and says nothing to indicate that one group depends on that communication more than the other group does. Choice C is incorrect. The text states that all members of the species can emit the same sounds but that the function and context of sounds differ depending on whether the fish live in rivers or caves, but it does not indicate that river fish produce sounds that cave fish do not or vice versa. Choice D is incorrect because it contradicts the text. The text says that there is little genetic difference between the river fish and the cave fish, not that the river fish and cave fish are so genetically distinct that they can be considered separate species.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Psychologists Dacher Keltner and Jonathan Haidt have argued that experiencing awe—a sensation of reverence and wonder typically brought on by perceiving something grand or powerful—can enable us to feel more connected to others and thereby inspire us to act more altruistically. Keltner, along with Paul K. Piff, Pia Dietze, and colleagues, claims to have found evidence for this effect in a recent study where participants were asked to either gaze up at exceptionally tall trees in a nearby grove (reported to be a universally awe-inspiring experience) or stare at the exterior of a nearby, nondescript building. After one minute, an experimenter deliberately spilled a box of pens nearby.

Which finding from the researchers’ study, if true, would most strongly support their claim?

Answer

- A. Participants who had been looking at the trees helped the experimenter pick up significantly more pens than did participants who had been looking at the building.
- B. Participants who helped the experimenter pick up the pens used a greater number of positive words to describe the trees and the building in a postexperiment survey than did participants who did not help the experimenter.
- C. Participants who did not help the experimenter pick up the pens were significantly more likely to report having experienced a feeling of awe, regardless of whether they looked at the building or the trees.
- D. Participants who had been looking at the building were significantly more likely to notice that the experimenter had dropped the pens than were participants who had been looking at the trees.

Correct Answer: A

Rationale

Choice A is the best answer because it presents a finding that, if true, would most strongly support the researchers’ claim that they found evidence that experiencing awe can make people feel more connected to others and thus more likely to behave altruistically (with beneficial and unselfish concern for others). According to the text, the researchers tested for this effect by first having participants look at either something known to be awe-inspiring (very tall trees) or something ordinary (a plain building) and then purposely spilling pens near the participants. The finding that participants who had looked at the trees helped pick up significantly more pens than did participants who had looked at the building would support the researchers’ claim by demonstrating that the people who had experienced awe behaved more altruistically when the experimenter needed help than the other participants did.

Choice B is incorrect because a finding about helpful participants using positive words to describe the trees and the building after the experiment was over wouldn’t have any bearing on the researchers’ claim that experiencing awe increases altruistic behavior. The text doesn’t address the use of positive words to describe things or suggest any connection between using such words and having experienced awe, so that behavior wouldn’t serve as evidence that experiencing awe played a role in promoting helpful behavior. Choice C is incorrect because a finding that participants who didn’t help the experimenter were significantly more likely than others to report having experienced awe whether they had looked at the building or the trees would weaken the researchers’ claim that experiencing awe increases altruistic behavior by suggesting that the opposite might be true—that experiencing awe is in fact linked to choosing not to act in a way that benefits someone else. Choice D is incorrect because a finding about participants noticing that the experimenter had dropped the pens wouldn’t have any bearing on the researchers’ claim about people behaving altruistically. Being aware of a challenge or problem isn’t necessarily beneficial on its own and isn’t the same as offering help, so the finding wouldn’t support the idea that experiencing awe increases altruistic behavior.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

To address the susceptibility of materials used in components of high-performance machinery, such as aircraft engines, to creep (deformation that is induced by persistent mechanical stress and that often occurs at elevated temperatures), materials researchers have developed silicon carbide (SiC) fibers for producing aerospace composites. Testing the thermomechanical properties of several commercially available SiC fibers, Ramakrishna T. Bhatt et al. found that in comparison with two polymer-derived SiC fibers, a nitrogen-treated SiC fiber exhibited a lower minimum creep rate, a measure of the rate at which a stress-exposed material deforms at a constant temperature and uniaxial load. The finding suggests that _____

Which choice most logically completes the text?

Answer

- A. unlike the two polymer-derived SiC fibers, the nitrogen-treated SiC fiber can substantially inhibit creep, provided that temperatures and loads are consistent.
- B. the two polymer-derived SiC fibers likely hold similar potential for reducing the creep resistance of materials exposed to stress and elevated temperatures, thus prolonging the life span of aerospace machinery.
- C. composites based on the two polymer-derived SiC fibers have chemical properties that may improve the mechanical and thermal stability of aerospace equipment to a greater extent than do composites based on the nitrogen-treated SiC fiber.
- D. aerospace composites containing the nitrogen-treated SiC fiber may have the ability to withstand mechanical stress for a longer period of time than can aerospace composites containing either of the two polymer-derived SiC fibers.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of silicon carbide (SiC) fibers and creep, or deformation related to ongoing mechanical stress and elevated temperatures. The text states that Bhatt et al. found that a nitrogen-treated SiC fiber had a lower minimum creep rate than two polymer-derived SiC fibers did. Because having a lower creep rate means that the material is slower to deform with exposure to stress, as the text explains, this finding suggests that aerospace composites made with the nitrogen-treated SiC fiber may be able to withstand mechanical stress for a longer period than those made with the other two polymer-derived SiC fibers can.

Choice A is incorrect because it overstates the implications of the study’s findings, which have to do with the rate of a material’s deformation under stress, not the absolute degree of deformation. The text states that Bhatt et al. observed that a nitrogen-treated SiC fiber had a lower minimum creep rate than two polymer-derived SiC fibers did, meaning only that it deformed more slowly over time under constant stress, not that it underwent less deformation overall. Choice B is incorrect because the text doesn’t establish any similarity between the two polymer-derived SiC fibers other than that both had a higher creep rate than the nitrogen-treated SiC fiber did in Bhatt et al.’s study. Moreover, reducing a material’s resistance to creep would mean that the material becomes *more* susceptible to deformation with exposure to stress and elevated temperatures, which would be expected to shorten rather than prolong the lifespan of machinery made with that material. Choice C is incorrect because the text suggests that the stability of aerospace equipment may be better improved by composites containing nitrogen-treated SiC fiber than by composites containing the two polymer-derived SiC fibers, not the other way around. The text indicates that Bhatt et al. observed that the nitrogen-treated SiC fiber had a lower minimum creep rate than the other two fibers did, meaning that it was slower to degrade under exposure to mechanical stress and elevated temperatures—suggesting that it may remain stable for longer periods.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Many governments that regularly transfer money to individuals—to provide supplemental incomes for senior citizens, for example—have long done so electronically, but other countries typically have distributed physical money and have only recently developed electronic transfer infrastructure. Researchers studied the introduction of an electronic transfer system in one such location and found that recipients of electronic transfers consumed a different array of foods than recipients of physical transfers of the same amount did. One potential explanation for this result is that individuals conceive of and allocate funds in physical money differently than they conceive of and allocate funds in electronic form.

Which finding from the study, if true, would most directly weaken the potential explanation?

Answer

- A. Recipients of electronic transfers typically spent their funds at a slower rate than recipients of physical transfers did.
- B. Nearly every recipient of an electronic transfer withdrew the entire amount in physical money shortly after receiving the transfer.
- C. Recipients of physical transfers tended to purchase food about as frequently as recipients of electronic transfers did.
- D. Some recipients of physical transfers received small amounts of money relatively frequently, while others received large amounts relatively infrequently.

Correct Answer: B

Rationale

Choice B is the best answer. This would weaken the explanation. If the recipients of electronic money immediately withdrew it all as physical money, then both kinds of recipients ended up spending physical money on food. So there must be some other explanation why those who initially received electronic money ate different kinds of food.

Choice A is incorrect. This wouldn't weaken the explanation. If anything, it actually supports it: it demonstrates that recipients of electronic money and recipients of physical money have different spending habits. Choice C is incorrect. This wouldn't weaken the explanation. The explanation we're testing this choice against is about the way that people might "conceive of and allocate" physical and electronic funds differently. This choice only focuses on the timing, not what they spend the money on. Choice D is incorrect. This would have no impact on the explanation. It doesn't tell us anything about possible differences between the spending habits of those who spend physical money and those who spend money electronically.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Icebergs generally appear to be mostly white or blue, depending on how the ice reflects sunlight. Ice with air bubbles trapped in it looks white because much of the light reflects off the bubbles. Ice without air bubbles usually looks blue because the light travels deep into the ice and only a little of it is reflected. However, some icebergs in the sea around Antarctica appear to be green. One team of scientists hypothesized that this phenomenon is the result of yellow-tinted dissolved organic carbon in Antarctic waters mixing with blue ice to produce the color green.

Which finding, if true, would most directly weaken the team’s hypothesis?

Answer

- A. White ice doesn’t change color when mixed with dissolved organic carbon due to the air bubbles in the ice.
- B. Dissolved organic carbon has a stronger yellow color in Antarctic waters than it does in other places.
- C. Blue icebergs and green icebergs are rarely found near each other.
- D. Blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would weaken the scientists’ hypothesis about icebergs that appear to be green. The text indicates that most icebergs are either mostly white or blue in color but that some icebergs in Antarctica appear to be green. The text goes on to say that the scientists hypothesized that this green color occurs when yellow-tinted dissolved organic carbon in ocean waters mixes with blue ice. A finding that both blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon would suggest that something other than yellow-tinted organic carbon causes some icebergs’ green color, since the blue icebergs that contain yellow-tinted organic carbon remained blue instead of turning green.

Choice A is incorrect because, according to the text, the scientists’ hypothesis was that blue icebergs, not white ones, change color when their ice mixes with yellow-tinted dissolved organic carbon. A finding that white ice, because of its air bubbles, doesn’t change color when it’s mixed with dissolved organic carbon would therefore have no bearing on the scientists’ hypothesis. Choice B is incorrect because the text focuses only on Antarctic icebergs that appear to be green. It doesn’t indicate that icebergs in locations other than Antarctica have been found to have a green hue. A finding that dissolved organic carbon has a stronger yellow color in Antarctic waters than in other places would therefore have no bearing on the scientists’ hypothesis that green color in icebergs in Antarctica is caused by yellow-tinted dissolved organic carbon mixing with blue ice. Choice C is incorrect because, according to the text, the scientists’ hypothesis was that blue icebergs turn green when their ice mixes with yellow-tinted dissolved organic carbon in the sea around them. If that’s correct, one would expect blue icebergs and green icebergs to be located at a distance from each other since all blue icebergs in an area where the waters contain yellow-tinted dissolved organic carbon would take on a green hue. A finding that blue icebergs and green icebergs are rarely found near each other would therefore strengthen, not weaken, the researchers’ hypothesis.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Boldly mixing elements of poetry, fiction, drama, philosophy, and manifesto, Puerto Rican writer Giannina Braschi creates cross-genre literature that explores themes such as immigration and independence. Her works have inspired responses from individuals across different fields and in a wide range of formats, from musical compositions and a comic book to architecture and furniture design. In an essay, a student asserts that the production of these diverse creations by others is reflective of Braschi’s own approach to crafting literature.

Which quotation from a scholarly review of Braschi’s work best supports the student’s claim?

Answer

- A. “Braschi is the focus of a 2020 collection of essays in which fifteen scholars from seven different countries delved into the linguistic and structural patterns of her writings.”
- B. “Braschi’s eagerness to push boundaries and blend genres within literature invites us to consider how other art forms might also engage with literature.”
- C. “Before settling in New York City, where she would go on to become a college professor, Braschi studied both literature and philosophy in several cities around the world.”
- D. “In addition to her creative literary works, Braschi has produced academic pieces analyzing writings by Miguel de Cervantes, Federico García Lorca, and other authors.”

Correct Answer: B

Rationale

Choice B is the best answer because it presents the quotation that best supports the student’s claim about Braschi. By describing how Braschi’s blending of genres invites her audience to think about how other art forms could also engage with literature, the quotation supports the idea that the diversity of responses to Braschi’s work reflects Braschi’s own approach to creating literature.

Choice A is incorrect because the quotation describes scholars from different countries writing essays about Braschi’s use of language in her writings; it doesn’t address how Braschi’s creation of cross-genre literature inspires diverse types of responses, which is the claim the student makes. Choice C is incorrect because the quotation focuses on the fact that Braschi studied in several different cities, which doesn’t address the student’s claim that Braschi’s creation of cross-genre literature inspires diverse types of responses. Choice D is incorrect because the quotation lists some of the authors who Braschi has written academic works about, which is irrelevant to the student’s claim that Braschi’s creation of cross-genre literature inspires diverse types of responses.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Early Earth is thought to have been characterized by a stagnant lid tectonic regime, in which the upper lithosphere (the outer rocky layer) was essentially immobile and there was no interaction between the lithosphere and the underlying mantle. Researchers investigated the timing of the transition from a stagnant lid regime to a tectonic plate regime, in which the lithosphere is fractured into dynamic plates that in turn allow lithospheric and mantle material to mix. Examining chemical data from lithospheric and mantle-derived rocks ranging from 285 million to 3.8 billion years old, the researchers dated the transition to 3.2 billion years ago.

Which finding, if true, would most directly support the researchers’ conclusion?

Answer

- A. Among rocks known to be older than 3.2 billion years, significantly more are mantle derived than lithospheric, but the opposite is true for the rocks younger than 3.2 billion years.
- B. Mantle-derived rocks older than 3.2 billion years show significantly more compositional diversity than lithospheric rocks older than 3.2 billion years do.
- C. There is a positive correlation between the age of lithospheric rocks and their chemical similarity to mantle-derived rocks, and that correlation increases significantly in strength at around 3.2 billion years old.
- D. Mantle-derived rocks younger than 3.2 billion years contain some material that is not found in older mantle-derived rocks but is found in older and contemporaneous lithospheric rocks.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers’ conclusion that the transition from a stagnant lid regime to a tectonic plate regime occurred around 3.2 billion years ago. The text explains that early in Earth’s history, Earth exhibited a stagnant lid regime in which there’s no interaction between the lithosphere and the underlying mantle. The text further explains that, by contrast, once Earth began to exhibit a tectonic plate regime, its lithospheric and mantle material began to mix. If mantle-derived rocks younger than 3.2 billion years contain material not found in older mantle-derived rocks, that material must have originated somewhere other than the mantle. And if this material is found in both older and contemporaneous lithospheric rocks, that would imply that the lithosphere was able to mix with mantle material beginning around 3.2 billion years ago, as the researchers concluded.

Choice A is incorrect. The text gives no basis for comparing the quantities of lithospheric and mantle-derived rocks. Choice B is incorrect. The text gives no basis for comparing the material makeup of lithospheric rocks to that of mantle-derived rocks. Choice C is incorrect. A positive correlation between the age of lithospheric rocks and these rocks’ chemical similarity to mantle-derived rocks would mean that the oldest rocks would be the most similar, which contradicts the text’s claim that lithospheric and mantle-derived rocks were completely separate until 3.2 billion years ago. If the researchers’ conclusion about the onset of tectonics on Earth is correct, then younger lithospheric rocks would show greater chemical similarity to mantle-derived rocks than older lithospheric rocks do.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is from Thomas Mann’s 1924 novel *The Magic Mountain*, translated by John E. Woods in 1995.

The story of Hans Castorp that we intend to tell here—not for his sake (for the reader will come to know him as a perfectly ordinary, if engaging young man), but for the sake of the story itself, which seems to us to be very much worth telling (although in Hans Castorp’s favor it should be noted that it is *his* story, and that not every story happens to everybody)—is a story that took place long ago, and is, so to speak, covered with the patina of history and must necessarily be told with verbs whose tense is that of the deepest past.

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What does the text most strongly suggest about the story of Hans Castorp?

Answer

- A. Though it is true that stories of even the most uninteresting people are themselves interesting because all people are unique, the reason this story is interesting is nonetheless difficult to understand because of the passage of time.
- B. Even though it is a story of a person of no particular importance, its age and the manner in which it therefore must be told are both indicators that the story itself is important.
- C. Like all stories about the lives of inconsequential people, this story must necessarily be related in a particular way if the reason the story is consequential is to be made evident to the audience.
- D. It is a remarkable story that happened to an unremarkable person, though one could plausibly argue that because the story is valuable, some of its value accrues to the person at its center.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a statement about Hans Castorp’s story that is suggested by the text. The narrator of the text indicates that the story about Hans Castorp will be told not because there is something particularly notable about him, since he is pleasant but "perfectly ordinary," but because the story itself is remarkable ("very much worth telling"). The narrator then notes that there is a benefit in being at the heart of the story—that it is "in Hans Castorp's favor" that the story is his, and maybe uniquely so ("not every story happens to everybody"). Thus, the text suggests both that the story that will be told is a remarkable one that happened to an unremarkable person and that it is reasonable to argue that the person at the center of a valuable story takes on some of the story's value.

Choice A is incorrect. Although the narrator of the text makes the point that "not every story happens to everybody," the narrator doesn’t state that stories are interesting simply because the people they are about are unique. Rather, the narrator suggests that one particular story is "very much worth telling" on its own and that Hans Castorp benefits from the fact that the story is remarkable and may be unique to him. Further, the narrator never suggests that the story will be hard to understand even though it is old. Choice B is incorrect. Although the narrator of the text suggests that Hans Castorp is of no particular importance, since he is a "perfectly ordinary" person, the narrator never reveals what makes the story of Castorp important, just that "the story itself" is "very much worth telling." The narrator states that the story "took place long ago," is "covered with the patina of history," and can be told only "with verbs whose tense is that of the deepest past," but the story’s age and the way it must be told aren’t presented as reasons the story is important; the narrator is simply providing details about how the story will be told. Choice C is incorrect because the narrator of the text doesn’t suggest that all stories about people who are "perfectly ordinary" (like Hans Castorp) must be told in particular ways to make it clear why those stories are consequential. Further, the narrator suggests that Hans Castorp’s story must be told "with

verbs whose tense is that of the deepest past" because it took place so long ago, not because telling it that way will convey the story's importance.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Barchester Towers is an 1857 novel by Anthony Trollope. In the novel, Trollope’s portrayal of Dr. Proudie underscores the character’s exaggerated sense of his own abilities: _____

Which quotation from *Barchester Towers* most effectively illustrates the claim?

Answer

- A. “It must not...be taken as proved that Dr. Proudie was a man of great mental powers, or even of much capacity for business, for such qualities had not been required in him.”
- B. “[Dr. Proudie] was comparatively young, and had, as he fondly flattered himself, been selected as possessing such gifts, natural and acquired, as must be sure to recommend him to a yet higher notice.”
- C. “[Dr. Proudie’s] residence in the metropolis, rendered necessary by duties thus entrusted to him, his high connexions, and the peculiar talents and nature of the man, recommended him to persons in power.”
- D. “[Dr. Proudie] was certainly possessed of sufficient tact to answer the purpose for which he was required without making himself troublesome.”

Correct Answer: B

Rationale

Choice B is the best answer. In this quotation, Dr. Proudie is described as "fondly flatter[ing] himself" that he has gifts that "must be sure to recommend him to a yet higher notice." In other words, he expects his skills to push him to greater fame and success. This implies an exaggerated sense of his own abilities, which matches the claim we’re trying to support.

Choice A is incorrect. This quotation doesn’t describe Proudie’s view of himself, nor does it paint him in an especially flattering light. Instead, by saying his mental powers and business skill are not "proved," it implies that he is actually dim-witted and bad at business. Choice C is incorrect. This choice describes Proudie’s closeness to power and importance, but it doesn’t show what Proudie thinks of himself. Proudie is not describing himself or his abilities here. The narrator is. Choice D is incorrect. While this quotation offers Proudie very mild praise, it doesn’t show what Proudie thinks of himself or his own abilities, which is what the claim focuses on.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

For its 1974 work *Instant Mural*, the Chicano art collective Asco taped members Patssi Valdez and Humberto Sandoval to an outdoor wall in East Los Angeles. The work is manifestly a commentary on constraint, but many critics focus on Valdez and the social constraints women faced at the time, which is understandable but leaves the presence of Valdez’s male collaborator Sandoval unexplained. We should instead consider that in 1974, the art establishment’s recognition of Chicano artists was (and had long been) restricted to sociohistorical muralists, leaving nonmuralist Chicano artists—like Asco’s members—struggling to even exhibit their work; attending to this context opens an interpretation that accounts for all the evidence, allowing us to conclude that _____

Which choice most logically completes the text?

Answer

- A. while Valdez’s presence in *Instant Mural* represents the social constraints placed on women at the time, Sandoval’s presence represents Chicano muralists’ frustration at their lack of recognition by the art establishment.
- B. the main subject of *Instant Mural* is female Chicano artists’ experience of being doubly constrained by gender-role expectations and the marginalization of certain types of art.
- C. *Instant Mural* is a reflection on the constraining aesthetic expectations placed on Chicano artists in general rather than on the social constraints placed on women specifically.
- D. *Instant Mural* is best understood not as a critique of the social constraints placed on women but rather as a critique of sociohistorical muralists’ depictions of Chicano culture.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of *Instant Mural*. According to the text, the 1974 work *Instant Mural* involved taping two Asco members (Patssi Valdez and Humberto Sandoval) to an outdoor wall as a direct commentary on constraint (limitation or restriction). The text suggests that instead of focusing on Valdez’s role in the work and on social limitations women faced at the time—an approach that fails to explain the role of Sandoval, a man—people should consider that in 1974 (and long before) the art establishment limited recognition of Chicano artists to those who created murals with social and historical subjects, making it challenging for nonmuralist Chicano artists (such as Asco’s members) to show their work. The text suggests that considering the situation with the art establishment leads to a conclusion that accounts for the fact that *Instant Mural* is explicitly about constraint and features both a woman and a man taped to the wall: *Instant Mural* isn’t a reflection of the social constraints placed on women specifically but instead is a reflection on the constraining aesthetic expectations placed on Chicano artists in general (that is, the limits resulting from the lack of recognition of Chicano artists of any gender who did not create sociohistorical murals).

Choice A is incorrect. The text indicates that the art establishment in 1974 (and earlier) did recognize Chicano muralists to the exclusion of other Chicano artists, so there’s no reason to assume Asco’s work *Instant Mural* was intended to represent Chicano muralists’ frustration; it’s far more likely that Asco’s members—described as Chicano nonmuralists—were representing their own frustration at being unrecognized by the art establishment. Choice B is incorrect. The text emphasizes that while a common approach to *Instant Mural* focuses on Valdez and the social limitations women faced in 1974 but fails to account for the presence of a man in the work, there is a conclusion that explains Sandoval’s role. But concluding that *Instant Mural* represents a double constraint experienced specifically by female Chicano artists would still fail to account for the inclusion of a man in the work. Choice D is incorrect. Although the text suggests that *Instant Mural* can be understood as something other than a commentary on social constraints women faced in 1974, it doesn’t suggest that the work is best understood as a critique of the content of any other artworks. The text doesn’t address how Chicano culture is depicted in sociohistorical murals; it discusses such murals only to make

the point that in 1974 (and earlier) the art establishment recognized Chicano artists who produced them to the exclusion of Chicano nonmuralists, indicating a constraint all of Asco's members faced, regardless of gender.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

An archaeological team led by Piotr Bieliński and Sultan al-Bakri found remnants of a 4,000-year-old Bronze Age board game at a site in Oman. Little is left of the game except a stone board, which is carved with a grid and has places to hold game pieces. Some scholars claim that the game was largely played by traders.

Which finding, if true, would most directly support the scholars’ claim?

Answer

- A. Other examples of the game dating to the same period have been found in the remains of several homes in the region, including in one home that may have belonged to a trader.
- B. Similar games have been found in other sites dating to the same period that were connected to the site in Oman via trade routes.
- C. The other known examples of the game dating to the same period have been found along routes that seem to have been used primarily by traders at the time.
- D. Remnants of other goods have been found at the site in Oman that probably also reached the location through trade.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most directly support the scholars’ claim about the board game. The text explains that the remains of a 4,000-year-old board game were found in Oman. The text then states that scholars claim this board game was played mostly by traders. If the other known examples of this board game from the same time period were discovered along routes that seem to have been used primarily by traders, this finding would directly support the scholars’ claim because it suggests that the game was largely played by traders who brought it with them for entertainment as they traveled.

Choice A is incorrect because this finding would suggest only that a single trader may have possessed examples of the board game, perhaps for the purpose of trading or selling the game to residents of Oman. For this reason, the finding wouldn’t directly support the scholars’ claim that the majority of the game’s players were traders. Choice B is incorrect because this finding doesn’t mention the board game at all, referring only to similar games found at other sites, and would therefore provide no direct support for the scholars’ claim about the board game. Choice D is incorrect because this finding doesn’t mention the board game at all, referring only to the remains of other goods found at the site in Oman, and would therefore provide no direct support for the scholars’ claim about the board game.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Dutch painters in the sixteenth and seventeenth centuries often showed tables filled with large wheels of cheese or carved shards of butter. Some art historians, noting that dairy products were a major component of the Dutch diet, interpret these depictions as reflections of everyday Dutch eating habits. However, a group of researchers recently reviewed hundreds of food-related paintings and found that lemons—which could only be acquired in the Netherlands at great cost, since they had to be imported from warmer climates—feature in Dutch paintings of the period more than three times as frequently as dairy products do, thereby casting doubt on the idea that _____

Which choice most logically completes the text?

Answer

- A. dairy products were a more significant component of the Dutch diet of the period than lemons were.
- B. food was a more popular subject among Dutch painters than it was among painters from other countries at the time.
- C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.
- D. Dutch painters of the period may have depicted foods for symbolic reasons rather than to show what Dutch people typically ate.

Correct Answer: C

Rationale

Choice C is the best answer. The text tells us that lemons—an expensive imported product—feature in Dutch paintings of the period more frequently than dairy products do. Since it’s unlikely lemons were eaten more often than dairy, this fact casts doubt on the theory that these paintings realistically depicted Dutch diets.

Choice A is incorrect. In fact, the text suggests the opposite: it says that dairy products were a "major component of the Dutch diet," while lemons were an expensive import item, so we can infer that lemons were eaten much more rarely than dairy products. Choice B is incorrect. The passage doesn’t mention painters from other countries, so there’s no basis to make this inference. Choice D is incorrect. In fact, the text suggests that Dutch painters do have reasons for depicting foods other than to show what Dutch people typically ate. For example, lemons in a painting could indicate wealth or prosperity.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

In a three-year study of parasitic infections by *Anomotaenia brevis* tapeworms in *Temnothorax nylander* ants, entomologist Susanne Foitzik and colleagues found something unexpected: rather than reducing its host’s fitness, as is typical of parasites, *A. brevis* greatly extends the lifespan of a *T. nylander* worker ant and seems to halt the effects of aging. Furthermore, those infected receive special treatment, ceasing their share of labor to sustain the colony and remaining in the nest as uninfected workers feed, groom, and transport them. By contrast, the researchers observed that uninfected workers in parasitized colonies have shortened lifespans, most likely because the _____

Which choice most logically completes the text?

Answer

- A. uninfected workers are at high risk for direct exposure to *A. brevis* in the course of providing social care to the infected workers in the nest.
- B. need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.
- C. high level of activity maintained by the uninfected workers makes them better able than infected workers to quickly disperse when the nest is attacked by a predator.
- D. average lifespan of *T. nylander* worker ants in colonies without parasitic activity typically falls well below three years, the range covered by the study.

Correct Answer: B

Rationale

Choice B is the best answer. The text describes a study examining the relationship between a species of parasitic tapeworm, *A. brevis*, and its host insect, the *T. nylander* ant. According to the text, researchers were surprised to find that the tapeworm extends the life of its ant host, rather than reducing it. The text goes on to state that the infected ants end up doing less work to sustain the colony and that as a result, the uninfected ants take on the infected ants’ share of labor in addition to caring for them in their infected state. The study’s researchers also observed that the uninfected ants have shorter lifespans than expected. If the infected ants, who are doing less work in the colony, have longer lifespans, it can be inferred that the less an ant works, the longer it will live. The opposite of this statement can also be inferred: the more an ant works, the shorter its life. So, since the workload within the colony is being redistributed so that the infected ants work less while the uninfected ants work more (as they take on the neglected duties of the infected ants and also care for those ants), then it can be inferred that the lifespans of the uninfected ants are shortened because the need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.

Choice A is incorrect because the text does not indicate how *A. brevis* is transmitted to the ants or assert that uninfected ants are more likely to be directly exposed to *A. brevis* while caring for infected ants. Choice C is incorrect because the text makes no mention of the relative abilities of infected and uninfected ants to escape predators: in fact, predators are not mentioned in the text at all. Choice D is incorrect because the text does not supply any information about the average lifespans of the ants in colonies without parasitic activity; the text only indicates factors that lengthen and shorten the lifespans of ants in parasitized colonies.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Sir Gawain and the Green Knight is a circa 1400 poem written in Middle English—an archaic form of the English spoken today. Over the last several centuries, the English language has undergone such transformations in vocabulary, spelling, and grammar that most readers now rely on translations to read *Sir Gawain*. In the introduction to his 2007 translation, Simon Armitage remarks that the sonic patterns of the poem, which was written in alliterative verse (a verse form featuring extensive repetition of initial consonant sounds), are essential to its structure. Because many Modern English words begin with different sounds than their Middle English equivalents do, a strictly literal translation of *Sir Gawain* would therefore likely _____

Which choice most logically completes the text?

Answer

- A. preserve much of the original text’s meaning at the expense of other qualities that are also integral to the experience of reading the poem.
- B. appeal more to modern readers than would translations like Armitage’s that instead prioritize the original text’s sonic and structural qualities.
- C. be more faithful to the original intentions of the poem than most strictly literal translations of alliterative Middle English poems are to their originals.
- D. be preferable to modern readers who are primarily interested in learning what the poem reveals about historical conditions during the time it was originally written.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of translating *Sir Gawain and the Green Knight*. The text states that the poem was written in alliterative verse, a form that features extensive repetition of initial consonant sounds, and that, according to Armitage, the patterns of sound in the poem are essential to the poem’s structure. The text also notes that many Modern English words begin with different sounds than their Middle English equivalents (that is, the words with the same meanings) do. This means that if a translator were to render the poem’s Middle English words into their direct Modern English equivalents, the initial consonant sounds of many words would change and the original alliterative patterns wouldn’t hold. It follows, then, that a strictly literal (word for word) translation would preserve much of the original text’s meaning at the expense of sonic and structural qualities that are also integral to the experience of reading the poem.

Choice B is incorrect because the text doesn’t address modern readers’ preferences among translation approaches. Although the text implies that Armitage may have prioritized the sonic patterns and structure of the original poem in his translation, it doesn’t give any indication that this approach would be less appealing to modern readers than an approach that directly translates the words but loses the original patterns of sounds in the poem. Choice C is incorrect. The text actually suggests that a strictly literal Modern English translation of *Sir Gawain and the Green Knight* wouldn’t be especially faithful to the original intentions of the poem because the Modern English words would change the sonic patterns Armitage notes are essential to the alliterative poem’s structure—that is, such a translation would change patterns of sound that are a fundamental part of the original poem’s composition. Moreover, there’s no reason to think that a strictly literal translation of that poem would be more faithful to the original intentions than literal translations of other alliterative Middle English poems would be, since they would presumably all face the issue of a change in the patterns of sounds. Choice D is incorrect because the end of the text makes a point about sounds, not meaning. Modern readers interested in gathering historical information from the poem might prefer a strictly literal translation (one that directly translates the words without making additional changes), but that idea doesn’t logically follow from the fact that many Modern English words start with different sounds than their Middle English counterparts; a change in the sounds wouldn’t have any effect on the information the poem conveys.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In the “language nest” model of education, Indigenous children learn the language of their people by using it as the medium of instruction and socialization at pre-K or elementary levels. In their 2016 study of a school in an Anishinaabe community in Ontario, Canada, scholars Lindsay Morcom and Stephanie Roy (who are Anishinaabe themselves) found that the model not only imparted fluency in the Anishinaabe language but also enhanced students’ pride in Anishinaabe culture overall. Given these positive effects, Morcom and Roy predict that the model increases the probability that as adults, former students of the school will transmit the language to younger generations in their community.

Which finding, if true, would most strongly support the researchers’ prediction?

Answer

- A. Anishinaabe adults who didn’t attend the school feel roughly the same degree of cultural pride as the former students of the school feel.
- B. After transferring to the school, new students experience an increase in both fluency and academic performance overall.
- C. As adults, former students of the school are just as likely to continue living in their community as individuals who didn’t attend the school.
- D. As they complete secondary and higher education, former students of the school experience no loss of fluency or cultural pride.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support the researchers’ prediction about the language nest model of education. The text states that Morcom and Roy studied the effects of the language nest model of education on students at an Anishinaabe school, and they found that the model—which is used with students during pre-K or elementary school—increased students’ fluency in the Anishinaabe language and pride in Anishinaabe culture. The researchers predicted that the students’ positive early experiences with the Anishinaabe language would lead them to be more likely to later share the language with younger generations. If former students maintain full fluency and cultural pride after finishing secondary and higher education, it follows that they would be both able and motivated to share what they know with others; this would likely result in a higher probability of transmitting the language to younger generations, as the researchers predict.

Choice A is incorrect because finding that Anishinaabe adults who didn’t attend the school feel approximately the same degree of cultural pride as those adults who did attend wouldn’t support the researchers’ prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people. Choice B is incorrect because finding that new students experience increased performance in language fluency and academics would suggest that the school has a positive effect on students when they attended but wouldn’t reveal anything about those students’ later actions as adults (such as their likelihood of sharing their knowledge with younger generations). Choice C is incorrect because finding that Anishinaabe adults who attended the school are equally likely to stay in the community as adults who didn’t attend the school wouldn’t support the researchers’ prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

When digging for clams, their primary food, sea otters damage the roots of eelgrass plants growing on the seafloor. Near Vancouver Island in Canada, the otter population is large and well established, yet the eelgrass meadows are healthier than those found elsewhere off Canada’s coast. To explain this, conservation scientist Erin Foster and colleagues compared the Vancouver Island meadows to meadows where otters are absent or were reintroduced only recently. Finding that the Vancouver Island meadows have a more diverse gene pool than the others do, Foster hypothesized that damage to eelgrass roots increases the plant’s rate of sexual reproduction; this, in turn, boosts genetic diversity, which benefits the meadows’ health overall.

Which finding, if true, would most directly undermine Foster’s hypothesis?

Answer

- A. At some sites in the study, eelgrass meadows are found near otter populations that are small and have only recently been reintroduced.
- B. At several sites not included in the study, there are large, well-established sea otter populations but no eelgrass meadows.
- C. At several sites not included in the study, eelgrass meadows’ health correlates negatively with the length of residence and size of otter populations.
- D. At some sites in the study, the health of plants unrelated to eelgrass correlates negatively with the length of residence and size of otter populations.

Correct Answer: C

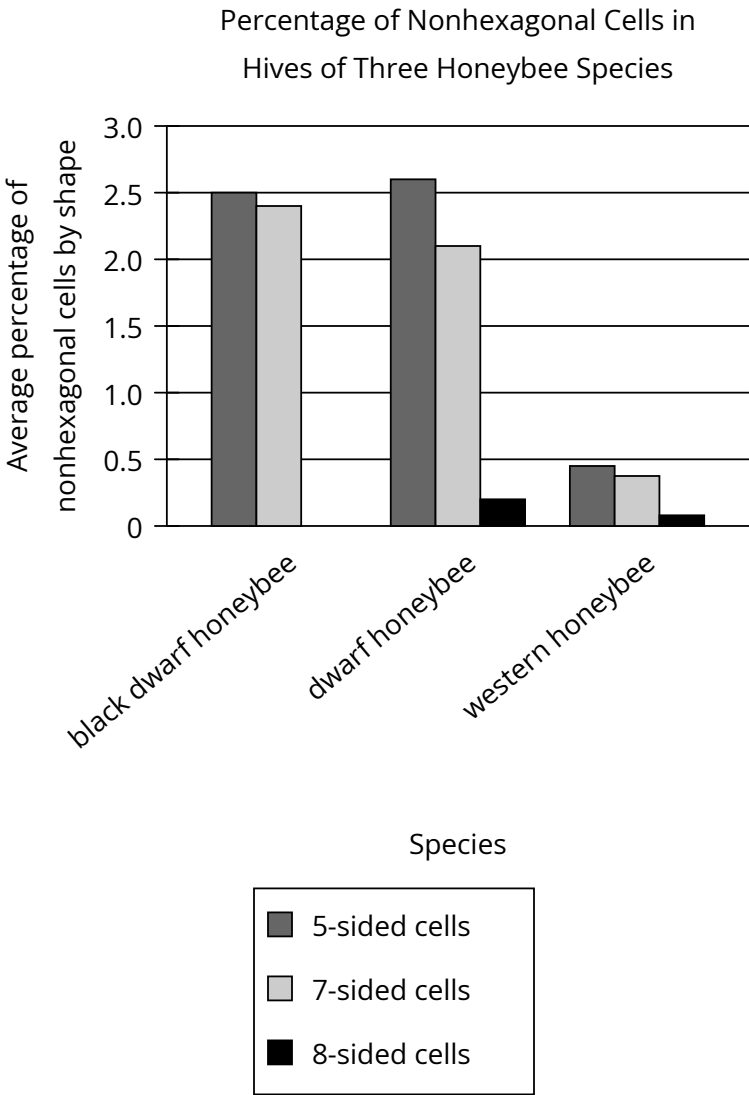
Rationale

Choice C is the best answer because it presents a finding that, if true, would weaken Foster’s hypothesis that damage to eelgrass roots improves the health of eelgrass meadows by boosting genetic diversity. The text indicates that sea otters damage eelgrass roots but that eelgrass meadows near Vancouver Island, where there’s a large otter population, are comparatively healthy. When Foster and her colleagues compared the Vancouver Island eelgrass meadows to those that don’t have established otter populations, the researchers found that the Vancouver Island meadows are more genetically diverse than the other meadows are. This finding led Foster to hypothesize that damage to the eelgrass roots encourages eelgrass reproduction, thereby improving genetic diversity and the health of the meadows. If, however, other meadows not included in the study are less healthy the larger the local otter population is and the longer the otters have been in residence, that would suggest that damage to the eelgrass roots, which would be expected to increase with the size and residential duration of the otter population, isn’t leading meadows to be healthier. Such a finding would therefore weaken Foster’s hypothesis.

Choice A is incorrect because finding that small, recently introduced otter populations are near other eelgrass meadows in the study wouldn’t weaken Foster’s hypothesis. If otter populations were small and only recently established, they wouldn’t be expected to have caused much damage to eelgrass roots, so even if those eelgrass meadows were less healthy than the Vancouver Island meadows, that wouldn’t undermine Foster’s hypothesis. In fact, it would be consistent with Foster’s hypothesis since it would suggest that the greater damage caused by larger, more established otter populations is associated with healthier meadows. Choice B is incorrect because the existence of areas with otters but without eelgrass meadows wouldn’t reveal anything about whether the damage that otters cause to eelgrass roots ultimately benefits eelgrass meadows. Choice D is incorrect because the health of plants other than eelgrass would have no bearing on Foster’s hypothesis that damage to eelgrass roots leads to greater genetic diversity and meadow health. It would be possible for otters to have a negative effect on other plants while nevertheless improving the health of eelgrass meadows by damaging eelgrass roots.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Honeybee hives consist mainly of hexagonal (six-sided) units called cells, in which queens lay eggs. Hexagonal cells for eggs that develop into nonreproductive workers are smaller than those for eggs that develop into reproductive drones, though the size difference varies by species. Difference in cell size results in a construction problem—it’s hard to neatly connect sections of small cells to sections of large cells—that worsens as the difference increases. To fill in gaps between the sections when building a hive, bees rely on cells that have more or fewer than six sides. A student studying beehive structure consults data on three species, concluding that _____

Which choice most effectively uses data from the graph to complete the student’s conclusion?

Answer

- A. cells for worker eggs are probably closer in size to cells for drone eggs in the hives of the western honeybee than in the hives of the dwarf honeybee and the black dwarf honeybee.
- B. both the western honeybee and the black dwarf honeybee probably reserve eight-sided cells for drone eggs, while the dwarf honeybee likely deposits drone eggs in seven-sided cells.

- C. the western honeybee probably relies on many more geometrical shapes when constructing cells than either the dwarf honeybee or the black dwarf honeybee does.
- D. the percentage of hexagonal cells is probably slightly lower in the hives of the western honeybee than in the hives of the dwarf honeybee and the black dwarf honeybee.

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the graph to complete the student's conclusion about beehive structure. The text explains that in the hives of honeybees, the hexagonal cells housing drone eggs are larger than the hexagonal cells housing worker eggs, and that this size difference results in a construction problem that the bees address by using nonhexagonal cells to fill gaps between sections of drone-egg cells and worker-egg cells. The text also states that the size difference between drone-egg cells and worker-egg cells varies by species of honeybee. The graph displays data on the percentage of nonhexagonal cells in the hives of three species. In the hives of the western honeybee, the percentages of five-sided, seven-sided, and eight-sided cells are all less than 0.5%. But in the hives of the black dwarf honeybee, the percentages of five-sided and seven-sided cells are higher than those for the western honeybee: about 2.5% for both. And for the dwarf honeybee, the percentages of five-sided and seven-sided cells are also higher than those for the western honeybee: slightly over 2.5% and slightly over 2.0%, respectively; additionally, the dwarf honeybee possesses a higher percentage of eight-sided cells than the western honeybee does. Taken altogether, the graph shows that the hives of the western honeybee consist of a smaller percentage of nonhexagonal cells than the hives of the two other species do. Since the nonhexagonal cells exist only to solve the construction problem arising from the difference in size between drone-egg cells and worker-egg cells, a smaller percentage of nonhexagonal cells would be associated with a smaller size difference between the two types of cells. Therefore, it can be concluded from the data that worker-egg cells are probably closer in size to drone-egg cells in the hives of the western honeybee than in the hives of the other two species.

Choice B is incorrect because, as the text states, honeybee species deposit their eggs in hexagonal cells, not in nonhexagonal ones. Thus, the western honeybee and black dwarf honeybee wouldn't deposit drone eggs in eight-sided cells, and the dwarf honeybee wouldn't deposit drone eggs in seven-sided cells. Choice C is incorrect. The text explains that honeybees rely mainly on one geometric shape, the hexagon, when constructing their hives, and the graph shows that the western honeybee relies on the same nonhexagonal shapes as the dwarf honeybee does: five-sided, seven-sided, and eight-sided cells. In other words, the western honeybee and dwarf honeybee rely on the same number of geometric shapes. For the black dwarf honeybee, the graph displays data only for five-sided and seven-sided cells, which suggests a total absence of eight-sided cells. Yet this would be only one less nonhexagonal shape than is seen in the western honeybee. Thus, based on the graph, it would be inaccurate to say that the western honeybee relies on "many more" geometrical shapes than the other two species do. Choice D is incorrect. As the text explains, honeybee hives consist mainly of hexagonal cells, and sections of nonhexagonal cells are used to connect sections of hexagonal cells of different sizes. Since the graph indicates that the percentage of nonhexagonal cells is lower for the western honeybee than it is for the dwarf honeybee or black dwarf honeybee, the western honeybee would conversely have a *higher* percentage of hexagonal cells than the either the dwarf honeybee or black dwarf honeybee does, not a lower percentage.

Question ID: 8d88740e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is adapted from William Shakespeare's 1597 play *The Tragedy of King Richard III*. Richard is reflecting on the recent arrest of his brother, the Duke of Clarence, on suspicion of treason against King Edward IV. Derby, Hastings, Buckingham, Rivers, Dorset, and Grey are also members of the English nobility.

RICHARD: I do the wrong, and first begin to brawl.

The secret mischiefs that I set [flowing]

I lay unto the grievous charge of others.

Clarence, whom I indeed have cast in darkness,

I do bewEEP to many simple [gullible people],

Namely, to Derby, Hastings, Buckingham;

And tell them 'tis the Queen and her allies

That stir the King against the Duke my brother.

Now they believe it, and withal whet me

To be revenged on Rivers, Dorset, Grey.

Which choice best describes what happens in the text?

Answer

- A. Richard describes having wept as he informed Derby, Hastings, and Buckingham that the queen and her allies convinced the king to act against Clarence, and says that the earnestness of his grief caused them to accept his version of events.
- B. Richard attributes Clarence's troubles to both his own secret plotting and the distrust of Clarence that the queen and her allies Derby, Hastings, and Buckingham have planted in the king's mind.
- C. Richard indicates that he has pretended to be aggrieved about Clarence's situation and has proclaimed it to be the fault of the queen and her allies, but in reality, he has caused the hostility the king feels toward Clarence.
- D. Richard acknowledges that his mischievous nature has spurred him to commit misdeeds in the past, including instigating enmity between the king and Clarence, but he reports that he has hitherto not lost the trust of the queen and her allies.

Correct Answer: C

Rationale

Choice C is the best answer because it accurately describes what happens in the text. Richard admits he committed “the wrong” and blames others for his own “secret mischiefs.” He reveals that he has “cast [Clarence] in darkness” (that is, caused his imprisonment) but feigns grief—“I do bewEEP”—before “many simple [gullible people],” namely Derby, Hastings, and Buckingham. He tells them that the queen and her allies are the

ones who turned the king against Clarence. Richard's use of "beweep" and his labeling of his audience as gullible confirm that his grief is a performance. Thus, Richard pretends to be aggrieved and blames the queen while actually being responsible for Clarence's predicament.

Choice A is incorrect because it characterizes Richard's grief as earnest, claiming that his sincere grief moved others to accept his claims. However, Richard admits his wrongdoing and calls his audience "simple [gullible people]," making clear that they were deceived by a calculated performance rather than persuaded by genuine emotion. Rather than being sincere, Richard's weeping is part of a deliberate scheme to blame the queen and her allies for what he did. Choice B is incorrect because Richard does not attribute any portion of Clarence's troubles to the queen; he reveals that he "cast [Clarence] in darkness" and merely tells others the queen is responsible. Additionally, Derby, Hastings, and Buckingham are not identified as the queen's allies; rather, Rivers, Dorset, and Grey are implied to be the queen's allies. Choice D is incorrect because the text does not address whether Richard has maintained the trust of the queen and her allies. Instead, Richard tells Derby, Hastings, and Buckingham that the queen is to blame for Clarence's troubles, and they in turn urge him to seek revenge on Rivers, Dorset, and Grey, who are implied to be the queen's allies. The speech focuses on Richard's manipulation of others, not on his standing with the queen.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is from Mick Herron’s 2023 novel *The Secret Hours*. The narrator is describing members of a government committee.

Finally, and adding much-needed gravitas, was Sir Winston Day, whose features seemed moulded to adorn a bust, or possibly a stamp, and whose forehead was so evidently bulging with grey matter that it would have been impertinent to inquire too closely into the actual achievements his half century of public service had produced. His recently published memoirs possibly cast light on this enigma, but given that such details were not provided until after the thirty-page mark, they might as well have remained state secrets.

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Based on the text, which choice best describes Sir Winston Day?

Answer

- A. He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction.
- B. He looks like a person worthy of respect, but his memoirs reveal that some of his actions were dishonorable.
- C. He would be a celebrated public figure if his achievements did not have to be kept secret.
- D. He has maintained a modest profile even though he has served the public capably for many years.

Correct Answer: A

Rationale

Choice A is the best answer because it accurately describes Sir Winston Day as he is portrayed in the text. The text indicates that Sir Winston’s “features seemed moulded to adorn a bust, or possibly a stamp,” suggesting that he appeared distinguished. But the text also indicates that his “actual achievements” during his long public service constitute an “enigma,” or mystery, highlighting the point with a joke about Sir Winston’s uninteresting memoirs (any information about his achievements past the first thirty pages “might as well have remained state secrets” since, it’s implied, no one bothered to read that far). Thus, the text suggests that Sir Winston appears distinguished even though his known accomplishments, if any, are relatively few.

Choice B is incorrect because nothing in the text indicates that Sir Winston behaved dishonorably. Rather, the text indicates that his “actual achievements” during his long public service constitute an enigma, or mystery, suggesting that he may have done very little, not that what he has done was disreputable. Choice C is incorrect. Although the text says that Sir Winston’s achievements that appear “after the thirty-page mark” in his memoirs “might as well have remained state secrets,” this is a joke about the public’s lack of interest in the memoirs, not a sincere claim that secrecy was intended. Choice D is incorrect because the text does not associate Sir Winston with modesty, nor does it suggest that he was a capable public servant. Rather, by indicating that Sir Winston’s “features seemed moulded to adorn a bust, or possibly a stamp,” the text suggests that Sir Winston’s appearance communicated importance, not modesty. The text further indicates that his “actual achievements” during his long public service constitute an enigma, or mystery, suggesting that he may have done very little, not that he was particularly capable.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Average Nitrate and Phosphate Concentrations in Seawater after Volcanic Eruption

Nutrient	Seawater in lava-affected area, 5–45 meters below surface	Seawater in lava-affected area, 75–125 meters below surface	Seawater outside of lava-affected area, 5–45 meters below surface	Seawater outside of lava-affected area, 75–125 meters below surface
Nitrate (micromoles per liter)	3.1	0.4	≤0.03	≤0.01
Phosphate (micromoles per liter)	0.17	0.09	0.14	0.06

After a volcanic eruption spilled lava into North Pacific Ocean waters, a dramatic increase of diatoms (a kind of phytoplankton) near the surface occurred. Scientists assumed the diatoms were thriving on nutrients such as phosphate from the lava, but analysis showed these nutrients weren't present near the surface in forms diatoms can consume. However, there was an abundance of usable nitrate, a nutrient usually found in much deeper water and almost never found in lava. Microbial oceanographer Sonya Dyhrman and colleagues believe that as the lava plunged nearly 300 meters below the surface it dislodged pockets of this nutrient, releasing it to float upward, given that _____

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. at 5–45 meters below the surface, the average concentration of phosphate was about the same in the seawater in the lava-affected area as in the seawater outside of the lava-affected area.
- B. for both depth ranges measured, the average concentrations of nitrate were substantially higher in the seawater in the lava-affected area than in the seawater outside of the lava-affected area.
- C. for both depth ranges measured in the seawater in the lava-affected area, the average concentrations of nitrate were substantially higher than the average concentrations of phosphate.
- D. in the seawater outside of the lava-affected area, there was little change in the average concentration of nitrate from 75–125 meters below the surface to 5–45 meters below the surface.

Correct Answer: B

Rationale

Choice B is the best answer. The claim is that the lava freed the deep nitrate, allowing it to float upward. The table supports this by showing that there was more nitrate in the lava-affected seawater at various depths above 300 meters (the depth to which the lava plunged) than in unaffected seawater.

Choice A is incorrect. This choice doesn't complete the statement in a way that supports the claim. The claim is about nitrate, not phosphate. Choice C is incorrect. This choice doesn't complete the statement in a way that supports the claim. The claim is only about nitrate—the concentrations of phosphate aren't relevant. Choice D is incorrect. This choice doesn't complete the statement in a way that supports the claim. It doesn't say anything about the seawater inside the lava-affected area.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

If some artifacts recovered from excavations of the settlement of Kuulo Kataa, in modern Ghana, date from the thirteenth century CE, that may lend credence to claims that the settlement was founded before or around that time. There is other evidence, however, strongly supporting a fourteenth century CE founding date for Kuulo Kataa. If both the artifact dates and the fourteenth century CE founding date are correct, that would imply that _____

Which choice most logically completes the text?

Answer

- A. artifacts from the fourteenth century CE are more commonly recovered than are artifacts from the thirteenth century CE.
- B. the artifacts originated elsewhere and eventually reached Kuulo Kataa through trade or migration.
- C. Kuulo Kataa was founded by people from a different region than had previously been assumed.
- D. excavations at Kuulo Kataa may have inadvertently damaged some artifacts dating to the fourteenth century CE.

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of artifacts and Kuulo Kataa’s founding date. If it were true both that Kuulo Kataa was founded in the fourteenth century CE and that artifacts found in excavations of the settlement are from the thirteenth century CE, it would be reasonable to conclude that the artifacts weren’t created in the Kuulo Kataa settlement. That would suggest, then, that the artifacts originated somewhere else and eventually reached the settlement through trading or as people migrated.

Choice A is incorrect because the existence of thirteenth-century CE artifacts recovered during excavations of a settlement founded in the fourteenth century CE isn’t logically connected to artifacts from one century being more commonly recovered than artifacts from another century. Rather than suggesting anything about how frequently artifacts from different times are found, the existence of artifacts confirmed as predating the settlement’s founding suggests that those items arrived in Kuulo Kataa during or after its establishment. Choice C is incorrect because the text focuses on time periods and says nothing about which region the founders of Kuulo Kataa have been thought to come from; similarly, the text doesn’t suggest anything about where the thirteenth-century CE artifacts originated other than not from Kuulo Kataa. Therefore, it isn’t logical to conclude that the mere existence of artifacts confirmed as predating the Kuulo Kataa settlement suggests that the founders of the settlement came from a particular region other than one previously assumed. Choice D is incorrect because the existence of artifacts from the thirteenth century CE at a site dated to the fourteenth century CE doesn’t imply that fourteenth-century objects were damaged during excavations. There’s nothing in the text to suggest that any objects were damaged; rather, the existence of artifacts confirmed as predating the settlement’s founding suggests that those items were brought to Kuulo Kataa during or after its establishment.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Effects of Mycorrhizal Fungi on 3 Plant Species

Plant species	Mycorrhizal host	Average mass of plants grown in soil containing mycorrhizal fungi (in grams)	Average mass of plants grown in soil treated to kill fungi (in grams)
Corn	yes	15.1	3.8
Marigold	yes	10.2	2.4
Broccoli	no	7.5	7

Mycorrhizal fungi in soil benefits many plants, substantially increasing the mass of some. A student conducted an experiment to illustrate this effect. The student chose three plant species for the experiment, including two that are mycorrhizal hosts (species known to benefit from mycorrhizal fungi) and one nonmycorrhizal species (a species that doesn't benefit from and may even be harmed by mycorrhizal fungi). The student then grew several plants from each species both in soil containing mycorrhizal fungi and in soil that had been treated to kill mycorrhizal and other fungi. After several weeks, the student measured the plants' average mass and was surprised to discover that _____

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. broccoli grown in soil containing mycorrhizal fungi had a slightly higher average mass than broccoli grown in soil that had been treated to kill fungi.
- B. corn grown in soil containing mycorrhizal fungi had a higher average mass than broccoli grown in soil containing mycorrhizal fungi.
- C. marigolds grown in soil containing mycorrhizal fungi had a much higher average mass than marigolds grown in soil that had been treated to kill fungi.
- D. corn had the highest average mass of all three species grown in soil that had been treated to kill fungi, while marigolds had the lowest.

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to complete the statement. The text explains that mycorrhizal hosts are plants that benefit from the presence of mycorrhizal fungi in the soil and that some such plants produce more mass when grown in the presence of these fungi, while for nonmycorrhizal species the fungi either have no effect or may be harmful. The experiment included two mycorrhizal hosts (corn and marigold) and one nonmycorrhizal species (broccoli). Given the claim in the text that nonmycorrhizal species will see either no difference or a decrease in mass when exposed to mycorrhizal fungi, the student would likely have been surprised by the higher average mass for broccoli grown in the presence of the fungi than the broccoli grown in the soil treated to kill fungi.

Choice B is incorrect. Although this choice accurately describes the corn data from the table, the fact that the mycorrhizal host corn is more massive in the presence of the fungi likely fits with what the student expected and would therefore not be surprising. Choice C is incorrect. Although this choice accurately describes the marigold data from the table, the fact that the mycorrhizal host marigold is more massive in the presence of the fungi is likely what the student expected and thus would not be surprising. Choice D is incorrect because it does not accurately represent the data in the table—when grown in soil treated to kill fungi, corn had an average mass of 3.8 g while broccoli had an average mass of 7g—and because making comparisons among the plants in the no-fungi condition, by itself, does not provide a basis to compare the average mass of mycorrhizal hosts and nonmycorrhizal species grown in the presence of the fungi with those grown in the soil treated to kill fungi.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Indigenous cultures possess unique knowledge of the medicinal uses of plants. According to a 2021 study, 73 percent of the medicinal uses of plants native to North America are reflected in the vocabulary of a single Indigenous language. However, as more and more Indigenous people exclusively speak a globally dominant language, such as English, their ancestral languages fade from daily use. These facts lend added importance to tribal nations’ efforts to preserve their languages. By ensuring the continued use of Cherokee, Ojibwe, and the hundreds of other Indigenous languages in what is now the United States, tribal nations are also _____

Which choice most logically completes the text?

Answer

- A. increasing the number of medicinal plants represented in the vocabularies of Indigenous languages.
- B. transmitting terms for medicinal plants from Indigenous languages to globally dominant languages.
- C. preserving knowledge about the medicinal value of plants native to the tribal nations’ lands.
- D. ensuring that citizens of tribal nations have physical access to medicinal plants.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of the relationship between Indigenous languages and knowledge of the medicinal uses of plants. The text states that Indigenous cultures possess special knowledge of the medicinal uses of plants, which is reflected in their vocabulary. The text then discusses how tribal nations are working to preserve their languages, whose daily use is declining as globally dominant languages become increasingly dominant in Indigenous communities. Given that the languages of tribal nations in what is now the United States function as repositories of knowledge about plants’ medicinal uses, it logically follows that continued use of those languages will assist with passing on knowledge about the medicinal value of plants native to the tribal nations’ lands.

Choice A is incorrect because the text states that preserving Indigenous languages will increase the knowledge, not the number, of medicinal plants. Choice B is incorrect because the text is concerned with how vocabulary about the medicinal value of plants can be preserved through the continued daily use of Indigenous languages, not with how such vocabulary can be incorporated into globally dominant, non-Indigenous languages. Moreover, the text explains that the exclusive use of globally dominant languages in Indigenous communities comes at an expense to the continued daily use of those communities’ languages. Given this relationship, it is unlikely globally dominant languages would borrow Indigenous vocabulary pertaining to plants’ medicinal uses. Choice D is incorrect because the text doesn’t discuss physical access to medicinal plants, instead focusing on Indigenous knowledge and language surrounding the medicinal uses of plants.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Conservationists worldwide are working to protect ecosystems from habitat destruction and biodiversity loss, and in many cases, initiatives that rely on natural features or processes can help address such challenges. In response to a rapidly dwindling population of blueback salmon, the Quinault Indian Nation (a tribe in Washington State) partnered with the conservation organization Wild Salmon Center to restore naturally occurring logjams in the Quinault River. The logjams create shady pools where the blueback salmon can rest and spawn, thus promoting blueback population recovery.

Which choice best states the main idea of the text?

Answer

- A. A partnership between the Quinault Indian Nation and Wild Salmon Center shows the importance of collaborative approaches to preserving biodiversity.
- B. Nature-based approaches can be effective ways to achieve conservation goals.
- C. As indicated by a recent project, logjams help the blueback salmon thrive and reproduce.
- D. Scientists now realize that nature-based conservation methods offer better long-term solutions to environmental issues than methods that are not nature-based do.

Correct Answer: B

Rationale

Choice B is the best answer because it best states the main idea of the text: that nature-based approaches can be effective for achieving conservation goals. The text indicates that in many cases where conservationists are trying to protect ecosystems, their methods depend on natural processes or features. The text then gives an example of this phenomenon, a project with the Quinault Indian Nation that allowed logjams to form naturally in a river, creating spawning habitats for blueback salmon.

Choice A is incorrect. Although the text does suggest that the partnership with the Quinault Indian Nation was beneficial, this is not the central aim of the text; the text primarily argues that nature-based approaches to conservation can be effective. Choice C is incorrect. Although the text indicates that logjams are helpful to blueback salmon, the example of the blueback salmon project is included to illustrate the larger point made earlier in the text: that nature-based approaches to conservation are often effective. Choice D is incorrect. There is no evidence in the text to support a direct comparison of the efficacy of nature-based conservation approaches to other types of approaches. The text merely indicates that nature-based approaches can often be effective.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Whistler waves are low-frequency plasma waves that on Earth are typically generated by lightning. Numerous recordings of whistler waves on Venus have led many scientists to suggest that the planet’s atmosphere is host to extensive amounts of lightning, and, in fact, Venusian whistler waves have similar energy signatures to those of whistler waves generated by lightning on Earth. The majority of Venusian whistler wave data come from two spacecraft missions—the Pioneer Venus Orbiter (PVO) and the Venus Express (VEX)—which have included few observations of other phenomena consistent with lightning occurrences (such as flashes of light), leading other scientists to suggest that _____

Which choice most logically completes the text?

Answer

- A. similarities in the energy signatures of Venusian and Earth whistler waves may reflect imprecisions in the PVO and VEX data.
- B. the purported Venusian whistler waves must actually be some other type of atmospheric activity than whistler waves.
- C. Venusian lightning has properties that make it unlikely to generate whistler waves.
- D. there are geophysical characteristics of Venus not shared with Earth that promote the generation of whistler waves.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of whistler waves on Venus and the question of their origin. The text explains that whistler waves are plasma waves and that on Earth they are generated by lightning. The text goes on to state that numerous recordings of whistler waves on Venus have led many scientists to suggest Venus hosts lightning. However, the text goes on to explain that data from the PVO and VEX missions have included little evidence of the other phenomena that would be expected if lightning were occurring in Venus’s atmosphere. Because there is a lack of additional evidence of lightning, it is reasonable to conclude that some scientists would be skeptical that lightning is the source of the whistler waves on Venus and would suggest unique geophysical characteristics of Venus as an alternative explanation for the waves’ generation.

Choice A is incorrect because the text indicates that the other scientists are responding to the PVO and VEX data simply not showing additional evidence of lightning on Venus, not to flaws in the data. The text presents the similar energy signatures of Venusian and Earth whistler waves as an established observation and doesn’t suggest the PVO and VEX data are imprecise. Choice B is incorrect because the text presents the similar energy signatures of Venusian and Earth whistler waves as an established observation. The text indicates that the other scientists are concerned with the lack of corroborating evidence of lightning on Venus, suggesting not that they doubt that the waves are whistler waves but rather that they doubt that the waves are generated by lightning. Choice C is incorrect because the text indicates that the other scientists are responding to the data providing little corroborating evidence of lightning on Venus. This suggests that instead of making a suggestion about properties of Venusian lightning, the other scientists likely would suggest that there may not even be lightning on Venus.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

The variety of species with adaptations to produce toxins is matched by the variety of uses of those toxins: northern stargazers, for example, use toxins for defense, whereas tiger snakes use toxins for predation and skeleton shrimp use toxins for intraspecific competition. In fact, a species may have adaptations enabling it to produce a toxin with multiple uses. Finding that the venom used by the Panamanian scorpion *Centruroides granosus* to subdue prey also inhibits growth of the pathogenic bacteria *Escherichia coli*, Dumas Gálvez and colleagues conclude that the particular form of venom production observed in *C. granosus* may have propagated through the species because it mitigates risk during feeding in addition to enhancing predation success.

Which finding, if true, would most directly support Gálvez and colleagues’ conclusion?

Answer

- A. *E. coli* does not appear to be virulent for *C. granosus* even when transmitted from prey captured without the use of venom.
- B. *E. coli* is frequently found in species preyed on by *C. granosus* and can survive exposure to the digestive juices of *C. granosus*.
- C. *C. granosus* appears to be chemically sensitive to prey infected with *E. coli* and tends to favor uninfected individuals.
- D. Exposure to *C. granosus* venom also inhibits the growth of nonpathogenic bacteria species common in the native environment of *C. granosus*.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support Dumas Gálvez and colleagues’ conclusion that the venom produced by the scorpion species *Centruroides granosus* helps mitigate, or reduce, risk during feeding in addition to enhancing predation success. According to the text, Gálvez and colleagues found that the venom used by *C. granosus* both subdues prey and inhibits the growth of *Escherichia coli*, a pathogen that, if ingested by *C. granosus*, would presumably cause disease unless neutralized in some way. If it were true that *E. coli* is commonly found in species preyed on by *C. granosus* and, moreover, can withstand *C. granosus*’s digestive juices after ingestion, this would suggest that *C. granosus* likely relies on another mechanism to neutralize *E. coli* to make *E. coli*-infected prey safe for consumption by the scorpion species. Given that, as the text states, *C. granosus*’s venom was found to inhibit the pathogen’s growth, it therefore follows that the venom provides protection for *C. granosus* against *E. coli* that its digestive system wouldn’t otherwise provide, making it reasonable to conclude that the trait may have spread through the species because it mitigates risk during feeding.

Choice A is incorrect because a finding that *E. coli* doesn’t appear to be virulent, or dangerous, for *C. granosus* even when venom isn’t used would weaken rather than strengthen Gálvez and colleagues’ conclusion that a particular form of venom production spread in *C. granosus* in part because it helps reduce risk during feeding, since this finding would suggest that *E. coli* isn’t actually a risk to *C. granosus* when consuming prey. Choice C is incorrect because a finding that *C. granosus* can detect and avoid consuming *E. coli*-infected prey in the first place would suggest that an ability other than venom production is the primary factor that reduces *C. granosus*’s risk when feeding, which would suggest there has been less evolutionary pressure to develop venom that provides protection; thus, this finding wouldn’t clearly support the conclusion that *C. granosus*’s form of venom production spread in the species in part because it helps reduce risk during feeding. Choice D is incorrect because Gálvez and colleagues’ conclusion focuses on *C. granosus* venom in relation to the risk of *E. coli*, and a finding that the venom also inhibits nonpathogenic (not disease-causing) bacteria species, which presumably don’t pose a risk if consumed, wouldn’t be relevant to the conclusion that *C. granosus*’s particular form of venom production spread in the species in part because it helps reduce risk during feeding.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Across brown bears—omnivores with high dietary plasticity—there is wide variety in dietary mix, which may reflect genetics, local resource availability, or social learning (cubs stay with their mothers for two years or more). Evaluating these possibilities, Anne Hertel et al. analyzed 30 years of data on trophic position (indicative of dietary mix) for female brown bears. After separation, daughters, who tended to settle near their mothers, occupied the same trophic positions as their mothers for two years, but the correlation disappeared by year five. Trophic correlation with unrelated individuals in similar habitats was modest, while habitat-independent correlation with nonmaternal relatives (e.g., cousins) was no different than with unrelated individuals. These findings suggest that _____

Which choice most logically completes the text?

Answer

- A. growing dissimilarity between mothers and their daughters with regard to dietary mix may reflect changes in the resources available in maternal habitats, though social learning could also contribute to the trend.
- B. dietary mix among females may reflect a social learning effect that eventually diminishes, though environmental constraints cannot be ruled out as a contributing factor.
- C. female dietary mix is best understood as changeable and contingent on fluctuating environmental conditions rather than as the result of social learning or genetic factors.
- D. social learning and resource fluctuations may both play a role in dietary mix among females, at least temporarily, though genetic factors appear to make a significant contribution as well.

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of Hertel et al.’s study of dietary mix variation among female brown bears. According to the text, the researchers evaluated genetics, local resource availability, and social learning as factors in female brown bears’ trophic position, or position in the food web. According to the text, daughters tended to settle near their mothers after separating and held the same trophic positions as the mothers for two years, but no longer did by the fifth year—a temporary similarity that may reflect a socially learned behavior that faded over time after separation. Additionally, the text states that the degree of similarity in trophic position between nonmaternal relatives (e.g., cousins) was the same as that between unrelated individuals, suggesting that genetics don’t significantly influence dietary mix, and the text also states that there was modest similarity in the trophic positions of unrelated individuals in similar habitats, suggesting that environmental conditions (which would include resource availability) may have some influence on dietary mix. Taken together, the findings suggest that while environmental constraints can’t be ruled out as a contributing factor, dietary mix among female brown bears may reflect a social learning effect that eventually diminishes.

Choice A is incorrect because the text notes that daughters tended to settle near their mothers after separating—that is, that they often didn’t leave the maternal habitat. Therefore, changes in resource availability in the mothers’ habitats would likely have affected the daughters similarly instead of causing their dietary mix to differ from that of their mothers. Further, the form of social learning addressed by the text is that between mothers and daughters; although the daughters no longer held the same trophic position as their mothers five years after separating from them, the text doesn’t address the possibility that the daughters’ divergence from the dietary behaviors of their mothers is the result of learning from other bears or suggest that the daughters even interacted with other adult bears in a substantive way. Choice C is incorrect. Instead of suggesting that environmental conditions are the most important factor in dietary mix variation and that social learning isn’t a factor, the findings discussed in the text suggest that environmental conditions may have some effect (since there was modest similarity in dietary mix between unrelated bears in similar habitats, or similar environmental conditions) and that social learning may be a factor (since daughters held the same

trophic positions as their mothers for two years after separating but not for five years, which may reflect daughters maintaining dietary behaviors they learned from their mothers but changing their behaviors over time). Choice D is incorrect because the findings presented in the text argue against, not for, genetics as a significant factor in variation in dietary mix. The text indicates that independent of their habitats, the degree of similarity in dietary mix between nonmaternal relatives, like cousins, was the same as that between unrelated individuals. This means that genetic relatedness didn't have a detectable effect on dietary mix independent of shared habitat.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

To investigate the history of plate subduction—when one of Earth’s tectonic plates slides beneath another—Sarah M. Aarons and colleagues compared ancient rocks from the Acasta Gneiss Complex in Canada to modern rocks. Using isotope analysis, the researchers found that Acasta rocks dating to about 4.02 billion years ago (bya) most strongly resemble modern rocks formed in a plume setting (an area in which hot rocks from Earth’s mantle flow upward into the crust). By contrast, they found that Acasta rocks dating to about 3.75 bya and 3.6 bya have an isotope composition that is similar to that of modern rocks formed in a subduction setting. Aarons’s team therefore concluded that _____

Which choice most logically completes the text?

Answer

- A. subduction-like processes began occurring in some locations no later than 3.75 bya.
- B. subduction replaced mantle plume formation as the most common geological process by about 4.02 bya.
- C. the majority of the rocks in the Acasta Gneiss Complex formed through subduction.
- D. the rocks in the Acasta Gneiss Complex are of a more recent origin than scientists previously thought.

Correct Answer: A

Rationale

Choice A is the best answer. Because researchers found “Acasta rocks dating to about 3.75 bya” were similar to “modern rocks formed in a subduction setting,” we can infer that subduction-like processes must have begun occurring in the Acasta Gneiss Complex by this time.

Choice B is incorrect. We only know about geological processes at the Acasta Gneiss Complex, so we do not have information to make inferences about geological processes in general. Also, notice that the rocks dated to 4.02 bya were found to still be formed in a plume setting, so the transition must have happened after this time. Choice C is incorrect. There’s no mention in the passage of what proportion of the rocks within Acasta Gneiss Complex were formed through subduction, so there’s no basis for this inference. Choice D is incorrect. The passage discusses various rocks in the Acasta Gneiss Complex that are dated to different periods of time, but nothing in the passage suggests that these dates are or were wrong.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Data Sources for Neptune Temperature Analysis

Instrument	Observatory	Data type	Observation years
TEXES (Texas Echelon Cross Echelle Spectrograph)	Gemini Observatory	spectroscopy	2007, 2019
T-ReCS (Thermal-Region Camera Spectrograph)	Gemini Observatory	infrared imaging	2007, 2010
LWS (Long Wavelength Spectrometer)	Keck Observatory	infrared imaging	2003
VISIR (VLT Imager and Spectrometer for mid-InfraRed)	European Southern Observatory	spectroscopy	2006

Julianne I. Moses and colleagues have reported that Neptune may have cooled significantly between 2003 and 2020. The team reached this conclusion by analyzing existing infrared imaging and spectroscopy data about the planet obtained from various instruments in different years. Of the team’s sources listed in the table, the earliest example of spectroscopy data included in the analysis was obtained in _____

Which choice most effectively uses data from the table to complete the text?

Answer

- A. 2007 using TEXES at the Gemini Observatory.
- B. 2007 using T-ReCS at the Gemini Observatory.
- C. 2006 using VISIR at the European Southern Observatory.
- D. 2003 using LWS at the W.M. Keck Observatory.

Correct Answer: C

Rationale

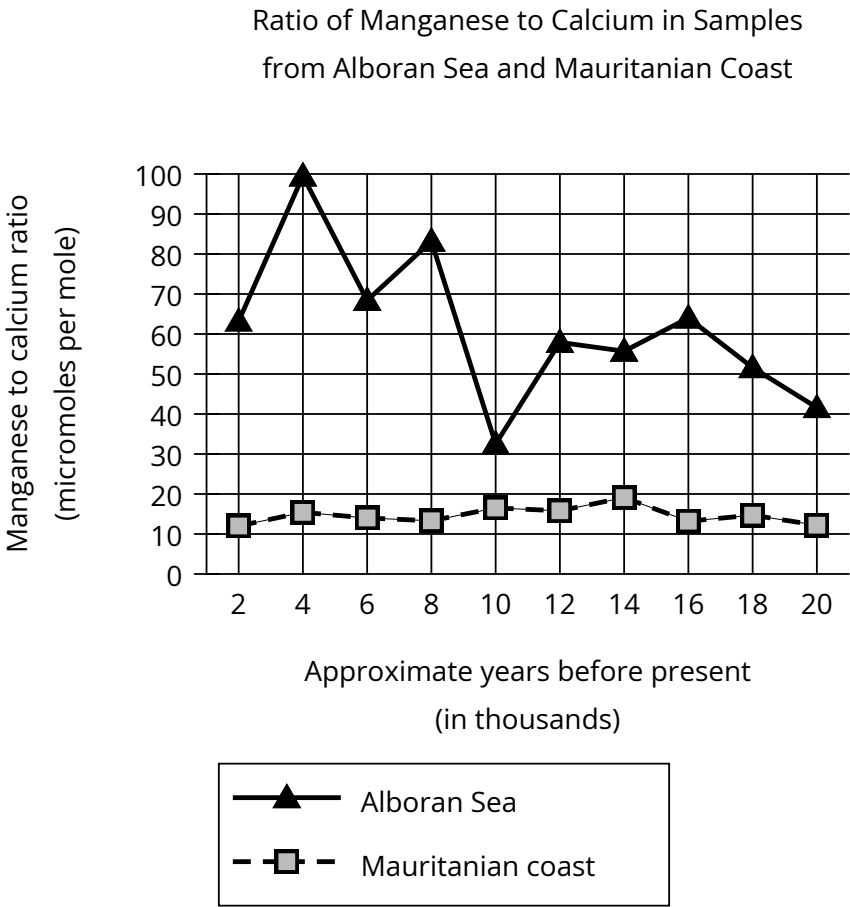
Choice C is the best answer because it effectively completes the text’s discussion of Moses and colleagues’ analysis of changes in Neptune’s temperature by using data from the table to identify the source of the earliest example of spectroscopy data included in this analysis. The text indicates that Moses and colleagues analyzed infrared imaging and spectroscopy data about Neptune obtained in different years by various instruments. The table lists the observatories where these instruments are housed, the type of data each instrument collected (infrared imaging or spectroscopy), and the years in which the instruments’ observations were made. According to the table, only two of these instruments obtained spectroscopy data: VISIR at the European Southern Observatory, which made its observations in 2006, and TEXES at the Gemini Observatory, which made its observations in later years (2007 and 2019). Although two other instruments are also included in the table, they obtained infrared imaging data, not spectroscopy data. Therefore, the earliest example of spectroscopy data included in the researchers’ analysis is the data obtained by VISIR at the European Southern Observatory in 2006.

Choice A is incorrect. Although the table indicates that TEXES at the Gemini Observatory collected spectroscopy data in 2007 and 2019, the table also indicates that this type of data was collected in 2006 by VISIR at the European Southern Observatory. Because the VISIR data were collected in the previous year, the data obtained in 2007 by TEXES are not the earliest example of spectroscopy data included in Moses and team’s analysis of Neptune’s temperature. Choice B is incorrect because the table indicates that T-ReCS at the Gemini Observatory collected infrared imaging data, not spectroscopy data. T-ReCS therefore cannot be the source of the earliest example of spectroscopy data that is included in Moses and colleagues’ analysis of Neptune’s temperature. Additionally, the T-ReCS data were collected in 2007, meaning that these data were obtained later than any type of data collected in 2006. Choice D is incorrect. Although the table indicates that data from LWS at the W.M. Keck

Observatory were obtained in 2003, making these data the earliest example of any type of data included in the analysis, the table also indicates that LWS collected infrared imaging data, not spectroscopy data. LWS therefore cannot be the source of the earliest example of spectroscopy data included in Moses and team's analysis of Neptune's temperature.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



The population of the coral *Lophelia pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Using the ratio of manganese to calcium, which inversely correlates with ocean oxygenation levels, marine scientist Rodrigo da Costa Portilho-Ramos and colleagues evaluated whether oxygenation played a role in the declines of *L. pertusa*. The researchers concluded that oxygenation may have been important in the Alboran Sea but not near the Mauritanian coast, since _____

Which choice most effectively uses data from the graph to complete the statement?

Answer

- A. a substantial increase in oxygenation in the Alboran Sea corresponded with the local decline in *L. pertusa*, but the opposite relationship between oxygenation and *L. pertusa* was found near the Mauritanian coast.
- B. *L. pertusa* declined in the Alboran Sea during a period of substantial local decline in oxygenation, but *L. pertusa* declined near the Mauritanian coast during a period of little local change in oxygenation.
- C. oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after the decline, whereas oxygenation near the Mauritanian coast was relatively low both before and after the decline in *L. pertusa*.
- D. oxygenation in the Alboran Sea tended to be substantially higher than oxygenation near the Mauritanian coast during the period studied.

Correct Answer: B

Rationale

Choice B is the best answer because it effectively uses data from the graph to complete the statement about Rodrigo da Costa Portilho-Ramos and colleagues' conclusion. The graph shows the ratio of manganese to calcium in *L. pertusa* coral samples from the Alboran Sea and the Mauritanian coast. The graph reflects time in approximate years before present: in other words, the greater the number in years noted on the graph's horizontal axis, the farther that moment is in the past. The text indicates that the researchers tested the samples to determine whether oxygenation played a role in the decline of *L. pertusa*. The text goes on to note that a change in the ratio of manganese to calcium would signal an inverse, or opposite, change in oxygenation. According to the graph, the ratio of manganese to calcium in samples from the Alboran Sea increased from about 30 micromoles per mole 10,000 years ago to about 80 micromoles per mole 8,000 years ago, which means that oxygenation decreased between 10,000 and 8,000 years ago. Meanwhile, there was almost no discernible change in the ratio of manganese to calcium in samples from the Mauritanian coast between 12,000 and 10,000 years ago. According to the text, the population of *L. pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Thus, the increase in the ratio of manganese to calcium around 9,000 years ago in the Alboran Sea coincides with the decline in the *L. pertusa* population, suggesting an association between the decrease in oxygenation and the decline in population of the coral. No such relationship is suggested around 11,000 years ago near the Mauritanian coast. So, oxygenation likely played a role in the *L. pertusa* decline in the Alboran Sea but not in the coral's decline near the Mauritanian coast.

Choice A is incorrect because it asserts the opposite of what the graph indicates regarding oxygenation in the Alboran Sea, and it misrepresents what the graph indicates about oxygenation near the Mauritanian coast. The graph indicates that at the time of the decline in *L. pertusa* (approximately 9,000 years ago), the samples from the Alboran Sea contained a ratio of manganese to calcium that was increasing. According to the text, this ratio inversely correlates with ocean oxygenation levels, so if the ratio was increasing, oxygenation was decreasing, not substantially increasing. Furthermore, the graph shows that the ratio of manganese to calcium remained relatively stable in coral samples from the Mauritanian coast during the period studied, which suggests that there was no discernible relationship between oxygenation and the coral's population decline in that location, not that there was a substantial decrease in oxygenation corresponding to the coral's decline. Choice C is incorrect. Although the graph suggests that the level of oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after—because the ratio of manganese to calcium inversely correlates with ocean oxygenation levels and this ratio was lower before the decline than after—the graph doesn't support the claim that oxygenation near the Mauritanian coast was consistently low before and after the coral's decline there. Rather, the graph indicates that relative to coral samples from the Alboran Sea, the ratio of manganese to calcium in samples from near the Mauritanian coast was consistently low, which suggests that oxygenation levels were relatively high both before and after the decline of *L. pertusa*. Choice D is incorrect because it states the opposite of what the graph indicates: the graph shows that throughout the period studied, the ratio of manganese to calcium was higher in coral samples from the Alboran Sea than it was in samples from near the Mauritanian coast. Since the text indicates that the ratio of manganese to calcium inversely correlates with ocean oxygenation levels, oxygenation in the Alboran Sea was therefore lower than, not higher than, oxygenation near the Mauritanian coast during the period studied. Moreover, even if choice D did accurately represent the graph, it wouldn't effectively complete the statement since a comparison of the ocean oxygenation levels at the two locations is not relevant to the claim that a decline in oxygenation levels was associated with the decline of *L. pertusa* in the Alboran Sea but not near the Mauritanian coast.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is from Milan Kundera’s 1984 novel *The Unbearable Lightness of Being* (translated by Michael Henry Heim in 1984). Karenin is a dog that belongs to Tomas and Tereza.

Karenin was not overjoyed by the move to Switzerland [from Prague]. Karenin hated change. Dog time cannot be plotted along a straight line; it does not move on and on, from one thing to the next. It moves in a circle like the hands of a clock, which—they, too, unwilling to dash madly ahead—turn round and round the face, day in and day out following the same path. In Prague, when Tomas and Tereza bought a new chair or moved a flower pot, Karenin would look on in displeasure. It disturbed his sense of time. It was as though they were trying to dupe the hands of the clock by changing the numbers on its face.

©1984 by Milan Kundera. Translation ©1984 by Harper & Row, Publishers, Inc.

Which choice best states the main idea of the text?

Answer

- A. As a dog, Karenin possesses a sense of time that involves a strong preference for predictability and an aversion to disruption.
- B. After he’s moved to a new home, Karenin’s negative response to changes has become more pronounced.
- C. Similar to Tomas and Tereza, Karenin comprehends time as circular rather than as a straightforward progression.
- D. As is the case for other dogs, Karenin’s sense of time seems to accelerate depending on the objects and places that surround him.

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main idea of the text. The text begins by stating that Tomas and Tereza’s dog Karenin felt disrupted by a recent move because of his dislike of change. The text then goes on to suggest that this is because the way a dog experiences time differs from the way humans experience time: time for a dog doesn’t move linearly, going "on and on, from one thing to the next," but instead moves circularly, "like the hands of a clock." That is, time for a dog is experienced as a cyclical pattern characterized by routine and predictability, with each day "following the same path." The text then concludes by providing examples of seemingly insignificant changes in routine that profoundly "disturbed [Karenin’s] sense of time," causing him to feel displeasure. Thus, the main idea of the text is that Karenin’s sense of time as a dog involves a strong preference for predictability and an aversion to disruption.

Choice B is incorrect. Although the text emphasizes Karenin’s displeasure with the recent move to a new home, it doesn’t suggest that the move has made his negative responses more pronounced than they once were. Rather, in accounting for Karenin’s displeasure with the move to Switzerland, the text explains that Karenin generally has a negative response to any kind of change. Choice C is incorrect because the text doesn’t suggest that Karenin comprehends time similarly to how Tomas and Tereza comprehend it. On the contrary, the text strongly implies a contrast between dogs’ circular experience of time with the way humans experience time as a straightforward progression that can be "plotted on a straight line." Choice D is incorrect because the text provides no indication that a change in the places and objects surrounding Karenin causes him to feel as though time is accelerating. Although the text does use the language of "dash[ing] madly ahead" in relation to time, the phrase appears in the context of a comparison illustrating how dogs experience time: time for a dog moves just as the hands of a clock do, in a circle and "unwilling to dash madly ahead"—that is, always in a regular and predictable way.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Several studies of sediment (e.g., dirt, pieces of rock, etc.) in streams have shown an inverse correlation between sediment grain size and downstream distance from the primary sediment source, suggesting that stream length has a sorting effect on sediment. In a study of sediment sampled at more than a dozen sites in Alpine streams, however, geologists Camille Litty and Fritz Schlunegger found that cross-site variations in grain size were not associated with differences in downstream distance, though they did not conclude that downstream distance is irrelevant to grain size. Rather, they concluded that sediment influx in these streams may have been sufficiently spatially diffuse to prevent the typical sorting effect from being observed.

Which finding about the streams in the study, if true, would most directly support Litty and Schlunegger’s conclusion?

Answer

- A. The streams regularly experience portions of their banks collapsing into the water at multiple points upstream of the sampling sites.
- B. The streams contain several types of sediment that are not typically found in streams where the sorting effect has been demonstrated.
- C. The streams mostly originate from the same source, but their lengths vary considerably due to the different courses they take.
- D. The streams are fed by multiple tributaries that carry significant volumes of sediment and that enter the streams downstream of the sampling sites.

Correct Answer: A

Rationale

Choice A is the best answer. This finding would support the conclusion. If stream banks are collapsing into the water at multiple points, then sediment is getting into the water at those various points. This supports the conclusion that the inflow of sediment is very spread out.

Choice B is incorrect. This finding wouldn’t support the conclusion. The conclusion is about the influx of sediment being “spatially diffuse,” meaning spread out over a large area. The type of sediment wouldn’t have an impact on the conclusions. Choice C is incorrect. This finding wouldn’t support the conclusion. It doesn’t say anything about the influx of sediment being “spatially diffuse” (spread out). Choice D is incorrect. This finding wouldn’t support the conclusion. Any sediment that enters downstream of the sampling sites wouldn’t end up in the samples, so it wouldn’t affect the findings or the conclusion.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Matthew D. Rocklage and team examined whether consumer ratings of movies can predict box office success. The team considered both numeric star ratings and written reviews in their research. To analyze the written reviews, the team measured the emotionality—the degree to which a written review expresses an emotional reaction—of user reviews on a movie rating website, assigning each review an emotionality score. After reviewing this research, a student argues that the emotionality of movie reviews is unrelated to a movie’s success at the box office.

Which finding, if true, would most directly weaken the student’s conclusion?

Answer

- A. Movies that had the highest average emotionality scores received the lowest average star ratings on the movie rating website.
- B. The average emotionality score of a movie’s reviews was a positive predictor of that movie’s box office earnings.
- C. More than half of the movies that the team examined received an average star rating of 3 out of 5 stars.
- D. The movies that were most successful at the box office tended to have high average star ratings.

Correct Answer: B

Rationale

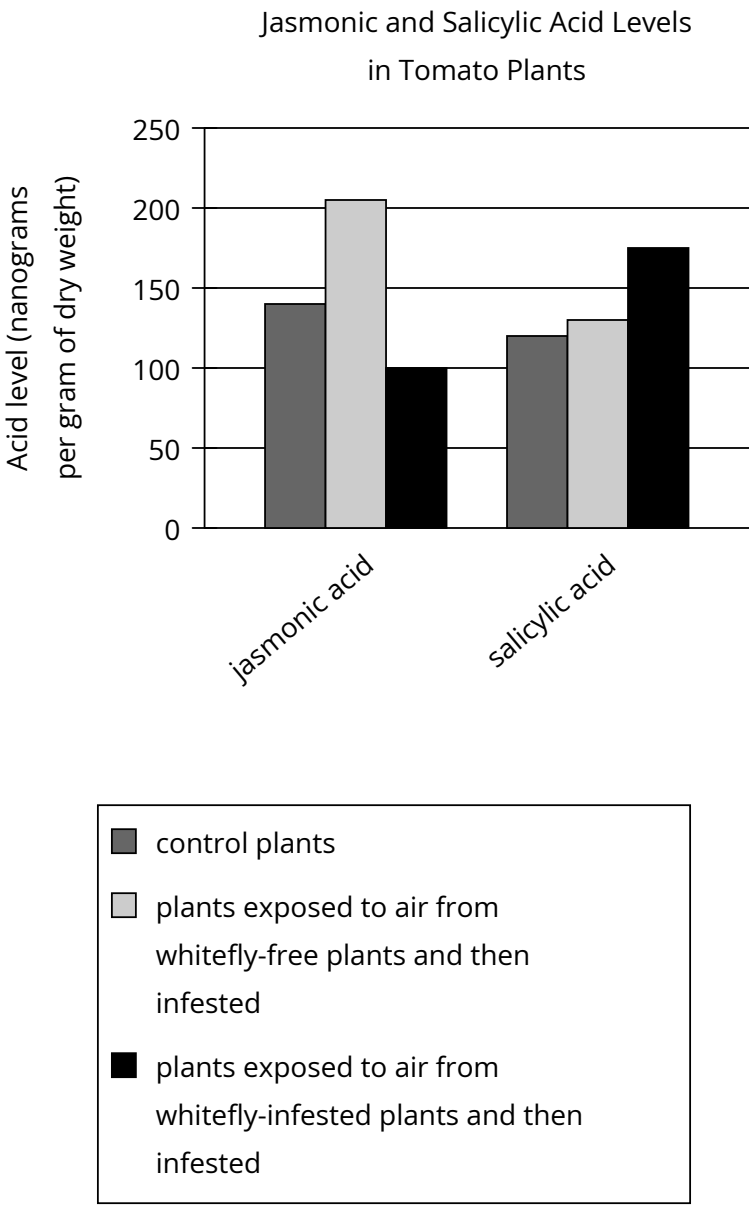
Choice B is the best answer. This choice weakens the student’s conclusion by suggesting that the emotionality of movie reviews is related to box office success: the higher the emotionality score, the better the movie performs at the box office.

Choice A is incorrect. While this choice does mention emotionality scores, it only connects them to star ratings, not to box office success.

Choice C is incorrect. The fact that many movies received an average star rating doesn’t tell us anything about the relationship between emotionality and box office success. Choice D is incorrect. While this choice suggests that star ratings can predict box office success, it doesn’t address the issue of emotionality in written reviews, which is the focus of the student’s conclusion.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



In tomato plants, herbivory induces defensive production of jasmonic acid, while microbial infection induces defensive production of salicylic acid; plants also emit airborne chemicals to initiate the appropriate defense in nearby tomato plants. Researchers investigated the poor resistance tomato plants show to whitefly herbivory by exposing some plants to airborne chemicals from whitefly-free plants and others to airborne chemicals from whitefly-infested plants, then infesting both groups of plants with whiteflies. The researchers concluded that whiteflies induce tomato plants to emit chemicals that cause other tomato plants to preferentially defend against microbial infection even when under herbivorous attack.

Which choice best describes data from the graph that support the researchers’ conclusion?

Answer

- A. When plants exposed to air from whitefly-free plants were infested, they produced more jasmonic acid than did control plants, whereas when plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid and more salicylic acid than did control

plants.

- B. When plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid than salicylic acid, whereas when plants exposed to air from whitefly-free plants were infested, they produced about the same amount of jasmonic acid and salicylic acid.
- C. When plants exposed to air from whitefly-free plants were infested, they produced both jasmonic acid and salicylic acid, whereas when plants exposed to air from whitefly-infested plants were infested, they exclusively produced salicylic acid.
- D. When plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid than did control plants, whereas when plants exposed to air from whitefly-free plants were infested, they produced more jasmonic acid and salicylic acid than did control plants.

Correct Answer: A

Rationale

Choice A is the best answer because it best describes data from the graph that support the researchers' conclusion about whitefly-induced defenses in tomato plants. According to the text, tomato plants produce airborne chemicals that prompt nearby tomato plants to increase their production of appropriate defensive chemicals, such as jasmonic acid in response to herbivory and salicylic acid in response to microbial infection. The text goes on to indicate that researchers concluded that whiteflies induce tomato plants to emit chemicals that cause nearby plants to defend against microbial infection instead of herbivory from whiteflies. The graph shows that control tomato plants (plants that weren't exposed to any chemicals from nearby tomato plants and weren't subsequently infested with whiteflies) produced about 140 nanograms of jasmonic acid per gram of dry weight and about 120 nanograms of salicylic acid per gram of dry weight, which indicates the baseline levels of these defensive chemicals in tomato plants. The graph also shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when subsequently infested by whiteflies, which is more than the amount produced by the control plants. This suggests that when infested with whiteflies, tomato plants that weren't previously exposed to air from nearby whitefly-infested plants begin producing higher levels of jasmonic acid to defend against the infestation. Finally, the graph also shows that plants exposed to air from whitefly-infested plants and then subsequently infested by whiteflies produced approximately 100 nanograms of jasmonic acid per gram of dry weight (which is less than control plants did) and approximately 175 nanograms of salicylic acid per gram of dry weight (which is more than control plants did). This suggests that whiteflies cause tomato plants to emit chemical signals that decrease nearby plants' defenses against whitefly herbivory relative to baseline levels and redirect their defenses toward microbial infection, thereby supporting the researchers' conclusion that whiteflies cause tomato plants to emit airborne chemical signals that induce nearby plants to preferentially defend against microbes even when experiencing herbivory from whiteflies.

Choice B is incorrect. Although the graph shows that tomato plants exposed to whitefly-infested plants produced less jasmonic acid than salicylic acid (about 100 nanograms of jasmonic acid per gram of dry weight vs. about 175 nanograms of salicylic acid per gram of dry weight), and although this might suggest that those plants are therefore better defended against microbial attack than against herbivory, this wouldn't be sufficient to conclude that this is the case: without reference to the relative levels of defensive chemicals produced by the plants in the other experimental conditions, it would be impossible to conclude that exposure to air from whitefly-infested plants either induces or diminishes these plants' defenses to different kinds of threats. Furthermore, the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight and about 130 nanograms of salicylic acid per gram of dry weight, not that they produced the same amount of the two acids. Choice C is incorrect because it inaccurately describes the data in the graph. The graph shows that plants in all three conditions produced both jasmonic and salicylic acid, not that plants exposed to air from whitefly-infested plants produce no jasmonic acid at all. Choice D is incorrect. Although the graph shows that plants exposed to air from whitefly-infested plants produced about 100 nanograms of jasmonic acid per gram of dry weight when infested, which is less than the amount produced by control plants (about 140 nanograms per gram of dry weight), and although the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when infested (which is more than the amount produced by control plants) and about 130 nanograms of salicylic acid per gram of dry weight (which is more than the approximately 120 nanograms per gram of dry weight produced by control plants), this wouldn't support the researchers' conclusion that plants exposed to air from whitefly-infested plants preferentially defend against microbial infection—that is, that they increase their levels of salicylic acid and decrease their levels of jasmonic acid relative to control plants. Rather, the data would suggest that exposure to air from whitefly-infested plants reduces plants' defenses against herbivory, but these data wouldn't indicate that whiteflies induce plants to redirect their defenses to guard against microbial infection. Furthermore, the data concerning plants exposed to air from whitefly-free plants merely indicate that these plants increase their defenses against both herbivory and microbial infection when infested by whiteflies.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Fiber Characteristics of Mouflon, Navajo-Churro, and Spanish Merino Sheep

Type of sheep	Diameter of outer coat fibers (in microns)	Diameter of inner coat fibers (in microns)
Spanish Merino	19–24	17–21
Navajo-Churro	35 or higher	10–35
Mouflon	150	15

Domestic sheep’s wild ancestor, the mouflon, has a coarse outer coat and an inner coat of wool fiber that is finer in diameter and therefore much softer. In some domestic breeds, such as the Spanish Merino, the outer fiber is only marginally coarser than the inner, and the wool is soft overall. Thus, Merino wool is ideal for delicate garments worn against the skin. Meanwhile, the Navajo-Churro has been selected to retain the marked distinction between outer and inner fiber that the Merino has lost. Being coarser than Merino wool overall, Navajo-Churro wool yields a more durable yarn, which Diné (Navajo) weavers use in their celebrated rugs. Yet a comparison of the fiber characteristics of all three sheep reveals that _____

Which choice most effectively uses data from the table to complete the text?

Answer

- A. the selection process that enabled the Navajo-Churro to retain its somewhat coarse outer fiber also resulted in inner fiber that, at its softest, is softer than either the mouflon’s or the Merino’s inner fiber.
- B. the Navajo-Churro more closely resembles its ancestor, the mouflon, in the uniform softness of its inner fiber, while the Merino more closely resembles the mouflon in the variable diameter of its outer fiber.
- C. domestication resulted in a counterintuitive increase in the inner fiber’s minimum diameter, making the inner fiber of the Merino and the Navajo-Churro less suitable for delicate garments than the mouflon’s inner fiber is.
- D. the domestication of the mouflon and the subsequent selection process that produced the Merino and the Navajo-Churro resulted in greater softness of outer and inner fiber alike.

Correct Answer: A

Rationale

Choice A is the best answer because it accurately uses data from the table to complete the statement about the fibers of three types of sheep. The text discusses some qualities of outer and inner fibers of mouflons, Spanish Merino sheep, and Navajo-Churro sheep and indicates that sheep fibers with finer, or smaller, diameters are softer than fibers with larger diameters. The text then sets up a conclusion that can be made by comparing the fibers of all three sheep types. The table shows that the minimum diameter of the Navajo-Churro’s inner fibers is 10 microns, which is smaller than the minimum diameters of the inner fibers of both the mouflon (15 microns) and the Spanish Merino (17 microns). Thus, comparing the fiber characteristics of all three sheep reveals that at its softest, the inner fiber of the Navajo-Churro is softer than either the mouflon’s or the Merino’s inner fiber.

Choice B is incorrect because it doesn’t accurately describe the data in the table. The table shows that the softness of Navajo-Churro inner fiber isn’t uniform (since the diameter of Navajo-Churro inner fiber ranges from 10 microns to 35 microns) and that the diameter of the outer fiber of the mouflon isn’t variable (rather, it is a uniform 150 microns). Choice C is incorrect because it doesn’t accurately describe the data in the table. The text indicates that both Spanish Merino sheep and Navajo-Churro sheep were domesticated from mouflon sheep. The table doesn’t show

that domestication increased the minimum diameter of inner fibers for both Spanish Merino sheep and Navajo-Churro sheep from the 15-micron diameter of mouflon inner fibers: the minimum diameter of Spanish Merino increased to 17 microns, but the minimum diameter of Navajo-Churro inner fibers *decreased* to 10 microns. Choice D is incorrect because it doesn't accurately describe the data in the table. The text explains that sheep fibers with finer, or smaller, diameters are softer than fibers with larger diameters. The table does show that Spanish Merino outer fibers are finer, and thus softer, than mouflon outer fibers and that at their finest, Navajo-Churro outer fibers are softer than mouflon outer fibers, but it also shows that Spanish Merino inner fibers are less soft than mouflon inner fibers, since their minimum diameter is greater than 15 microns, and that Navajo-Churro inner fibers can be less soft than mouflon inner fibers, since the diameter ranges up to 35 microns.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Ratified by more than 90 countries, the Nagoya Protocol is an international agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge of wild plants and animals are utilized by agricultural corporations. However, the protocol has shortcomings. For example, it allows corporations to insist that their agreements with communities to conduct research on the commercial uses of the communities’ resources and knowledge remain confidential. Therefore, some Indigenous advocates express concern that the protocol may have the unintended effect of _____

Which choice most logically completes the text?

Answer

- A. diminishing the monetary reward that corporations might derive from their agreements with Indigenous communities.
- B. limiting the research that corporations conduct on the resources of the Indigenous communities with which they have signed agreements.
- C. preventing independent observers from determining whether the agreements guarantee equitable compensation for Indigenous communities.
- D. discouraging Indigenous communities from learning new methods for harvesting plants and animals from their corporate partners.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the argument about an unintended effect of the Nagoya Protocol. The text explains that the Nagoya Protocol is an agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge are used by corporations. The text then states that the protocol allows corporations to keep their agreements with Indigenous communities confidential, about which some Indigenous advocates express concern. Choice C, when inserted into the blank, gives a good justification for the advocates’ concern: such secrecy could mean that the public is unable to determine whether participating Indigenous communities were properly compensated under these agreements.

Choice A is incorrect. The text suggests that because corporations can keep their agreements with Indigenous communities confidential, Indigenous communities, not corporations, might not be compensated fairly. Choice B is incorrect because the text doesn’t suggest that the ability of corporations to keep their agreements with Indigenous communities confidential would place limits on how much research corporations can undertake. Choice D is incorrect because the text doesn’t indicate that Indigenous communities aim to learn new harvesting methods from their corporate partners. Rather, the text suggests that corporations use the knowledge of Indigenous communities for their research.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Political scientists who favor the traditional view of voter behavior claim that voting in an election does not change a voter’s attitude toward the candidates in that election. Focusing on each US presidential election from 1976 to 1996, Ebonya Washington and Sendhil Mullainathan tested this claim by distinguishing between subjects who had just become old enough to vote (around half of whom actually voted) and otherwise similar subjects who were slightly too young to vote (and thus none of whom voted). Washington and Mullainathan compared the attitudes of the groups of subjects toward the winning candidate two years after each election.

Which finding from Washington and Mullainathan’s study, if true, would most directly weaken the claim made by people who favor the traditional view of voter behavior?

Answer

- A. Subjects’ attitudes toward the winning candidate two years after a given election were strongly predicted by subjects’ general political orientation, regardless of whether subjects were old enough to vote at the time of the election.
- B. Subjects who were not old enough to vote in a given election held significantly more positive attitudes towards the winning candidate two years later than they held at the time of the election.
- C. Subjects who voted in a given election held significantly more polarized attitudes toward the winning candidate two years later than did subjects who were not old enough to vote in that election.
- D. Two years after a given election, subjects who voted and subjects who were not old enough to vote were significantly more likely to express negative attitudes than positive attitudes toward the winning candidate in that election.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would weaken the claim made by people who favor the traditional view of voter behavior. According to the text, people who favor that view believe that voting in an election doesn’t change a voter’s attitude toward the candidates in that election. If Washington and Mullainathan found that two years after an election, attitudes toward the winning candidate were significantly more polarized among subjects who had voted than among subjects who had been too young to vote, that would suggest that the act of voting did have an effect on the voters’ attitudes toward the candidates, which would undermine the claim that voting doesn’t change voters’ attitudes.

Choice A is incorrect because a finding about links between subjects’ attitudes and general political orientation, regardless of age and ability to vote, wouldn’t address the presence or absence of changes in attitudes among those subjects who did actually vote. Therefore, the finding wouldn’t have any bearing on the claim that voting in an election doesn’t change a voter’s attitude toward the candidates in that election. Choice B is incorrect because a finding that positive attitudes toward a winning candidate significantly increased in the two years after the election among subjects who had been too young to vote would involve only people who didn’t vote; therefore, the finding wouldn’t have any bearing on the claim that when people do vote, the act of voting doesn’t change their attitudes toward the candidates. Choice D is incorrect because the finding that subjects in both groups were more likely to have negative attitudes than positive attitudes toward the winning candidate two years after an election would reflect all subjects’ attitudes at one particular time whether they voted or not, rather than the presence or absence of a change in voters’ attitudes after voting. Therefore, the finding would neither weaken nor strengthen the claim that voting in an election doesn’t change a voter’s attitude toward the candidates.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Under normal atmospheric pressure at Earth’s surface, water molecules form a tetrahedral network stabilized by hydrogen bonds between adjacent molecules. Extreme high pressure, such as can be found in deep ocean waters, destabilizes these bonds and compresses water’s structure, allowing water molecules within organisms to permeate proteins and impede crucial biological functions; yet deep-sea organisms known as piezophiles have adapted to extreme pressure. Studies have found a positive correlation between the depths that various piezophiles inhabit and concentrations of a compound called trimethylamine N-oxide (TMAO) in their muscle tissues, which has led a team of researchers to hypothesize that TMAO reduces water’s compressibility.

Which finding, if true, would most directly support the researchers’ hypothesis?

Answer

- A. Water molecules are found to be impervious to TMAO even when the water molecules’ tetrahedral configuration has been distorted by high pressure.
- B. Examination of TMAO’s molecular structure shows that TMAO molecules retain their shape even as pressure increases.
- C. A positive correlation is found between concentrations of TMAO and the rate at which water’s molecular structure compresses as pressure increases.
- D. Analysis of water’s molecular structure under high pressure reveals that hydrogen bonds are more stable when TMAO is present than when it is not.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support the researchers’ hypothesis that TMAO reduces water’s compressibility. The text explains that at great depths in the ocean, extreme pressure compresses the molecular structure of water by destabilizing the hydrogen bonds between adjacent molecules, thereby allowing water to penetrate proteins and harm the associated organisms. However, deep-sea organisms called piezophiles have adapted to live at these depths and previous studies show a positive correlation between the depth at which a piezophile species lives and the species’ level of the compound TMAO. Because this hypothesis links TMAO levels with reduced compressibility of water’s tetrahedral molecular structure, a finding that TMAO helps maintain the hydrogen bonds between water molecules under high pressure would strongly support that hypothesis.

Choice A is incorrect. Although the researchers’ hypothesis suggests a relationship between TMAO and water molecules’ tetrahedral molecular structure, that relationship involves TMAO helping maintain water’s tetrahedral molecular structure under high pressure; as presented in the text, the hypothesis doesn’t contend that water molecules are impervious to, or incapable of being penetrated by, TMAO. Choice B is incorrect because the text discusses how the molecular structure of water, not TMAO, is compressed under extreme pressure and never addresses how TMAO might be affected by such pressure. Choice C is incorrect because the researchers’ hypothesis holds that water under extreme pressure is more resistant, not less, to being compressed when TMAO concentrations are higher. Moreover, the positive correlation mentioned in the text is between TMAO concentrations and the depths at which piezophiles live, not between concentrations of TMAO and the rate at which water’s molecular structure compresses as pressure increases.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Researchers hypothesized that a decline in the population of dusky sharks near the mid-Atlantic coast of North America led to a decline in the population of eastern oysters in the region. Dusky sharks do not typically consume eastern oysters but do consume cownose rays, which are the main predators of the oysters.

Which finding, if true, would most directly support the researchers’ hypothesis?

Answer

- A. Declines in the regional abundance of dusky sharks’ prey other than cownose rays are associated with regional declines in dusky shark abundance.
- B. Eastern oyster abundance tends to be greater in areas with both dusky sharks and cownose rays than in areas with only dusky sharks.
- C. Consumption of eastern oysters by cownose rays in the region substantially increased before the regional decline in dusky shark abundance began.
- D. Cownose rays have increased in regional abundance as dusky sharks have decreased in regional abundance.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers’ hypothesis about the connection between the dusky shark population decline and the eastern oyster population decline. The text indicates that although dusky sharks don’t usually eat eastern oysters, they do consume cownose rays, which are the main predators of eastern oysters. An increase in the abundance of cownose rays in the region in response to a decline in the abundance of dusky sharks would directly support the researchers’ hypothesis: a higher number of cownose rays would consume more eastern oysters, driving down the oyster population.

Choice A is incorrect because a finding that there’s an association between a decline in the regional abundance of some of dusky sharks’ prey and the regional abundance of dusky sharks wouldn’t directly support the researchers’ hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. Although such a finding might help explain why shark abundance has declined, it would reveal nothing about whether the shark decline is related to the oyster decline. Choice B is incorrect because a finding that eastern oyster abundance tends to be greater when dusky sharks and cownose rays are present than when only dusky sharks are present wouldn’t support the researchers’ hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. The text indicates that the sharks prey on the rays, which are the main predators of the oysters; if oyster abundance is found to be greater when rays are present than when rays are absent, that would suggest that rays aren’t keeping oyster abundance down, and thus that a decline in rays’ predators, which would be expected to lead to an increase in the abundance of rays, wouldn’t bring about a decline in oyster abundance as the researchers hypothesize. Choice C is incorrect because a finding that consumption of eastern oysters by cownose rays increased substantially before dusky sharks declined in regional abundance wouldn’t support the researchers’ hypothesis that the decline in dusky sharks has led to a decline in eastern oysters in the region. Such a finding would suggest that some factor other than shark abundance led to an increase in rays’ consumption of oysters and thus to a decrease in oyster abundance, thereby weakening the researchers’ hypothesis.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

King Lear is a circa 1606 play by William Shakespeare. In the play, the character of King Lear attempts to test his three daughters’ devotion to him. He later expresses regret for his actions, as is evident when he _____

Which choice most effectively uses a quotation from *King Lear* to illustrate the claim?

Answer

- A. says of himself, “I am a man / more sinned against than sinning.”
- B. says during a growing storm, “This tempest will not give me leave to ponder / On things would hurt me more.”
- C. says to himself while striking his head, “Beat at this gate that let thy folly in / And thy dear judgement out!”
- D. says of himself, “I will do such things— / What they are yet, I know not; but they shall be / The terrors of the earth!”

Correct Answer: C

Rationale

Choice C is the best answer because it most effectively uses a quotation from *King Lear* to illustrate the claim that King Lear expresses regret for his actions. In the quotation, Lear describes striking himself on the head—the same act he’s engaged in as he speaks, and one that suggests he’s deeply upset with himself. Referring to himself in the second person (with “thy”), the character exclaims “Beat at this gate that let thy folly in / And thy dear judgement out!” Lear refers metaphorically to his own mind as a gate that has allowed folly, or poor judgement, to enter and good judgement to escape. This suggests that Lear regrets his attempts to test his three daughters’ devotion to him, regarding those attempts as examples of the folly that has entered the gate of his mind.

Choice A is incorrect because this quotation doesn’t express King Lear’s sense of regret over his own actions; instead, it expresses his belief that the harm that others have done to him (or the extent to which they have “sinned against” him) outweighs whatever harm he himself has caused by “sinning.” Choice B is incorrect because this quotation doesn’t express King Lear’s sense of regret over his own actions; instead, it expresses his thoughts about an approaching storm (“this tempest”), which he believes “will not give [him] leave to ponder,” or time to consider, the harm that he will continue to experience (“things” that “would hurt [him] more”). Choice D is incorrect because this quotation expresses King Lear’s vow to commit terrible actions (or “things” that “shall be / The terrors of the earth”) in the future, not his regret over actions that he’s already taken.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In the twentieth century, ethnographers made a concerted effort to collect Mexican American folklore, but they did not always agree about that folklore’s origins. Scholars such as Aurelio Espinosa claimed that Mexican American folklore derived largely from the folklore of Spain, which ruled Mexico and what is now the southwestern United States from the sixteenth to early nineteenth centuries. Scholars such as Américo Paredes, by contrast, argued that while some Spanish influence is undeniable, Mexican American folklore is mainly the product of the ongoing interactions of various cultures in Mexico and the United States.

Which finding, if true, would most directly support Paredes’s argument?

Answer

- A. The folklore that the ethnographers collected included several songs written in the form of a *décima*, a type of poem originating in late sixteenth-century Spain.
- B. Much of the folklore that the ethnographers collected had similar elements from region to region.
- C. Most of the folklore that the ethnographers collected was previously unknown to scholars.
- D. Most of the folklore that the ethnographers collected consisted of *corridos*—ballads about history and social life—of a clearly recent origin.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support Paredes’s argument that Mexican-American folklore is mostly the result of cultural interactions in Mexico and the United States rather than an adaptation of Spanish folklore. The text describes a disagreement among scholars about whether Mexican-American folklore mostly derived from the folklore of Spain or originated in Mexico and the United States as cultures there have interacted. The latter view is the argument that Paredes puts forward. If Mexican-American folklore collected in the twentieth century mostly consists of ballads about history and social life that originated recently, then that would support Paredes’s argument, since it would suggest that the folklore mostly arose after Spanish rule ended in the early nineteenth century and that the folklore reflects cultural interactions in Mexico and the United States rather than traditions from Spain.

Choice A is incorrect because the inclusion of songs influenced by sixteenth-century Spanish poetry among Mexican-American folklore collected in the twentieth century would not support Paredes’s view that the folklore was the result of cultural interactions in Mexico and the United States rather than an offshoot of Spanish folklore. If anything, the presence of such songs among the folklore collected in the twentieth century would weaken Paredes’s argument, since it would reflect the influence of Spanish culture on the folklore. Choice B is incorrect because the mere presence of similarities in Mexican-American folklore across regions would not be sufficient to draw a conclusion about where the folklore originated, let alone to support Paredes’s argument that the folklore reflects cultural interactions in Mexico and the United States. In fact, since Paredes argued that Mexican-American folklore is the product of various cultures interacting in Mexico and the United States, he would likely expect there to be regional variations in folklore as different cultures have interacted in different places. Choice C is incorrect because scholars’ previous ignorance of the folklore would have no bearing on Paredes’s argument that Mexican-American folklore mostly reflects cultural interactions in Mexico and the United States. The folklore’s origins are independent of scholars’ knowledge of the folklore.

Question ID: 4d3e3c52

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell’s electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as “highly reproducible, concise, and practical.”

What does the text most strongly suggest about conventional ETL fabrication?

Answer

- A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.
- B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.
- C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.
- D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

Correct Answer: A

Rationale

Choice A is the best answer. Conventional solar cell fabrication increases “the cost of manufacturing at scale,” but spray coating might get rid of that problem.

Choice B is incorrect. This is not completely supported by the text. While it’s true that conventional ETL fabrication is expensive at scale, there’s nothing in the text that mentions other perovskite solar cell architectures. Choice C is incorrect. This choice does not match the text. Only one conventional method of ETL fabrication is described, so we can’t compare the solution loss in this method to that of other conventional methods. Choice D is incorrect. This choice isn’t supported by the text. The text never suggests that the effectiveness of solar cells changes based on their method of fabrication.

Question ID: d1b8a9ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Disco remains one of the most ridiculed popular music genres of the late twentieth century. But as scholars have argued, the genre is far less superficial than many people believe. Take the case of disco icon Donna Summer: she may have been associated with popular songs about love and heartbreak (subjects hardly unique to disco, by the way), but like many Black women singers before her, much of her music also reflects concerns about community and identity. These concerns are present in many of the genre’s greatest songs, and they generally don’t require much digging to reveal.

What does the text most strongly suggest about the disco genre?

Answer

- A. It has been unjustly ignored by most scholars despite the importance of the themes addressed by many of the genre’s songs.
- B. It evolved over time from a superficial genre focused on romance to a genre focused on more serious concerns.
- C. It has been unfairly dismissed for the inclusion of subject matter that is also found in other musical genres.
- D. It gave rise to a Black women’s musical tradition that has endured even though the genre itself faded in the late twentieth century.

Correct Answer: C

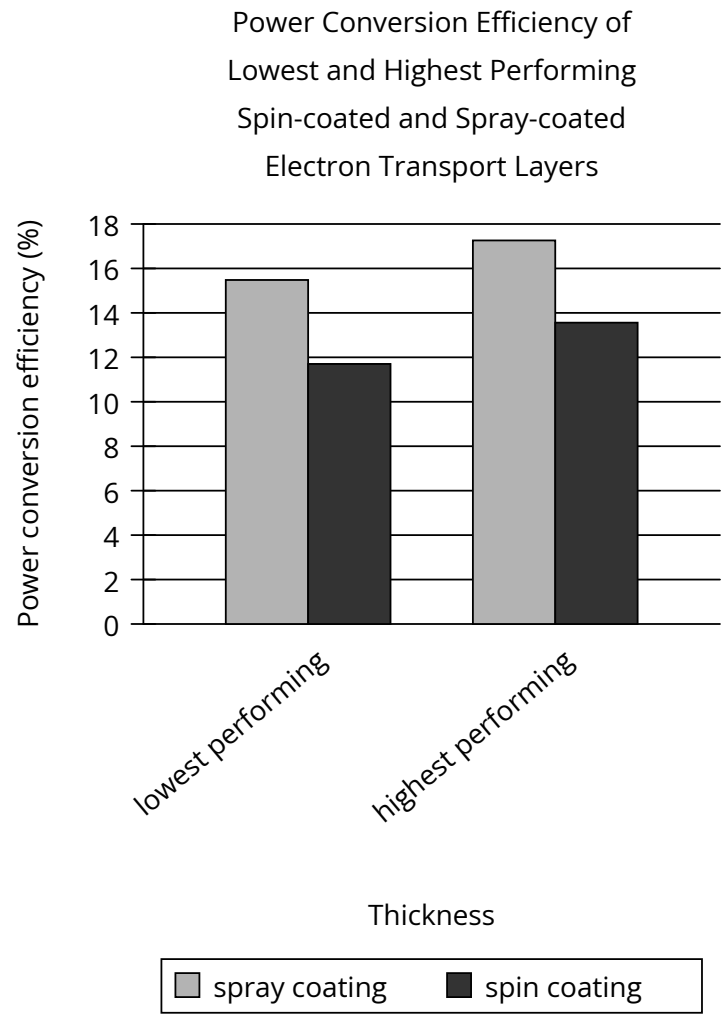
Rationale

Choice C is the best answer. The text argues that disco is "far less superficial" than its popular perception might indicate, and that love and heartbreak are "subjects hardly unique to disco."

Choice A is incorrect. This choice conflicts with the text, which says that scholars argue that disco "is far less superficial than many people believe." Choice B is incorrect. This choice says the opposite of what the text suggests. The writer argues that the genre is not as superficial as commonly believed, but that it always reflected "concerns about community and identity." Choice D is incorrect. The text doesn’t support this choice. There’s nothing in the text about disco giving rise to an enduring Black women’s musical tradition.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Perovskite solar cells convert light into electricity more efficiently than earlier kinds of solar cells, and manufacturing advances have recently made them commercially attractive. One limitation of the cells, however, has to do with their electron transport layer (ETL), through which absorbed electrons must pass. Often the ETL is applied through a process called spin coating, but such ETLs are fairly inefficient at converting input power to output power. André Taylor and colleagues tested a novel spray coating method for applying the ETL. The team produced ETLs of various thicknesses and concluded that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells.

Which choice best describes data from the graph that support Taylor and colleagues’ conclusion?

Answer

- A. Both the ETL applied through spin coating and the ETL applied through spray coating showed a power conversion efficiency greater than 10% at their lowest performing thickness.
- B. The lowest performing ETL applied through spray coating had a higher power conversion efficiency than the highest performing ETL applied through spin coating.

- C. The highest performing ETL applied through spray coating showed a power conversion efficiency of approximately 13%, while the highest performing ETL applied through spin coating showed a power conversion efficiency of approximately 11%.
- D. There was a substantial difference in power conversion efficiency between the lowest and highest performing ETLs applied through spray coating.

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the graph that support Taylor and colleagues' conclusion that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells. The text explains that perovskite solar cells' efficiency at converting light into electricity is diminished by their electron transport layer (ETL), which is applied through spin coating, but that Taylor's team devised a new spray coating method for applying the ETL that improves its power conversion efficiency. The graph displays data on the power conversion efficiency of solar cells in tests conducted by Taylor's team, with bars for both the highest- and lowest-performing ETLs in two data categories: spray coating and spin coating. According to the graph, the lowest-performing ETL applied through spray coating had a power conversion efficiency of between 14% and 16%, while the highest-performing ETL applied through spin coating had a power conversion efficiency of less than 14%. These data confirm that ETLs applied through novel spray coating are more efficient than those applied through traditional spin coating. Thus, the data support Taylor and colleagues' conclusion about spray coating's potential value.

Choice A is incorrect. Although this claim correctly describes the data in the graph by stating that both the lowest-performing ETL applied through spin coating and the lowest-performing ETL applied through spray coating had a power conversion efficiency greater than 10%, this relationship in the data doesn't support or relate to Taylor and colleagues' conclusion that spray coating promises greater efficiency for solar cells than traditional spin coating does. Choice C is incorrect. This claim does address the greater power conversion efficiency of the highest-performing ETL applied through spray coating, compared with the highest-performing ETL applied through spin coating. However, it also incorrectly cites the value for the efficiency of the highest-performing ETL applied through spray coating as approximately 13%, instead of a value between 14% and 16%, and the value for the efficiency of the highest-performing ETL applied through spin coating as approximately 11%, instead of a value between 12% and 14%, as shown in the graph. Choice D is incorrect because Taylor and colleagues' conclusion is based on the difference in the power conversion efficiency of ETLs applied through spray coating and that of ETLs applied through spin coating, not on the difference between the highest- and lowest-performing ETLs applied through just spray coating.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Ablation Rates for Three Elements in Cosmic Dust, by
Dust Source

Element	SPC	AST	HTC	OCC
iron	20%	28%	90%	98%
potassium	44%	74%	97%	100%
sodium	45%	75%	99%	100%

Earth’s atmosphere is bombarded by cosmic dust originating from several sources: short-period comets (SPCs), particles from the asteroid belt (ASTs), Halley-type comets (HTCs), and Oort cloud comets (OCCs). Some of the dust’s material vaporizes in the atmosphere in a process called ablation, and the faster the particles move, the higher the rate of ablation. Astrophysicist Juan Diego Carrillo-Sánchez led a team that calculated average ablation rates for elements in the dust (such as iron and potassium) and showed that material in slower-moving SPC or AST dust has a lower rate than the same material in faster-moving HTC or OCC dust. For example, whereas the average ablation rate for iron from AST dust is 28%, the average rate for _____

Which choice most effectively uses data from the table to complete the example?

Answer

- A. iron from SPC dust is 20%.
- B. sodium from OCC dust is 100%.
- C. iron from HTC dust is 90%.
- D. sodium from AST dust is 75%.

Correct Answer: C

Rationale

Choice C is the best answer because it most effectively completes the example regarding the ablation rate of iron. The table shows the ablation rates for three elements—iron, potassium, and sodium—found in cosmic dust that comes from one of four sources. The text says that the ablation rate for a given element in slower-moving SPC and AST dust was lower than the ablation rate for that same element in faster-moving HTC or OCC dust. The text then presents the first part of an example of this pattern, describing an ablation rate of 28% for iron in AST dust. The information that iron from HTC dust had an ablation rate of 90% is therefore the most effective way to complete this example—the comparison of a relatively low ablation rate for iron in slower-moving AST dust with a relatively high ablation rate for iron in faster-moving HTC dust illustrates the tendency of ablation rates for a given element to be lower in slower-moving dust than in faster-moving dust.

Choice A is incorrect because the text indicates that SPC dust, like AST dust, moves relatively slowly; a comparison of the ablation rates of iron from two slower-moving dust sources could not be an example of the difference between ablation rates in slower-moving dust and faster-moving dust, which is the pattern that the example is supposed to illustrate. Choice B is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Choice D is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of

iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Additionally, any ablation rate from AST dust would be ineffective in this example since AST dust is referenced in the first part of the example and thus additional data focused on AST dust would not illustrate a variation across dust types.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Percentage of Available Eggs Eaten by Cane Toad Tadpoles

Amphibian species (common name)	Percentage of eggs eaten	Native to Australia	Produces bufadienolide
Little red tree frog	1%	yes	no
Cane toad	90%	no	yes
Short-footed frog	7%	yes	no
Striped burrowing frog	10%	yes	no
Dainty green tree frog	1%	yes	no

Native to Latin America, the cane toad was introduced to Australia in the 1930s. In recent decades, tadpoles in the Australian population have been shown to consume eggs of their own species. A 2022 study showed that when presented with cane toad eggs as well as eggs of native Australian amphibians, cane toad tadpoles disproportionately consumed eggs of their own species. This behavior results from their attraction to bufadienolide, a chemical produced by the eggs of cane toads but not by the eggs of native amphibians. However, using data from this study, a student wishes to argue that the presence of bufadienolide doesn’t entirely explain the cane toad tadpoles’ preference for certain eggs over others.

Which choice best describes data from the table that support the student’s argument?

Answer

- A. The tadpoles consumed a higher percentage of the striped burrowing frog eggs than they did of the eggs of the dainty green tree frog.
- B. The tadpoles left a certain percentage of the eggs of each of the five species unharmed, thus ultimately allowing them to hatch.
- C. The tadpoles consumed a lower percentage of the short-footed frog eggs than they did of the eggs of their own species.
- D. The tadpoles consumed the same percentage of the dainty green tree frog eggs as they did of the little red tree frog eggs.

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to support the student’s argument about the role of bufadienolide in the egg preferences of cane toad tadpoles. For each of five amphibian species included in the 2022 study, the table gives the percentage of available eggs that the cane toad tadpoles ate. According to the table, the tadpoles ate 10% of striped burrowing frog eggs and 1% of dainty green tree frog eggs, which suggests a preference for striped burrowing frog eggs over dainty green tree frog eggs. The table also indicates that neither of these species’ eggs produces bufadienolide. Thus, these data suggest that something other than the presence or absence of bufadienolide is needed to adequately explain the tadpoles’ egg preferences.

Choice B is incorrect. Although the table shows that for each of the five amphibian species, the cane toad tadpoles ate less than 100% of that species’ eggs, which demonstrates that the tadpoles did indeed leave some eggs for each species unharmed, this fact alone is irrelevant to the tadpoles’ preferences for some species’ eggs over other species’ eggs. Choice C is incorrect. Although the table indicates that the cane toad tadpoles ate 90% of the cane toad eggs and 7% of the short-footed frog eggs, which suggests that they prefer cane toad eggs over short-footed frog eggs, the table also indicates that cane toad eggs produce bufadienolide, whereas short-footed frog eggs do not. Therefore, these data are not sufficient to exclude that bufadienolide alone could explain the tadpoles’ preference for some species’ eggs over other species’ eggs. Choice

D is incorrect. Although the table shows that for both dainty green tree frog eggs and little red tree frog eggs, the cane toad tadpoles ate 1% of those species' eggs, it also indicates that neither produces bufadienolide. Thus, these data alone don't indicate bufadienolide's role in the tadpoles' egg preferences.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In vertical inheritance, parents pass genes to their offspring, but in horizontal transfer (HT), one species, often bacteria, passes genetic material to an unrelated species. In a 2022 study, herpetologist Atsushi Kurabayashi and his team investigated HT in multicellular organisms—namely, snakes and frogs in Madagascar. The team detected *BovB*—a gene transmitted vertically in snakes—in many frog species. The apparent direction of gene transfer seems counterintuitive because frogs usually don’t survive encounters with snakes and so wouldn’t be able to transmit the newly acquired gene to offspring, but the team concluded that *BovB* is indeed transmitted from snakes to frogs, either directly or indirectly, via HT.

Which finding, if true, would most directly support the team’s conclusion?

Answer

- A. *BovB* can be transmitted across frog species through HT.
- B. Parasites known to feed on species of snakes and frogs in which the *BovB* gene occurs also carry *BovB*.
- C. *BovB* cannot be reliably transmitted from a snake species to bacteria that are usually encountered by frog species.
- D. Frog species with *BovB* show few discernible advantages as compared with frog species that do not carry *BovB*.

Correct Answer: B

Rationale

Choice B is the best answer. If there are parasites that feed on both snakes and frogs, they could carry *BovB* from snakes to frogs. HT could occur “indirectly” through those encounters, which frogs are more likely to survive than snake encounters.

Choice A is incorrect. The team’s conclusion specifically argues that *BovB* is transmitted from snakes to frogs via HT, and this choice doesn’t mention snakes. Choice C is incorrect. The team argues that *BovB* is transmitted from snakes to frogs via HT, and this answer choice discusses a scenario in which the gene is not transmitted. Choice D is incorrect. Whether or not frog species with *BovB* are advantaged isn’t relevant to the team’s conclusion about how the gene is transmitted.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The increased integration of digital technologies throughout the process of book creation in the late 20th and early 21st centuries lowered the costs of book production, but those decreased costs have been most significant in the manufacturing and distribution process, which occurs after the authoring, editing, and design of the book are complete. This suggests that in the late 20th and early 21st centuries, _____

Which choice most logically completes the text?

Answer

- A. digital technologies made it easier than it had been previously for authors to write very long works and get them published.
- B. customers generally expected the cost of books to decline relative to the cost of other consumer goods.
- C. publishers increased the variety of their offerings by printing more unique titles but also printed fewer copies of each title.
- D. the costs of writing, editing, and designing a book were less affected by the technologies used than were the costs of manufacturing and distributing a book.

Correct Answer: D

Rationale

Choice D is the best answer because it presents the conclusion that most logically follows from the text’s discussion of how digital technologies affected the process of book creation. The text explains that in the late 20th and early 21st centuries digital technologies lowered book production costs most significantly in manufacturing and distribution. The text goes on to point out that authoring, editing, and book design are distinct steps in the process that occur before manufacturing and distribution. Because the savings connected to digital technologies have been most significant in manufacturing and distribution, it’s reasonable to infer that those technologies had less of an effect on writing, editing, and designing books.

Choice A is incorrect because the text focuses on lowered book production costs that occur after authoring has taken place; there’s no indication in the text whether digital technologies made writing and publishing lengthy books easier. Choice B is incorrect. Although it’s logical to conclude that customers would expect the cost of books to decline if production costs have declined, the text doesn’t address customer expectations for the cost of books or any other consumer goods. Choice C is incorrect because the text focuses broadly on how digital technologies have affected the cost of the publishing process; it doesn’t address the kinds of books being published or how many copies are printed.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Biologist Valentina Gómez-Bahamón and her team have investigated two subspecies of the fork-tailed flycatcher bird that live in the same region in Colombia, but one subspecies migrates south for part of the year, and the other doesn't. The researchers found that, due to slight differences in feather shape, the feathers of migratory forked-tailed flycatcher males make a sound during flight that is higher pitched than that made by the feathers of nonmigratory males. The researchers hypothesize that fork-tailed flycatcher females are attracted to the specific sound made by the males of their own subspecies, and that over time the females' preference will drive further genetic and anatomical divergence between the subspecies.

Which finding, if true, would most directly support Gómez-Bahamón and her team's hypothesis?

Answer

- A. The feathers located on the wings of the migratory fork-tailed flycatchers have a narrower shape than those of the nonmigratory birds, which allows them to fly long distances.
- B. Over several generations, the sound made by the feathers of migratory male fork-tailed flycatchers grows progressively higher pitched relative to that made by the feathers of nonmigratory males.
- C. Fork-tailed flycatchers communicate different messages to each other depending on whether their feathers create high-pitched or low-pitched sounds.
- D. The breeding habits of the migratory and nonmigratory fork-tailed flycatchers remained generally the same over several generations.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support Gómez-Bahamón and her team's hypothesis about fork-tailed flycatchers. The text indicates that although two subspecies of the birds live in the same region, the tail feathers of the migrating males make a higher-pitched sound than the tail feathers of the nonmigrating males do. Gómez-Bahamón and her team hypothesize that female fork-tailed flycatchers are attracted to the particular sound made by the tail feathers of males of their own subspecies, which will bring about additional "genetic and anatomical divergence" between the two subspecies. If it were found that the pitch generated by the tail feathers of migrating males is getting higher over successive generations, it would indicate that the shape of the migrating subspecies' tail feathers is diverging further from that of the nonmigrating subspecies. And if females continue to prefer the sounds of the males of their own subspecies, the females of the migrating subspecies will become acclimated to increasingly higher pitches over subsequent generations, causing further divergence between the subspecies. Thus, if it were found that migrating males' tail feathers were producing higher pitches over time, that would support the researchers' hypothesis.

Choice A is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding is about the shape of wing feathers and how that affects long-distance flight, whereas the hypothesis is about the shape of tail feathers and how that relates to female mate preference. Choice C is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding focuses on how the tail feather sounds communicate different messages, which doesn't address differences between the subspecies or female preferences. Choice D is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. The finding that breeding habits haven't changed for either subspecies does not, by itself, suggest anything about female preferences or divergence between the two subspecies.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is adapted from William Shakespeare's 1598 play *Henry IV, Part 1*. King Henry is addressing several noblemen who are rumored to have been disloyal to him.

KING HENRY: My blood hath been too cold and temperate,
Unapt to stir at these indignities,
And you have found me, for accordingly
You tread upon my patience; but be sure
I will from henceforth rather be myself,
Mighty and to be fear'd, than my [disposition],
Which hath been smooth as oil, soft as young down,
And therefore lost that title of respect
Which the proud soul ne'er pays but to the proud.

Which statement best describes how King Henry presents himself in the text?

Answer

- A. He declares that he has judged the noblemen's conduct in an impartial manner, and that despite their attempts to regain his trust through professions of submissiveness, they have irrevocably lost his respect.
- B. He acknowledges to the noblemen that his tolerance of their conduct has undermined their respect for him, and he resolves to display his genuine nature, which is more forceful and compels deference.
- C. He defends his calm disposition, which he sees as fundamental to his personality, and vows to maintain his tranquil demeanor even if it causes him to lose the respect of overly proud noblemen.
- D. He concedes that he has treated the noblemen with indifference, which in part explains why they have defied him, but he asserts that their disrespectful behavior is primarily driven by their excessive pride.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately describes King Henry's self-presentation in the text. Henry states that his temperament has been "too cold and temperate, / Unapt to stir at these indignities" (that is, too calm and slow to react to insults), and he asserts that the noblemen have taken advantage of this: "accordingly / You tread upon my patience." He then resolves to change, declaring he will "from henceforth rather be myself, / Mighty and to be fear'd," since being lenient ("smooth as oil, soft as young down") has caused him to lose "that title of respect / Which the proud soul ne'er pays but to the proud." In other words, Henry recognizes that proud people like the noblemen respect only strength and resolves to show his more forceful, true nature.

Choice A is incorrect because the text doesn't describe Henry as having judged the noblemen impartially, nor does it mention any attempts by the noblemen to regain his trust through submissiveness. Rather, Henry describes the noblemen as taking advantage of his patience and declares he will adopt a more commanding demeanor in the future. Choice C is incorrect. Although Henry concedes that he has perhaps been too calm and mild, Henry does not defend or vow to maintain his calm disposition. On the contrary, he explicitly states he will "from henceforth rather be myself, / Mighty and to be fear'd," resolving to abandon the mild manner that has cost him the respect of the noblemen. Choice D is incorrect because Henry characterizes his manner as patient and mild—"smooth as oil, soft as young down"—not indifferent. Indifference implies

a lack of concern, whereas Henry describes himself as being overly lenient. Additionally, although the final lines mention “the proud soul,” the focus is not on the noblemen’s excessive pride but rather on how Henry’s apparent *lack* of pride has caused him to lose the noblemen’s respect.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Some ethicists hold that the moral goodness of an individual’s actions depends solely on whether the actions themselves are good, irrespective of the context in which they are carried out. Philosopher L. Sebastian Purcell has shown that surviving works of Aztec (Nahua) philosophy express a very different view. Purcell reveals that these works posit an ethical system in which an individual’s actions are judged in light of how well they accord with the individual’s role in society and how well they contribute to the community. To the extent that these works are representative of Aztec thought, Purcell’s analysis suggests that _____

Which choice most logically completes the text?

Answer

- A. the Aztecs would have disputed the idea that the morality of an individual’s actions can be assessed by appealing to standards of behavior that are independent of the individual’s social circumstances.
- B. the Aztecs would not have accepted the notion that the morality of an individual’s actions can be fairly evaluated by people who do not live in the same society as that individual.
- C. actions by members of Aztec society who contributed a great deal to their community could be judged as morally good even if those actions were inconsistent with behaviors the Aztecs regarded as good in all contexts.
- D. similar actions performed by people in different social roles in Aztec society would have been regarded as morally equivalent unless those actions led to different outcomes for the community.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion about Aztec (Nahua) ethics. The text indicates that, according to Purcell’s interpretation of available Aztec philosophical works, the Aztec ethical system views an individual’s actions in relation to that individual’s societal role and how the actions affect the community. The text contrasts this view with another held by some ethicists, namely that actions are morally good or bad regardless of the context in which they occur. Thus, Purcell’s analysis suggests that the Aztecs would have asserted that the morality of an individual’s actions are rooted in that person’s position in the community and the actions’ effects and therefore cannot be determined in the absence of that context.

Choice B is incorrect. Although the text indicates that morally judging an action according to Aztec ethics requires an understanding of the action’s effects and the individual’s social circumstances, it does not specify that only members of that society can acquire this information. Choice C is incorrect because it implies that the Aztecs considered some actions good or bad regardless of the surrounding context, which contradicts the text’s claim that the Aztecs believed that the morality of an individual’s action is dependent on the action’s effects on the community and the person’s specific circumstances. Choice D is incorrect. Although the text indicates that in Aztec ethics the morality of an action depends in part on how it affects the community, this is only one of the two factors—the other being the person’s societal role—that need to be considered. Therefore, it is possible that two actions with the same effect on the community could be considered morally distinct if they are performed by individuals in different social roles.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Philosophers note that many people have an intuitive sense that while we ought not to lie, there may be circumstances in which lying is permissible. If this intuition is correct and we lack an inviolable duty to speak truthfully, what grounds opposition to lying in the first place? Japa Pallikkathayil has advanced one answer by appealing to a duty to respect others’ agential interests: the possession of false beliefs constrains agency, and thus we ought not to impede the formation of true beliefs unless doing so prevents a greater constraint on someone’s agency or an otherwise impermissible end.

Which choice best states the main idea of the text?

Answer

- A. One potential means of justifying opposition to lying is Pallikkathayil’s argument that we have an obligation to respect other people’s agency that entails a commitment to truthfulness except in certain circumstances.
- B. Many people have an intuitive sense that lying is permissible in some circumstances but lack a principled way to identify those circumstances, and Pallikkathayil’s argument may provide a means of resolving that problem.
- C. Pallikkathayil’s argument suggests that if we have a duty to respect other people’s agential interests and if possession of false beliefs constrains agency, then we have an inviolable duty to speak truthfully.
- D. Pallikkathayil’s argument shows that if our intuition that circumstances may make lying permissible is correct, then it is unclear whether there are any grounds for an opposition to lying in the first place.

Correct Answer: A

Rationale

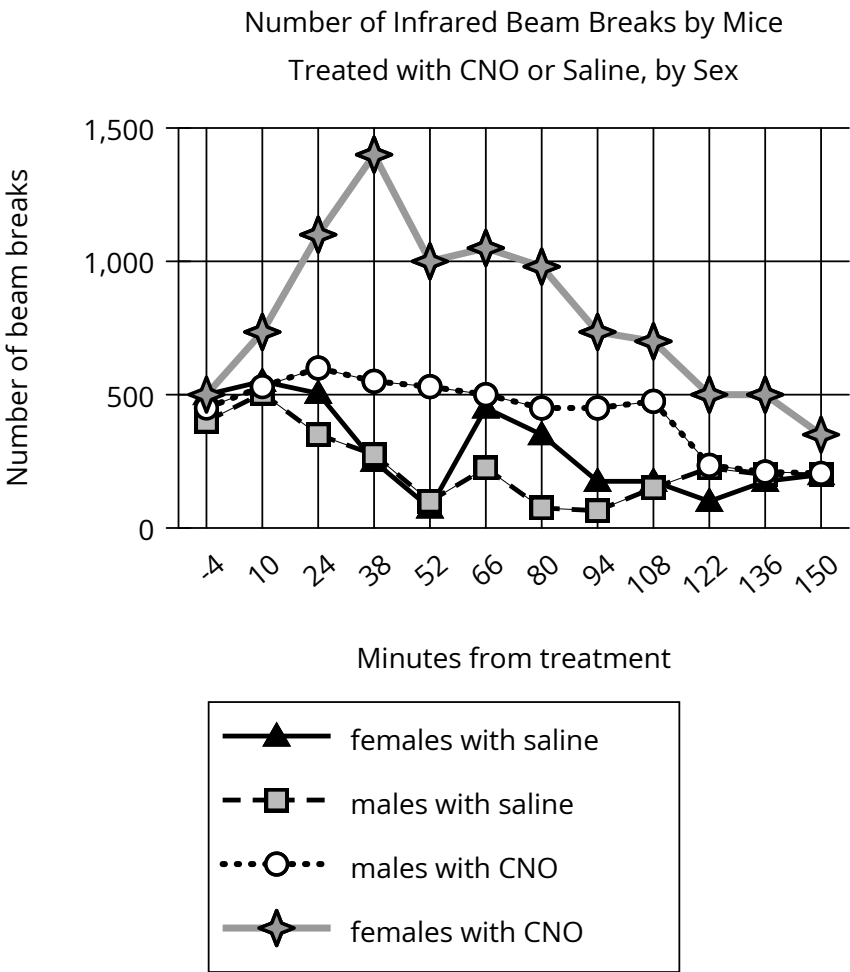
Choice A is the best answer because it most accurately states the main idea of the text. The text begins by noting a common intuition that lying might sometimes be permissible, which raises the question of what is the reason for being against lying. The text then presents Pallikkathayil’s answer to this question: we have a duty to respect others’ agential interests (that is, we have an obligation to allow people to exercise their free will in order to best serve their own ends), false beliefs constrain agency (the implication here being that people who are deceived about what is true are impeded in acting in their own best interests), and so we should not hinder the formation of true beliefs (we should not deliberately deceive a person) unless doing so prevents a greater limitation on someone’s agency or prevents some otherwise impermissible end (for example, it might be morally permissible to lie to someone in the rare circumstance where a false belief would lead a person to act in their own best interest, and where they would not do so if they knew the truth). Thus, the main idea of the text is that Pallikkathayil offers a potential justification for opposing lying based on respecting agency, while still allowing for exceptions in specific circumstances.

Choice B is incorrect because the text doesn’t focus on people’s lack of a principled way to identify circumstances where lying is permissible or indicate that Pallikkathayil’s argument resolves this specific problem. Rather, the text presents Pallikkathayil’s argument that lies limit agency but may be permitted in very specific situations (to prevent a greater limitation on someone’s agency or an otherwise unacceptable outcome) as one of a number of possible arguments about why lies are undesirable but sometimes permissible. Choice C is incorrect because it directly contradicts the text. Pallikkathayil’s argument, as described in the text, wouldn’t lead to an "inviolable duty to speak truthfully." On the contrary, the text explicitly states that Pallikkathayil’s position is that interfering with people developing accurate beliefs is wrong, except when such interference prevents either a more significant limitation on someone’s agency or an outcome that would be unacceptable for other reasons. This indicates that Pallikkathayil’s argument allows for some exceptions to the duty to speak truthfully, making it not inviolable. Choice D is incorrect because Pallikkathayil’s argument doesn’t suggest that if it is permissible to lie in at least some circumstances, then it is unclear whether there are good reasons for opposing lying in any circumstances. Instead, the text presents Pallikkathayil as offering a specific ground

for opposing lying (respecting others' ability to act in their own best interests) while qualifying that assertion by presenting a situation in which lying might sometimes be morally permissible.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



To investigate the influence of certain estrogen-responsive neurons on energy expenditure, biologist Stephanie Correa et al. treated female and male mice with either saline solution or clozapine-N4-oxide (CNO), which activates the neurons. Monitoring the activity levels of the mice by measuring how frequently the animals broke infrared beams crossing their enclosures, Correa et al. found that the mice in their study showed sex-specific differences in response to neuron activation: _____

Which choice most effectively uses data from the graph to complete the assertion?

Answer

- A. the four groups of mice differed greatly in their activity levels before treatment but showed identical activity levels at the end of the monitoring period.
- B. saline-treated females showed substantially more activity at certain points in the monitoring period than saline-treated males did.
- C. CNO-treated females showed more activity relative to saline-treated females than CNO-treated males showed relative to saline-treated males.
- D. CNO-treated females showed a substantial increase and then decline in activity over the monitoring period, whereas CNO-treated males showed a substantial decline in activity followed by a steep increase.

Correct Answer: C

Rationale

Choice C is the best answer. The graph shows that the CNO-treated females were way more active than the CNO-treated males, while the saline-treated males and females (the control groups) had very similar activity levels. This supports the claim that there were sex-specific differences in the mice's response to neuron activation.

Choice A is incorrect. This choice misreads the graph. All four groups of mice started at nearly the same activity level before treatment (see how all four points are very close together at -4 minutes, meaning four minutes before treatment). Choice B is incorrect. This choice doesn't complete the assertion. The assertion is about the mice's response to neuron activation, so we need to include the data about the CNO-treated females and males. Choice D is incorrect. This choice misreads the graph. The line for the CNO-treated males does not show a "substantial decline" until around 122 minutes, and there is no "steep increase" afterward.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Neurobiologists Laura Cuaya, Raúl Hernández-Pérez, and colleagues investigated the language detection abilities of eighteen dogs. The researchers monitored the brain activity of Joey (an Australian shepherd), Mini (a mixed breed), and other dogs while the animals listened to three recordings: one of *The Little Prince* being read in Spanish, the second in Hungarian, and a third made up of short, randomly selected fragments of the first two, scrambled so that they didn't resemble human speech. Each dog was familiar with either Spanish or Hungarian, but not both. The team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech.

Which finding from the study, if true, would most directly support the team's conclusion?

Answer

- A. Long-headed dogs accustomed to hearing Spanish tended to show more brain activity when hearing Spanish than long-headed dogs accustomed to hearing Hungarian showed when hearing Hungarian.
- B. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing either Spanish or Hungarian than when hearing the scrambled recording.
- C. The pattern of brain activity that long-headed dogs showed when hearing the scrambled recording was different from the pattern of brain activity that short-headed dogs showed when hearing the language they were accustomed to.
- D. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing the language they were accustomed to than when hearing the other language.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the research team's conclusion about anatomical features and speech detection in dogs. The text explains that a team of researchers monitored the brain activity of dogs while the dogs listened to three recordings: one of spoken Spanish, one of spoken Hungarian, and one of scrambled fragments that weren't recognizable as human speech. The text then states that the team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech. The finding that longer-headed dogs exhibited a greater difference in brain activity when listening to the speech recordings (in Spanish or Hungarian) versus the nonspeech (scrambled) recording compared with shorter-headed dogs would establish an association between an anatomical feature (head length) and responses (as measured by brain activity) to speech versus nonspeech. This observed relationship between head length and brain activity patterns during exposure to speech and nonspeech would support the team's conclusion.

Choice A is incorrect because it describes a finding about dogs' responses (as indicated by brain activity) to hearing their respective familiar languages rather than dogs' ability to distinguish speech from nonspeech, which is what the team's conclusion specifically addresses. Moreover, this finding pertains to only one anatomical type (long-headed dogs), so it wouldn't support the conclusion that anatomical differences may affect dogs' ability to distinguish speech from nonspeech. Choice C is incorrect because it describes a finding involving a comparison between long-headed dogs listening to nonspeech and short-headed dogs listening to speech, which wouldn't provide enough information to support the researchers' conclusion that anatomical differences may affect dogs' ability to distinguish speech from nonspeech. To support that conclusion, the finding would need to show how dogs with different anatomical features (for example, long-headed and short-headed dogs) respond to recordings of speech as well as recordings of nonspeech. Choice D is incorrect because it describes a finding about dogs' ability to distinguish between a familiar language and an unfamiliar one, not between speech and nonspeech. While this finding does compare dogs with different anatomical features (longer-headed vs. shorter-headed ones), it focuses on language recognition (Spanish vs. Hungarian) rather than the ability to distinguish speech from nonspeech, which is what the team's conclusion specifically addresses.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Mean Time (in Seconds) Spent per Flower for Four Pollinator Genera

Pollinator genus	Seconds per intact pin flower	Seconds per damaged pin flower	Seconds per intact thrum flower	Seconds per damaged thrum flower
<i>Habropoda</i>	2.7	5.4	4.1	9.5
<i>Osmia</i>	5.2	8.2	7.1	8.3
<i>Pierid</i>	2.6	4.0	2.4	1.9
<i>Xylocopa</i>	2.3	2.8	2.5	2.2

To study how floral damage affects the behavior of pollinators, such as bees, a team of researchers punched holes in the floral tissue of flowers from the vine yellow jessamine (*Gelsemium sempervirens*), a plant that produces flowers that have either a long pistil and a short stamen (pin morphs) or a short pistil and a long stamen (thrum morphs). The researchers then compared the time different insect pollinators spent visiting intact pin and thrum flowers to the time such pollinators spent visiting the artificially damaged pin and thrum flowers. The researchers concluded that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator.

Which choice best describes data from the table that support the researchers’ conclusion?

Answer

- A. For pin flowers, damage led to longer times per flower in all pollinator genera, whereas for thrum flowers, damage led to longer times per flower only in *Habropoda* and *Osmia*.
- B. Compared with pollinators belonging to the genus *Osmia*, pollinators belonging to the genus *Xylocopa* spent less time on damaged pin flowers but more time on damaged thrum flowers.
- C. Damage led to shorter times per thrum flower in three pollinator genera (*Osmia*, *Pierid*, and *Xylocopa*), whereas it led to longer times per thrum flower in one pollinator genus (*Habropoda*).
- D. Pollinators belonging to the genus *Habropoda* spent 2.7 seconds on intact pin flowers and 4.1 seconds on intact thrum flowers.

Correct Answer: A

Rationale

Choice A is the best answer because it describes data from the table that support the researchers’ conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator. The table presents the average time pollinators spent per floral morph. The data in the table shows that for pin flowers, the average time spent per flower by all pollinator genera was higher for flowers that had been artificially damaged than for intact flowers. By contrast, for thrum flowers, the difference in time spent on intact and damaged flowers is seen for only some pollinator genera. This supports the researchers’ conclusion that both floral morph and the genus of the pollinator are factors involved in the effect that floral damage has on time pollinators spend per flower.

Choice B is incorrect. The table shows that pollinators belonging to the genus *Xylocopa* spent less time on both damaged pin flowers and damaged thrum flowers than pollinators belonging to the genus *Osmia* did. Choice C is incorrect. The table shows that the artificial damage to the thrum flowers led to shorter average times spent by pollinators on those flowers for only two of the four pollinator genera represented in the table, not three. Furthermore, this choice doesn’t address the effect the artificial damage had on pin flowers; thus, even if accurate, this evidence

doesn't support the researchers' conclusion that both floral morph and pollinator genus affect time spent per flower when a flower has been damaged. Choice D is incorrect. While this statement is true based on the information in the table, it doesn't provide evidence that supports the researchers' conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator because this choice provides information only about how much time one pollinator genus spent on intact flowers.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Many animals, including humans, must sleep, and sleep is known to have a role in everything from healing injuries to encoding information in long-term memory. But some scientists claim that, from an evolutionary standpoint, deep sleep for hours at a time leaves an animal so vulnerable that the known benefits of sleeping seem insufficient to explain why it became so widespread in the animal kingdom. These scientists therefore imply that _____

Which choice most logically completes the text?

Answer

- A. prolonged deep sleep is likely advantageous in ways that have yet to be discovered.
- B. most traits perform functions that are hard to understand from an evolutionary standpoint.
- C. it is more important to understand how widespread prolonged deep sleep is than to understand its function.
- D. many traits that provide significant benefits for an animal also likely pose risks to that animal.

Correct Answer: A

Rationale

Choice A is the best answer. It most logically completes the text. The text says that some scientists can't explain why prolonged deep sleep is so widespread, given that the known benefits of sleep don't seem to make up for how vulnerable it leaves an animal. This suggests that prolonged deep sleep probably has unknown evolutionary benefits that make up for the vulnerability.

Choice B is incorrect. It doesn't logically complete the text. The text only discusses the benefits and risks of one trait: sleep. So there's no basis for an inference about "most traits." Choice C is incorrect. It doesn't logically complete the text. The text says that it is already known that prolonged deep sleep is widespread in the animal kingdom. Rather, what some scientists can't explain is WHY prolonged deep sleep is so widespread, given that the known benefits of sleep don't seem to make up for how vulnerable it leaves an animal. Choice D is incorrect. It doesn't logically complete the text. The text only discusses the benefits and risks of one trait: sleep. So there's no basis for an inference about "many traits."

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Richard II is a play from the 1590s by William Shakespeare. Although King Richard has been vanquished by his cousin Henry Bolingbroke, he intimates that he is not entirely ready to show subservience to his cousin, saying, _____

Which quotation from *Richard II* most effectively illustrates the claim?

Answer

- A. "I have no name, no title,— / No, not that name was given me at the font,— / But 'tis usurp'd:— Alack the heavy day, / That I have worn so many winters out, / And know not now what name to call myself!"
- B. "Alack, why am I sent for to a king, / Before I have shook off the regal thoughts / Wherewith I reign'd? I hardly yet have learn'd / To insinuate, flatter, bow, and bend my knee."
- C. "Still my griefs are mine. / You may my glories and my state depose, / But not my griefs; still am I king of those."
- D. "Am I both priest and clerk? Well then, amen. / God save the King! although I be not he; / And yet, amen, if heaven do think him me."

Correct Answer: B

Rationale

Choice B is the best answer because it most effectively presents a quotation that illustrates the claim that Richard intimates he is not entirely ready to show subservience to Bolingbroke. The quotation conveys that Richard has not yet shaken off "the regal thoughts / Wherewith [he] reign'd"—that is, he still feels like a king—and that he has "hardly yet have learn'd / To insinuate, flatter, bow, and bend [his] knee," directly listing acts of subservience he is not prepared to perform. By stating his unreadiness to behave subserviently while still inhabiting a regal mindset, Richard intimates exactly what the claim describes.

Choice A is incorrect because this quotation centers on Richard's anguish over the loss of his name and title; it expresses confusion about his identity rather than illustrating that he is not yet prepared to show subservience to Bolingbroke. Choice C is incorrect because in this quotation Richard claims that his griefs remain his own even after his "glories" and "state" have been deposed; it concerns ownership of his suffering and identity rather than any unreadiness to behave subserviently toward Bolingbroke. Choice D is incorrect because the quotation depicts Richard as willing to go along with the assertion of others that Bolingbroke is the rightful king, although God, who in this case serves as an objective arbiter, may have a different opinion on who the rightful king is ("if heaven do think him me"). So the quotation suggests that Richard is willing to behave subserviently even if he thinks he has been wronged; it does not suggest that he is at present unwilling to show subservience.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

An analysis by Alain Elayi and colleagues of coins minted in Sidon in the fifth and fourth centuries BCE reveals a change in their composition over time: while a coin from circa 450 BCE contains about 98% silver and 1% copper, a coin from 367 BCE (the end of Ba'alšillem II's reign) contains 74.2% silver and 24.7% copper, giving it a relatively yellowish appearance that traders would have noticed. Because coins with a silver content below 80% were widely considered unsuitable for trade, Elayi et al. speculate that a crisis in confidence in the currency occurred in Sidon around 367 BCE, which was likely relieved—despite Sidon's persistent oppressive financial obligations—as a result of Ba'alšillem II's successor Abd'aštar I's decision to _____

Which choice most logically completes the text?

Answer

- A. proclaim that the percentage of silver in coins suitable for trade would be raised to a threshold higher than 80%.
- B. keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but decrease their weight.
- C. begin minting heavier coins with a proportion of silver to copper similar to that in coins minted in 367 BCE.
- D. fund the mining of some copper deposits that were not available to Ba'alšillem II.

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text's discussion of Sidonian coins. As the text explains, researchers determined that Sidonian coins were made of silver and copper and that from 450 BCE to 367 BCE, the percentage of silver in each coin decreased from 98% to 74.2% while the percentage of copper increased from 1% to 24.7%. The text indicates that because the coins containing less than 80% silver weren't considered suitable for trade (suggesting that copper was less valuable than silver) and looked different from coins containing more silver, the researchers suspect there was a serious loss in confidence in the currency in Sidon in 367 BCE when the copper content was high. It's reasonable to assume that it wasn't possible to boost confidence simply by devoting a greater amount of valuable silver to the currency, since Sidon was under significant and ongoing financial pressure; however, keeping the total amount of silver the same and reducing the amount of copper in the coins would have resulted in smaller coins with a higher percentage of silver. Therefore, it makes sense to suggest that Abd'aštar I (the ruler after 367 BCE) likely restored confidence in the currency by deciding to keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but to decrease the coins' weight.

Choice A is incorrect because the text conveys that a crisis in confidence in the currency of Sidon likely occurred around 367 BCE because the percentage of silver in coins had fallen below 80% (presumably because Sidon's financial pressures meant that less silver was available for currency), making the coins unsuitable for trade. Thus, announcing that the threshold for the percentage of silver in coins would be raised—that is, that coins would need to contain even more than 80% silver to be suitable for trade—likely would have worsened the crisis rather than relieved it. Choice C is incorrect because the text strongly suggests that a crisis in confidence in the currency of Sidon was caused by the proportion of silver to copper in the coins in 367 BCE, with 74.2% being too little silver for the coins to be considered suitable for trade; therefore, it's unlikely that minting coins with a similar proportion of silver to copper (that is, still around 74.2% silver) would have restored confidence, even if the coins were heavier. Choice D is incorrect because the text gives no indication that funding the mining of more copper would have relieved a crisis in confidence in the currency of Sidon. The text establishes that Sidonian coins that visibly contained copper weren't considered suitable for trade, so Abd'aštar I wouldn't have wanted to add even more copper to them, and it's unclear how else copper mining would affect views of the currency.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is adapted from William Shakespeare’s 1609 poem “Sonnet 27.” The poem is addressed to a close friend as if he were physically present.

Weary with toil, I [hurry] to my bed,

The dear repose for limbs with travel tired;

But then begins a journey in my head

To work my mind, when body’s work’s expired:

For then my thoughts—from far where I abide—

[Begin] a zealous pilgrimage to thee,

And keep my drooping eyelids open wide,

What is the main idea of the text?

Answer

- A. The speaker is asleep and dreaming about traveling to see the friend.
- B. The speaker is planning an upcoming trip to the friend’s house.
- C. The speaker is too fatigued to continue a discussion with the friend.
- D. The speaker is thinking about the friend instead of immediately falling asleep.

Correct Answer: D

Rationale

Choice D is the best answer it most accurately states the main idea of the text. The speaker describes going to bed for “repose” (rest) but finding that his thoughts are focused on the friend the speaker is addressing, and the thoughts are keeping the speaker awake.

Choice A is incorrect because the speaker isn’t asleep; the thoughts about the friend are keeping the speaker awake. Choice B is incorrect because the speaker isn’t talking about taking a literal trip; rather, the speaker uses the metaphor of a journey to describe internal thoughts. Choice C is incorrect because the speaker isn’t having a discussion with the friend.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

“The Poet Walt Whitman” is an 1887 essay by José Martí, a Cuban author and political activist, originally written in Spanish. In the essay, Martí explores the value of literature, arguing that a society’s spiritual well-being depends on the character of its literary culture: _____

Which quotation from a translation of “The Poet Walt Whitman” most effectively illustrates the claim?

Answer

- A. “Poetry, which brings together or separates, which fortifies or brings anguish, which shores up or demolishes souls, which gives or robs men of faith and vigor, is more necessary to a people than industry itself, for industry provides them with a means of subsistence, while literature gives them the desire and strength for life.”
- B. “Every society brings to literature its own form of expression, and the history of the nations can be told with greater truth by the stages of literature than by chronicles and decades.”
- C. “Where will a race of men go when they have lost the habit of thinking with faith about the scope and meaning of their actions? The best among them, those who consecrate Nature with their sacred desire for the future, will lose, in a sordid and painful annihilation, all stimulus to alleviate the ugliness of humanity.”
- D. “Listen to the song of this hardworking and satisfied nation; listen to Walt Whitman. The exercise of himself exalts him to majesty, tolerance exalts him to justice, and order to joy.”

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively illustrates the claim that Martí argues that a society’s spiritual well-being depends on the character of its literary culture. In the quotation, Martí asserts that poetry is “more necessary to a people than industry itself” and that it has the power to provide people with “faith and vigor.” He also adds that literature gives people “the desire and strength for life.” Therefore, this quotation shows that Martí believes that literature is a societal necessity because it uplifts people and nourishes their spiritual well-being.

Choice B is incorrect. Although this quotation emphasizes the importance of literature, it focuses on how the nature of a society is reflected in that society’s literature rather than on literature’s value for people’s spiritual well-being. Choice C is incorrect. Although this quotation involves an element of spirituality, it doesn’t discuss literature. The quotation instead focuses on humanity’s actions. Choice D is incorrect because this quotation mainly focuses on the importance of Walt Whitman rather than on the value of literature in general.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

The linguistic niche hypothesis (LNH) posits that the exotericity of languages (how prevalent non-native speakers are) and grammatical complexity are inversely related, which the LNH ascribes to attrition of complex grammatical rules as more non-native speakers adopt the language but fail to acquire those rules. Focusing on two characteristics that are positive indices of grammatical complexity, fusion (when new phonemes arise from the merger of previously distinct ones) and informativity (languages’ capacity for meaningful variation), Olena Shcherbakova and colleagues conducted a quantitative analysis for more than 1,300 languages and claim the outcome is inconsistent with the LNH.

Which finding, if true, would most directly support Shcherbakova and colleagues’ claim?

Answer

- A. Shcherbakova and colleagues’ analysis showed a slightly negative correlation between grammatical complexity and fusion and between grammatical complexity and informativity.
- B. Shcherbakova and colleagues’ analysis showed a slightly negative correlation between grammatical complexity and exotericity.
- C. Shcherbakova and colleagues’ analysis showed a slightly positive correlation between grammatical complexity and fusion.
- D. Shcherbakova and colleagues’ analysis showed a slightly positive correlation between fusion and exotericity and between informativity and exotericity.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support Shcherbakova and colleagues’ claim that the outcome of their study is inconsistent with the linguistic niche hypothesis (LNH). The text explains that the LNH holds that there is an inverse relationship between the prevalence of non-native speakers of a language (exotericity) and the grammatical complexity of that language—that is, that as the number of non-native speakers increases, grammatical complexity decreases, and vice versa. According to the text, Shcherbakova and colleagues focused on two positive indications of grammatical complexity—fusion and informativity—and analyzed their occurrence in over 1,300 languages. If the researchers found a slightly positive correlation between fusion and exotericity and between informativity and exotericity—meaning that to some extent, grammatical complexity increases as the number of non-native speakers of a language increases—their outcome would not be consistent with the assumption that exotericity and grammatical complexity are inversely related (the LNH).

Choice A is incorrect because it wouldn’t be possible to say that a finding of a slightly negative correlation between grammatical complexity and both fusion and informativity is inconsistent or consistent with the LNH, since the finding would address only grammatical complexity (given that fusion and informativity are aspects of grammatical complexity) and wouldn’t move beyond that factor to address its relationship to the prevalence of non-native speakers of a language (exotericity), which is the relationship the LNH focuses on. Choice B is incorrect because a finding of a slightly negative correlation between grammatical complexity and the prevalence of non-native speakers of a language (exotericity)—meaning that as the number of non-native speakers increases, grammatical complexity somewhat decreases, and vice versa—would be consistent, not inconsistent, with the LNH, since the text indicates that according to the LNH, there is an inverse relationship between grammatical complexity and exotericity; a negative correlation reflects an inverse relationship. Choice C is incorrect because it wouldn’t be possible to say that a finding of a slightly positive correlation between grammatical complexity and fusion is inconsistent or consistent with the LNH, since the finding would address only grammatical complexity (given that fusion is a positive indication of grammatical complexity) and wouldn’t move beyond that factor to address its relationship to the prevalence of non-native speakers of a language (exotericity), which is the relationship the LNH focuses on.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Some astronomers searching for extraterrestrial life have proposed that atmospheric NH₃ (ammonia) can serve as a biosignature gas—an indication that a planet harbors life. Jingcheng Huang, Sara Seager, and colleagues evaluated this possibility, finding that on rocky planets, atmospheric NH₃ likely couldn't reach detectably high levels in the absence of biological activity. But the team also found that on so-called mini-Neptunes—gas planets smaller than Neptune but with atmospheres similar to Neptune's—atmospheric pressure and temperature can be high enough to produce atmospheric NH₃.

Based on the text, Huang, Seager, and colleagues would most likely agree with which statement about atmospheric NH₃?

Answer

- A. Its presence is more likely to indicate that a planet is a mini-Neptune than that the planet is a rocky planet that could support life.
- B. Its absence from a planet that's not a mini-Neptune indicates that the planet probably doesn't have life.
- C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.
- D. It doesn't reliably reach high enough concentrations in the atmospheres of rocky planets or mini-Neptunes to be treated as a biosignature gas.

Correct Answer: C

Rationale

Choice C is the best answer because it states a conclusion the researchers likely agree with, given the details in the text. The text explains that a biosignature gas is a gas that can be used as an indicator that a planet harbors some form of life and some astronomers have proposed that NH₃ could serve as a biosignature gas. The researchers evaluating this claim found that the atmosphere of rocky planets would be unlikely to reach "detectably high levels" of NH₃ without biological activity, which would support the proposal of NH₃ serving as a biosignature gas. However, the text also states that mini-Neptune planets can produce NH₃ in the absence of biological activity. Thus, the text is structured to lead to the conclusion that detectable levels of NH₃ in the atmospheres of rocky planets could constitute a biosignature, but that is not the case for detectable levels of the gas in the atmospheres of mini-Neptune planets.

Choice A is incorrect because the text indicates that biological activity likely accounts for detectable levels of NH₃ in the atmospheres of rocky planets but mini-Neptune planets can have detectable levels of NH₃ in their atmospheres in the absence of biological activity. Therefore, both rocky planets and mini-Neptune planets can have detectable levels of atmospheric NH₃. Choice B is incorrect because the text states that for NH₃ to reach detectable levels in the atmospheres of rocky planets likely means they harbor biological activity, meaning that rocky planets with detectable NH₃ usually harbor biological activity. However, that does not entail that every rocky planet with biological activity will have detectable levels of NH₃ in their atmospheres. Choice D is incorrect because the text claims only that some astronomers have proposed using NH₃ as a biosignature gas without mentioning a minimum concentration of atmospheric NH₃ that must be met for it to function as a biosignature gas.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In 2019, 20 previously unknown moons were confirmed to be orbiting Saturn. Three of the moons have prograde orbits (orbiting in the direction the planet spins), and the other 17 have retrograde orbits (orbiting in the opposite direction of the planet’s spin). All but one of the 20 moons are thought to be remnants of bodies that orbited Saturn until they broke apart in collisions. Although the one exceptional moon orbits in the same direction as the planet’s spin, its orbit is highly eccentric compared to the rest, which may suggest that it has a different origin than the other 19 moons.

Based on the text, which choice best describes the moon with the eccentric orbit?

Answer

- A. It doesn’t have a retrograde orbit, but it likely has the same origin as the moons with retrograde orbits.
- B. Its orbit is so tilted with respect to the other moons’ orbits that it’s neither prograde nor retrograde.
- C. It has a prograde orbit that is likely the result of having collided with another body orbiting Saturn.
- D. It has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.

Correct Answer: D

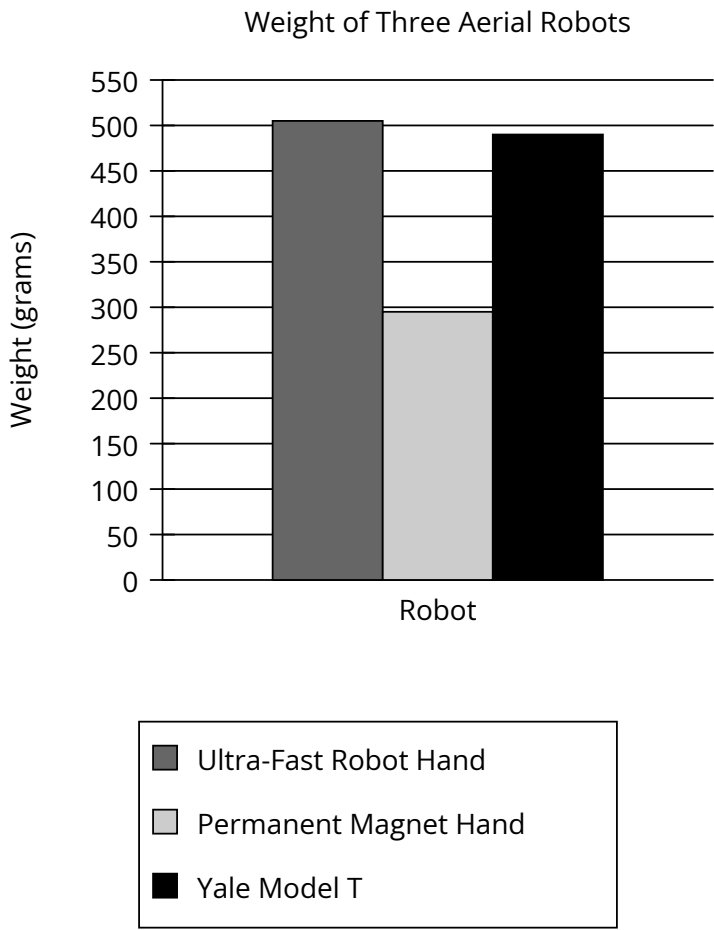
Rationale

Choice D is the best answer because it most accurately describes the moon with the eccentric orbit. The text indicates that three of the 20 newly discovered moons have prograde orbits, meaning that they orbit Saturn in the same direction as the planet’s spin, while the other 17 moons have retrograde orbits, meaning that they orbit Saturn in the opposite direction of the planet’s spin. The text then states that 19 of the 20 moons appear to be the remains of earlier bodies that orbited Saturn but were broken apart in collisions. The one exception is a moon that orbits Saturn in the same direction as the planet’s spin, meaning that the exceptional moon’s orbit is prograde. The text goes on to state that the exceptional moon’s orbit is so eccentric that the moon may have formed through a different process than the other 19 moons. The moon with the eccentric orbit, therefore, has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.

Choice A is incorrect because nothing in the text supports the idea that the moon with the eccentric orbit likely has the same origin as the moons with retrograde orbits. Although it’s true that the moon has a prograde orbit (and thus doesn’t have a retrograde orbit), the only information the text provides about the moon’s origin is that it may be different than the origin of the other 19 moons. Choice B is incorrect because the text states that the moon in question orbits Saturn in the same direction as the planet’s spin, meaning that the moon’s orbit is prograde, not that its orbit is neither prograde nor retrograde. Choice C is incorrect because the text merely notes that the moon in question has a prograde orbit without giving any indication of what likely caused that orbit.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Aerial robots vary considerably in their holding force; the Ultra-Fast Robot Hand, for example, has a holding force of 56 newtons, more than twice that of the Permanent Magnet Hand and more than four times that of the Yale Model T. Since an aerial robot must lift its own weight along with its cargo, engineer Jiawei Meng and colleagues used a ratio of each robot’s holding force to the robot’s weight to calculate payload capacity, with higher ratios corresponding to greater capacity, concluding that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Which choice best describes data in the graph that support Meng and colleagues’ conclusion?

- Answer**
- A. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than 450 grams.
 - B. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than the Permanent Magnet Hand does.
 - C. The Yale Model T has a lower holding force than the Permanent Magnet Hand despite weighing more.
 - D. The Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T does.

Correct Answer: D

Rationale

Choice D is the best answer because it describes data in the graph that support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. According to the text, payload capacity is calculated by using a ratio of a robot's holding force to the robot's weight, and higher ratios indicate a greater payload capacity. The Ultra-Fast Robot Hand has a holding force of 56 newtons, four times greater than that of the Yale Model T. Additionally, the graph shows that the Ultra-Fast Robot Hand has a weight of approximately 500 grams, slightly more than the Yale Model T's weight of approximately 480 grams. Therefore, the Ultra-Fast Robot Hand has a higher ratio of holding force to weight than the Yale Model T. Since higher ratios correspond to greater payload capacity, the information from the graph indicating that the Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T combined with the information in the text ultimately supports the conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Choice A is incorrect. Although, according to the graph, it's true that both the Ultra-Fast Robot Hand and the Yale Model T weigh more than 450 grams, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the two robots. Choice B is incorrect. Although, according to the graph, it's true that the Ultra-Fast Robot Hand and the Yale Model T both weigh more than the Permanent Magnet Hand does, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the Ultra-Fast Robot Hand and the Yale Model T. Furthermore, the comparison to the Permanent Magnet Hand is irrelevant to the claim about the relative ratios and payload capacities of the Ultra-Fast Robot Hand and the Yale Model T. Choice C is incorrect. Although the text states that the Yale Model T has a lower holding force than the Permanent Magnet Hand, the graph provides no information about holding force. Moreover, information about the Permanent Magnet Hand is irrelevant to the conclusion by Meng and colleagues, which only concerns the Ultra-Fast Robot Hand and the Yale Model T.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Overgrazing by purple sea urchins has caused many kelp forests along North America’s west coast to be replaced by urchin barrens—areas stripped of vegetation and covered in purple sea urchins. Urchins in barrens persist in a state of starvation that lessens their nutritional value—and thus their appeal—to many predators. Sarah Gravem and colleagues placed sunflower sea stars, a once-abundant predator species suffering massive population declines in recent years, in aquariums that each contained a nutritionally poor and a nutritionally rich purple sea urchin. The researchers found that the sea stars selected the nutritionally rich urchin in 42.7% of trials and the nutritionally poor urchin in 37.5% of trials, suggesting that _____

Which choice most logically completes the text?

Answer

- A. sunflower sea stars are willing to hunt sea urchins, but if given a choice, they will prey on other more nutritious marine animals instead.
- B. sunflower sea stars are reluctant to feed on both nutritionally poor and nutritionally rich sea urchins and are therefore unlikely to thrive in kelp forests.
- C. sunflower sea stars are less likely to consume sea urchins in barrens than other species of sea stars are, putting sunflower sea stars at a high risk of extinction.
- D. sunflower sea stars do not always avoid foraging on nutritionally poor sea urchins, making sunflower sea star population recovery a potentially important tool for controlling urchin barrens.

Correct Answer: D

Rationale

Choice D is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the researchers’ findings about purple sea urchins and sunflower sea stars. The text explains that urchin barrens are areas that used to be kelp forests but are now covered by purple sea urchins. The text suggests that because there is no more vegetation to be consumed in those areas, the urchins exist in a state of starvation that makes them less nutritional for many predators. The text goes on to explain that in a study with a choice between two purple sea urchins, sunflower sea stars (a predator species that has been substantially declining) consumed a nutritionally rich urchin 42.7% of the time and a nutritionally poor urchin 37.5% of the time. Because the sunflower sea stars didn’t always avoid consuming nutritionally poor urchins, even when nutritionally rich ones were available, it follows that helping sunflower sea star populations to grow could help control urchin barrens by increasing the number of sea stars that may consume and thus remove nutritionally poor purple sea urchins from barrens.

Choice A is incorrect because the text indicates only that when presented with purple sea urchins, the sunflower sea stars in the study consumed both nutritionally rich and nutritionally poor ones. It doesn’t suggest that sunflower sea stars generally prefer other marine animals that are more nutritious; there’s no mention of other marine animals. Choice B is incorrect because the text doesn’t suggest that sunflower sea stars are generally reluctant to feed on sea urchins. In fact, the text indicates that the sunflower sea stars in the study did consume sea urchins, feeding on both nutritionally poor and nutritionally rich ones. Choice C is incorrect because the text addresses only the willingness of sunflower sea stars to consume the type of sea urchins found in barrens (nutritionally poor sea urchins), not how likely other species of sea stars are to consume them.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Henry Ossawa Tanner’s 1893 painting *The Banjo Lesson*, which depicts an elderly man teaching a boy to play the banjo, is regarded as a landmark in the history of works by Black artists in the United States. Scholars should be cautious when ascribing political or ideological values to the painting, however: beliefs and assumptions that are commonly held now may have been unfamiliar to Tanner and his contemporaries, and vice versa. Scholars who forget this fact when discussing *The Banjo Lesson* therefore _____

Which choice most logically completes the text?

Answer

- A. risk judging Tanner’s painting by standards that may not be historically appropriate.
- B. tend to conflate Tanner’s political views with those of his contemporaries.
- C. forgo analyzing Tanner’s painting in favor of analyzing his political activity.
- D. wrongly assume that Tanner’s painting was intended as a critique of his fellow artists.

Correct Answer: A

Rationale

Choice A is the best answer. It most logically completes the text. The text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. This suggests that scholars who attribute those modern values to Tanner’s painting are risking judging the painting by standards that are not historically accurate.

Choice B is incorrect. It doesn’t logically complete the text. The text argues that Tanner AND his contemporaries may have been unfamiliar with modern views. It never suggests that Tanner’s views were different from his contemporaries’ views. Choice C is incorrect. It doesn’t logically complete the text. The text never suggests that scholars should analyze Tanner’s political activity instead of his painting. Rather, the text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. Choice D is incorrect. It doesn’t logically complete the text. The text never suggests that Tanner wanted to critique his contemporaries with his painting. Rather, the text argues that Tanner AND his contemporaries may have been unfamiliar with modern beliefs and values.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

“The Young Girl” is a 1920 short story by Katherine Mansfield. In the story, the narrator takes an unnamed seventeen-year-old girl and her younger brother out for a meal. In describing the teenager, Mansfield frequently contrasts the character’s pleasant appearance with her unpleasant attitude, as when Mansfield writes of the teenager, _____

Which quotation from “The Young Girl” most effectively illustrates the claim?

Answer

- A. “I heard her murmur, ‘I can’t bear flowers on a table.’ They had evidently been giving her intense pain, for she positively closed her eyes as I moved them away.”
- B. “While we waited she took out a little, gold powder-box with a mirror in the lid, shook the poor little puff as though she loathed it, and dabbed her lovely nose.”
- C. “I saw, after that, she couldn’t stand this place a moment longer, and, indeed, she jumped up and turned away while I went through the vulgar act of paying for the tea.”
- D. “She didn’t even take her gloves off. She lowered her eyes and drummed on the table. When a faint violin sounded she winced and bit her lip again. Silence.”

Correct Answer: B

Rationale

Choice B is the best answer because it most effectively illustrates the claim in the text that in describing the teenaged girl, Mansfield contrasts the character’s pleasant appearance with her unpleasant attitude. In the quotation, Mansfield describes the teenager as having a “lovely nose” (a compliment about her appearance) but also as treating her makeup puff “as though she loathed it” (a judgment suggesting her unpleasant attitude).

Choice A is incorrect because the teenager’s reaction to the flowers doesn’t make it clear that she has an unpleasant attitude, and nothing in the quotation indicates that any part of her appearance is pleasant. Choice C is incorrect because the quotation suggests that the teenager has an unpleasant attitude (being upset with the location and leaving the table before the narrator has paid for the meal) but doesn’t give any indication that she has a pleasant appearance. Choice D is incorrect because the quotation suggests that the teenager may have an unpleasant attitude (lowering her eyes, wincing, and sitting in silence) but doesn’t give any indication that any part of her appearance is pleasant.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

A student is writing a paper about *One Night in Miami...*, a 2020 film directed by Regina King and written by Kemp Powers. Powers adapted the film’s screenplay from his 2013 play, which he wrote after learning about a 1964 meeting that took place in Miami, Florida, between four prominent figures of the Civil Rights movement: Malcolm X, Muhammad Ali, Jim Brown, and Sam Cooke. The student claims that although Powers was inspired by this meeting, the film is best understood not as a precise retelling of historical events but rather as a largely imagined but informed representation of them.

Which quotation from an article about *One Night in Miami...* would be the most effective evidence for the student to include in support of this claim?

Answer

- A. “When Powers learned of the meeting, he initially planned to write a much longer work about its four famous participants rather than focusing on the meeting itself.”
- B. “*One Night in Miami...* received numerous awards and nominations, including an Academy Award nomination for Powers for Best Adapted Screenplay.”
- C. “Powers has described *One Night in Miami...* as the story of four friends encouraging and supporting one another while engaged in a crucial political debate about how best to achieve equality for Black people in the United States.”
- D. “Powers could find only the most superficial historical details about the meeting, so he read extensively about the four individuals and their thinking at the time in an effort to portray what might have happened between them.”

Correct Answer: D

Rationale

Choice D is the best answer because it provides a quotation that effectively supports the student’s claim about the film *One Night in Miami...* The quotation states that in researching the play on which the film was based, Kemp Powers only found superficial details about what actually happened during the 1964 meeting in Miami between four leading Civil Rights leaders, meaning that there is very little information about the meeting in the historical record. In the absence of greater details, it wouldn’t have been possible for the film to be a precise retelling of the historical events it depicts. The quotation explains that to compensate for this lack of information about the meeting, Powers did extensive research into the four figures and how they thought at the time in order to speculate in an informed way about what they might have said or what might have occurred between them. Therefore, the quotation effectively supports the claim that the film is best understood not as a precise retelling of a historical event but as a deeply informed imaginative rendering of that event.

Choice A is incorrect. Although the quotation discusses how on learning about the 1964 meeting in Miami, Powers was inspired to write a play and, later, to adapt it into a screenplay, it doesn’t discuss Powers’s approach to representing what had occurred in the meeting. Instead, it states that Powers didn’t initially plan to write a story only “focusing on the meeting itself” but rather had considered writing a “much longer” and more expansive work about the meeting’s four participants. Choice B is incorrect because the quotation doesn’t discuss Powers’s approach to representing historical events in his play and in the film; instead, the quotation focuses on the film’s positive critical reception by mentioning that it received numerous awards and nominations. Choice C is incorrect. Although the quotation references historical events that are discussed directly in the play and film by explaining how the four historical figures featured in the story engage in political debates about contemporary issues, it doesn’t specify to what extent Powers’s representation of what occurred during the 1964 meeting in Miami is a factual retelling of what happened and how much is an imaginative rendering of what might have happened. Rather, the quotation focuses on Powers’s description of the film’s basic premise and how the characters engage with the historical context of its setting.

Question ID: 5632ffb4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

In a study of the cognitive abilities of white-faced capuchin monkeys (*Cebus imitator*), researchers neglected to control for the physical difficulty of the tasks they used to evaluate the monkeys. The cognitive abilities of monkeys given problems requiring little dexterity, such as sliding a panel to retrieve food, were judged by the same criteria as were those of monkeys given physically demanding problems, such as unscrewing a bottle and inserting a straw. The results of the study, therefore, _____

Which choice most logically completes the text?

Answer

- A. could suggest that there are differences in cognitive ability among the monkeys even though such differences may not actually exist.
- B. are useful for identifying tasks that the monkeys lack the cognitive capacity to perform but not for identifying tasks that the monkeys can perform.
- C. should not be taken as indicative of the cognitive abilities of any monkey species other than *C. imitator*.
- D. reveal more about the monkeys’ cognitive abilities when solving artificial problems than when solving problems encountered in the wild.

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the study of capuchin monkeys’ cognitive abilities. The text explains that the study failed to distinguish between outcomes for the tasks performed by the capuchin monkeys, such that simpler tasks requiring less dexterity, or skill, were judged by the same criteria as tasks that demanded more dexterity. Because the study didn’t account for this discrepancy, the researchers might have assumed that observed differences in performance were due to the abilities of the monkeys rather than the complexity of the tasks. In other words, the results may suggest cognitive differences among the monkeys even though such differences may not really exist.

Choice B is incorrect because the text focuses on the fact that the tasks assigned to the capuchin monkeys in the study varied in difficulty and that the variety wasn’t taken into consideration. The text doesn’t suggest that the capuchin monkeys couldn’t perform certain tasks, just that some tasks were more difficult to do. Choice C is incorrect because the text doesn’t suggest that the study’s results are indicative of the abilities of capuchin monkeys but not of other monkey species; in fact, the text suggests that the results may not even be an accurate reflection of capuchin monkeys’ abilities. Choice D is incorrect because the text doesn’t indicate that the researchers compared results for artificial tasks with those for tasks encountered in the wild, although the tasks described in the text—sliding a panel and putting a straw in a bottle—are presumably artificial.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In subtropical Asia, *Apis dorsata* (giant honeybee) plays an essential role in pollinating a wide variety of crops and wild plants. To study how different agricultural land covers affect the species, Rika Raffiudin and colleagues monitored the foraging activity of the bees as well as the pollen content of the honey from *A. dorsata* colonies at two sites in Indonesia: Kampar, characterized by its surrounding monoculture farms (growing a single crop), and Kerinci, a forest-agriculture site where multiple crops, including hot peppers and coffee, are grown nearby. The researchers concluded that a lack of crop variety may reduce total pollen collection by *A. dorsata*.

Which finding, if true, would most directly support the researchers’ conclusion?

Answer

- A. Pollen in honey samples from Kampar bee colonies was predominantly sourced from a single plant species, whereas pollen in honey samples from Kerinci bee colonies was sourced from multiple different plant species.
- B. In one Kerinci bee colony, a greater proportion of bees returned to their nests with pollen than returned without pollen, whereas the inverse was observed in a second Kerinci bee colony.
- C. Significantly fewer bees were observed engaging in foraging activities with the crops surrounding Kerinci than with the crops surrounding Kampar.
- D. Honey samples from Kerinci bee colonies contained significantly higher concentrations of pollen than honey samples from Kampar bee colonies did.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers’ conclusion that a lack of crop variety may reduce total pollen collection by *A. dorsata*. The text explains that the researchers compared two sites: Kampar, which is surrounded by farms all growing the same crop, and Kerinci, which is a forest-agriculture site with multiple crops. If honey samples from Kerinci bee colonies contained significantly higher pollen concentrations than honey samples from Kampar bee colonies did, this finding would suggest that bees in the multiple-crop environment collected more pollen overall, thereby supporting the conclusion that the low crop variety at the Kampar site corresponds with reduced total pollen collection.

Choice A is incorrect because this finding addresses the diversity of pollen sources—single species at the Kampar site versus multiple species at the Kerinci site—not the total quantity of pollen collected. Choice B is incorrect because this finding compares two colonies within Kerinci rather than comparing the two sites and it concerns whether bees returned with pollen, which doesn’t address the total quantity of pollen collected. Choice C is incorrect because the finding that fewer bees foraged at Kerinci—the higher-variety site—than at Kampar doesn’t address the conclusion about total pollen collected. If anything, a lower number of foraging bees would suggest reduced activity and hence a lower amount of pollen collected in the variety-rich environment, which would undermine rather than support the researchers’ conclusion.

Question ID: b32c4b3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

The Intertropical Convergence Zone (ITCZ), a band of clouds that encircles Earth in the tropics and is a major rainfall source, shifts position in response to temperature variations across Earth’s hemispheres. Data from Huagapo Cave in Peru suggest the ITCZ shifted south during the Little Ice Age (circa 1300–1850), but a shift as far into South America as Huagapo should have led to dry conditions in Central America, which is inconsistent with climate models. To resolve the issue, geologist Yemane Asmerom and colleagues collected data from Yok Balum Cave in Central America and compared them with the Huagapo data. They concluded that during the Little Ice Age, the ITCZ may have expanded northward and southward rather than simply shifted.

Which finding from Asmerom and colleagues’ study, if true, would most directly support their conclusion?

Answer

- A. Neither the Yok Balum data nor the Huagapo data show significant local variations in temperature during the Little Ice Age.
- B. Both the Yok Balum data and the Huagapo data show increased temperatures and prolonged dry conditions during the Little Ice Age.
- C. The Yok Balum data show prolonged dry conditions during the same portions of the Little Ice Age in which the Huagapo data show heightened levels of rainfall.
- D. The Yok Balum data and the Huagapo data show strongly correlated patterns of high rainfall during the Little Ice Age.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support Asmerom and colleagues’ conclusion that the ITCZ may have expanded northward and southward rather than shifting south during the Little Ice Age. The text indicates that the ITCZ, a band of clouds in the tropics that is a significant rainfall source, can change position. Data from Peru’s Huagapo Cave suggest that the ITCZ shifted south during the Little Ice Age. But according to the text, if the ITCZ moved into South America in that way, then Central America should have been drier than climate models suggest it was. In other words, rainfall should have been reduced in Central America because the ITCZ, a significant rainfall source, had shifted into South America, but climate models do not show such a reduction in Central America. The text goes on to say that Asmerom and colleagues tried to resolve this apparent conflict by collecting data from Yok Balum cave in Central America and comparing them with data from Huagapo, which led the researchers to conclude that the ITCZ may have expanded both northward and southward rather than simply shifting south. If it is true that Yok Balum in Central America and Huagapo in South America show strongly correlated patterns of high rainfall during the Little Ice Age, such a finding would support Asmerom and colleagues’ conclusion by suggesting that the two areas were affected by the same rainfall source, and thus that the ITCZ may have expanded rather than shifted.

Choice A is incorrect because there is no information in the text about how, if at all, the ITCZ affects temperature in areas where it is located. Rather, the text states that temperature variations across Earth’s hemispheres can shift the position of the ITCZ. Finding that neither Yok Balum nor Huagapo data show evidence of significant local variations in temperature during the Little Ice Age would have no clear bearing on Asmerom and colleagues’ claim. Choice B is incorrect because finding that both Yok Balum and Huagapo experienced prolonged dry conditions during the Little Ice Age would not support Asmerom and colleagues’ conclusion that the ITCZ, a major source of rainfall, may have expanded northward and southward rather than simply shifting south. Dry conditions in both locations would suggest that the ITCZ did not cover either location. Additionally, finding that temperatures were elevated in both locations would have no clear bearing on Asmerom and colleagues’ conclusion, since there is no information in the text that indicates how, if at all, the ITCZ affects temperature. Choice C is incorrect because finding that Yok Balum experienced prolonged dry conditions at the same time that Huagapo experienced high rainfall would weaken Asmerom and colleagues’ conclusion, not strengthen it. Such a finding would suggest that the ITCZ shifted south and left Central America dry rather than expanding both northward and southward.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Estimates of Tyrannosaurid Bite Force

Study	Year	Estimation method	Approximate bite force (newtons)
Cost et al.	2019	muscular and skeletal modeling	35,000–63,000
Gignac and Erickson	2017	tooth-bone interaction analysis	8,000–34,000
Meers	2002	body-mass scaling	183,000–235,000
Bates and Falkingham	2012	muscular and skeletal modeling	35,000–57,000

The largest tyrannosaurids—the family of carnivorous dinosaurs that includes *Tarbosaurus*, *Albertosaurus*, and, most famously, *Tyrannosaurus rex*—are thought to have had the strongest bites of any land animals in Earth’s history. Determining the bite force of extinct animals can be difficult, however, and paleontologists Paul Barrett and Emily Rayfield have suggested that an estimate of dinosaur bite force may be significantly influenced by the methodology used in generating that estimate.

Which choice best describes data from the table that support Barrett and Rayfield’s suggestion?

Answer

- A. The study by Meers used body-mass scaling and produced the lowest estimated maximum bite force, while the study by Cost et al. used muscular and skeletal modeling and produced the highest estimated maximum.
- B. In their study, Gignac and Erickson used tooth-bone interaction analysis to produce an estimated bite force range with a minimum of 8,000 newtons and a maximum of 34,000 newtons.
- C. The bite force estimates produced by Bates and Falkingham and by Cost et al. were similar to each other, while the estimates produced by Meers and by Gignac and Erickson each differed substantially from any other estimate.
- D. The estimated maximum bite force produced by Cost et al. exceeded the estimated maximum produced by Bates and Falkingham, even though both groups of researchers used the same method to generate their estimates.

Correct Answer: C

Rationale

Choice C is the best answer because it accurately describes data from the table that support Barrett and Rayfield’s suggestion about bite force estimates. According to the text, Barrett and Rayfield believe that estimates of dinosaur bite force may be strongly influenced by the methods used to produce them—that is, that different methods may produce significantly different results. The table shows that the studies by Bates and Falkingham and by Cost et al. used the same estimation method (muscular and skeletal modeling) and produced similar bite force estimates (approximately 35,000–57,000 newtons and 35,000–63,000 newtons, respectively). The study by Meers, however, used body-mass scaling and produced a much higher bite force estimate (183,000–235,000 newtons), while the study by Gignac and Erickson used tooth-bone interaction analysis and produced a much lower bite force estimate (8,000–34,000 newtons). The fact that one method produced similar estimates in two different studies and that two different methods used in other studies produced substantially different estimates supports the idea that dinosaur bite force estimates are significantly influenced by the methodology used to produce them.

Choice A is incorrect because it inaccurately describes data from the table. The table does show that the studies by Meers and by Cost et al. used different estimation methods and produced very different ranges of estimated dinosaur bite force, which would support Barrett and

Rayfield's suggestion that different methodologies may produce significantly different estimates. However, the table doesn't show that the study by Meers produced the lowest estimated maximum bite force while the study by Cost et al. produced the highest. In fact, the study by Meers estimated a maximum bite force of approximately 235,000 newtons, which is the highest of all the estimated maximums. Choice B is incorrect. Although the data from Gignac and Ericson's study are accurately described, a single set of findings from one study using only one methodology can't show that different methodologies may produce significantly different dinosaur bite force estimates, as Barrett and Rayfield suggest. Choice D is incorrect. Although the table shows that the maximum bite force estimated by Cost et al. was higher than that estimated by Bates and Falkingham, the difference is relatively small; in fact, both teams estimated a minimum bite force of approximately 35,000 newtons and a maximum bite force close to approximately 60,000 newtons. Because these findings demonstrate that a single methodology (muscular and skeletal modeling) produced similar overall results in two studies, the findings don't support Barrett and Rayfield's suggestion that different methodologies may produce significantly different dinosaur bite force estimates.

Question ID: 1b9b29f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

A team of biologists led by Jae-Hoon Jung, Antonio D. Barbosa, and Stephanie Hutin investigated the mechanism that allows *Arabidopsis thaliana* (thale cress) plants to accelerate flowering at high temperatures. They replaced the protein ELF3 in the plants with a similar protein found in another species (stiff brome) that, unlike *A. thaliana*, displays no acceleration in flowering with increased temperature. A comparison of unmodified *A. thaliana* plants with the altered plants showed no difference in flowering at 22° Celsius, but at 27° Celsius, the unmodified plants exhibited accelerated flowering while the altered ones did not, which suggests that _____

Which choice most logically completes the text?

Answer

- A. temperature-sensitive accelerated flowering is unique to *A. thaliana*.
- B. *A. thaliana* increases ELF3 production as temperatures rise.
- C. ELF3 enables *A. thaliana* to respond to increased temperatures.
- D. temperatures of at least 22° Celsius are required for *A. thaliana* to flower.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of accelerated flowering in *A. thaliana* plants. The text indicates that *A. thaliana* plants show accelerated flowering at high temperatures. To investigate the mechanism for this accelerated flowering, biologists replaced the ELF3 protein in one group of *A. thaliana* plants with a similar protein found in another plant species that doesn’t show accelerated flowering. The team then compared these modified plants to *A. thaliana* plants that retained their original ELF3 protein. The text states that the two samples of plants showed no difference in flowering at 22° Celsius, but at 27° Celsius the unaltered plants with ELF3 showed accelerated flowering while the plants without ELF3 didn’t. If accelerated flowering at the higher temperature occurred in the *A. thaliana* plants with ELF3 but not in the plants without the protein, then ELF3 likely enables *A. thaliana* to respond to increased temperatures.

Choice A is incorrect because the text doesn’t mention whether any plants other than *A. thaliana* and stiff brome show temperature-sensitive flowering, so there is no support for the idea that this type of flowering is unique to *A. thaliana*. Choice B is incorrect because the text discusses the effects of ELF3 and not the production of it. There’s nothing in the text to suggest that the amount of ELF3 in *A. thaliana* varies with temperature. Choice D is incorrect. While the text states that there was no difference in the flowering of modified and unmodified *A. thaliana* plants at 22° Celsius, there’s no suggestion that *A. thaliana* only begins to flower at 22° Celsius; the text doesn’t mention a specific temperature threshold required for *A. thaliana* flowering.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Jürgen Kocka and other historians of capitalism rarely discuss domestic capitalism in Africa before the period of European colonization, implicitly presenting capitalism as external to and imposed on Africa. Crislayne Alfagali and other Africanist scholars have shown, however, that in parts of Africa, returns-focused investment, the establishment of open markets for wage labor, and other features of capitalism predated colonization. One reason for this discrepancy is that historians of capitalism tend to focus on longitudinal economic data drawn from archival records, which do not exist for much of precolonial Africa.

Which statement about Alfagali and other Africanist scholars is best supported by information in the text?

Answer

- A. They likely make use of different types of evidence than historians of capitalism typically rely on.
- B. They likely differ from historians of capitalism in the methods they use to derive longitudinal economic data from archival records.
- C. They likely have a different view about which activities should be considered capitalist in nature than historians of capitalism do.
- D. They likely view capitalism as having been more beneficial for Africa than historians of capitalism do.

Correct Answer: A

Rationale

Choice A is the best answer because it presents a statement about Alfagali and other Africanist scholars that is supported by the text. The last sentence of the text explains that historians of capitalism generally rely on a type of data that requires "archival records, which do not exist for much of precolonial Africa." Nevertheless, Alfagali and other Africanist scholars have demonstrated that features of capitalism including "returns-focused investment,...open markets for wage labor, and other features of capitalism" existed in parts of Africa before colonization. Because archival records typically used by historians of capitalism aren't available for precolonial Africa, Alfagali and other Africanist scholars of precolonial capitalism in Africa likely used atypical sources for their studies.

Choice B is incorrect because the text states that archival records containing longitudinal economic data "do not exist for much of precolonial Africa." If these records don't exist for precolonial Africa, then Alfagali and other Africanist scholars couldn't derive data from them by any method. Choice C is incorrect because the text doesn't suggest that Alfagali and other Africanist scholars define capitalism differently than other historians of capitalism do. On the contrary, the text simply asserts that specific features they found, such as "returns-focused investment [and] open markets for wage labor," are "features of capitalism," which implies scholarly agreement that these activities are capitalist in nature. Choice D is incorrect because the text doesn't indicate any effects of capitalism in Africa, positive or negative. Rather, the text explains why precolonial Africa has been omitted from mainstream histories of capitalism and presents work by Alfagali and other Africanist scholars that challenges and corrects the historical record of precolonial capitalism.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Electronic music pioneer Wendy Carlos is credited with the music for three feature films: *A Clockwork Orange* (1971), *The Shining* (1980), and *Tron* (1982). However, her musical score for *A Clockwork Orange* is mostly made up of her arrangements of Ludwig van Beethoven’s work. Also, almost all the music that she and Rachel Elkind composed for *The Shining* was unused by director Stanley Kubrick. It did not appear in the film. Of the three films, *Tron* is the one in which audiences can hear the most of Carlos’s original compositions.

Which choice best states the main idea of the text?

Answer

- A. Of the three films for which Carlos is credited, *Tron* features the most original music from her.
- B. The director of *The Shining* used most of the music that Carlos composed for it.
- C. Beethoven is widely considered to be a more important composer than Carlos.
- D. Carlos is a notable innovator among film composers in the 1970s and 1980s.

Correct Answer: A

Rationale

Choice A is the best answer because it best states the main idea of the text. The text indicates that Wendy Carlos scored three feature films: *A Clockwork Orange*, *The Shining*, and *Tron*. It also indicates that Carlos’s work on *A Clockwork Orange* consisted primarily of electronic arrangements of Beethoven compositions and that very little of what she and Rachel Elkind composed for *The Shining* was used in the film. But the soundtrack for *Tron* consists largely of music composed by Carlos, so it contains more of her original music than do the soundtracks for the other two films. Thus, the main idea of the text is that of the three films for which Carlos is credited, *Tron* features the most original music from her.

Choice B is incorrect because it directly contradicts the text’s claim that “very little of what she and Rachel Elkind composed for *The Shining* was used in the film.” Choice C is incorrect because the text doesn’t address the perceived quality of different composers, instead, it discusses how much of Carlos’s original work was used in each of the three films she scored. Choice D is incorrect because it doesn’t relate to any of the major themes of the text: the work Wendy Carlos did for three feature films, how much of that work was composed by Carlos, and how much of her original work was ultimately used in the film.

Question ID: af9e3240

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Electra is a circa 420–410 BCE play by Sophocles, translated in 1870 by R.C. Jebb. Electra, who is in mourning for her dead father and her long-absent brother, is aware of the intensity of her grief but believes it to be justified: _____

Which quotation from *Electra* most effectively illustrates the claim?

Answer

- A. “O thou pure sunlight, and thou air, earth’s canopy, how often have ye heard the strains of my lament, the wild blows dealt against this bleeding breast, when dark night fails!”
- B. “Send to me my brother; for I have no more the strength to bear up alone against the load of grief that weighs me down.”
- C. “I know my own passion, it escapes me not; but, seeing that the causes are so dire, will never curb these frenzied complaints, while life is in me.”
- D. “But never will I cease from dirge and sore lament, while I look on the trembling rays of the bright stars, or on this light of day.”

Correct Answer: C

Rationale

Choice C is the best answer. Electra states that she “knows her own passion,” which shows that she’s aware of the intensity of her grief. But she also claims that the “causes are so dire”—meaning the reasons for her grief are so awful—that she can’t let it go, which shows that she believes her grief is justified.

Choice A is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice B is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice D is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of the intensity of her grief, but it doesn’t suggest that she has a legitimate reason for feeling that way.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Belonging to neither the minimalist nor the abstract art movements but heavily influencing both, Constantin Brâncuși spent his sculpting career returning to the same few themes—a kiss, a sleeping head, and a bird in flight—each rendered in pristine, simple, almost abstract forms. Perhaps the most famous example is *Bird in Space* (1923), a tall, slender bronze sculpture that arcs upward in a manner suggestive of flight but that otherwise lacks any identifiable characteristics of a bird. Despite the seeming simplicity of his works, the exacting standards to which Brâncuși held his work meant that he produced relatively few pieces over his career. There is thus something of a disparity between _____

Which choice most logically completes the text?

Answer

- A. the nature of much of Brâncuși’s work and the abstraction of *Bird in Space*.
- B. Brâncuși’s relatively limited productivity and the diversity of his sculptures.
- C. the themes Brâncuși explored and the themes favored by artists he inspired.
- D. Brâncuși’s importance to the history of art and his total artistic output.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of the sculptor Constantin Brâncuși. The text establishes Brâncuși’s significance by noting that although he belonged to neither the minimalist nor the abstract art movements, he heavily influenced both—a detail that signals his importance to the history of art. At the same time, the text notes that the exacting standards to which Brâncuși held his work meant that he produced relatively few pieces over his career. Because the text presents both Brâncuși’s major artistic influence and his limited output, it is reasonable to infer that the disparity the text identifies is between his importance to the history of art and his total artistic output.

Choice A is incorrect because the text explicitly identifies Brâncuși’s *Bird in Space* as “a famous example” of how the artist conveyed his preferred themes—in this case, a bird in flight—through simplicity bordering on abstraction. Thus, this particular sculpture is consistent with the nature of much of Brâncuși’s work instead of showing a disparity, or marked difference, with it. Choice B is incorrect because the text indicates that Brâncuși returned to the same few themes, suggesting limited thematic diversity. Since his productivity and his thematic diversity were both limited, these two aspects of Brâncuși’s work are consistent with each other instead of showing a disparity, or a marked difference. Choice C is incorrect because although the text mentions that Brâncuși heavily influenced both the minimalist and abstract art movements, it doesn’t discuss the specific themes favored by artists in those movements. There is therefore no basis for identifying a disparity between Brâncuși’s themes and those of artists he inspired.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The widespread use of social media enables linguists to study changes in language usage in real time. A notable recent example is the proliferation of the affix *meng-* among speakers of Bahasa Indonesia, the official language of Indonesia. Linguists observed *meng-* originate as an onomatopoetic tag that social-media users applied to images of cats they posted; over time, users increasingly applied it as a prefix to existing words (e.g., *mengsedih* affixes *meng-* to the word for *sad*) in text that they posted. From there, it has begun to move into spoken Bahasa Indonesia. Linguists have noted many similar examples of this phenomenon occurring in other languages, suggesting that social media _____

Which choice most logically completes the text?

Answer

- A. is more useful for studying informal language than for studying formal or official language.
- B. appears to be exerting an exceptionally strong influence on the evolution of Bahasa Indonesia.
- C. may give linguists a somewhat misleading sense of how languages are changing.
- D. does not merely register changes in language usage but can facilitate such changes.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of linguists using social media to study changes in language usage in real time, providing the specific example of the affix *meng-* in Bahasa Indonesia. The text states that linguists first observed *meng-* being used as an onomatopoeic tag on social media, which then spread to being affixed to existing words in text posted on social media; from there, it has begun to move into spoken Bahasa Indonesia. As presented in the text, this progression from online usage to spoken language suggests that social media does more than just register or reflect changes in language—it can actively drive such changes. In the case of *meng-*, the text suggests, social media facilitated the movement of the affix from an online tag to part of spoken Bahasa Indonesia.

Choice A is incorrect because the text doesn’t differentiate between social media’s usefulness for studying informal versus formal or official language; it merely notes that Bahasa Indonesia is an official language. Choice B is incorrect because while the example given in the text focuses on changes in Bahasa Indonesia, the text doesn’t provide any evidence that social media is exerting an exceptionally strong influence on the evolution of this particular language compared to others. In fact, the text states that "linguists have noted many similar examples of this phenomenon occurring in other languages," suggesting that social media’s influence is evident across multiple languages and not unique to Bahasa Indonesia. Choice C is incorrect because the text never suggests that social media provides a somewhat misleading sense of how languages are changing. Rather, the text suggests that by allowing linguists to directly observe linguistic changes, social media offers a clear window into language evolution.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Outi Tervo and team studied the effect of human-caused noise on narwhals (*Monodon monoceros*), arctic marine mammals that are sensitive to acoustic changes in their environment. Hypothesizing that elevated sound levels affect foraging among narwhals, Tervo’s team compared narwhal diving behaviors in natural sound conditions with those behaviors in two human-caused sound exposure conditions—ship sounds and ship sounds coupled with sonic pulses. Both exposure conditions resulted in significant decreases in the number and target depth of deep dives (associated with foraging) relative to natural conditions. However, differences between diving behaviors in the two exposure types were negligible, a finding that could be attributed to the fact that _____

Which choice most logically completes the text?

Answer

- A. sonic pulses can be heard at significantly greater ocean depths than ship sounds can.
- B. ship sounds contribute so much to the overall sound level that the addition of sonic pulses has little effect on the narwhals’ auditory environment.
- C. narwhals forage at shallower depths in the presence of ship sounds alone than in the presence of ship sounds coupled with sonic pulses.
- D. the narwhals weren’t as sensitive to human-caused sounds as the researchers had predicted.

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of human-caused noise and narwhal behavior. The text establishes that Outi Tervo and team found that while both ship sounds alone and ship sounds coupled with sonic pulses significantly affected narwhal foraging behaviors compared to natural conditions, there were negligible differences between the narwhals’ responses to the two sound exposure types (a somewhat surprising finding because one might expect that adding sonic pulses would have an even greater impact). It logically follows that if ship sounds are already so loud that adding sonic pulses to them doesn’t substantially increase the level of human-caused sound exposure, the two conditions would produce essentially the same behavioral responses because the additional sonic pulses are essentially irrelevant.

Choice A is incorrect because the fact that sonic pulses are audible at greater depths than are ship sounds would be more likely to help explain a difference—not a similarity—in how the narwhals responded to the two types of sound exposure. If at some points during the study the narwhals were at depths where only sonic pulses (but not ship sounds) could be heard, then they might be expected to alter their diving behavior more dramatically in response to those pulses. But since the text reports negligible differences between diving behaviors in the two exposure types, the fact that sonic pulses reach greater depths than do ship sounds has little explanatory power in this context. Choice C is incorrect because the idea that the narwhals foraged at shallower depths in one of the two exposure conditions directly contradicts the information given in the text. The text states that there were negligible differences in the narwhals’ diving behaviors between the two conditions, meaning that the behaviors were essentially the same. Choice D is incorrect. Although the text states that two conditions with human-caused sounds (ship sounds alone and ship sounds coupled with sonic pulses) had essentially the same effect on narwhals’ deep dives, this indicates only that sonic pulses didn’t meaningfully change the effect of ship sounds, not that the narwhals were less sensitive to human-caused sounds than predicted. Indeed, the text makes it clear that the sounds significantly altered the narwhals’ behavior.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Archaeologist Weiwei Wang and her colleagues analyzed footed grinding slabs and other food-preparation tools excavated from Óc Eo, a Southeast Asian port city that flourished between the first and sixth centuries CE. Wang and colleagues recovered microscopic remnants of turmeric and other spices from the surfaces of the tools. Turmeric is native to South Asia, more than a thousand miles west of Óc Eo, and the researchers showed that the footed grinding slabs at Óc Eo are very similar to footed grinding slabs common throughout South Asia from around 500 BCE to 300 CE. Wang and colleagues’ findings therefore indicate that there must have been a trade link, whether direct or indirect, between the two regions.

Which finding, if true, would directly weaken the conclusion about Wang and her colleagues’ findings that is presented in the text?

Answer

- A. Some of the spices recovered from Óc Eo are native to the Maluku Islands, which are located approximately 2,000 miles southeast of Óc Eo.
- B. In the first through third centuries CE, there was a significant migration of people from South Asia to Southeast Asia.
- C. Other types of artifacts originating in South Asia and dating to the first through sixth centuries CE have been found throughout Southeast Asia.
- D. The people of Óc Eo and several communities in South Asia regularly traded with people in the region that is now the Southeast Asian country of Malaysia no later than the first century CE.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly weaken the conclusion that Wang and colleagues’ findings indicate a trade link between Óc Eo and South Asia. The text explains that turmeric (native to South Asia) and footed grinding slabs of a style common in South Asia between roughly 500 BCE and 300 CE were found at Óc Eo, a Southeast Asian port that flourished between the first and sixth centuries CE. From this evidence, the text concludes that these items must have reached Óc Eo through trade. If, however, there was a significant migration of people from South Asia to Southeast Asia during the first through third centuries CE, that would offer an alternative explanation: South Asian people could have brought turmeric and grinding slabs with them when they migrated. Thus, the finding would weaken the conclusion that trade must account for the items’ presence at Óc Eo.

Choice A is incorrect. A finding involving spices from a different region—the Maluku Islands, southeast of Óc Eo—wouldn’t undermine the conclusion because it wouldn’t reveal how the items from South Asia, a region to the west, reached Óc Eo. If anything, such a finding might be further evidence of broad trade networks involving Óc Eo. Choice C is incorrect. If it were true that other South Asian artifacts from the period when Óc Eo flourished have been found throughout Southeast Asia, that would suggest only that South Asian items other than turmeric and footed grinding slabs were present in the region at the time; it wouldn’t reveal anything about how any of those items reached Southeast Asia from South Asia. If anything, the widespread presence of South Asian artifacts in Southeast Asia during this period could be seen as further support for a trade network connecting the regions. Choice D is incorrect because the finding that Óc Eo and some South Asian communities traded with people in another Southeast Asian location (what is now Malaysia) would suggest that this location could have been an intermediary for the movement of items from South Asia to Óc Eo. Such a finding would support, not weaken, the plausibility of an indirect trade link that’s presented in the conclusion.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The Uto-Aztecan language family is divided into a northern branch, which includes the Shoshone language of present-day Idaho and Utah, and a southern one, whose best-known representative is Nahuatl, the language of the Aztec Empire in Mexico. Lexical similarities across the family, including of botanical terms, confirm descent from a single language spoken millennia ago, and the family’s geographical distribution suggests an origin in what is now the US Southwest. However, vocabulary pertaining to maize isn’t shared between northern and southern branches, despite the crop’s universal cultivation among Uto-Aztecan tribes. Given archaeological evidence that maize originated in Mexico and diffused northward into what became the US Southwest, some linguists reason that _____

Which choice most logically completes the text?

Answer

- A. northern Uto-Aztecan tribes likely obtained the crop directly from a southern Uto-Aztecan tribe rather than from a non-Uto-Aztecan tribe.
- B. variation in maize-related vocabulary within each branch of the Uto-Aztecan family likely reflects regionally specific methods for cultivating the crop.
- C. southern Uto-Aztecan tribes likely acquired maize at roughly the same time as northern Uto-Aztecan tribes did, though from different sources.
- D. the family’s division into northern and southern branches likely preceded the acquisition of the crop by the Uto-Aztecan tribes.

Correct Answer: D

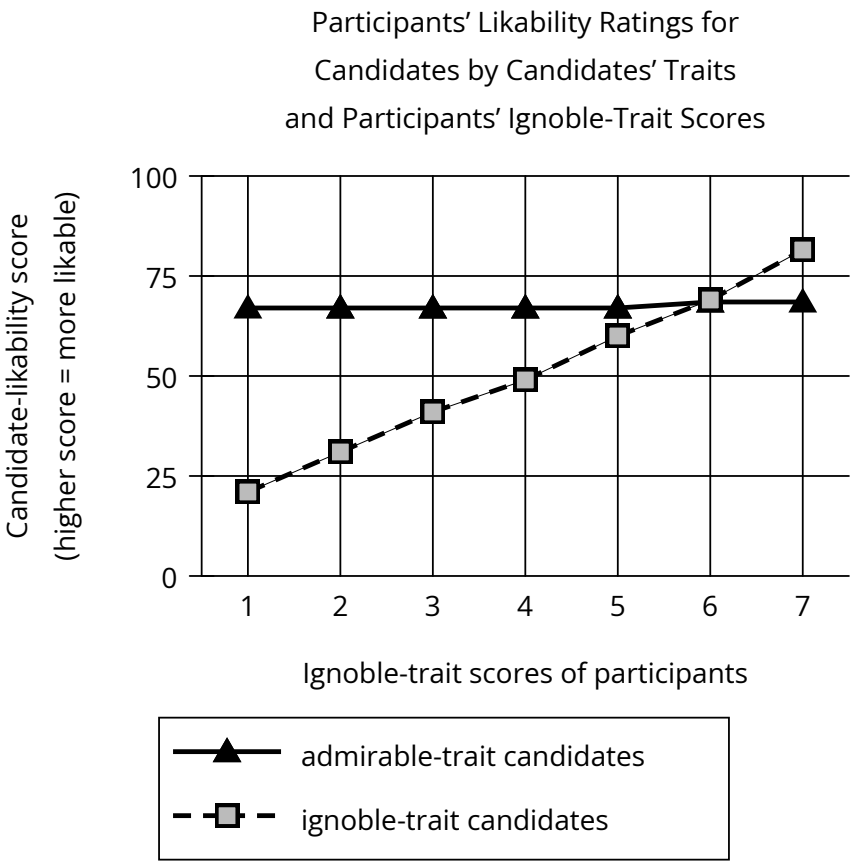
Rationale

Choice D is the best answer because it most logically completes the discussion of Uto-Aztecan languages. The text explains that the northern and southern branches of the Uto-Aztecan language family descended from a single language (believed to have originated in what is now the US Southwest), resulting in similarities across the family’s languages; however, the branches don’t have similar vocabulary for maize, even though maize has been cultivated by all Uto-Aztecan tribes. The text also indicates that maize originated in Mexico and spread northward into what is now the US Southwest—the area where the Uto-Aztecan language family originated. It follows, then, that the language family had already divided into northern and southern branches before maize reached that area; if maize had been present before the division occurred, the family’s origin language would have had terminology for it that likely would have been reflected in the branches, meaning they would have had similar vocabulary for maize. If maize arrived after the division occurred, however, the tribes in the two regions likely would have developed vocabulary pertaining to maize separately, at the times when they acquired the crop.

Choice A is incorrect because the text focuses on vocabulary pertaining to maize in the branches of the Uto-Aztecan language family, and referring only to how some Uto-Aztecan tribes obtained maize wouldn’t directly address the role of language. Moreover, if northern Uto-Aztecan tribes had acquired maize from a southern Uto-Aztecan tribe, it’s reasonable to assume that the northern tribes might have also picked up southern Uto-Aztecan terminology for maize in that exchange. Choice B is incorrect because the text discusses the fact that the northern and southern branches of the Uto-Aztecan language family don’t have shared vocabulary pertaining to maize, not the idea that there are variations in such vocabulary within each branch—that is, the text focuses on differences between the two branches, not on differences between languages within a branch. Choice C is incorrect because the text focuses on vocabulary pertaining to maize in the branches of the Uto-Aztecan language family, and referring only to the timing and source of maize acquisition wouldn’t directly address the role of language. Furthermore, the text implies that southern Uto-Aztecan tribes probably acquired maize before the northern tribes did, given the evidence that maize originated in Mexico—the location of the best-known representative of the southern branch of the Uto-Aztecan language family—before spreading to the north.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Alessandro Nai et al. presented study participants with vignettes about fictive political candidates, portraying them as embodying a personality trait widely considered admirable (e.g., agreeableness) or one considered ignoble (e.g., cynicism). A survey recorded participants' ratings of the candidates' likability and showed that across participants, ignoble-trait candidates were less likable than admirable-trait candidates. However, when the researchers factored in the participants' own personality-trait scores, on a scale of 1 (least ignoble) to 7 (most ignoble), they concluded that this relative ranking of candidates persisted except among the participants with high ignobility scores.

Which choice best describes data from the graph that support the researchers' conclusion?

Answer

- A. There was a strong positive correlation between participants' ignobility scores and admirable-trait candidates' likability ratings, but there was no correlation between ignobility scores and ignoble-trait candidates' likability ratings.
- B. Participants with an ignobility score of 5 or less rated admirable-trait candidates as more likable than ignoble-trait candidates, whereas participants with an ignobility score of 6 or more rated ignoble-trait candidates as equally likable as or even more likable than admirable-trait candidates.
- C. Overall, participants rated admirable-trait candidates as quite likable, and that rating was not significantly affected by the participants' ignobility scores.
- D. Unlike participants with an ignobility score of 6, participants with an ignobility score either greater or less than 6 gave admirable-trait candidates and ignoble-trait candidates different likability ratings.

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the graph that support the researchers' conclusion that the trend of admirable-trait candidates being rated as more likable than ignoble-trait candidates held true when participants' own personality-trait scores were factored in, except among participants with high ignobility scores. The values on the x-axis represent survey participants grouped by their own ignobility scores, from low ignobility (1) to high ignobility (7), while the values on the y-axis represent the likability scores given to the political candidates. The graph shows that the full range of participants (from least to most ignoble) gave the admirable-trait candidates (represented by the line with triangles) a likability rating of approximately 70 out of 100; that is, regardless of their own level of ignobility, participants generally found admirable-trait candidates quite likable. However, the graph shows that participants varied in their views of ignoble-trait candidates (represented by the line with squares); likability ratings increased as the participants' own ignobility scores increased. Participants with low to medium-high ignobility scores (1 to 5) still rated the ignoble-trait candidates as less likable than the admirable-trait candidates, with all ratings falling below approximately 70, but participants with high ignobility scores (6 and 7) gave ratings equal to or higher than approximately 70. In other words, the previously observed trend of ranking admirable-trait candidates as more likable than ignoble-trait candidates persisted for participants with low to medium-high ignobility but not for participants with high ignobility.

Choice A is incorrect because it describes the opposite of what the graph shows. The graph shows a positive correlation between participants' ignobility scores and ignoble-trait candidates' likability ratings (as participants' ignobility scores increased, so did their ratings for ignoble-trait candidates' likability) and no correlation between ignobility scores and admirable-trait candidates' likability ratings (all participants gave admirable-trait candidates a rating of approximately 70 out of 100). Choice C is incorrect. The graph does show that regardless of their own ignobility scores, participants rated admirable-trait candidates as quite likable (a rating of approximately 70 out of 100). However, this doesn't support the researchers' conclusion because the conclusion has to do with how participants rated both types of candidates, not just the admirable-trait ones; moreover, the conclusion is that relative ratings were actually affected by the participants' ignobility scores. Choice D is incorrect. The graph does show that only participants with an ignobility score of 6 gave the same likability score to both admirable- and ignoble-trait candidates while participants with other ignobility scores gave a different rating for each candidate, but this doesn't support the researchers' conclusion. The conclusion isn't just that participants gave different ratings to the two types of candidates—it's that participants with low to medium-high ignobility scores specifically gave higher likability ratings to admirable-trait candidates than to ignoble-trait candidates and that participants with high ignobility scores didn't.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Eighteenth-century economist Adam Smith is famed for his metaphor of the invisible hand, which he putatively used to illustrate a robust model of how individuals produce aggregate benefits by pursuing their own economic interests. Note “putatively”: as Gavin Kennedy has shown, Smith deploys this metaphor only once in his economic writings—to make a narrow point about the then-dominant economic theory of mercantilism—and it was largely ignored until some twentieth-century economists eager to secure an intellectual pedigree for their views elevated it to a fully-fledged paradigm.

Which choice best states the main idea of the text?

Answer

- A. Although Smith is famed for his metaphor of the invisible hand, the metaphor was largely ignored until economists in the twentieth century came to realize that the metaphor was a robust model that anticipated their own views.
- B. Some twentieth-century economists gave Smith’s metaphor of the invisible hand a significance it does not have in Smith’s work, but it is nevertheless a useful model of how individuals produce aggregate benefits by pursuing their own economic interests.
- C. Smith’s metaphor of the invisible hand has been interpreted as a model of how individuals acting in their own interest produce aggregate benefits, but it was intended as a subtle critique of the economic theory of mercantilism.
- D. The reputation of Smith’s metaphor of the invisible hand is not due to the importance of the metaphor in Smith’s work but rather to the promotion of the metaphor by some later economists for their own ends.

Correct Answer: D

Rationale

Choice D is the best answer because it most accurately states the main idea of the text. The text explains that economist Adam Smith’s famous metaphor of the invisible hand was putatively (that is, widely assumed but not proven) intended to illustrate a robust model (a consistently accurate generalization) of how individuals pursuing their own economic interests can create broader benefits for the population. The text then emphasizes the lack of affirmative evidence for this idea by calling out the term “putatively,” and explaining that, according to Gavin Kennedy, Smith used the metaphor only once in his works, in reference to specific circumstances related to the now-outdated economic view known as mercantilism, and that the metaphor didn’t garner much attention until economists in the twentieth century held it up as a paradigm (a theoretical framework in the field) and thereby implied that Smith shared some of their views on economics. By emphasizing “putatively,” the text implies that there is no independent reason to believe that Smith would agree with the metaphor’s use outside of the specific context for which he wrote it and that, therefore, the twentieth-century economists who used it did so to support their own views without regard for the metaphor’s importance to Smith’s work.

Choice A is incorrect. Although the text indicates that Smith’s metaphor was largely ignored until some twentieth-century economists revived it and bolstered its status, the text suggests that the later economists used Smith’s metaphor to self-servingly boost their own work while ignoring the original context in which Smith wrote it. Moreover, the statement in this choice fails to reflect the text’s emphasis on Smith’s limited use of the metaphor in his work. Choice B is incorrect. Although the text indicates that some twentieth-century economists altered the significance of Smith’s metaphor, the text doesn’t suggest that the metaphor is a “useful model” of how aggregate benefits arise from individuals’ selfish actions, let alone that this usefulness is unaffected by taking the metaphor out of its original context. Choice C is incorrect. Although the text indicates that Smith’s metaphor was intended as a model of how individuals acting in their own interest produce aggregate benefits and it was written within the context of the now-outdated economic theory of mercantilism, these points are subordinate to the primary idea in the text, which is that Smith’s use of the metaphor was tightly constrained but twentieth-century economists ignored the original context so that they could use the metaphor to suggest, without support, that Smith would agree with their economic views.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In the mountains of Brazil, *Barbacenia tomentosa* and *Barbacenia macrantha*—two plants in the Velloziaceae family—establish themselves on soilless, nutrient-poor patches of quartzite rock. Plant ecologists Anna Abrahão and Patricia de Britto Costa used microscopic analysis to determine that the roots of *B. tomentosa* and *B. macrantha*, which grow directly into the quartzite, have clusters of fine hairs near the root tip; further analysis indicated that these hairs secrete both malic and citric acids. The researchers hypothesize that the plants depend on dissolving underlying rock with these acids, as the process not only creates channels for continued growth but also releases phosphates that provide the vital nutrient phosphorus.

Which finding, if true, would most directly support the researchers’ hypothesis?

Answer

- A. Other species in the Velloziaceae family are found in terrains with more soil but have root structures similar to those of *B. tomentosa* and *B. macrantha*.
- B. Though *B. tomentosa* and *B. macrantha* both secrete citric and malic acids, each species produces the acids in different proportions.
- C. The roots of *B. tomentosa* and *B. macrantha* carve new entry points into rocks even when cracks in the surface are readily available.
- D. *B. tomentosa* and *B. macrantha* thrive even when transferred to the surfaces of rocks that do not contain phosphates.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would support the researchers’ hypothesis about the plants’ dependence on dissolving rock. The text indicates that the roots of the two plant species grow directly into quartzite rock, where hairs on the roots secrete acids that dissolve the rock. The researchers hypothesize that the plants depend on this process because dissolving rock opens spaces for the roots to grow and releases phosphates that provide the plants with phosphorus, a vital nutrient. If the plants carry out this process of dissolving rock even when the rock already has spaces into which the roots could grow, that would support the researchers’ hypothesis because it suggests that the plants are getting some advantage—such as access to phosphorus—from the action of dissolving rock. If the plants don’t benefit from dissolving rock, they would be expected to grow in the cracks that already exist, as doing so would mean that the plants don’t have to spend energy creating and secreting acids; if, however, the plants create new entry points by dissolving rock even when cracks already exist, that would support the hypothesis that they depend on dissolving rock for some benefit.

Choice A is incorrect because the existence of soil-inhabiting members of the Velloziaceae family with similar root structures to those of the two species discussed in the text wouldn’t support the researchers’ hypothesis that the species discussed in the text depend on dissolving rock. If other such members exist, that might suggest that the root structures can serve more functions than secreting acids to dissolve rock (since dissolving rock may not be necessary for plants living in soil), but that wouldn’t suggest anything about whether the species discussed in the text benefit from dissolving rock. Choice B is incorrect because differences in the proportions of citric and malic acid secreted by the two species would be irrelevant to the hypothesis that the plants depend on dissolving rock. There’s no information in the text to suggest that the proportion of each acid has any bearing on the process of dissolving rock or on any benefits the plants might receive from that process. Choice D is incorrect because if the two species thrive on rocks without phosphates, that would weaken the researchers’ hypothesis that the plants depend on dissolving rock partly because dissolving rock gives them access to phosphates. If the plants can survive on rocks without getting a vital nutrient by dissolving those rocks, then either the nutrient isn’t actually vital for those plants or they can get the nutrient in some way other than by dissolving rocks.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Tatiana R. Feuerborn and colleagues analyzed the genomes of more than a hundred domesticated dogs from sites in Siberia dating from 11,000 years ago to the present. They found that the dogs constituted a genetically isolated population of Arctic breeds until approximately 2,000 years ago, at which point there was substantial interbreeding with Near Eastern dog breeds. Furthermore, beginning around 2,000 years ago, some sites contain artifacts consistent with a Near East origin, like glass beads, but the people show no evidence of having traveled extensively outside Siberia. From this, Feuerborn and colleagues concluded that around 2,000 years ago _____

Which choice most logically completes the text?

Answer

- A. dogs and artifacts like glass beads began to be transported from the Near East to Siberia.
- B. people from Siberia began to reach the Near East, where they acquired dogs and artifacts such as glass beads.
- C. glass beads and other artifacts from the Near East began to be exchanged for dogs from Siberia.
- D. dogs from the Near East began to be exchanged for glass beads and other artifacts from Siberia.

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically follows from the text’s discussion of Feuerborn and colleagues’ findings about sites in Siberia. The text states that genomic analysis of many domesticated dogs from Siberian sites dating from the last 11,000 years revealed that the dogs represented a genetically isolated population of breeds (that is, they didn’t reproduce with dogs of any outside breeds) until about 2,000 years ago, when there is evidence that the dogs began interbreeding with breeds from the Near East. The text also indicates that around the same time, the sites began to contain glass beads and other artifacts seeming to be from the Near East, even though there is no evidence that the people of those sites traveled widely outside Siberia. Because there is evidence of Near East dog breeds and Near East artifacts at Siberian sites beginning about 2,000 years ago but no evidence of the people’s travel to the Near East around that time, it’s reasonable to infer that Feuerborn and colleagues concluded that dogs and artifacts like glass beads were transported to Siberia from the Near East, starting around 2,000 years ago.

Choice B is incorrect because the text explicitly states that there is no evidence that the people of the Siberian sites 2,000 years ago traveled extensively beyond Siberia, which they would’ve had to do to reach the Near East; therefore, it wouldn’t make sense to conclude that the presence of Near East dog breeds and artifacts like glass beads suggests that people from Siberia began arriving in the Near East around 2,000 years ago. Choice C is incorrect because the text addresses the presence of Near East dog breeds (and evidence of interbreeding with Siberian dogs) and artifacts like glass beads at Siberian sites starting about 2,000 years ago, suggesting only that the dogs and artifacts started to arrive in Siberia around that time. The text doesn’t suggest the purpose of the arrival of the Near East dogs and artifacts or give any indication of Siberian dog breeds being transported elsewhere through trade. Choice D is incorrect because the text addresses the presence of Near East dog breeds and artifacts like glass beads at Siberian sites starting about 2,000 years ago, suggesting only that the dogs and artifacts started to arrive in Siberia around that time; it doesn’t suggest the purpose of this arrival and makes no mention of Siberian glass beads or other artifacts of Siberian origin.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In countries with right-hand traffic, drivers who want to make a left turn at a traffic intersection with stoplights have to wait for either a gap in oncoming traffic or a designated left-turn signal to turn green. At busy intersections, this often causes a backup of vehicles waiting to turn left or being prevented from proceeding by left-turning vehicles in front of them. Transportation researcher Vikash V. Gayah claims that in urban areas eliminating the option to turn left at busy intersections—both with and without dedicated left-turn signals—would improve traffic flow and, as a result, reduce overall travel times even if such a restriction would require drivers to sometimes travel a slightly longer distance.

Which finding, if true, would most directly support the researcher’s claim?

Answer

- A. In a town that installed left-turn signals at all busy intersections, seven out of ten survey respondents agreed with the statement “the streets in my community are easier to navigate by motor vehicle than before.”
- B. A traffic study of intersections in a large city shows that on average drivers wait longer to make a left turn at intersections without left-turn signals than at intersections with such signals.
- C. After a city eliminated left turns at busy intersections, a package-delivery company reports that its drivers have been able to reach more addresses in the city daily, on average, and therefore deliver more packages there annually.
- D. Statistics reveal that school buses in a city that eliminated left turns at most intersections took on average two minutes longer to complete their routes after the restriction took effect than they did before.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most directly support Vikash V. Gayah's claim that eliminating the option to turn left at busy intersections would improve traffic flow and reduce overall travel times. The text begins by describing a problem encountered by drivers in countries with right-hand traffic—namely that drivers wanting to make a left turn must wait for either gaps in oncoming traffic or for designated left-turn signals before proceeding. The resulting backup of vehicles causes increased traffic congestion at busy intersections that slows overall travel times. According to Gayah, eliminating left turns from busy intersections in urban areas would ease the congestion caused by vehicles waiting to turn left. If vehicles spend less time waiting at intersections for left turns, faster overall travel times would result even if some drivers would have to drive slightly longer distances to make the desired left turn. Drivers for package-delivery companies, who presumably spend most of the day driving to destinations across the city in which they are based, would likely provide a good indication of overall traffic patterns across the city. A finding that after a city had eliminated left turns at busy intersections, package-delivery companies were able to complete more daily deliveries on average—which implies faster travel times between package destinations—would therefore support the claim that overall travel times would decrease if left turns were eliminated at busy intersections.

Choice A is incorrect because a finding that a majority of survey respondents agreed with the statement that implementing left-turn signals at all busy intersections made navigating streets in their communities easier wouldn’t support the claim that eliminating the option to turn left altogether at some busy intersections—both with and without dedicated left-turn signals—would improve the flow of traffic and overall traffic times. In fact, the text emphasizes that designated left-turn signals contribute to increased congestion because they require drivers to wait to turn left, which results in a backup of vehicles. Installing such signals at all busy intersections would thus potentially compound the problem of congestion, not improve it. Moreover, although making streets easier for drivers to navigate might indicate that left-turn signals are beneficial to drivers, it doesn’t indicate that overall travel times would necessarily be reduced. Choice B is incorrect because a study concluding that drivers wait longer to make left turns at intersections without dedicated left-turn signals than at intersections with such signals wouldn’t support the claim that eliminating the option to turn left altogether would reduce overall travel times. In fact, the finding would merely support the idea that

installing left-turn signals would likely reduce the time drivers spend waiting at busy intersections that didn't previously have left-turn signals, not that overall travel times would be reduced. Choice D is incorrect because the finding that after a restriction eliminating left turns at most intersections took effect, school buses took longer to complete their routes than they did before the restriction was implemented would contradict rather than support the researcher's claim that eliminating left turns would reduce overall travel times.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The fynbos shrubland is a diverse habitat found only in South Africa. It is adjacent to the Afro-temperate forest, with almost no transition space between the two distinct habitats. Plants in the fynbos have uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil. Ecologists transplanted tree seedlings from the forest into plots in the fynbos. Seedlings in plots isolated from the roots of fynbos plants exhibited a growth rate five times greater than that of the seedlings in plots in close proximity to the roots of fynbos plants.

Based on the text, what role do fynbos roots most likely have in maintaining the border between the fynbos shrubland and the Afro-temperate forest habitats?

Answer

- A. Fynbos roots damage the root systems of forest plants, leaving those plants unable to acquire sufficient nutrients.
- B. Fynbos roots extend close enough to the forest plants’ roots that they constitute a physical barrier that forest plants’ roots cannot pass.
- C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.
- D. The root systems of fynbos plants enhance the soil immediately surrounding the plants, allowing them to thrive in an otherwise harsh habitat.

Correct Answer: C

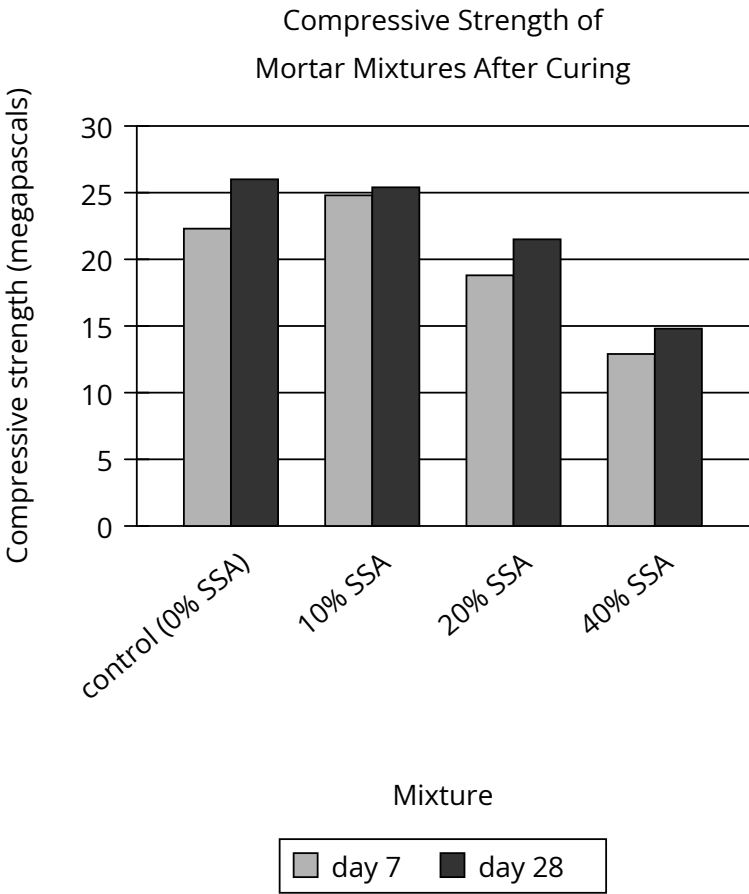
Rationale

Choice C is the best answer because it presents a conclusion about the role of fynbos roots in maintaining the border with the Afro-temperate forest that can be reasonably inferred from the text. The text states that plants in the fynbos have "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." The text then describes an experiment where forest tree seedlings grown in plots isolated from fynbos roots grew at a rate five times greater than seedlings in areas near fynbos roots. This strongly suggests that fynbos roots are depleting soil nutrients to such an extent that forest plants can't get enough nutrients from that soil to flourish, thereby hindering the formation of any significant "transition space" between the habitats and helping keep the border between them sharp.

Choice A is incorrect because there is no evidence in the text that fynbos roots physically damage the root systems of forest plants. Instead, the text explicitly describes fynbos plants as having "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." This description establishes that fynbos roots are specialized for efficient nutrient absorption. The experiment then confirms this function of the roots by showing that forest seedlings "isolated from the roots of fynbos plants exhibited a growth rate five times greater" than those near fynbos roots. This result demonstrates that when forest seedlings don't have to compete with the nutrient-absorbing fynbos roots, they grow much better. Choice B is incorrect. Although the text does indicate that the fynbos and the Afro-temperate forest are distinct from, and adjacent to, each other and that there is "almost no transition space" (area of mixed fynbos and forest plants), the experiment shows that forest plants can grow in soil occupied by fynbos roots, just not very well. So, the idea that the fynbos roots physically obstruct forest plants is unsupported by the text. Choice D is incorrect because it directly contradicts the text's claim that fynbos roots absorb nutrients from, and therefore deplete, the soil they're in, and furthermore, nothing in the text suggests that the habitat of the fynbos is harsh in general.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Mollusk shells are mainly composed of calcium carbonate, a chemical compound used in mortar and other construction materials. A research team led by Ariane da Silva Cardoso investigated pulverizing discarded Sururu mussel shells to create a substitute for fine aggregates (such as sand) in mortar mixtures. The team prepared mixtures with Sururu shell aggregate (SSA) replacing varying percentages of sand and then tested the mortars’ compressive strength on days 7 and 28 of curing. Based on their findings, the team recommended SSA substitution in relatively small proportions to avoid compromising strength.

Which choice most effectively uses data from the graph to support the researchers’ recommendation?

Answer

- A. The 10% SSA mixture’s compressive strength was greater than or approximately equal to that of the control mixture on day 7 and on day 28, but on both days the 20% and 40% SSA mixtures had less compressive strength than the control mixture did.
- B. The compressive strength of the 10% SSA mixture remained nearly the same from day 7 to day 28, while that of the 20% and 40% SSA mixtures changed substantially.
- C. The 10% SSA mixture’s compressive strength was greater than that of the control mixture on day 7, but by day 28 the 10% SSA mixture’s compressive strength had decreased while that of the control mixture had increased.
- D. The compressive strength of the 10% SSA mixture never surpassed the compressive strength measured for the control mixture on day 28.

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the graph to support the researchers' recommendation that Sururu shell aggregate (SSA) substitution be done in relatively small proportions to avoid compromising the compressive strength of mortar mixtures. The text explains that a research team investigated whether SSA can serve as a substitute for sand in mortar mixtures. The graph presents the compressive strength (in megapascals) of mortar mixtures with SSA replacing 0% (control), 10%, 20%, and 40% of the sand on day 7 and day 28 of curing. The graph shows that the compressive strength of the 10% SSA mixture was approximately 25 megapascals on day 7, which is greater than the control mixture's approximately 22 megapascals on the same day. The graph also shows that the 10% SSA mixture's compressive strength was slightly more than 25 megapascals on day 28, which is only slightly less than the control's compressive strength on the same day (about 26 megapascals). Therefore, the graph shows that the compressive strength of the 10% SSA mixture was greater than that of the control on day 7 and roughly equal to that of the control on day 28, suggesting that substituting only 10% of the sand with SSA avoids compromising the mortar's compressive strength. The graph also shows that on day 7, the 20% and 40% SSA mixtures had compressive strengths of about 19 and 13 megapascals, respectively (both lower than the control's), and on day 28, they had compressive strengths of about 22 and 15 megapascals, respectively (once again lower than the control's). Thus, the data support the recommendation by indicating that substituting relatively small proportions of sand with SSA in mortar mixtures doesn't compromise strength compared with the control, whereas substituting larger proportions does.

Choice B is incorrect because the comparison it makes isn't relevant to the researchers' recommendation. Although the graph does show that the bars representing the compressive strength of the 10% SSA mixture on day 7 and day 28 are roughly the same height (suggesting that the mixture's compressive strength changed by only a small amount), and although there are relatively large gaps between the bars showing the compressive strength of the 20% and 40% SSA mixtures on day 7 and day 28 (suggesting that their compressive strength changed over time by a greater amount), this comparison concerns only how each SSA mixture's strength changed over time and doesn't compare the SSA mixtures with the control mixture. Supporting the recommendation that SSA substitution be done in small proportions requires evidence that small substitutions don't compromise strength relative to the control while larger substitutions do. Choice C is incorrect because it doesn't accurately represent the data in the graph, nor is the comparison it makes relevant to the researchers' recommendation. While the graph does show that the compressive strength of the 10% SSA mixture was greater than that of the control on day 7 (about 25 megapascals vs. about 22 megapascals) and, moreover, that the compressive strength of the control increased from day 7 to day 28 (from about 22 megapascals to about 26 megapascals), the graph shows that the compressive strength of the 10% SSA mixture increased by a small amount (from about 25 megapascals to slightly more than 25 megapascals), not that it decreased. Moreover, the researchers recommended substituting relatively small proportions (compared with larger proportions) of the mixtures with SSA; comparing the compressive strength of the 10% SSA mixture with that of the control without also considering how the compressive strength of the 10% SSA mixture compared with that of the 20% and 40% SSA mixtures wouldn't support the recommendation. Choice D is incorrect. Though the graph indicates that the 10% SSA mixture's compressive strength on day 7 and day 28 (about 25 megapascals and slightly more than 25 megapascals, respectively) didn't surpass the control mixture's compressive strength on day 28 (about 26 megapascals), this observation doesn't support the recommendation. The recommendation concerns the trade-off between SSA proportion and compressive strength; supporting it requires not just evidence about how the 10% SSA mixture compared with the control but also evidence about how the mixtures with larger proportions of SSA (20% and 40%) compared with the 10% SSA mixture and the control.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Ancestral Puebloans, the civilization from which present-day Pueblo tribes descended, emerged as early as 1500 B.C.E. in an area of what is now the southwestern United States and dispersed suddenly in the late 1200s C.E., abandoning established villages with systems for farming crops and turkeys. Recent analysis comparing turkey remains at Mesa Verde, one such village in southern Colorado, to samples from modern turkey populations in the Rio Grande Valley of north central New Mexico determined that the latter birds descended in part from turkeys cultivated at Mesa Verde, with shared genetic markers appearing only after 1280. Thus, researchers concluded that _____

Which choice most logically completes the text?

Answer

- A. conditions of the terrains in the Rio Grande Valley and Mesa Verde had greater similarities in the past than they do today.
- B. some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried farming practices with them.
- C. Indigenous peoples living in the Rio Grande Valley primarily planted crops and did not cultivate turkeys before 1280.
- D. the Ancestral Puebloans of Mesa Verde likely adopted the farming practices of Indigenous peoples living in other regions.

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text’s discussion of Ancestral Puebloans’ migration to the Rio Grande Valley. The text states that in the late 1200s C.E., the Ancestral Puebloan civilization abandoned villages in its original homeland, which included the Mesa Verde site. The text goes on to say that recent genetic analysis has demonstrated that the modern turkey population in the Rio Grande Valley descends partly from the ancient turkeys raised at Mesa Verde, and that the genetic markers shared by the two turkey populations first appeared at Mesa Verde only after 1280 C.E. Therefore, it can reasonably be concluded that some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried their agricultural practices—including the farming of turkeys—to their new home.

Choice A is incorrect because the text never compares the condition of the Rio Grande Valley’s terrain to that of Mesa Verde’s terrain, either in the present or in the past. Choice C is incorrect. Although genetic analysis has demonstrated that the modern turkey population in the Rio Grande valley descended in part from the turkey population raised by the Ancestral Puebloans of Mesa Verde before their migration to the valley in 1280, this finding doesn’t eliminate the possibility that Indigenous peoples living in the valley before 1280 might also have farmed turkeys. Choice D is incorrect. The text doesn’t consider the possibility that before their migration to the Rio Grande Valley after 1280, the Ancestral Puebloans of Mesa Verde might have adopted turkey farming from an outside Indigenous civilization in another region; instead, the text provides evidence suggesting that the Ancestral Puebloans brought turkey farming to another region—the Rio Grande Valley—after 1280.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Percentages of New Year’s Resolution Makers Who Make Certain Kinds of Resolutions

Type of resolution	Age 18-29	Age 30-49	Age 50-64	Age 65+
Health and exercise	79	80	79	76
Finances	68	63	56	47
Personal relationships	63	53	58	52
Hobbies	65	53	51	45

A Pew Research Center survey conducted in January 2024 found that three out of ten US adults make at least one New Year’s resolution (a promise for the year ahead), while half of those who make a resolution make more than one. The survey asked participants what kinds of resolutions they made and separated them into several categories. The table presents percentages of people who make particular kinds of New Year’s resolutions among those who choose to make them, indexed by age bracket.

Which choice best presents a conclusion about the habits of New Year’s resolution makers that is best supported by information in the text and the table?

Answer

- A. Resolution makers between the ages of 50 and 64 are more likely to make resolutions related to personal relationships and less likely to make resolutions related to finances than resolution makers between the ages of 30 and 49 are.
- B. Resolution makers between the ages of 18 and 29 are more likely to make resolutions about health and exercise than resolution makers between the ages of 30 and 49 are.
- C. The majority of US adults who make resolutions related to health and exercise also make resolutions in multiple additional categories.
- D. Among all US adults, people become less likely to make New Year’s resolutions as they age, regardless of the type of resolution.

Correct Answer: A

Rationale

Choice A is the best answer because it presents a conclusion about the habits of New Year’s resolution makers that is best supported by information in the text and the table. The table presents percentages of resolution makers who make particular kinds of resolutions, indexed by age bracket. According to the table, 58 percent of resolution makers ages 50 to 64 made resolutions related to personal relationships, compared to 53 percent of resolution makers ages 30 to 49—indicating that the older group is more likely to make such resolutions. The table also shows that 56 percent of resolution makers ages 50 to 64 made resolutions related to finances, compared to 63 percent of resolution makers ages 30 to 49—indicating that the older group is less likely to make financial resolutions. Thus, the data support the conclusion that resolution makers between the ages of 50 and 64 are more likely to make resolutions related to personal relationships and less likely to make resolutions related to finances than resolution makers between the ages of 30 and 49 are.

Choice B is incorrect because according to the table, 79 percent of resolution makers ages 18 to 29 made resolutions about health and exercise, while 80 percent of resolution makers ages 30 to 49 made such resolutions. Therefore, resolution makers ages 18 to 29 are slightly less likely, not more likely, to make resolutions about health and exercise than resolution makers ages 30 to 49 are. Choice C is incorrect because neither

the text nor the table provides information about whether resolution makers who make resolutions related to health and exercise also make resolutions in multiple additional categories. The text indicates that half of those who make a resolution make more than one, but it doesn't specify how many or in which categories, and the table presents only the percentages of resolution makers in each age bracket who made particular kinds of resolutions. Choice D is incorrect because the table presents data only about people who chose to make resolutions, not about all US adults; the table therefore can't support a conclusion about whether US adults overall become less likely to make resolutions as they age. Furthermore, although percentages of resolution makers generally decline as age bracket increases in some resolution categories, the table shows that the percentage of resolution makers who made health and exercise resolutions remains relatively stable across age brackets; thus, even among resolution makers, there isn't a consistent age-related decline across all types of resolutions.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In their meta-analysis of research on advergames (video games developed to promote products or services), Zeph M.C. van Berlo et al. confirm that such games, though they can elicit player interest, may not facilitate subsequent recall of product and brand information. This phenomenon can be explained by the finite nature of cognitive capacity as it is articulated in Annie Lang’s limited capacity model of motivated mediated message processing. In this case, players’ cognitive resources are directed foremost toward the advergence’s mechanics, leaving little or no capacity for encoding and storing the information the advertiser intends to be salient.

Which choice best states the main idea of the text?

Answer

- A. The limited capacity model of motivated mediated message processing developed by Lang provides a means of explaining the finding by van Berlo et al. that players may not readily recollect the brand and product information embedded in advergames.
- B. Research by van Berlo et al. corroborates Lang’s conclusion that because people predominantly focus on game mechanics when playing video games, it is difficult for advergames to communicate brand and product information in ways that are highly memorable.
- C. The meta-analysis by van Berlo et al. reveals that higher engagement in advergence mechanics is linked to lower effectiveness in persuading players to purchase particular brands and products.
- D. Although the limited capacity model of motivated mediated message processing developed by Lang suggests otherwise, advergames can succeed as marketing tools, provided that they achieve a balance between game mechanics and the promotion of a brand or product.

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main idea of the text. The text begins by presenting van Berlo et al.’s finding that advergames—video games intended to promote a specific product or service—can be engaging to players but “may not facilitate subsequent recall of product and brand information,” ultimately limiting their effectiveness as advertisements. The text then suggests that Lang’s limited capacity model of motivated mediated message processing, which characterizes cognitive capacity as finite, can explain or account for van Berlo et al.’s finding: specifically, players’ attention is “directed foremost toward the advergence’s mechanics” instead of toward the advertising information. The main idea of the text is therefore that Lang’s model provides a means of explaining van Berlo et al.’s finding.

Choice B is incorrect because it mischaracterizes the relationship between Lang’s model and van Berlo et al.’s finding. The text presents Lang’s model as a broad theoretical framework about cognitive capacity that can explain van Berlo et al.’s finding. The text doesn’t indicate that Lang drew conclusions specific to video games that van Berlo et al. then corroborated. Choice C is incorrect because it mischaracterizes the findings of the meta-analysis by van Berlo et al. Although Lang’s model suggests that engagement in advergence mechanics occurs at the expense of recalling advertising information, the text doesn’t indicate that van Berlo et al. offered this explanation for their finding. The text merely states that van Berlo et al. found that advergames “can elicit player interest” but “may not facilitate subsequent recall of product and brand information.” In addition, the text doesn’t discuss whether players’ ability to recall advertising content ultimately affects their purchasing behavior. Choice D is incorrect. Although the text presents Lang’s model as an explanation of why advergames may not succeed in promoting recall of advertising information, the text doesn’t suggest that advergames can succeed as marketing tools if they balance mechanics and promotion. The text doesn’t propose a design solution to the limitations of advergames or suggest that these limitations can be overcome.

Question ID: 5eda42a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is from Maggie Pogue Johnson’s 1910 poem “Poet of Our Race.” In this poem, the speaker is addressing Paul Laurence Dunbar, a Black author.

Thou, with stroke of mighty pen,
Hast told of joy and mirth,
And read the hearts and souls of men
As cradled from their birth.

The language of the flowers,
Thou hast read them all,
And e’en the little brook
Responded to thy call.

Which choice best states the main purpose of the text?

Answer

- A. To praise a certain writer for being especially perceptive regarding people and nature
- B. To establish that a certain writer has read extensively about a variety of topics
- C. To call attention to a certain writer’s careful and elaborately detailed writing process
- D. To recount fond memories of an afternoon spent in nature with a certain writer

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main purpose of the text. In the first part of the text, the speaker addresses Paul Laurence Dunbar’s ability to understand people (he has “read the hearts and souls of men” and written of their “joy and mirth”). In the second part of the text, the speaker describes Dunbar’s thorough understanding of the natural world (he has read “the language of the flowers” and engaged with “the little brook”). Thus, the text mainly praises Dunbar for being especially perceptive about people and nature.

Choice B is incorrect because the speaker describes Dunbar as having read the “hearts and souls of men” and the “language of flowers” to convey Dunbar’s ability to comprehend people and nature, not to suggest that Dunbar has literally read any of these things or has read a great deal about them. Choice C is incorrect because the text notes how well Dunbar has made sense of the topics he’s written about but doesn’t address any specific parts of Dunbar’s writing process beyond the suggestion that he used a pen. Choice D is incorrect because the text focuses on Dunbar’s understanding of people and nature as expressed in his writing. Nothing in the text suggests that the speaker is recalling a particular afternoon actually spent in nature with Dunbar; even if there had been a shared experience, the text isn’t focused on reminiscing.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Simulated Change in Annual Aquifer Input and Irrigation Output if Precipitation Concentration Increases as Climate Models Predict

Baseline concentration of annual precipitation	% change in water entering aquifers	% change in surface water used for irrigation	% change in groundwater used for irrigation
Precipitation is currently somewhat concentrated	4.9	0.4	0.9
Precipitation is currently evenly distributed	11.0	9.0	7.9

Some climate models for the western United States predict that while total annual precipitation may remain unchanged from the present level, precipitation will become concentrated into fewer but more intense rain and snow events. University of Texas climate scientist Geeta Persad and her colleagues simulated how the amount of water entering aquifers and the amount being used for irrigation purposes would change if this were to occur. Persad and her colleagues concluded that concentration of precipitation into fewer events would result in a higher number of dry days, triggering more irrigation, but that this change in irrigation output is highly sensitive to the baseline concentration of precipitation that currently exists in an area.

Which choice best describes data from the table that support Persad and her colleagues’ conclusion?

Answer

- A. If baseline precipitation is somewhat concentrated, the amount of water being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater, whereas the amount of water entering aquifers will increase 11.0% if baseline precipitation is evenly distributed.
- B. If baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas it will increase 9.0% for surface water and 7.9% for groundwater if baseline precipitation is evenly distributed.
- C. If baseline precipitation is somewhat concentrated, the amount of water entering aquifers will increase 4.9%, while the amount being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater.
- D. If baseline precipitation is somewhat concentrated, water use for irrigation will decline by a small amount, whereas it will increase 11.0% for surface water and 9.0% for groundwater if baseline precipitation is evenly distributed.

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the table that support Persad and her colleagues’ conclusion. The text explains that, according to some climate models, precipitation in the western United States will become concentrated into fewer, more intense rain and snow events. According to the text, Persad and her colleagues concluded that more irrigation will consequently be needed but that the change in irrigation output will be highly sensitive to, or greatly affected by, the baseline concentration of precipitation in an area. This conclusion is supported by data from the researchers’ simulations of changes in annual irrigation output in two different scenarios—one in which an area’s annual precipitation is already somewhat concentrated and one in which its annual precipitation is evenly distributed. The table shows that if baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas if baseline precipitation is evenly distributed, water use for irrigation will increase much more—9.0% for surface water and 7.9% for groundwater. This difference illustrates the researchers’ conclusion that the amount of additional water needed for irrigation will vary greatly depending on how concentrated or spread out the annual precipitation in an area already is.

Choice A is incorrect because it compares changes in the amount of water being used for irrigation to changes in the amount of water entering aquifers. Persad and her colleagues' conclusion doesn't focus on changes to the amount of water entering aquifers; rather, the researchers' conclusion focuses on changes to irrigation output relative to how concentrated or spread out the annual precipitation in an area is. Choice C is incorrect because it supports only part of Persad and her colleagues' conclusion. According to the text, the researchers concluded that the concentration of precipitation into fewer events will trigger more irrigation but that this change in irrigation output will be highly sensitive to an area's baseline concentration of annual precipitation. The data in this choice support the idea that more irrigation will be needed, but to support the rest of the researchers' conclusion, additional data from the table are required to show that the increases in water use for irrigation will vary depending on how concentrated or spread out the annual precipitation in an area already is. Choice D is incorrect because data in the table indicate no declines in water use for irrigation, showing only increases in the form of positive values.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Studies of ocean wave breaking have predominantly focused on traveling waves (those propagating along the horizontal plane), so Mark McAllister et al. utilized a circular wave tank to produce and study spike waves, axisymmetric standing waves that can erupt vertically when traveling waves propagating in opposing directions intersect. Traveling waves break when wave steepness (height-to-length ratio) passes a critical threshold; breaking thus constrains wave height. McAllister et al. found that spike waves can exceed that constraint, as other factors than just steepness (e.g., jet stability and cavity shape) mediate spike-wave breaking.

Which choice best states the main idea of the text?

Answer

- A. Previous studies have suggested that steepness mediates breaking in traveling waves, but the study by McAllister et al. shows that jet stability and cavity shape may also influence breaking in such waves.
- B. The process of breaking limits the height of traveling waves, but the study by McAllister et al. suggests that spike waves can exceed those limits if their height-to-length ratio reaches a critical threshold.
- C. McAllister et al. suggest that spike waves can form when traveling waves propagating in opposing directions intersect and that spike waves tend to be higher than traveling waves.
- D. The study by McAllister et al. suggests that when traveling waves intersect in specific ways, the resulting wave may be higher than would be expected based on the properties of traveling waves.

Correct Answer: D

Rationale

Choice D is the best answer because it most accurately states the main idea of the text. The text begins by explaining that most studies of ocean wave breaking have focused on traveling waves (those that move horizontally) but that McAllister et al. studied spike waves, which form when traveling waves moving in opposing directions intersect. The text establishes that while traveling waves break when their steepness exceeds a critical threshold (limiting their height), McAllister et al. found that spike waves can exceed these constraints because factors like jet stability and cavity shape can mitigate the effects of steepness. Thus, the main idea of the text is that spike waves resulting from traveling waves intersecting in specific ways can reach heights greater than what would be expected based on the general properties of traveling waves.

Choice A is incorrect because it misrepresents the information in the text; the text indicates that McAllister et al. showed that jet stability and cavity shape (along with steepness) influence breaking in spike waves, not in traveling waves. Further, the information about specific influences is not the main focus of the text. Choice B is incorrect because it misrepresents the information in the text; the text indicates that McAllister et al. found that spike waves can pass the steepness threshold at which traveling waves break because factors like jet stability and cavity shape affect the breaking of spike waves, not because they’ve simply reached a critical threshold of steepness. Choice C is incorrect. Although the text does explain that spike waves can form when traveling waves intersect, it presents this as a known fact that McAllister et al.’s study was based on, not as something the researchers suggested. Further, the text indicates that spike waves can exceed a threshold that limits traveling wave height but doesn’t focus on the idea that spike waves typically are higher than traveling waves.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Birds of many species ingest foods containing carotenoids, pigmented molecules that are converted into feather coloration. Coloration tends to be especially saturated in male birds’ feathers, and because carotenoids also confer health benefits, the deeply saturated colors generally serve to communicate what is known as an honest signal of a bird’s overall fitness to potential mates. However, ornithologist Allison J. Shultz and others have found that males in several species of the tanager genus *Ramphocelus* use microstructures in their feathers to manipulate light, creating the appearance of deeper saturation without the birds necessarily having to maintain a carotenoid-rich diet. These findings suggest that _____

Which choice most logically completes the text?

Answer

- A. individual male tanagers can engage in honest signaling without relying on carotenoid consumption.
- B. feather microstructures may be less effective than deeply saturated feathers for signaling overall fitness.
- C. scientists have yet to determine why tanagers have a preference for mates with colorful appearances.
- D. a male tanager’s appearance may function as a dishonest signal of the individual’s overall fitness.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of Shultz’s finding about male tanagers. The text explains that because carotenoids both contribute to deeply saturated feathers and offer health benefits, having deeply saturated feathers is usually “an honest signal” (a true indication) that a bird is generally fit. However, Shultz and others have found that certain male tanagers can appear to have deeply saturated feathers even if they haven’t consumed a diet rich in carotenoids, thanks to microstructures in their feathers that manipulate light. If those birds aren’t necessarily eating carotenoid-rich diets, they may actually be less fit than other birds that appear to have similarly saturated feathers; this suggests that a male tanager’s appearance may function as a dishonest signal, or a false indication, of the bird’s overall fitness.

Choice A is incorrect because Shultz’s finding suggests that some tanagers can signal fitness without consuming the carotenoids that contribute to fitness, thereby making those signals dishonest, not that tanagers can give honest signals of their fitness without consuming carotenoids. Choice B is incorrect because Shultz’s finding suggests that the microstructures in certain tanagers’ feathers can give a dishonest signal of fitness, not that the microstructures are less effective than actual pigmentation for signaling fitness. Whether the signal of fitness is honest or dishonest has no bearing on how effective the signal is: a signal is effective if potential mates behave as though it’s true, regardless of whether it’s actually true. Since there’s no information in the text about how potential mates respond to the dishonest signals of some tanagers, there’s no support for the idea that the dishonest signals are less effective than the honest signals. Choice C is incorrect because Shultz’s finding suggests that certain male tanagers may appear to be fitter than they actually are, not that scientists haven’t determined why tanagers prefer mates with colorful appearances.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Tadpole Body Mass and Toxin Production after Three Weeks in Ponds

Population density	Average tadpole body mass (milligrams)	Average number of distinct bufadienolide toxins per tadpole	Average amount of bufadienolide per tadpole (nanograms)	Average bufadienolide concentration (nanograms per milligram of tadpole body mass)
High	193.87	22.69	5,815.51	374.22
Medium	254.56	21.65	5,525.72	230.10
Low	258.97	22.08	4,664.99	171.43

Ecologist Veronika Bókony and colleagues investigated within-species competition among common toads (*Bufo bufo*), a species that secretes various unpleasant-tasting toxins called bufadienolides in response to threats. The researchers tested *B. bufo* tadpoles’ responses to different levels of competition by creating ponds with different tadpole population densities but a fixed amount of food. Based on analysis of the tadpoles after three weeks, the researchers concluded that increased competition drove bufadienolide production at the expense of growth.

Which choice uses data from the table to most effectively support the researchers’ conclusion?

Answer

- A. The difference in average tadpole body mass was small between the low and medium population density conditions and substantially larger between the low and high population density conditions.
- B. Tadpoles in the low and medium population density conditions had substantially lower average bufadienolide concentrations but had greater average body masses than those in the high population density condition.
- C. Tadpoles in the high population density condition displayed a relatively modest increase in the average amount of bufadienolide but roughly double the average bufadienolide concentration compared to those in the low population density condition.
- D. Tadpoles produced approximately the same number of different bufadienolide toxins per individual across the population density conditions, but average tadpole body mass decreased as population density increased.

Correct Answer: B

Rationale

Choice B is the best answer. This data shows that the tadpoles in the high-density pond (meaning those with the most competition) didn’t grow as big as the other two groups but produced more bufadienolide.

Choice A is incorrect. This doesn’t fully support the conclusion. It doesn’t include any data about bufadienolide production. Choice C is incorrect. This doesn’t fully support the conclusion. It doesn’t include any data about growth. Choice D is incorrect. This doesn’t fully support the conclusion. It doesn’t demonstrate that the tadpoles in the high-density pond produced more bufadienolide overall. The fact that they didn’t produce more kinds of bufadienolide isn’t relevant to the conclusion.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Swahili Speakers in Three African Countries

Country	Approximate number of speakers (in millions)	Estimated % of population
Democratic Republic of the Congo	22	25
Kenya	55	100
Tanzania	61	100

Swahili is estimated to be the first language of up to 15 million people worldwide. It’s also an officially recognized language in Tanzania, Kenya, and the Democratic Republic of the Congo, which means these countries use Swahili in government documents and proceedings. But even in countries where almost everyone speaks Swahili, for many it isn’t their first language but is instead their second, third, or even fourth language.

Which choice most effectively uses data from the table to support the underlined claim?

Answer

- A. Tanzania has approximately 61 million Swahili speakers, which is much more than the estimated total number of people worldwide for whom Swahili is their first language.
- B. Tanzania is estimated to have at most 15 million Swahili speakers, while the country’s total population is approximately 61 million people.
- C. Approximately 100 percent of the people who speak Swahili as their first language live in Kenya, which has a total population of approximately 55 million people.
- D. Approximately 100 percent of Kenya’s population speaks Swahili, while only about 25 percent of the Democratic Republic of the Congo’s population speaks Swahili.

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to support the underlined claim. The text indicates that Swahili is the first language of up to 15 million people worldwide. The text goes on to claim, in the underlined portion, that even in countries where nearly everyone speaks Swahili, many of the language’s speakers don’t have Swahili as their first language. The table indicates that 61 million people in Tanzania, which amounts to 100 percent of the population, speak Swahili. If 61 million people in Tanzania speak Swahili, but only 15 million people worldwide have Swahili as their first language, that means there are many people in Tanzania who speak Swahili as a language other than their first language. This information about Swahili speakers in Tanzania therefore supports the claim that many Swahili speakers in countries where nearly everyone speaks Swahili speak it as a language other than their first language (such as their second, third, or fourth language).

Choice B is incorrect because it doesn’t accurately describe information in the table. According to the table, Tanzania has 61 million Swahili speakers, not at most 15 million Swahili speakers. Additionally, the table indicates that 100 percent of Tanzania’s population speak Swahili, which means that the number of Swahili speakers in the country and the country’s total population should be the same, not that they should differ by such a large amount. Choice C is incorrect because there’s no information in the table or the text that indicates where people who speak Swahili as their first language live. Although Kenya’s total population can be inferred from the table—if Kenya has 55 million Swahili speakers and 100% of Kenya’s population speak Swahili, then Kenya must have a population of 55 million people—nothing suggests that all the people who speak Swahili as their first language live in a single country, let alone that they all live in Kenya. Choice D is incorrect. Although the table does indicate that 100 percent of Kenya’s population and 25 percent of the Democratic Republic of the Congo’s population speak Swahili, this comparison is

irrelevant to the claim that Swahili isn't the first language of many of its speakers even in countries where almost everyone speaks Swahili. On its own, a difference in the proportions of the population who speak Swahili cannot reveal whether those Swahili speakers have Swahili as their first language or a subsequent language.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Microbial fuel cells (MFCs) capitalize on the ability of some species of bacteria to oxidize organic matter and transfer electrons extracellularly. The bacteria form a dense biofilm on the surface of an electron-collecting anode, but moving the electrons from the bacterial cytoplasm to an external electrode requires that the electrons pass through a series of inefficient oxidation-reduction (redox) reactions. Accordingly, MFC power output rarely exceeds a density of 0.30 milliwatts per square centimeter (mW/cm^2). In an experiment, researchers added silver nanoparticles to carbon paper covering the anode in an MFC. The resulting power density was $0.66 \text{ mW}/\text{cm}^2$. Since metals such as silver exhibit high electrical conductivity, the researchers hypothesized that _____

Which choice most logically completes the text?

Answer

- A. silver nanoparticles may allow electrons to bypass the series of redox reactions and transfer directly to the electrode.
- B. electrons may be conducted directly to the electrode before the silver nanoparticles catalyze the redox reactions.
- C. silver nanoparticles may increase the metabolic processes of the bacteria, thereby increasing the number of free electrons available to transfer to the electrode.
- D. as the density of the biofilm increases, the series of redox reactions may accelerate independent of the presence of the silver nanoparticles.

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of microbial fuel cells (MFCs) and the effect of silver nanoparticles on power output. The text explains that in MFCs, electrons are transferred from the bacterial cytoplasm to an external electrode through a series of inefficient redox reactions, which typically prevent power output from exceeding $0.30 \text{ mW}/\text{cm}^2$. When researchers added silver nanoparticles to carbon paper covering the anode, power density more than doubled to $0.66 \text{ mW}/\text{cm}^2$. Critically, the text notes that metals such as silver exhibit high electrical conductivity. Given that the inefficiency of the redox reactions accounted for the limited power output of MFCs and that the addition of a highly conductive material dramatically increased that output, it is reasonable to hypothesize that the silver nanoparticles allowed electrons to bypass the redox reactions and transfer directly to the electrode.

Choice B is incorrect because the researchers’ hypothesis, as set up by the last sentence of the text, is specifically grounded in the high electrical conductivity of metals such as silver. Nothing in the text supports the idea that the silver nanoparticles function as catalysts for the redox reactions, nor does the text suggest that electrons are being conducted to the electrode independently of any role played by the silver nanoparticles. The text points to the conductivity of silver as the explanation for the increased power density, not to any catalytic effect of the nanoparticles on the existing redox reactions. Choice C is incorrect because the text never discusses the metabolic processes of the bacteria in the biofilm coating the electron-collecting anodes, nor is there any suggestion that the high conductivity of silver nanoparticles would interact with the bacteria’s metabolism in any way. Choice D is incorrect because the text never suggests that in the experiment, the biofilm on the electron-collecting anodes varied in density. Moreover, the hypothetical statement begun in the last sentence of the text concerns the high electrical conductivity of metals such as silver. Thus, proposing that redox reactions—which, as the text establishes, conduct electricity—accelerated independently of a conductive metallic material (the silver nanoparticles) wouldn’t logically complete that hypothesis.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Through adaptive radiation, multiple species of *Tetragnatha*—a genus of orb-weaving spiders—have significantly diversified their web architecture over a relatively short timescale. To investigate whether the material properties of *Tetragnatha* silk have been similarly affected by this evolutionary process, Angela M. Alicea-Serrano et al. examined the chemical composition of both the radial (non-sticky threads extending from the center) and capture (sticky threads that hold prey) silks that form the web as well as tested the tensile strength and viscosity of samples collected from three *Tetragnatha* species at two sites in the Hawaiian archipelago. The team found significant interspecies variation in these biomaterials.

Which choice best states the main idea of the text?

Answer

- A. Research reveals that web architecture among species of *Tetragnatha* spiders has likely diversified more rapidly than the material properties of *Tetragnatha*'s silk have.
- B. Research suggests that the material properties of silk from species of *Tetragnatha* spiders have diversified during adaptive radiation.
- C. A study indicates that variations in web architecture among species of *Tetragnatha* spiders can likely be explained by corresponding variations in the material properties of their silks.
- D. A study shows that adaptive radiation can explain interspecies variation in the web architecture of *Tetragnatha* spiders but not in the material properties of the spiders' silk.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately states the main idea of the text. The text notes that *Tetragnatha* spiders have diversified their web architecture, or web design, through adaptive radiation and that Alicea-Serrano et al. investigated whether silk material properties were also affected by this process. After examining chemical composition, tensile strength, and viscosity of silks from three *Tetragnatha* species, the team found significant interspecies variation in these material properties, meaning that there were observable differences in these traits across the studied species. In the context of the study's research question, this finding suggests that the material properties of silk from species of *Tetragnatha* spiders have indeed diversified during adaptive radiation, which is the main idea of the text.

Choice A is incorrect because the text doesn't compare the rate of diversification in web architecture to the rate of diversification in silk properties. Although the text notes that web architecture diversified "over a relatively short timescale," it doesn't make a corresponding temporal claim about silk properties or suggest that one changed more rapidly than the other. Choice C is incorrect because the study investigated silk properties as a separate question, not as an explanation for differences in web architecture. The researchers discussed in the text investigated whether silk properties were "similarly affected" by adaptive radiation, not whether they caused the variations in web design. Choice D is incorrect because it contradicts the findings described in the text. The study found "significant interspecies variation" in silk biomaterials, suggesting that adaptive radiation did affect the properties of *Tetragnatha* spiders' silk rather than showing that adaptive radiation cannot explain silk variation.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In a study of perceptions of listening ability, participants explained their position on a sociopolitical issue and then received a listener’s written summary that expressed agreement or disagreement. Within selected summaries, researchers embedded markers of attentive listening (e.g., references to specific details), hypothesizing that such indications would positively influence perceptions of listening skill even in the context of disagreement. Instead, participants consistently rated listeners who expressed disagreement as less skilled, regardless of the other traits of the listeners’ summaries.

What does the text most strongly imply about how participants responded to expressions of disagreement?

Answer

- A. When participants felt personally invested in the topics they discussed, they were less likely to perceive listeners who expressed disagreement as attentive, regardless of evidence to the contrary in these listeners’ summaries.
- B. When participants encountered summaries from listeners who expressed disagreement with their views, participants tended to disregard evidence that the listeners had in fact been attentive.
- C. Although participants maintained their positions regardless of a summary’s level of detail, they tended to regard listeners who expressed disagreement as more attentive when these listeners provided more detailed summaries.
- D. Although participants were critical of expressions of disagreement, they gave higher ratings to listeners whose summaries included markers of attentiveness than to listeners whose summaries did not include these markers.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately describes what the text implies about participants’ responses to disagreement. The text describes a study in which participants were asked to explain their position on an issue to a listener, who then provided a written summary of that position that also expressed their level of agreement. The participants were then asked, based on these summaries, to rate listeners’ listening skill. The text indicates that researchers embedded markers of attentive listening, such as references to details, in some of the summaries, expecting that these markers would improve the participants’ perceptions of listening skill even when the listeners disagreed. Instead, participants rated disagreeing listeners as less skilled, or attentive, “regardless of the other traits of the listeners’ summaries.” This means that even when summaries contained evidence of attentive listening, participants still judged disagreeing listeners as less skilled—in effect, disregarding evidence that those listeners had been attentive.

Choice A is incorrect. Although the text does suggest that participants tended to disregard evidence of attentiveness when listeners disagreed with their position on the topic they discussed, it doesn’t address participants’ level of personal investment in these topics; therefore, the text can’t support any conclusions about how personal investment in a topic affected participants’ perceptions of the listening skill of disagreeing listeners. Choice C is incorrect because it directly contradicts the text. The text states that participants rated disagreeing listeners as less skilled regardless of the other traits of the listeners’ summaries, which means markers of attentiveness (specific details) didn’t lead participants to regard disagreeing listeners as more attentive. Instead, disagreement alone determined the participants’ lower ratings. Moreover, the text doesn’t address whether participants maintained their positions based on the listeners’ summaries. The text focuses on factors affecting participants’ perceptions of listeners’ attentiveness, not on how features of listeners’ summaries influenced the participants’ views of the sociopolitical issue they discussed. Choice D is incorrect because the text indicates the opposite: participants didn’t give higher ratings to disagreeing listeners whose summaries included markers of attentiveness than to disagreeing listeners whose summaries didn’t include these markers. Rather, the text states that participants rated disagreeing listeners as less skilled than agreeing listeners regardless of the other traits of the summaries, meaning attentiveness markers had no positive effect when listeners’ summaries expressed disagreement.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Among social animals that care for their young, such as chickens, macaque monkeys, and humans, newborns appear to show an innate attraction to faces and face-like stimuli. Elisabetta Versace and her colleagues used an image of three black dots arranged in the shape of eyes and a nose or mouth to test whether this trait also occurs in *Testudo* tortoises, which live alone and do not engage in parental care. They found that tortoise hatchlings showed a significant preference for the image, suggesting that _____

Which choice most logically completes the text?

Answer

- A. face-like stimuli are likely perceived as harmless by newborns of social species that practice parental care but as threatening by newborns of solitary species without parental care.
- B. researchers should not assume that an innate attraction to face-like stimuli is necessarily an adaptation related to social interaction or parental care.
- C. researchers can assume that the attraction to face-like stimuli that is seen in social species that practice parental care is learned rather than innate.
- D. newly hatched *Testudo* tortoises show a stronger preference for face-like stimuli than adult *Testudo* tortoises do.

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the study by Versace and colleagues. The text indicates that newborn animals of some species are attracted to faces and to stimuli that resemble faces. These species, the text says, share two characteristics: they’re social and they practice parental care, meaning that parents care for their young. The text goes on to describe Versace and colleagues’ experiment, which showed that *Testudo* tortoises, which aren’t social and don’t practice parental care, were attracted to a stimulus that resembles a face. Since Versace and colleagues have shown that a species that isn’t social and doesn’t practice parental care nevertheless has the innate characteristic of being attracted to face-like stimuli, it follows that this characteristic shouldn’t be assumed to be an adaptation related to social interaction or parental care.

Choice A is incorrect because the text indicates that the tortoise hatchlings, which are solitary and don’t practice parental care, were attracted to the face-like stimuli, not that they perceived the stimuli as threatening. Choice C is incorrect because the phenomenon discussed in the text is an attraction to faces and face-like stimuli on the part of newborn animals, which can’t show any learned characteristics since they were just born. Additionally, the text tells us that the tortoises Versace and colleagues studied aren’t social and don’t practice parental care, so any findings about those tortoises wouldn’t be relevant to the question of whether an attraction to faces in social species that practice parental care is innate or learned. Choice D is incorrect because the text gives no indication that adult tortoises were tested on face-like stimuli and, if adults were in fact tested, no information about how they responded is provided. Since no information about adult tortoises’ responses is provided, no conclusion comparing those responses to the responses of newly hatched tortoises can be supported.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Aerogels are highly porous foams consisting mainly of tiny air pockets within a solidified gel. These lightweight materials are often applied to spacecraft and other equipment required to withstand extreme conditions, as they provide excellent insulation despite typically being brittle and eventually fracturing due to degradation from repeated exposure to high heat. Now, Xiangfeng Duan of the University of California, Los Angeles, and colleagues have developed an aerogel with uniquely flexible properties. Unlike earlier aerogels, Duan’s team’s material contracts rather than expands when heated and fully recovers after compressing to just 5% of its original volume, suggesting that _____

Which choice most logically completes the text?

Answer

- A. the aerogel’s remarkable flexibility results from its higher proportion of air pockets to solidified gel as compared to other aerogels.
- B. the aerogel’s overall strength is greater than that of other insulators but its ability to withstand exposure to intense heat is lower.
- C. the aerogel will be more effective as an insulator for uses that involve gradual temperature shifts than for those that involve rapid heat increases.
- D. the aerogel will be less prone to the structural weakness that ultimately causes most other aerogels to break down with use.

Correct Answer: D

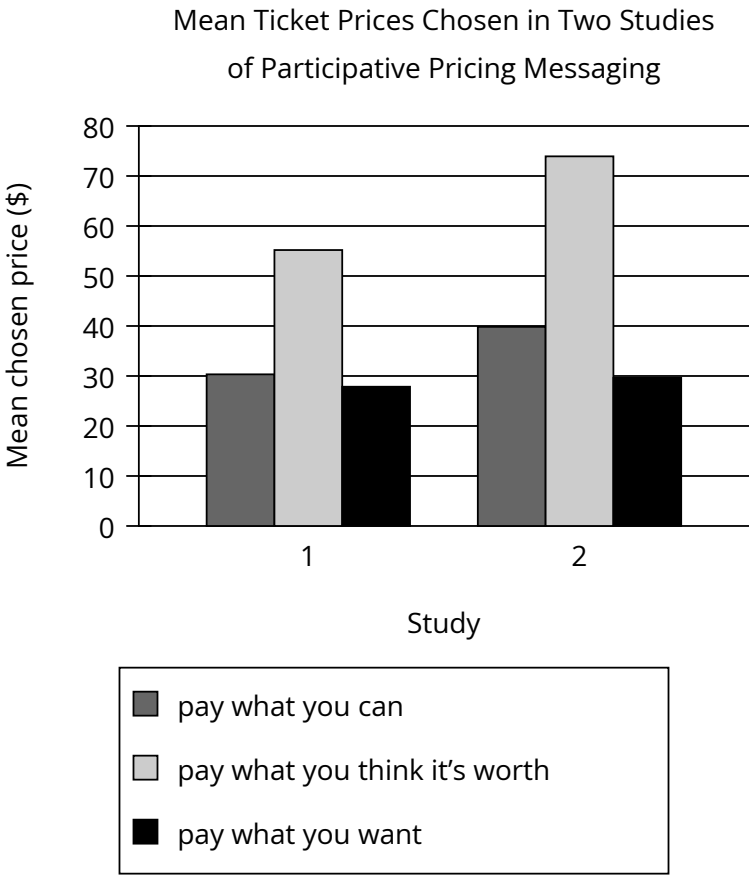
Rationale

Choice D is the best answer because it most logically completes the text’s discussion of aerogels. The text states that aerogels—highly porous foams—offer "excellent insulation" but typically break down after prolonged exposure to high heat. However, according to the text, Duan and colleagues developed an aerogel that "contracts rather than expands when heated" and recovers its original volume after this contraction. Thus, it is logical to conclude that Duan’s team’s aerogel material will be less prone to the structural weakness that caused earlier aerogels to break down.

Choice A is incorrect. Although the text indicates that aerogels consist "mainly of tiny air pockets within a solidified gel," it doesn’t mention the number or proportion of air pockets to solidified gel in typical aerogels or in the aerogel developed by Duan’s team. Choice B is incorrect because the text suggests that the aerogel developed by Duan’s team has a higher, not a lower, ability to withstand exposure to intense heat due to its contraction and subsequent recovery. Choice C is incorrect. Although the text discusses temperature tolerances of aerogels and says that they offer "excellent insulation despite typically being brittle and eventually fracturing," it doesn’t discuss how different rates of temperature change can affect aerogels.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Participative pricing, in which purchasers choose the prices they pay for products, can enable sellers to capitalize on the heterogeneous values consumers assign to the same goods and services, but doing so requires careful messaging. Annie Peng Cui and Jennifer Wiggins recruited 171 participants (ages 18–60) online for an initial study and 83 students (ages 18–31) at a state university for a second study to test the effect of three different messages—“pay what you can,” “pay what you think it’s worth,” and “pay what you want”—on how much participants would pay for concert tickets. Their results illustrate both the heterogeneity of consumer valuations and how sellers can benefit by prompting consumers to consider their own valuations: _____

Which choice most effectively uses data from the graph to complete the text?

Answer

- A. the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but when considering what they could afford to pay, the students tended to choose a lower price than did the other group.
- B. in all three messaging conditions, the group of participants recruited online tended to choose lower prices than did the students, but both groups tended to choose prices closest to the actual cost of the tickets when prompted to consider the tickets’ value.
- C. the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but both groups tended to choose a higher price when considering the value of the tickets than when considering what they could afford or wanted to pay.
- D. within each group of participants, there was wide variation in the value that individuals assigned to the concert tickets, but the students tended to assign a higher value to the tickets than did the more age-diverse group recruited online.

Correct Answer: C

Rationale

Choice C is the best answer because it most effectively uses data from the graph to complete the text about the effect of messaging on participative pricing. The graph shows mean ticket prices chosen by participants in response to three messages across two studies: Study 1, which the text indicates was conducted with an age-diverse group recruited online, and Study 2, which was conducted with student participants. The graph indicates that in the "pay what you think it's worth" condition, the mean price of the concert tickets in Study 2 was about \$74, which is greater than the mean price of about \$55 in Study 1. In other words, when participants were asked to consider their valuation of the tickets, the response was heterogeneous, or mixed. Moreover, according to the graph, both Study 1 and Study 2 show higher prices for the tickets under the "pay what you think it's worth" condition than they do under both the "pay what you can" and the "pay what you want" conditions. That is, the data suggest that both groups of participants named higher prices when considering the value of the tickets than when considering either what they could afford or wanted to pay, a finding that supports the idea that sellers can benefit when prompting consumers to consider their own valuations when they choose prices.

Choice A is incorrect because it contradicts information in the graph. Although the graph shows that students in Study 2 assigned a higher value to the tickets than did the age-diverse group in Study 1, which would support the idea that consumer valuations were heterogeneous, the graph shows that in the "pay what you can" (i.e., what you can afford) condition, the students in Study 2 assigned a higher price (about \$40), not a lower price, than the age-diverse group in Study 1 did (about \$30). Moreover, even if it were true that the students had assigned a lower price in this condition, it wouldn't support the result described in the text, only that the participants across the two studies had different ideas of what they can afford to pay. Choice B is incorrect. Although a finding that participants tended to choose prices that were closest to the actual ticket costs in the "pay what you think it's worth" condition would support the idea that sellers benefit by prompting consumers to think about their own valuations (since it's implied that sellers would lose money in the other conditions, where chosen prices were lower than the participants' valuations), neither the text nor the graph addresses how any of the prices chosen by the study participants relate to the tickets' actual market price. Choice D is incorrect. Although the wide variation in participant valuations would support the idea that consumer valuations tend to be heterogeneous, neither the text nor the graph provides any information from which to discern the relative levels of variance among the responses from participants in either study.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Art collectives, like the United States- and Vietnam-based collective The Propeller Group or Cuba’s Los Carpinteros, are groups of artists who agree to work together: perhaps for stylistic reasons, or to advance certain shared political ideals, or to help mitigate the costs of supplies and studio space. Regardless of the reasons, art collectives usually involve some collaboration among the artists. Based on a recent series of interviews with various art collectives, an arts journalist claims that this can be difficult for artists who are often used to having sole control over their work.

Which quotation from the interviews best illustrates the journalist’s claim?

Answer

- A. “The first collective I joined included many amazingly talented artists, and we enjoyed each other’s company, but because we had a hard time sharing credit and responsibility for our work, the collective didn’t last.”
- B. “We work together, but that doesn’t mean that individual projects are equally the work of all of us. Many of our projects are primarily the responsibility of whoever originally proposed the work to the group.”
- C. “Having worked as a member of a collective for several years, it’s sometimes hard to recall what it was like to work alone without the collective’s support. But that support encourages my individual expression rather than limits it.”
- D. “Sometimes an artist from outside the collective will choose to collaborate with us on a project, but all of those projects fit within the larger themes of the work the collective does on its own.”

Correct Answer: A

Rationale

Choice A is the best answer because it presents the quotation that best illustrates the journalist’s claim. By indicating that a collective didn’t continue because it was hard to share credit and responsibilities within the group even though the company was enjoyable, the quotation shows that working collaboratively can be difficult for artists who are used to having complete control over their work.

Choice B is incorrect because the quotation indicates that members of a collective are able to collaborate together and have agreed on a fair way to manage their responsibilities; this doesn’t demonstrate the challenge of sharing control among members of a collective. Choice C is incorrect because the quotation highlights the support and encouragement of individual expression an artist experiences due to working in a collective; these positive aspects don’t demonstrate the challenge of sharing control among members of a collective. Choice D is incorrect because the quotation doesn’t address any challenges of sharing control among members of a collective; it simply indicates that artists sometimes choose to work with collectives without having to be a member. Therefore, the quotation doesn’t illustrate the journalist’s claim.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Gorgets, or necklaces with large pendants, have been part of the ceremonial attire of tribes from the US Southeast for centuries. One of the oldest examples, the Fairfield Gorget, was found in Fairfield, Missouri, in 1958. Its overall design resembles that of other art from the region during the Mississippian period (900–1600 CE). Yet the image on the gorget is of a jaguar—a species whose range doesn’t extend to Missouri. Jaguar images are common in ancient Mexican art, and Mexico lies squarely in the species’ range. Therefore, some scholars argue that long-distance trade in Mexican art objects brought the imagery to Missouri, where a local artist could have adopted it and incorporated it into the Fairfield Gorget.

Which finding, if true, would most strongly support the underlined explanation?

Answer

- A. The Fairfield Gorget is dated not to the Mississippian period but instead to the earlier Woodland period, which ended around 900 CE.
- B. The range of the jaguar is shown to have expanded dramatically after the Mississippian period came to a close around 1600 CE.
- C. An ancient Mexican art object is found at a site that dates to the Mississippian period and is close to where the Fairfield Gorget was found.
- D. Certain works by present-day artists from Southeastern tribes reflect outside cultural influences, including contemporary Mexican art.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most strongly support the underlined explanation that the jaguar imagery on the Fairfield Gorget could have been inspired by art objects brought to Missouri from Mexico. The text describes the Fairfield Gorget as one of the oldest examples of these necklaces, noting that it’s similar to other art from the Mississippian period except for its image of a jaguar. The text goes on to explain that this is unusual because the jaguar’s range didn’t include Missouri. The Fairfield Gorget’s artist likely wouldn’t have encountered a live jaguar, but since such jaguar images were common in ancient Mexican art (and the jaguar’s range did include Mexico), perhaps some of that art inspired the gorget’s artist. Therefore, evidence that an ancient Mexican art object had been found at a site that was close to the Fairfield Gorget in both location and time period would strongly support the explanation that long-distance trade from Mexico took place during the Mississippian period and that jaguar imagery could have reached Missouri by those means.

Choice A is incorrect. The underlined explanation indicates that Mexican artistic influence preceded the creation of the Fairfield Gorget. An artifact from Mexico that appeared in Missouri only after the gorget was created cannot support such a claim. Choice B is incorrect. The underlined explanation pertains to whether the jaguar image on the Fairfield Gorget was suggested to the Mississippian artist who made it by encounters with depictions of jaguars in Mexican art. The appearance of jaguars in Missouri after the Mississippian period cannot provide evidence for such a claim. Choice D is incorrect. The underlined explanation pertains to influence that Mexican art might have had on Mississippian art, restricted to the period from 900 CE to 1600 CE. The actions of contemporary artists cannot provide evidence for centuries-old networks of artistic influence and exchange.

Question ID: 5cf5c0d3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Credited Film Output of James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr

Individual	Years active	Number of films known and commonly credited
James Young Deer	1909–1924	33 (actor), 35 (director), 10 (writer)
Dark Cloud	1910–1920	35 (actor), 1 (writer)
Edwin Carewe	1912–1934	47 (actor), 58 (director), 20 (producer), 4 (writer)
Lillian St. Cyr (Red Wing)	1908–1921	66 (actor)

Some researchers studying Indigenous actors and filmmakers in the United States have turned their attention to the early days of cinema, particularly the 1910s and 1920s, when people like James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr (known professionally as Red Wing) were involved in one way or another with numerous films. In fact, so many films and associated records for this era have been lost that counts of those four figures’ output should be taken as bare minimums rather than totals; it’s entirely possible, for example, that _____

Which choice most effectively uses data from the table to complete the example?

Answer

- A. Dark Cloud acted in significantly fewer films than did Lillian St. Cyr, who is credited with 66 performances.
- B. Edwin Carewe’s 47 credited acting roles includes only films made after 1934.
- C. Lillian St. Cyr acted in far more than 66 films and Edwin Carewe directed more than 58.
- D. James Young Deer actually directed 33 films and acted in only 10.

Correct Answer: C

Rationale

Choice C is the best answer because it uses data from the table to effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums—that is, that we should assume that the individuals actually had higher outputs than those recorded. The table presents the years during which the individuals were active and the number of known films the individuals are credited in. The table indicates that Lillian St. Cyr has 66 film credits as an actor and that Edwin Carewe has 58 film credits as a director; it follows that if some films and records for the era were lost, it’s possible that Lillian St. Cyr acted in far more than 66 films and that Edwin Carewe directed more than 58 films.

Choice A is incorrect because it doesn’t effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums. Rather than addressing the idea that the individuals likely had higher outputs than those presented in the table, this choice simply compares data from the table to make the point that Dark Cloud has fewer credited acting roles than Lillian St. Cyr (35 and 66, respectively). Choice B is incorrect because it misrepresents data from the table, even though it may exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums by implying that Edwin Carewe acted in more than 47 films. The table indicates that Edwin Carewe was active from 1912 to 1934, meaning that his 47 credited acting roles were in films made before or during 1934, not after that time. Choice D is incorrect because it doesn’t effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums. Instead of addressing the idea that the individuals likely had higher outputs than

those recorded, this choice suggests that James Young Deer actually acted in and directed fewer films than presented in the table (only 33 known films as a director instead of 35, and only 10 known films as an actor instead of 33).

Question ID: a12a83ce

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The following text is from William Shakespeare’s play *The Tempest*, first performed in 1611. Miranda has lived on an island with her father, Prospero, since she was three years old. Prospero has stated that Miranda likely does not remember anything other than her life on the island.

MIRANDA: 'Tis far off,

And rather like a dream than an assurance

That my remembrance warrants. Had I not

Four or five women once that tended me?

PROSPERO: Thou hadst, and more, Miranda. But how is it

That this lives in thy mind? What seest thou else

In the dark backward and abysm of time?

If thou remember'st ought ere thou camest here,

How thou camest here thou mayst.

In the text, which point does Prospero most directly make about Miranda and her memories?

Answer

- A. Miranda’s reminiscences about her early childhood have a melancholy quality that betrays her discontented view of her current circumstances.
- B. Miranda’s doubts about the accuracy of one recollection of a place other than the island are clouding her judgment and seem to be making her reluctant to explore her recollection of traveling to the island.
- C. Miranda’s ability to summon details of an experience she had before arriving on the island suggests that she may also be able to summon details of her arrival on the island.
- D. Miranda’s impression of a scene is vague because she is remembering a scenario she had daydreamed about as a child rather than a scenario that had occurred in reality.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a point that Prospero makes about Miranda. The text begins with Miranda responding to Prospero’s claim that she probably doesn’t remember her life before the island. She describes a distant memory, asking if she had "four or five women" caring for her. Prospero confirms this ("thou hadst") and then asks what else she can recall, stating that because she remembers one thing from the time before she lived on the island ("ought ere thou camest here"), she might also remember arriving on the island ("how thou camest here"). That is, Prospero indicates that Miranda’s ability to summon details of an experience from before her arrival on the island suggests that she may also be able to summon details of her arrival.

Choice A is incorrect because Prospero doesn't say anything about how Miranda presents her early childhood memory of having several people care for her—he indicates only his surprise that she remembers something from so long ago ("the dark backward and abysm of time") and his thought that she may also be able to recall another childhood event: her arrival on the island. Further, Prospero doesn't suggest that he believes Miranda is discontent in her current circumstances. Choice B is incorrect. Although Miranda suggests some doubt about her recollection of a place other than the island, in that she describes it as "like a dream" she can't be certain of, Prospero doesn't mention her uncertainty—though he himself wonders how it is possible that she correctly recalls something from so long ago ("abysm of time"). Rather than indicating that doubt is making Miranda reluctant, Prospero simply states that Miranda may have the ability to remember traveling to the island. Choice D is incorrect. Although Miranda describes a vague impression of several people who cared for her as a young child and states that it is "rather like a dream" instead of something she is certain is real, Prospero doesn't indicate that Miranda is remembering a childhood daydream. Instead, he confirms that what Miranda remembers actually happened, answering her question about having had caregivers with "thou hadst," and asks her what else she can remember from long ago.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

During their larval phase, numerous species of coral reef fish are drawn toward areas where light is present. To better understand how artificial light at night (ALAN) might affect some coral reef fish, researchers explored the effect of exposure to low levels of ALAN on the reproductive success of the common clownfish (*Amphiprion ocellaris*). While exposure to low levels of ALAN had no significant effect on spawning frequency and egg fertilization in *A. ocellaris*, incubation in the presence of ALAN completely inhibited hatching. These findings suggest that _____

Which choice most logically completes the text?

Answer

- A. *A. ocellaris* that settle in areas with low levels of ALAN have significantly higher rates of successful egg fertilization than *A. ocellaris* that settle in areas without ALAN do.
- B. the reproductive success of *A. ocellaris* would be at risk if they were to selectively settle in regions that are regularly exposed to low levels of ALAN.
- C. the reproductive success of *A. ocellaris* is more greatly affected by the presence of low levels of ALAN during incubation than the reproductive success of other species of coral reef fish is.
- D. the spawning frequency of *A. ocellaris* was more strongly affected by the presence of low levels of ALAN than egg fertilization was, though both were less affected than incubation.

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of artificial light at night (ALAN) and the common clownfish (*A. ocellaris*). The text indicates that researchers found that when *A. ocellaris* is exposed to low levels of ALAN, spawning frequency and egg fertilization aren’t significantly affected but hatching is completely inhibited—that is, the eggs never hatch. Therefore, if *A. ocellaris* were to settle only in regions where the fish would be regularly exposed to low levels of ALAN, their eggs would stop hatching and their reproductive success would be at risk.

Choice A is incorrect because the text indicates that exposure to low levels of ALAN had no significant effect on egg fertilization for *A. ocellaris*, so there’s no reason to expect there would be any significant difference in rates of successful egg fertilization between areas with low levels of ALAN and areas without ALAN. Choice C is incorrect because the text doesn’t discuss the particular effects of low levels of ALAN on any species of coral reef fish besides *A. ocellaris*. For this reason, there’s no support in the text for the idea that the reproductive success of *A. ocellaris* is more greatly affected by the presence of low levels of ALAN than the reproductive success of other species of coral reef fish is. Choice D is incorrect. The text does indicate that *A. ocellaris* incubation was most strongly affected by low levels of ALAN, but it doesn’t indicate that there was a greater effect on spawning frequency than on egg fertilization; in fact, the text states that there was no significant effect on either.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Correlations Between Congestion Ratings and Features of
the Crowd in Raters’ Immediate Vicinity

Crowd feature	Before obstacle	After obstacle	Overall
Density	0.8592	0.7308	0.7447
Velocity	−0.9357	−0.9518	−0.8587

Researcher Xiaolu Jia and colleagues monitored individuals’ velocity and the surrounding crowd density as a group of study participants walked through a space and navigated around an obstacle. Participants rated how congested it seemed before the obstacle, after the obstacle, and overall, and the researchers correlated those ratings with velocity and density. (Correlations range from −1 to 1, with greater distance from 0 indicating greater strength). The researchers concluded that the correlations with velocity are stronger than those with density.

Which choice best describes data from the table that support the researchers’ conclusion?

Answer

- A. The correlation between congestion ratings before the obstacle and density is further from 0 than the correlation between overall congestion rating and velocity is.
- B. The correlation between congestion ratings before the obstacle and velocity is further from 0 than the correlation between congestion overall and velocity is.
- C. For each of the three ratings, the correlation with velocity is negative while the correlation with density is positive.
- D. For each of the three ratings, correlations with velocity are further from 0 than the corresponding correlations with density are.

Correct Answer: D

Rationale

Choice D is the best answer. The text tells us that the farther the correlation is from 0, the “stronger” it is (doesn’t matter if it’s negative or positive). The table shows that the correlations with velocity are farther from zero than the correlations with density, which supports the conclusion that the correlations with velocity are stronger.

Choice A is incorrect. This choice doesn’t support the conclusion. It makes an “apples to oranges” comparison by comparing density and velocity correlations across features instead of comparing them for each feature. Choice B is incorrect. This choice doesn’t support the conclusion. It doesn’t include the density correlations for comparison. Choice C is incorrect. This choice doesn’t support the conclusion. The text tells us that the farther the correlation is from 0, the “stronger” it is: it doesn’t matter for “strength” whether it’s negative or positive.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Delta 15-N Values in Seagrass Samples from Four Sites on the Yucatan, 2016–2017

Site	February 2016	October 2016	February 2017	October 2017
Akumal Bay	no data available	3.3	2.0	6.3
Mahahual	0.7	no data available	2.5	3.4
Tulum	6.1	5.9	2.3	5.5
Xahuayxol	0.9	0.3	–0.9	1.4

Because water from natural, uncontaminated sources is less enriched with the stable nitrogen isotope ¹⁵N than wastewater from human activities is, the presence of such wastewater in nature can be detected by examining delta 15-N values (a measure of the ratio of ¹⁵N to ¹⁴N) in plants. Karla A. Camacho-Cruz and colleagues assessed delta 15-N values in the seagrass *Thalassia testudinum* from sites on Mexico’s Yucatan peninsula with intermediate tourism development, including Akumal Bay and Tulum, and low tourism development, including Mahahual and Xahuayxol, throughout 2016 and 2017. The data suggest that the intermediate-tourism sites experienced influxes of human wastewater. However, the researchers concluded that this happened intermittently.

Which choice best describes data from the table that support the underlined conclusion?

Answer

- A. Although delta 15-N values were generally higher in Akumal Bay and Tulum than in Mahahual and Xahuayxol, the values were lower in Akumal Bay than in Mahahual and Xahuayxol in February 2017.
- B. Delta 15-N values reached their lowest level in February 2017 in both Akumal Bay and Tulum, but no data were available for Akumal Bay in February 2016, when the values reached their highest level in Tulum.
- C. Although all sites showed considerable variation in delta 15-N values, the values remained relatively constant in Akumal Bay from October 2016 to February 2017 and in Tulum from February 2016 to October 2016.
- D. In Akumal Bay and Tulum, delta 15-N values fluctuated considerably across the three measurements made from October 2016 to October 2017.

Correct Answer: D

Rationale

Choice D is the best answer because it describes data from the table that support the underlined conclusion. The text accompanying the table states that delta 15-N values (a measure of the ratio of two nitrogen isotopes) can be used as an indirect means of measuring the ratio of human wastewater to water from uncontaminated sources—the higher the delta 15-N values, the greater the presence of human wastewater. The text goes on to mention an experiment conducted by Camacho-Cruz and colleagues measuring delta 15-N values in seagrass on the Yucatan peninsula at several sites, four of which are presented in the table accompanying the text. Two of the sites noted in the table, Akumal Bay and Tulum, had intermediate tourism development, and two others, Mahahual and Xahuayxol, had low tourism development. It is reasonable to infer that sites with intermediate tourism have a greater human presence than those with low tourism and therefore are subject to greater amounts of wastewater and exhibit higher delta 15-N values in seagrass. However, Camacho-Cruz and colleagues found that the increase in delta 15-N values was not constant. The table supports this conclusion because the two intermediate-tourism sites (Akumal Bay and Tulum) had delta 15-N

values that fluctuated considerably—both had higher delta 15-N values than the two low-tourism sites (Mahahual and Xahuayxol) in October 2016 and October 2017, but in February 2017 the delta 15-N values for the intermediate-tourism sites dropped significantly (in fact, Mahahual, a low-tourism site, had a higher delta 15-N value for that period). The fluctuations in values in Akumal Bay and Tulum support the conclusion that these sites experienced influxes of human wastewater intermittently.

Choice A is incorrect because it inaccurately describes data from the table: in February 2017 the delta 15-N value in Akumal Bay was higher than the delta 15-N value in Xahuayxol, not lower. Choice B is incorrect because the absence of data for delta 15-N levels in Akumal Bay in February 2016 would neither prove nor disprove that the presence of human wastewater at Akumal Bay and Tulum was not constant. Despite the unavailable data, there is still enough information to conclude that delta 15-N levels at both sites fluctuated significantly and that influxes of human wastewater therefore happened intermittently. Choice C is incorrect because a constancy in delta 15-N values would not indicate that the influx of human wastewater was intermittent, but rather the opposite. Moreover, the two periods selected are not indicative of general trends in the whole table (for example, the values fluctuated significantly in Akumal Bay between February 2017 and October 2017, and in Tulum between October 2016 and February 2017).

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Silicon-based photovoltaic cells account for 95% of the cells used in solar panels worldwide despite converting an average of only 18–22% of the sunlight that reaches them. In a study addressing this relative inefficiency, a team led by Laura Miranda-Pérez demonstrated that the addition of a thin layer of the mineral perovskite—which captures the blue range of light in the solar spectrum, whereas silicon captures the red range—allows the cells to convert 29.5% or more of the Sun’s energy into usable electricity. Cells made with only perovskite, however, are no more efficient than silicon-based ones. It’s reasonable to conclude, then, that _____

Which choice most logically completes the text?

Answer

- A. photovoltaic cells with both silicon and perovskite are more efficient because they make use of more of the solar spectrum.
- B. photovoltaic cells with only perovskite and no silicon would likely convert more than 29.5% of the Sun’s energy.
- C. solar power will remain elusive until photovoltaic cells are replaced with a more practical technology.
- D. researchers need to evaluate whether other minerals like perovskite are as effective as perovskite seems to be.

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically completes the text’s discussion of the efficiency of various photovoltaic cells. The text indicates that Miranda-Pérez and her team found that adding a layer of perovskite to silicon-based cells increased the percentage of sunlight converted to usable electricity relative to the percentage converted to electricity by silicon-based cells. Using cells with only perovskite and no silicon, however, resulted in no improvement in efficiency over silicon-based cells. The text also states that perovskite captures light in the blue range of the solar spectrum and silicon captures light in the red range of the spectrum. If perovskite and silicon capture different parts of the solar spectrum and the combination of perovskite and silicon results in greater efficiency than the use of either perovskite or silicon alone does, then it is reasonable to infer that the improved efficiency of perovskite-silicon cells could be attributable to their making use of more of the solar spectrum than perovskite-based or silicon-based cells do.

Choice B is incorrect because it is directly contradicted by the text, which says that silicon-based cells convert 18%–22% of the sunlight that reaches them to electricity and that cells with only perovskite are no more efficient than silicon-based cells are. Perovskite-only cells thus must convert 22% or less—not more than 29.5%—of the sunlight that reaches them to electricity. Choice C is incorrect because the text focuses on a possible improvement to photovoltaic cells and makes no mention of any alternatives to those cells, so there is no reason to conclude that photovoltaic cells must be replaced with some other technology. Additionally, the text presents solar power as something that currently exists, so it wouldn’t make sense to conclude that solar power will remain elusive. Choice D is incorrect because nothing in the text suggests that there is a need to evaluate the effectiveness of other minerals than perovskite. In fact, the text gives no indication that it is even possible to use other minerals than silicon and perovskite in photovoltaic cells. Instead, the text is focused on the effect of combining perovskite and silicon and the different parts of the solar spectrum that perovskite and silicon capture.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In 2018, scientists discovered an immense aggregation of *Muusoctopus robustus* (pearl octopuses) along a hydrothermal vent 3,200 meters beneath the ocean’s surface. Water temperatures at this site—named the Octopus Garden—climb as high as 11°C, much warmer than the ambient 1.6°C typical at this depth. Based on observations made over three years, scientists concluded that temperatures at the site likely confer reproductive benefits and that the site is used exclusively for reproduction—6,000 *M. robustus* adults, hatchlings, and eggs were observed at the garden, but no juveniles were present.

Which statement about *M. robustus* and the Octopus Garden is best supported by the text?

Answer

- A. *M. robustus* leave the Octopus Garden upon reaching an intermediary stage of development.
- B. The *M. robustus* population at the Octopus Garden remains stable despite variations in water temperature.
- C. *M. robustus* nests in the Octopus Garden contain on average fewer but larger eggs than nests at similar ocean depths.
- D. The Octopus Garden provides an ideal feeding ground for *M. robustus* hatchlings.

Correct Answer: A

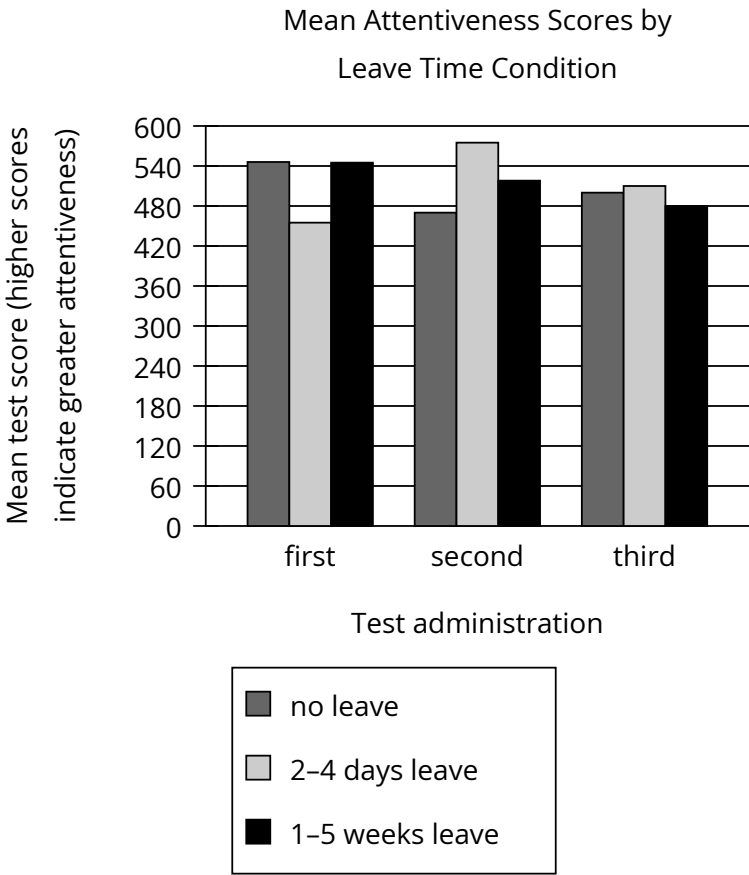
Rationale

Choice A is the best answer because it most logically completes the text’s discussion of the *M. robustus* population at the Octopus Garden. The text states that the scientists concluded that the site is likely used only for reproduction because over three years they saw many adults, freshly hatched octopuses, and eggs but didn’t see any juveniles. This suggests that the *M. robustus* octopuses that hatch at the Octopus Garden leave the site when they reach an intermediary state of development, returning only as adults for reproductive purposes.

Choice B is incorrect because the text never discusses the stability of the *M. robustus* population at the site, only that the scientists observed 6,000 adults, hatchlings, and eggs there. Further, the text presents the site’s temperatures as likely beneficial. Choice C is incorrect because the text doesn’t provide any details about the eggs at the site and makes no mention of nests; it indicates only that eggs are present along with hatchlings and adults. Choice D is incorrect because the text makes no mention of the hatchlings feeding at the Octopus Garden, indicating only that the temperatures at the site are probably beneficial and that the site is likely used for reproduction.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



To investigate potential cognitive benefits of taking leave from work, psychologist Jan Packer and colleagues conducted a six-month study of Australian university staff members who took no leave from work during the study, took 2–4 days of leave, or took 1–5 weeks of leave. Tests of attentiveness were administered to participants three times during the study: at random for the no-leave staff, and for the rest, one week before their leave, one week following their return to work, and one week after the second test administration. After analyzing the results, the researchers concluded that longer leave times might not confer a greater cognitive benefit than shorter leave times do.

Which choice best describes data from the graph that support the researchers’ conclusion?

Answer

- A. In the second test administration, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave, but in the third test administration, those who took no leave had higher average scores than those who took 1–5 weeks of leave.
- B. In the first test administration, participants who took 2–4 days of leave had lower average attentiveness scores than did those who took 1–5 weeks of leave and those who took no leave.
- C. In both the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did participants who took 1–5 weeks of leave.
- D. In the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave.

Correct Answer: C

Rationale

Choice C is the best answer because it describes data from the graph that support Jan Packer and colleagues' conclusion about the effect of leave time on the attentiveness of university employees. According to the text, the researchers' study design included a group of employees who took no leave, a group who took 2–4 days of leave, and a group who took 1–5 weeks of leave. The participants who took leave were tested for attentiveness one week before their leave (the first test administration), one week after their return to work (the second test administration), and two weeks after their return (the third test administration). The participants who took no leave were tested three times at random. The graph shows that at one week after their return to work, participants who took only 2–4 days of leave had an average attentiveness score of between 540 and 600, while participants who took 1–5 weeks of leave had an average score of between 480 and 540. At two weeks after their return to work, those who took only 2–4 days of leave had an average score of between 480 and 540, while those who took 1–5 weeks of leave had an average score of approximately 480. In other words, the graph shows that on both post-leave testing dates, participants with longer leave times had lower average attentiveness scores than those with shorter leave times. Since attentiveness is an indicator of cognitive functioning, these data confirm Packer and colleagues' conclusion that longer leave times might not confer a greater cognitive benefit than shorter leave times do.

Choice A is incorrect. The graph does show that in the second test administration, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave and also shows that in the third test administration, those who took no leave had higher average scores than those who took 1–5 weeks of leave. But neither of these findings has a direct bearing on the researchers' conclusion, which concerns a comparison of participants who took 2–4 days of leave with those who took 1–5 weeks, rather than a comparison of either group with participants who took no leave. Choice B is incorrect. Although the graph does show that in the first test administration, participants who took 2–4 days of leave had lower average attentiveness scores than did those who took 1–5 weeks of leave and those who took no leave, this test administration occurred before any participants went on leave; therefore, these results have no bearing on the researchers' conclusion about how the amount of leave taken by participants affected their cognitive functioning. Choice D is incorrect. Although the graph does show that in the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave, the researchers' conclusion is about the effects of short leave compared with the effects of long leave, not the effects of short leave compared with the effects of no leave. These results are therefore irrelevant to the conclusion.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Average Number and Duration of Torpor Bouts and Arousal Episodes for Alaska Marmots and Arctic Ground Squirrels, 2008–2011

Feature	Alaska marmots	Arctic ground squirrels
torpor bouts	12	10.5
duration per bout	13.81 days	16.77 days
arousal episodes	11	9.5
duration per episode	21.2 hours	14.2 hours

When hibernating, Alaska marmots and Arctic ground squirrels enter a state called torpor, which minimizes the energy their bodies need to function. Often a hibernating animal will temporarily come out of torpor (called an arousal episode) and its metabolic rate will rise, burning more of the precious energy the animal needs to survive the winter. Alaska marmots hibernate in groups and therefore burn less energy keeping warm during these episodes than they would if they were alone. A researcher hypothesized that because Arctic ground squirrels hibernate alone, they would likely exhibit longer bouts of torpor and shorter arousal episodes than Alaska marmots.

Which choice best describes data from the table that support the researcher’s hypothesis?

- Answer
- A. The Alaska marmots’ arousal episodes lasted for days, while the Arctic ground squirrels’ arousal episodes lasted less than a day.
 - B. The Alaska marmots and the Arctic ground squirrels both maintained torpor for several consecutive days per bout, on average.
 - C. The Alaska marmots had shorter torpor bouts and longer arousal episodes than the Arctic ground squirrels did.
 - D. The Alaska marmots had more torpor bouts than arousal episodes, but their arousal episodes were much shorter than their torpor bouts.

Correct Answer: C

Rationale

Choice C is the best answer because it describes data from the table that support the researcher’s hypothesis. According to the text, the researcher hypothesized that Arctic ground squirrels would exhibit longer torpor bouts and shorter arousal episodes than Alaska marmots do—or, put the other way, that the marmots would show shorter torpor bouts and longer arousal episodes than the ground squirrels do. The table shows data about torpor bouts and arousal episodes for the two species from 2008 to 2011. According to the table, the average duration of torpor bouts was 13.81 days for Alaska marmots, shorter than the average of 16.77 days for Arctic ground squirrels, and the average duration of arousal episodes was 21.2 hours for Alaska marmots, longer than the average of 14.2 hours for Arctic ground squirrels. Thus, the table supports the researcher’s hypothesis by showing that Alaska marmots had shorter bouts of torpor and longer arousal episodes than Arctic ground squirrels did.

Choice A is incorrect because it inaccurately describes data from the table and doesn’t support the researcher’s hypothesis. The table shows that the average duration of arousal episodes was less than a day for both Alaska marmots (21.2 hours) and Arctic ground squirrels (14.2 hours). Additionally, information about arousal episodes for Alaska marmots and Arctic ground squirrels isn’t sufficient to support a hypothesis involving comparisons of both arousal episodes and torpor bouts for those animals. Choice B is incorrect because it doesn’t support the

researcher’s hypothesis, which involves comparisons of arousal episodes as well as torpor bouts for Alaska marmots and Arctic ground squirrels. Noting that both animals had torpor bouts lasting several days, on average, doesn’t address arousal episodes at all, nor does it reveal how the animals’ torpor bouts compared. Choice D is incorrect because it doesn’t support the researcher’s hypothesis. Although the table does show that Alaska marmots had more torpor bouts (12) than arousal episodes (11) and that their arousal episodes were much shorter than their torpor bouts (21.2 hours and 13.81 days, respectively), comparing data across only Alaska marmot behaviors isn’t sufficient to support a hypothesis about torpor and arousal behaviors of both Alaska marmots and Arctic ground squirrels.

Question ID: 66fd248a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

“Poetry” is a 1919 poem by Marianne Moore. The poem highlights an ambivalence toward poetry as the speaker acknowledges its merits while also expressing a sense of displeasure, writing _____

Which quotation from “Poetry” most effectively illustrates the claim?

Answer

- A. “nor is it valid / to discriminate against ‘business documents and / school-books’; all these phenomena are important.”
- B. “One must make a distinction / however: when dragged into prominence by half poets, the result is not / poetry”
- C. “when [poems] become so derivative as to become unintelligible, the / same thing may be said for all of us—that we / do not admire what / we cannot understand.”
- D. “Reading [poetry], however, with a perfect contempt for it, one discovers that there is in / it after all, a place for the genuine.”

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses a quotation from "Poetry" to illustrate the claim that the poem highlights an ambivalence, or a conflicted attitude, toward poetry. In the quotation, the speaker suggests that one might read poetry with "contempt," or disdain, for it, but even with this negative attitude one will find "a place for the genuine." Because the quotation expresses conflicting attitudes toward poetry, it effectively illustrates the speaker’s ambivalence in discussing the merits and displeasure of reading poetry.

Choice A is incorrect because it doesn’t mention poetry or show ambivalence. Choice B is incorrect. Although the idea of "half poets" may seem to relate to ambivalence, the speaker mentions only negative attitudes toward certain works and the quotation therefore lacks a contrasting positive or neutral attitude that would be needed to indicate ambivalence. Choice C is incorrect because the speaker mentions only negative attitudes toward certain works and the quotation therefore lacks a contrasting positive or neutral attitude that would be needed to indicate ambivalence.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Hypothesizing that reliance on smartphones for information retrieval will lead a person to incorrectly remember instances of phone-assisted retrieval as instances of retrieval from memory, researchers asked participants general-knowledge trivia questions, instructing them to answer certain questions using only their smartphone and other questions using only their memory. When surveyed a week later, participants exhibited higher levels of misattribution for answers retrieved in the smartphone condition than for those retrieved in the memory condition, a finding that the researchers claimed supported their hypothesis.

Which question would be most useful to answer in determining the validity of the researchers’ claim as it is presented in the text?

Answer

- A. Were participants as likely to remember the source of their information when asked a week later as they would have been if asked immediately after answering the trivia questions?
- B. Were participants less confident about the accuracy of information they were instructed to retrieve from a smartphone than they were of that retrieved from their own memory?
- C. How likely were participants to answer all the trivia questions correctly in both the smartphone and the memory conditions?
- D. How did participants attribute information they were instructed to retrieve from a smartphone in instances when they already knew that information independently of their phone?

Correct Answer: D

Rationale

Choice D is the best answer because it identifies a question that directly addresses a potential confounding factor in the research. The text discusses a study investigating source misattribution for information retrieved with the assistance of a smartphone—that is, how well people remember whether they used a smartphone to look up information or retrieved the information from their memory. The researchers hypothesized that smartphone reliance causes people to mistakenly think that they retrieved the information from their memory when they in fact used a smartphone. To test this hypothesis, the researchers instructed participants to look up answers to some trivia questions on smartphones and to answer some questions using only their memory. The text then indicates that a week later, participants were more likely to misremember smartphone-assisted answers as coming from memory than to misremember memory-based answers as coming from smartphones. The researchers claimed that the higher misattribution rates for the smartphone-retrieval condition supported their hypothesis. However, the text doesn’t specify whether the study design accounted for the possibility that some participants might have known many—perhaps even all—of the trivia answers beforehand. If participants already knew some answers they were instructed to look up on their phones, their later attribution of those answers to memory might be accurate rather than a misattribution. Knowing how participants attributed information in such cases would help determine whether the observed effect reflects smartphone-induced misattribution or simply participants’ recall of prior knowledge. Thus, in evaluating the validity of the researchers’ claim, it would be most useful to know how participants attributed information they were instructed to retrieve from a phone in instances where they already knew the information.

Choice A is incorrect. Although survey timing is a methodological consideration, it doesn’t help distinguish between genuine misattribution caused by smartphone reliance and other explanations for the observed effect. Though it’s plausible that participants’ recollections of their information source may have been more accurate immediately after answering the trivia questions than they would have been a week later, the one-week delay would affect both conditions equally and doesn’t specifically challenge the researchers’ claim about smartphone-induced misattribution. Choice B is incorrect because confidence in the accuracy of retrieved information doesn’t directly pertain to source misattribution. The researchers’ claim concerns whether participants believed information came from their smartphone or their memory, not how confident they were in its correctness. Instead, a useful question would need to address the source-attribution process itself. Choice C is

incorrect because the accuracy of the participants' answers doesn't bear on the researchers' claim about source misattribution. The claim concerns the remembered origin of answers, not their correctness. Instead, the relevant issue is whether participants who looked up answers they already knew might have attributed those answers to memory regardless of which of the two experimental conditions they answered those questions under.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

A student performs an experiment testing her hypothesis that a slightly acidic soil environment is more beneficial for the growth of the plant *Brassica rapa parachinensis* (a vegetable commonly known as choy sum) than a neutral soil environment. She plants sixteen seeds of choy sum in a mixture of equal amounts of coffee grounds (which are highly acidic) and potting soil and another sixteen seeds in potting soil without coffee grounds as the control for the experiment. The two groups of seeds were exposed to the same growing conditions and monitored for three weeks.

Which finding, if true, would most directly weaken the student’s hypothesis?

Answer

- A. The choy sum planted in the soil without coffee grounds were significantly taller at the end of the experiment than the choy sum planted in the mixture of soil and coffee grounds.
- B. The choy sum grown in the soil without coffee grounds weighed significantly less at the end of the experiment than the choy sum grown in the mixture of soil and coffee grounds.
- C. The choy sum seeds planted in the soil without coffee grounds sprouted significantly later in the experiment than did the seeds planted in the mixture of soil and coffee grounds.
- D. Significantly fewer of the choy sum seeds planted in the soil without coffee grounds sprouted plants than did the seeds planted in the mixture of soil and coffee grounds.

Correct Answer: A

Rationale

Choice A is the best answer because it describes an experimental outcome that would most directly weaken the student’s hypothesis. According to the text, the student hypothesizes that *Brassica rapa parachinensis* (choy sum) will benefit more from acidic soil than it will from neutral soil. The text then explains that the student planted 16 choy sum seeds in potting soil with coffee grounds added to increase acidity and another 16 seeds in soil without coffee grounds as a control (a group identical to the experimental group except for the experimental modification being tested). If the hypothesis were correct, the plants in the more acidic soil-and-coffee-grounds mixture would grow faster than those in the control group. However, choice A proposes a scenario in which the plants in soil without coffee grounds were “significantly taller” than those in the more acidic mixture—an outcome that weakens the hypothesis that higher acidity is beneficial to the plants’ growth.

Choice B is incorrect. If the choy sum planted in the neutral soil produced less plant matter and therefore weighed less than the choy sum planted in the acidic soil-and-coffee-grounds mixture, this finding would strengthen the student’s hypothesis, not weaken it. Choice C is incorrect. If seeds planted in neutral soil (without coffee grounds) sprouted significantly later than seeds planted in the acidic soil-and-coffee-grounds mixture, this finding would strengthen, not weaken, the student’s hypothesis that acidic soil benefits choy sum. Choice D is incorrect. If seeds planted in the neutral soil (without coffee grounds) sprouted significantly fewer plants than seeds planted in the acidic soil-and-coffee-grounds mixture did, this finding would strengthen, not weaken, the student’s hypothesis that choy sum benefits from acidic soil.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Initially observed in 2017, the interstellar object ‘Oumuamua is the first object of its kind to be seen in our solar system. Researchers have been puzzled because its acceleration cannot be entirely explained by the gravitational pull of nearby bodies: there must be a nongravitational influence on its velocity and trajectory. Some previously suggested explanations for this nongravitational acceleration involve mechanisms that are unlikely or unrealistic, such as geometric effects from ‘Oumuamua being potentially composed of several spatially separated bodies. Now, Jennifer Bergner and colleagues propose that the nongravitational acceleration is due to the gaseous expulsion of entrapped hydrogen from ‘Oumuamua’s water-rich icy body.

Which statement, if true, would most strongly support the claim made by Bergner and colleagues about the cause of ‘Oumuamua’s acceleration?

Answer

- A. Existing proposed models of outgassing from ‘Oumuamua include the direct conversion of nitrogen or carbon monoxide from a solid to a gaseous state without becoming liquid, but these models have theoretical or observational inconsistencies.
- B. ‘Oumuamua’s trajectory is inconsistent with a nongravitational acceleration that would be caused by the release of hydrogen gas resulting from the processing of water ice (H₂O), but the interstellar object’s observable properties can be explained if it has a significant component of molecular hydrogen ice (H₂).
- C. Since nongravitational accelerations of interstellar objects are several orders of magnitude weaker than gravitational accelerations, deviation from behavior that could be fully attributed to gravitational pull has been detected on a limited number of objects similar to ‘Oumuamua.
- D. Exposure to interstellar cosmic radiation can result in the formation of embedded pockets of hydrogen gas in water ice; moreover, when traveling through the solar system, ‘Oumuamua experienced warming sufficient to alter its icy structure and allow for outgassing.

Correct Answer: D

Rationale

Choice D is the best answer because it presents findings that, if true, would support the claim made by Bergner and colleagues that the nongravitational acceleration of ‘Oumuamua is due to the expulsion of entrapped hydrogen. The text first introduces the observation of a unique interstellar object named ‘Oumuamua and goes on to explain that the object exhibited nongravitational acceleration that could not be fully attributed to the expected cause: gravitational pull of nearby celestial bodies. The text concludes by stating that Bergner and colleagues claim that the nongravitational acceleration is caused by expulsion of hydrogen gas from ‘Oumuamua’s water-rich icy body. To support this claim requires evidence that hydrogen gas could be present within ‘Oumuamua at all, which this answer choice presents: cosmic radiation can result in embedded pockets of hydrogen gas in water ice. Additionally, evidence that this gas can be released from such a body is required to fully support the claim, which this answer choice goes on to provide: ‘Oumuamua experienced sufficient warming as it traveled through the solar system to alter its icy structure and release the hydrogen gas. Thus, this answer choice provides the best evidence to support Bergner and colleagues’ claim.

Choice A is incorrect because this answer choice concerns faults with previous models of outgassing from ‘Oumuamua of carbon monoxide and nitrogen, which would not support a claim regarding hydrogen outgassing. Furthermore, inconsistencies in other models would not provide evidence in support of a different model or explanation. Choice B is incorrect. The evidence presented in this answer choice would weaken the claim proposed by Bergner and colleagues that the nongravitational acceleration of ‘Oumuamua is caused by the expulsion of hydrogen gas because this answer choice suggests that there is evidence that refutes this claim: ‘Oumuamua’s trajectory is inconsistent with a nongravitational acceleration caused by the release of hydrogen gas. Furthermore, the remaining portion of this answer choice is unrelated to the claim. Choice C is incorrect because the claim being made by Bergner and colleagues concerns the expulsion of entrapped hydrogen gas, but

this answer choice is concerned solely with the differences in magnitude of gravitational and nongravitational acceleration, which would not support Bergner's claim. Furthermore, this answer choice discusses interstellar objects similar to 'Oumuamua, but the text states that 'Oumuamua is the first observed object of its kind in our solar system, so evidence from other, similar bodies would not be available.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Volunteering, or giving time for a community service for free, is a valuable form of civic engagement because helping in a community is also good for society as a whole. In a survey of youths in the United States, most young people said that they believe volunteering is a way to help people on an individual level. Meanwhile, only 6% of the youths said that they think volunteering is a way to help fix problems in society overall. These replies suggest that _____

Which choice most logically completes the text?

Answer

- A. many young people think they can volunteer only within their own communities.
- B. volunteering may be even more helpful than many young people think it is.
- C. volunteering can help society overall more than it can help individual people.
- D. many young people may not know how to find ways to volunteer their time.

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of volunteering. The text asserts that volunteering benefits both the community in which one volunteers and society as a whole. It then states that in a survey of young people, a majority of respondents said that volunteering helps individuals, but only 6% of respondents said that volunteering helps society as a whole. If volunteering does in fact help society as a whole, as the text says, but only 6% of young people believe it does, then it’s reasonable to conclude that volunteering is more helpful than many young people think it is.

Choice A is incorrect because the text discusses young people’s beliefs about the benefits of volunteering, not where young people believe they are able to volunteer. Nothing in the text suggests that many young people believe they are only able to volunteer in their own communities.

Choice C is incorrect. Although the text indicates that volunteering is beneficial for society as a whole, nothing in the text suggests that volunteering can benefit society more than it can benefit individual people. The text doesn’t compare the benefits to society with the benefits to individuals. Choice D is incorrect because the text discusses young people’s beliefs about the benefits of volunteering, not how to find volunteering opportunities. There’s nothing in the text to suggest that many young people don’t know how to volunteer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Global Strontium
Seawater Curve

⁸⁷ Sr/ ⁸⁶ Sr	Age (Ma)
0.708980	6.20
0.709000	5.86
0.709020	5.40
0.709040	4.75
0.709060	3.00

The late Hemphillian (Hh) North American Land Mammal Age includes the subdivisions Hh3, 6.8 million years ago (Ma) to 6 Ma, and Hh4, 6 Ma to 4.75 Ma. While mammalian fossils have indicated that Florida’s Montbrook Fossil Site (MFS) and Palmetto Fauna of the Bone Valley Region (PFBV) date to Hh4, a more precise determination of the sites’ ages has proved challenging. Stephanie R. Killingsworth et al. compared average ratios of strontium-87 to strontium-86 (⁸⁷Sr/⁸⁶Sr) in fossil shark teeth from MFS and PFBV—0.709000 and 0.709028, respectively—to ⁸⁷Sr/⁸⁶Sr ratios in the global strontium seawater curve, a record that shows how ⁸⁷Sr/⁸⁶Sr ratios in seawater correspond to numerical ages and that is used to date fossils and, by extension, fossil sites. The researchers concluded that _____

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. mammalian fossil evidence offers less dating precision than do ⁸⁷Sr/⁸⁶Sr ratios in fossil shark teeth and that PFBV likely was deposited closer to the Hh3-Hh4 boundary than was MFS.
- B. the average ⁸⁷Sr/⁸⁶Sr ratios in the fossil shark teeth from MFS and PFBV only partially support the site age estimates previously established through mammalian fossil evidence.
- C. the average ⁸⁷Sr/⁸⁶Sr ratios in the fossil shark teeth from MFS and PFBV resolve previous uncertainty about the sites’ relative ages by indicating that both sites were deposited contemporaneously during the late Hh.
- D. the average ⁸⁷Sr/⁸⁶Sr ratios in the fossil shark teeth from MFS and PFBV corroborate that both MFS and PFBV fall within Hh4 but suggest that PFBV was likely deposited more recently than MFS.

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses data from the table to complete the statement about the researchers’ conclusion about fossil shark teeth. The text states that mammalian fossil evidence has placed both MFS and PFBV within Hh4 (6 Ma to 4.75 Ma) but “a more precise determination of the sites’ ages has proved challenging,” prompting Killingsworth and colleagues to attempt to date the sites by comparing average ⁸⁷Sr/⁸⁶Sr ratios for fossil shark teeth from MFS and PFBV to ratios in the global strontium seawater curve. Given that “⁸⁷Sr/⁸⁶Sr ratios in seawater correspond to numerical ages,” a fossil whose ⁸⁷Sr/⁸⁶Sr ratio corresponds to the ratio associated with a particular age in the global strontium seawater curve can be inferred to date to that age. According to the table, the ⁸⁷Sr/⁸⁶Sr ratio of 0.709000 for fossil shark

teeth from MFS corresponds to an age of 5.86 Ma, which falls within Hh4. By contrast, the table shows that the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.709028 for fossils from PFBV would fall between 0.709020 (5.40 Ma) and 0.709040 (4.75 Ma), indicating an age that falls within Hh4 but is more recent than MFS's age of 5.86 Ma. The data therefore corroborate the conclusion that both sites fall within Hh4 while suggesting that PFBV was likely deposited more recently than MFS was.

Choice A is incorrect because the table indicates that MFS was deposited closer to the Hh3-Hh4 boundary than PFBV was, not the reverse. The table indicates that PFBV's $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.709028 corresponds to an age between 4.75 Ma and 5.40 Ma, while MFS's $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.709000 corresponds to an age of 5.86 Ma. Because the Hh3-Hh4 boundary occurs at 6 Ma, MFS was deposited closer to that boundary than PFBV was. In addition, the text doesn't establish that mammalian fossil evidence offers less dating precision than $^{87}\text{Sr}/^{86}\text{Sr}$ ratios do; it states only that mammalian fossils have placed both MFS and PFBV in Hh4 and that more precise dating has proved challenging. Choice B is incorrect because it mischaracterizes the level of support for the site age estimates established through mammalian fossil evidence. The table indicates that the average $^{87}\text{Sr}/^{86}\text{Sr}$ ratios from MFS (0.709000) and PFBV (0.709028) both correspond to ages that fall within Hh4 (6 Ma to 4.75 Ma)—5.86 Ma for MFS and an age between 5.40 Ma and 4.75 Ma for PFBV. The $^{87}\text{Sr}/^{86}\text{Sr}$ ratios therefore fully support, rather than only partially support, the previous mammalian fossil evidence placing both sites within Hh4. Choice C is incorrect because the table indicates that the average $^{87}\text{Sr}/^{86}\text{Sr}$ ratios in fossil shark teeth from MFS (0.709000) and PFBV (0.709028) correspond to different ages—5.86 Ma for MFS and an age between 5.40 Ma and 4.75 Ma for PFBV. Because the ratios point to different ages, they don't support the conclusion that the fossils at the two sites were deposited contemporaneously, or at the same time.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

A student in a political science course is writing a paper on Aristotle’s *The Politics*, in which Aristotle offers his opinion on political instability and gives advice on how constitutions can be preserved. Aristotle observes that different forms of government can fall in different ways—for example, oligarchies might grant power to military leaders during wartime who refuse to relinquish that power during peacetime—but some methods of preserving order apply across all forms of government. The student claims that in particular Aristotle asserts that in a healthy state obedience to law must be as close to absolute as possible and that even minor infractions should not be ignored.

Which quotation from a philosopher’s analysis of *The Politics* would best support the student’s claim?

Answer

- A. “When constructing his argument regarding the characteristics of a well-functioning government, Aristotle asserts that ‘Transgression creeps in unperceived and at last ruins the state,’ illustrating this idea with a comparison to frequent small expenditures slowly and almost imperceptibly chipping away at a fortune until it is ultimately depleted.”
- B. “When Aristotle writes on the necessity of avoiding corruption in government, he proposes that ‘every state should be so administered and so regulated by law that its magistrates cannot possibly make money.’ In particular, he thinks oligarchies are particularly susceptible to corruption through bribery.”
- C. “When Aristotle considers the health of constitutions, he states that ‘Constitutions are preserved when their destroyers are at a distance, and sometimes also because they are near, for the fear of them makes the government keep in hand the constitution.’ He holds that rulers who wish to see constitutions preserved must continually remind the populace of the dangers that would result from a constitutional collapse.”
- D. “When contrasting different forms of government, Aristotle holds that ‘oligarchies may last, not from any inherent stability in such forms of government, but because the rulers are on good terms both with the unenfranchised and with the governing classes.’ That is, oligarchic leaders who wish to hold on to power will introduce members of disenfranchised classes into government in a participatory role.”

Correct Answer: A

Rationale

Choice A is the best answer because it presents the quotation that best supports the student’s claim that in *The Politics*, Aristotle gives advice on preserving constitutions—preventing governments from falling and maintaining order—and specifically asserts that in a healthy state, laws must be followed as strictly as possible and infractions should not be overlooked even if they are minor. The philosopher states that when Aristotle builds his argument about the characteristics of a well-functioning government, Aristotle asserts that transgression, or violation of law, will ruin the state if it “creeps in unperceived,” or goes unnoticed. The philosopher then adds that Aristotle illustrates this point by comparing the situation to one in which small but frequent expenses diminish a fortune almost unnoticeably until, eventually, the fortune is entirely gone. In other words, the philosopher indicates that Aristotle makes the point that total obedience to law preserves a healthy state while even small violations, if ignored, will undermine the health of the state.

Choice B is incorrect because the philosopher addresses Aristotle’s observation about corruption within the government (in particular, preventing the possibility that members of the government can take bribes), and although corruption can involve infractions, the observation is about a subset of people within the state and isn’t directly connected to the importance of upholding total obedience to the law throughout the state. Choice C is incorrect because the philosopher discusses Aristotle’s point about those who would intentionally destroy a constitution altogether and the need for rulers to remind the populace that it would be dangerous for a constitution to collapse, but neither idea is directly connected to the importance of upholding total obedience to the law. Choice D is incorrect because the philosopher explains that Aristotle makes the point that oligarchic leaders may retain power by having members of disenfranchised classes participate in government alongside governing classes, and this point doesn’t address the importance of ensuring obedience to the law and addressing even minor violations.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh’s interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs

Which choice most logically completes the text?

Answer

- A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.
- B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs’ location.
- C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.
- D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

Correct Answer: B

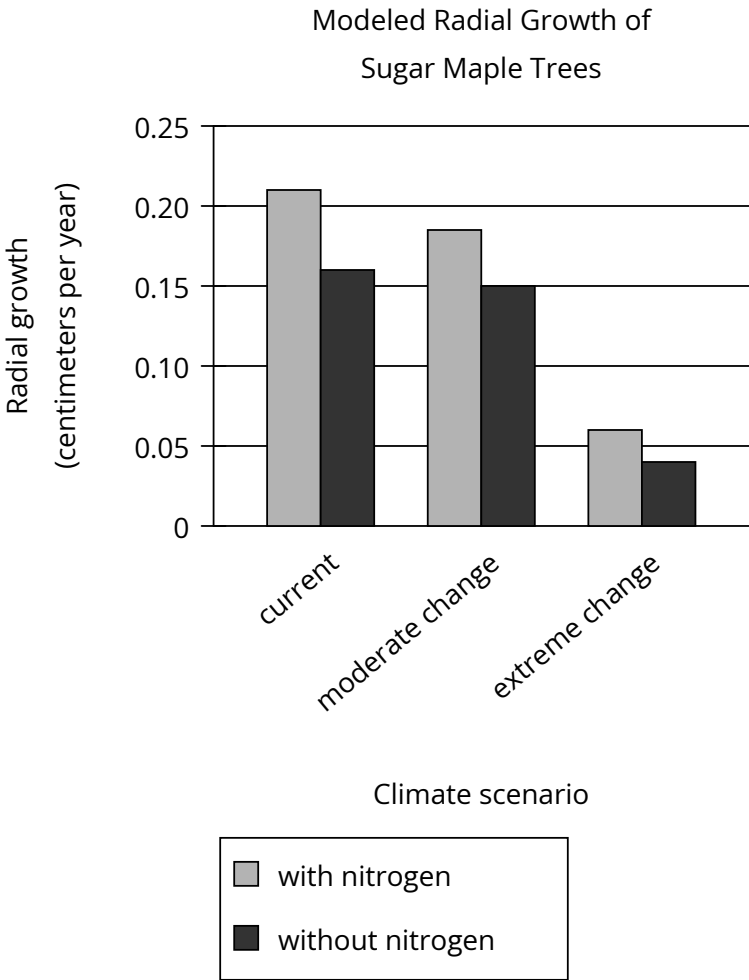
Rationale

Choice B is the best answer. The text says that crab burrowing in the pannes enables plants to grow there again. It also says that crab burrowing at the edges of the marsh speeds up marsh loss. This suggests that burrowing crabs can either help or hurt marshes, depending on where they’re located.

Choice A is incorrect. This inference isn’t supported. The text never discusses pannes of different sizes. Choice C is incorrect. This inference isn’t supported. The text never suggests that crabs do more burrowing in the pannes (the areas with standing water) than they do at the edges. Rather, the text says that the burrowing that happens in the pannes is beneficial, while the burrowing that happens at the edges is harmful. Choice D is incorrect. This inference isn’t supported. The text never discusses the intensity of crab burrowing, nor does it discuss the size of the panne relative to the size of the marsh.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Inés Ibáñez and colleagues studied a forest site in which some sugar maple trees receive periodic fertilization with nitrogen to mimic the broader trend of increasing anthropogenic nitrogen deposition in soil. Ibáñez and colleagues modeled the radial growth of the trees with and without nitrogen fertilization under three different climate scenarios (the current climate, moderate change, and extreme change). Although they found that climate change would negatively affect growth, they concluded that anthropogenic nitrogen deposition could more than offset that effect provided that change is moderate rather than extreme.

Which choice best describes data from the graph that support Ibáñez and colleagues’ conclusion?

Answer

- A. Growth with nitrogen under the current climate exceeded growth with nitrogen under moderate change, but the latter exceeded growth without nitrogen under extreme change.
- B. Growth without nitrogen under the current climate exceeded growth without nitrogen under moderate change, but the latter exceeded growth with nitrogen under extreme change.
- C. Growth with nitrogen under moderate change exceeded growth without nitrogen under moderate change, but the latter exceeded growth without nitrogen under extreme change.

D. Growth with nitrogen under moderate change exceeded growth without nitrogen under the current climate, but the latter exceeded growth with nitrogen under extreme change.

Correct Answer: D

Rationale

Choice D is the best answer because it describes data from the graph that support Ibáñez and colleagues' conclusion that increasing anthropogenic nitrogen deposition can compensate for the negative effect of climate change on tree growth if that change is moderate but not if it's extreme. The bar graph shows the growth of sugar maple trees with and without nitrogen fertilization under three different climate-change scenarios: current conditions, a moderate change, and an extreme change. According to the graph, radial growth without nitrogen fertilization is projected to be about 0.16 centimeters (cm) under current conditions, 0.15 cm under a moderate change, and 0.04 cm under an extreme change. The graph also shows that with nitrogen fertilization, growth is projected to be about 0.18 centimeters under a moderate change but only about 0.06 centimeters under an extreme change. Thus, the data in the graph support the researchers' conclusion by showing greater growth for a moderate change using nitrogen fertilization than they do either under current conditions without nitrogen fertilization or under an extreme change with nitrogen fertilization.

Choice A is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of anthropogenic nitrogen, but only if change to the climate is moderate and not if it's extreme. To support this would require comparing radial growth without nitrogen fertilization under current climate conditions to the growth with nitrogen fertilization under both moderate and extreme changes. This choice mentions only growth with nitrogen fertilization under current climate conditions and moderate change and growth without nitrogen fertilization under an extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change.

Choice B is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of atmospheric nitrogen, but only if change to the climate is moderate and not if it's extreme. The support needed would compare radial growth under current climate conditions without nitrogen fertilization to the growth with nitrogen fertilization under moderate and extreme changes. This choice mentions only growth without nitrogen fertilization under current conditions and moderate change and growth with nitrogen fertilization under extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change. Choice C is incorrect.

Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of atmospheric nitrogen, but only if change to the climate is moderate and not if it's extreme. The support needed would compare radial growth without adding nitrogen under current climate conditions to the growth with nitrogen fertilization under moderate and extreme changes. This choice mentions only the growth with and without nitrogen fertilization under moderate climate change and growth without nitrogen fertilization under extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change.

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Juvenile Plants Found Growing on Bare Ground and in Patches of Vegetation for Five Species

Species	Bare ground	Patches of vegetation	Total	Percent found in patches of vegetation
<i>T. moroderi</i>	9	13	22	59.1%
<i>T. libanitis</i>	83	120	203	59.1%
<i>H. syriacim</i>	95	106	201	52.7%
<i>H. squamatum</i>	218	321	539	59.6%
<i>H. stoechas</i>	11	12	23	52.2%

Alicia Montesinos-Navarro, Isabelle Storer, and Rocío Perez-Barrales recently examined several plots within a diverse plant community in southeast Spain. The researchers calculated that if individual plants were randomly distributed on this particular landscape, only about 15% would be with other plants in patches of vegetation. They counted the number of juvenile plants of five species growing in patches of vegetation and the number growing alone on bare ground and compared those numbers to what would be expected if the plants were randomly distributed. Based on these results, they claim that plants of these species that grow in close proximity to other plants gain an advantage at an early developmental stage.

Which choice best describes data from the table that support the researchers’ claim?

Answer

- A. For all five species, less than 75% of juvenile plants were growing in patches of vegetation.
- B. The species with the greatest number of juvenile plants growing in patches of vegetation was *H. stoechas*.
- C. For *T. libanitis* and *T. moroderi*, the percentage of juvenile plants growing in patches of vegetation was less than what would be expected if plants were randomly distributed.
- D. For each species, the percentage of juvenile plants growing in patches of vegetation was substantially higher than what would be expected if plants were randomly distributed.

Correct Answer: D

Rationale

Choice D is the best answer because it provides the most direct support from the table for the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. The table shows the total number of juvenile plants from five species that were found growing on bare ground and in patches of vegetation as well as the percentage of the total number of each species that were growing in patches of vegetation. For each of the five species, more than 50% of the juvenile plants were growing in patches of vegetation. The text notes, however, that a random distribution of plants across the landscape should result in only about 15% of the plants being found in patches of vegetation. In other words, for each of the five species, the percentage of juvenile plants found growing in patches of vegetation was substantially higher than could be explained by chance alone. This finding supports the claim in the text: if plants growing in patches are overrepresented among plants that have survived to the juvenile stage, as the data show they are, then it suggests that it’s advantageous for plants at an early stage of development to grow in patches of vegetation.

Choice A is incorrect because the statement that less than 75% of juvenile plants were found growing in patches of vegetation, while true, doesn't clearly support the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. Saying that less than 75% of plants were found in patches doesn't indicate how the percentage growing in patches compares with the percentage that would be expected to grow in patches on the basis of chance alone, which is the information necessary to evaluate whether the claim in the text has support in the table. Put another way, if the percentage of plants found growing in patches was 15% or less, it would be true that less than 75% were found in patches, but the data would in fact weaken the claim in the text, not strengthen it, since the data would show that growing in patches wasn't advantageous. Choice B is incorrect because only 12 plants of this species were found growing in patches, which was the lowest number of any species, not the greatest number. Additionally, even if it were true that this species had the greatest number of plants growing in patches, the finding would be irrelevant to the claim that plants of all five species gained an advantage by growing in close proximity to other plants. Choice C is incorrect because 59.1% of the plants of these species were found growing in patches, which is a far greater percentage, not a lower percentage, than what would be expected if plants were randomly distributed (around 15%). Additionally, if it were true that the percentage of plants growing in patches was lower for these species than what would be expected from chance alone, that finding would weaken, not strengthen, the claim that growing in patches is advantageous.